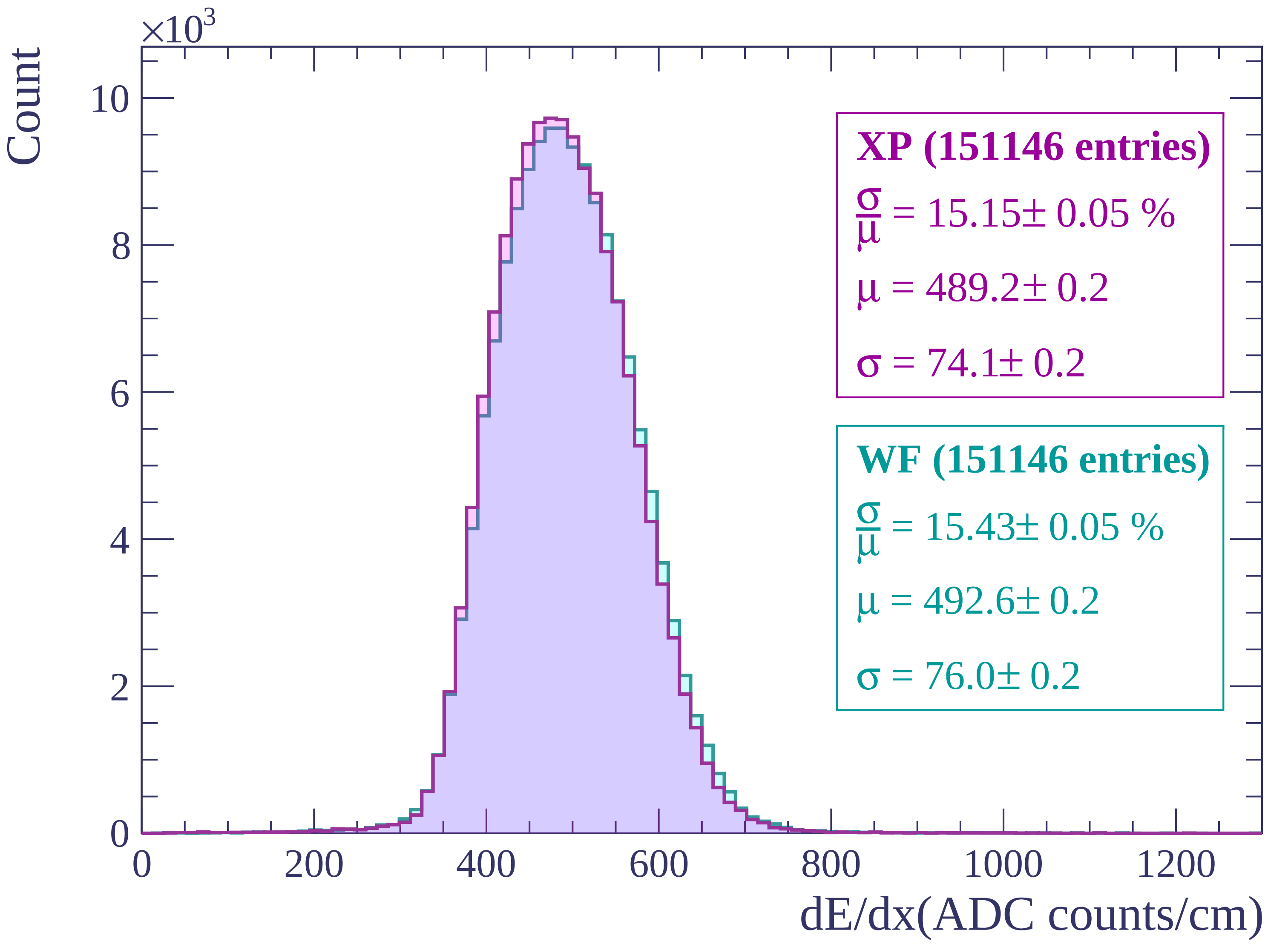
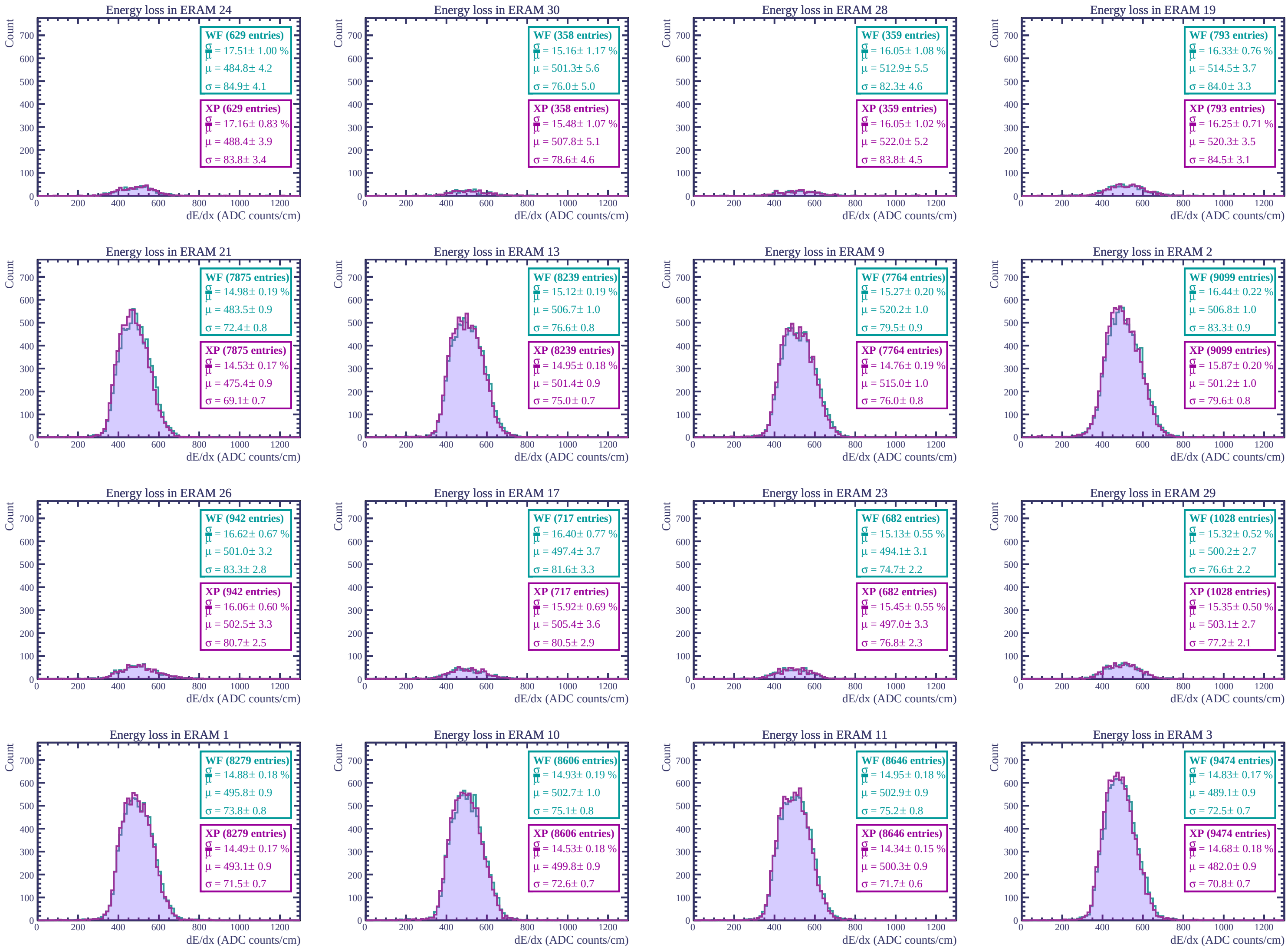
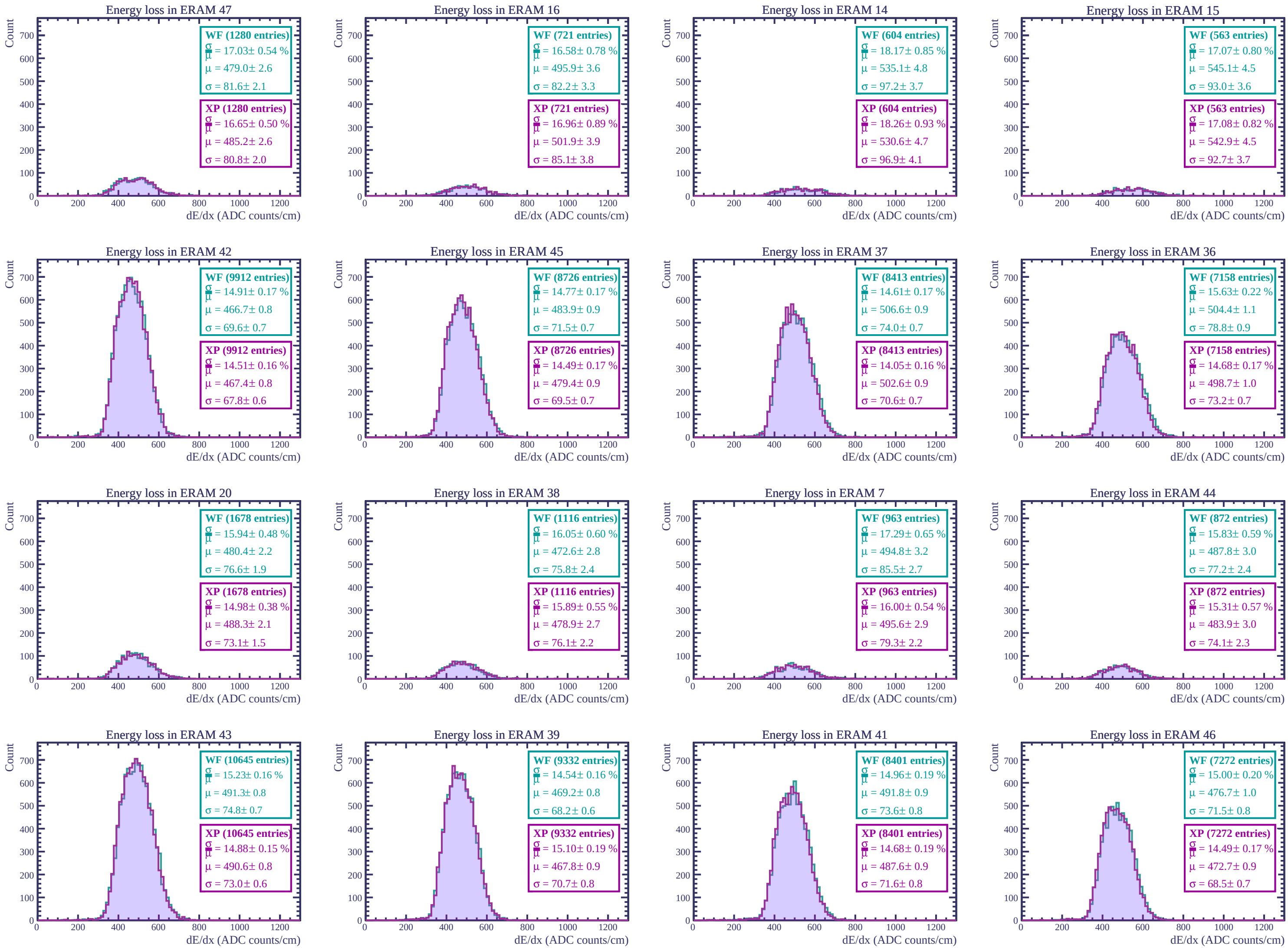
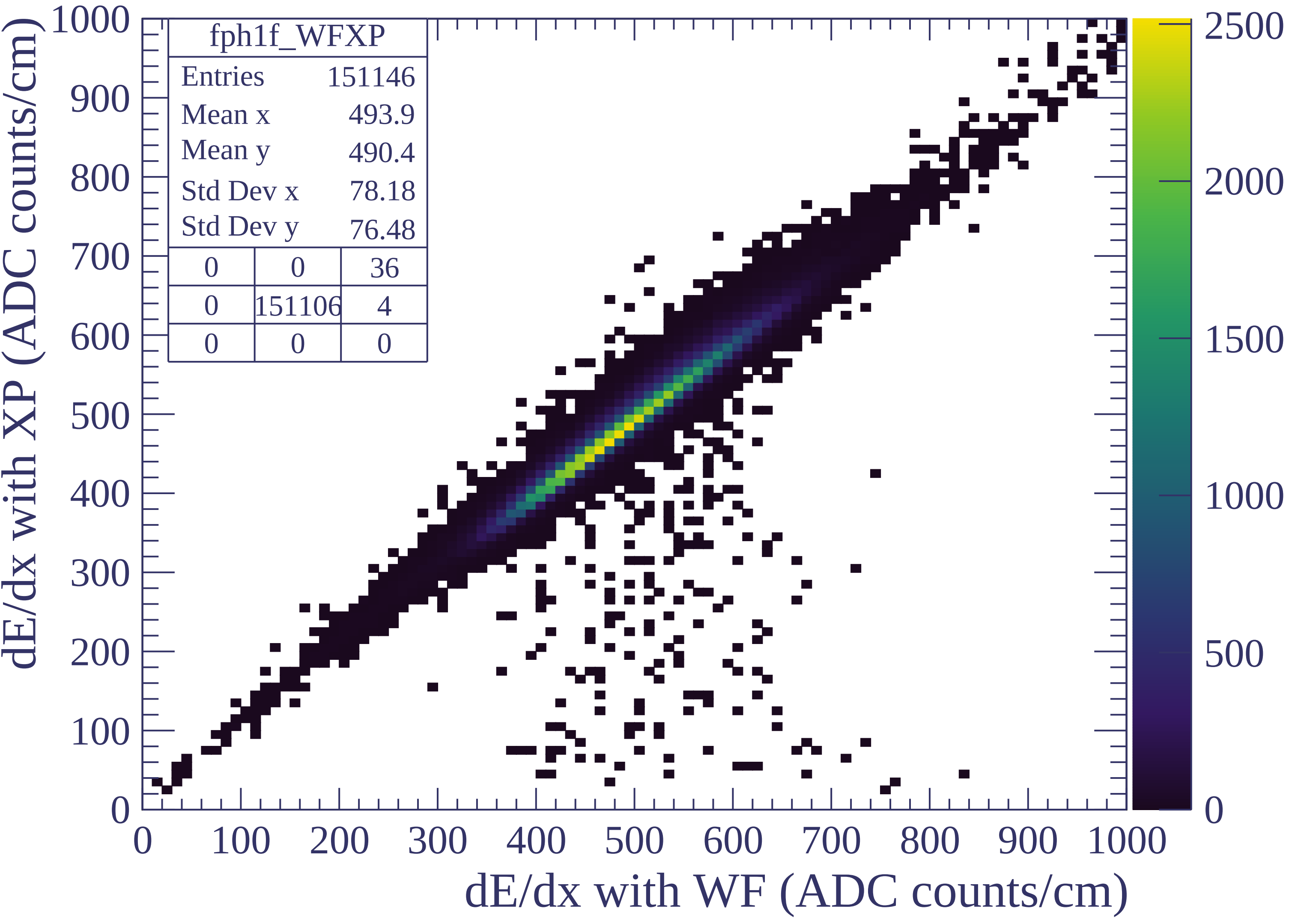


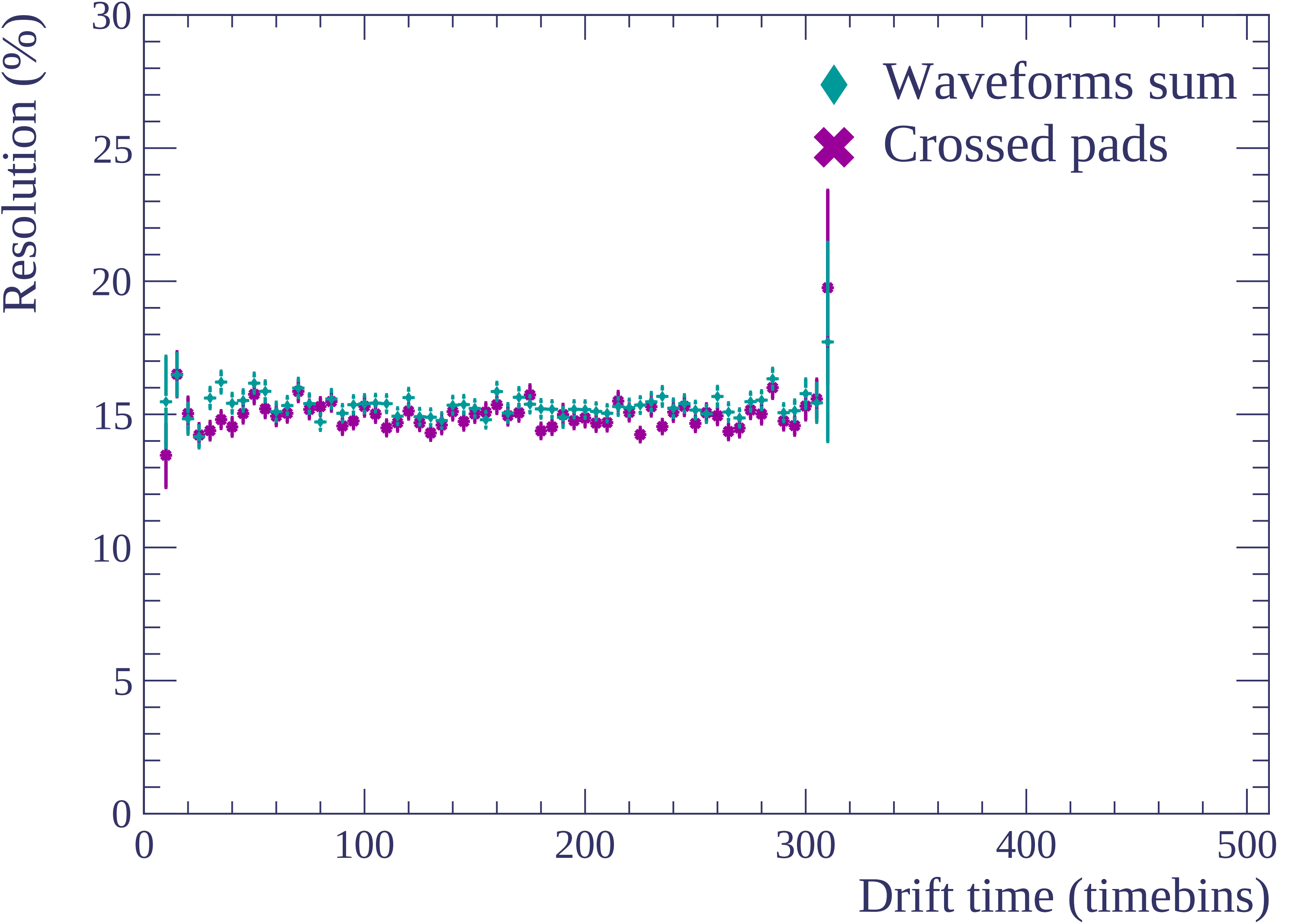
$\times 10^3$
Count
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (151146 entries)
$\frac{\sigma}{\mu}$ = 15.15± 0.05 %
μ = 489.2± 0.2
σ = 74.1± 0.2
WF (151146 entries)
$\frac{\sigma}{\mu}$ = 15.43± 0.05 %
μ = 492.6± 0.2
σ = 76.0± 0.2

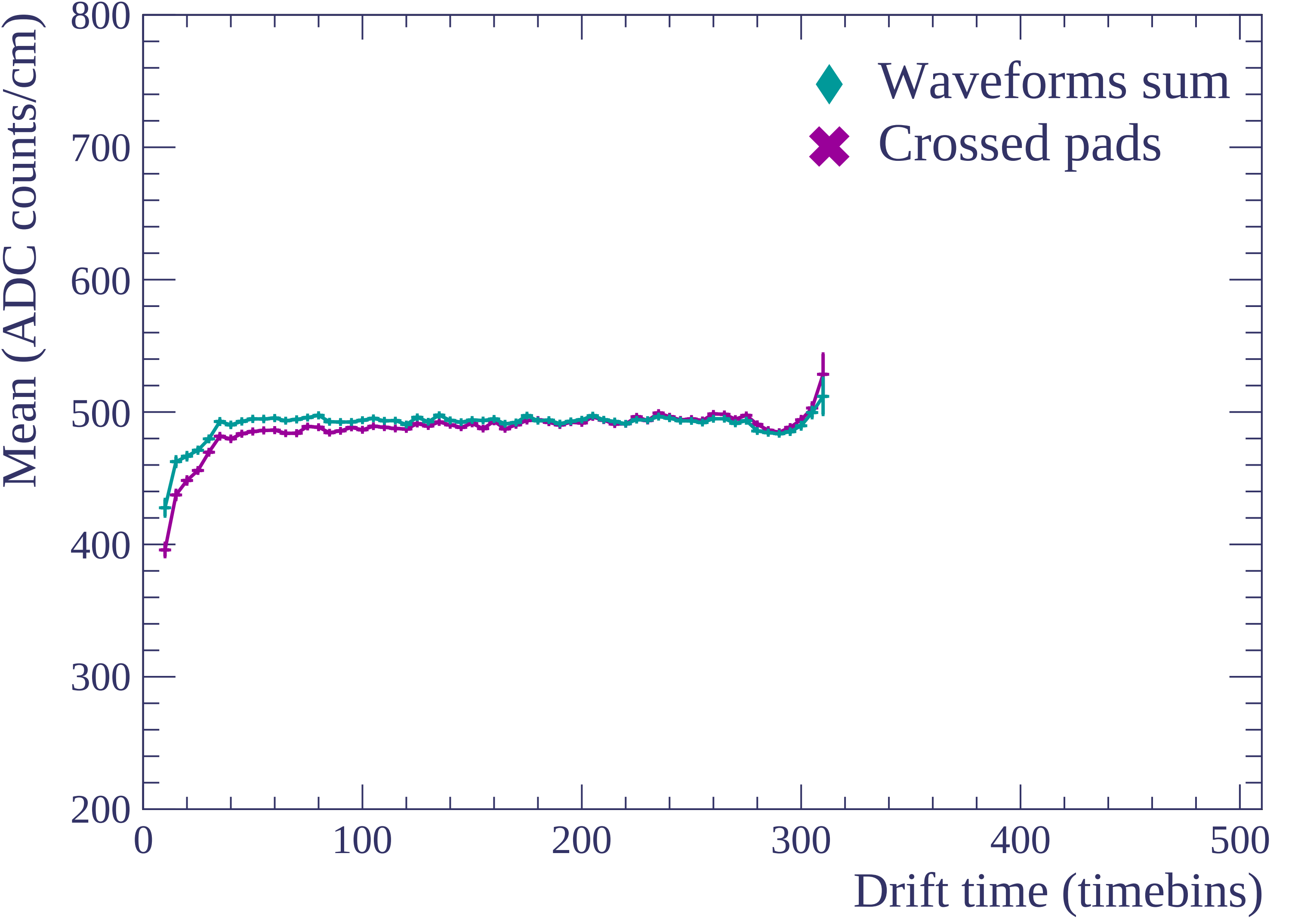


Energy loss in ERAM 24
WF (629 entries)
σ/μ = 17.51± 1.00 %
μ = 484.8± 4.2
σ = 84.9± 4.1
XP (629 entries)
σ/μ = 17.16± 0.83 %
μ = 488.4± 3.9
σ = 83.8± 3.4
Energy loss in ERAM 30
WF (358 entries)
σ/μ = 15.16± 1.17 %
μ = 501.3± 5.6
σ = 76.0± 5.0
XP (358 entries)
σ/μ = 15.48± 1.07 %
μ = 507.8± 5.1
σ = 78.6± 4.6
Energy loss in ERAM 28
WF (359 entries)
σ/μ = 16.05± 1.08 %
μ = 512.9± 5.5
σ = 82.3± 4.6
XP (359 entries)
σ/μ = 16.05± 1.02 %
μ = 522.0± 5.2
σ = 83.8± 4.5
Energy loss in ERAM 19
WF (793 entries)
σ/μ = 16.33± 0.76 %
μ = 514.5± 3.7
σ = 84.0± 3.3
XP (793 entries)
σ/μ = 16.25± 0.71 %
μ = 520.3± 3.5
σ = 84.5± 3.1
Energy loss in ERAM 21
WF (7875 entries)
σ/μ = 14.98± 0.19 %
μ = 483.5± 0.9
σ = 72.4± 0.8
XP (7875 entries)
σ/μ = 14.53± 0.17 %
μ = 475.4± 0.9
σ = 69.1± 0.7
Energy loss in ERAM 13
WF (8239 entries)
σ/μ = 15.12± 0.19 %
μ = 506.7± 1.0
σ = 76.6± 0.8
XP (8239 entries)
σ/μ = 14.95± 0.18 %
μ = 501.4± 0.9
σ = 75.0± 0.7
Energy loss in ERAM 9
WF (7764 entries)
σ/μ = 15.27± 0.20 %
μ = 520.2± 1.0
σ = 79.5± 0.9
XP (7764 entries)
σ/μ = 14.76± 0.19 %
μ = 515.0± 1.0
σ = 76.0± 0.8
Energy loss in ERAM 2
WF (9099 entries)
σ/μ = 16.44± 0.22 %
μ = 506.8± 1.0
σ = 83.3± 0.9
XP (9099 entries)
σ/μ = 15.87± 0.20 %
μ = 501.2± 1.0
σ = 79.6± 0.8
Energy loss in ERAM 26
WF (942 entries)
σ/μ = 16.62± 0.67 %
μ = 501.0± 3.2
σ = 83.3± 2.8
XP (942 entries)
σ/μ = 16.06± 0.60 %
μ = 502.5± 3.3
σ = 80.7± 2.5
Energy loss in ERAM 17
WF (717 entries)
σ/μ = 16.40± 0.77 %
μ = 497.4± 3.7
σ = 81.6± 3.3
XP (717 entries)
σ/μ = 15.92± 0.69 %
μ = 505.4± 3.6
σ = 80.5± 2.9
Energy loss in ERAM 23
WF (682 entries)
σ/μ = 15.13± 0.55 %
μ = 494.1± 3.1
σ = 74.7± 2.2
XP (682 entries)
σ/μ = 15.45± 0.55 %
μ = 497.0± 3.3
σ = 76.8± 2.3
Energy loss in ERAM 29
WF (1028 entries)
σ/μ = 15.32± 0.52 %
μ = 500.2± 2.7
σ = 76.6± 2.2
XP (1028 entries)
σ/μ = 15.35± 0.50 %
μ = 503.1± 2.7
σ = 77.2± 2.1
Energy loss in ERAM 1
WF (8279 entries)
σ/μ = 14.88± 0.18 %
μ = 495.8± 0.9
σ = 73.8± 0.8
XP (8279 entries)
σ/μ = 14.49± 0.17 %
μ = 493.1± 0.9
σ = 71.5± 0.7
Energy loss in ERAM 10
WF (8606 entries)
σ/μ = 14.93± 0.19 %
μ = 502.7± 1.0
σ = 75.1± 0.8
XP (8606 entries)
σ/μ = 14.53± 0.18 %
μ = 499.8± 0.9
σ = 72.6± 0.7
Energy loss in ERAM 11
WF (8646 entries)
σ/μ = 14.95± 0.18 %
μ = 502.9± 0.9
σ = 75.2± 0.8
XP (8646 entries)
σ/μ = 14.34± 0.15 %
μ = 500.3± 0.9
σ = 71.7± 0.6
Energy loss in ERAM 3
WF (9474 entries)
σ/μ = 14.83± 0.17 %
μ = 489.1± 0.9
σ = 72.5± 0.7
XP (9474 entries)
σ/μ = 14.68± 0.18 %
μ = 482.0± 0.9
σ = 70.8± 0.7
Count
dE/dx (ADC counts/cm)

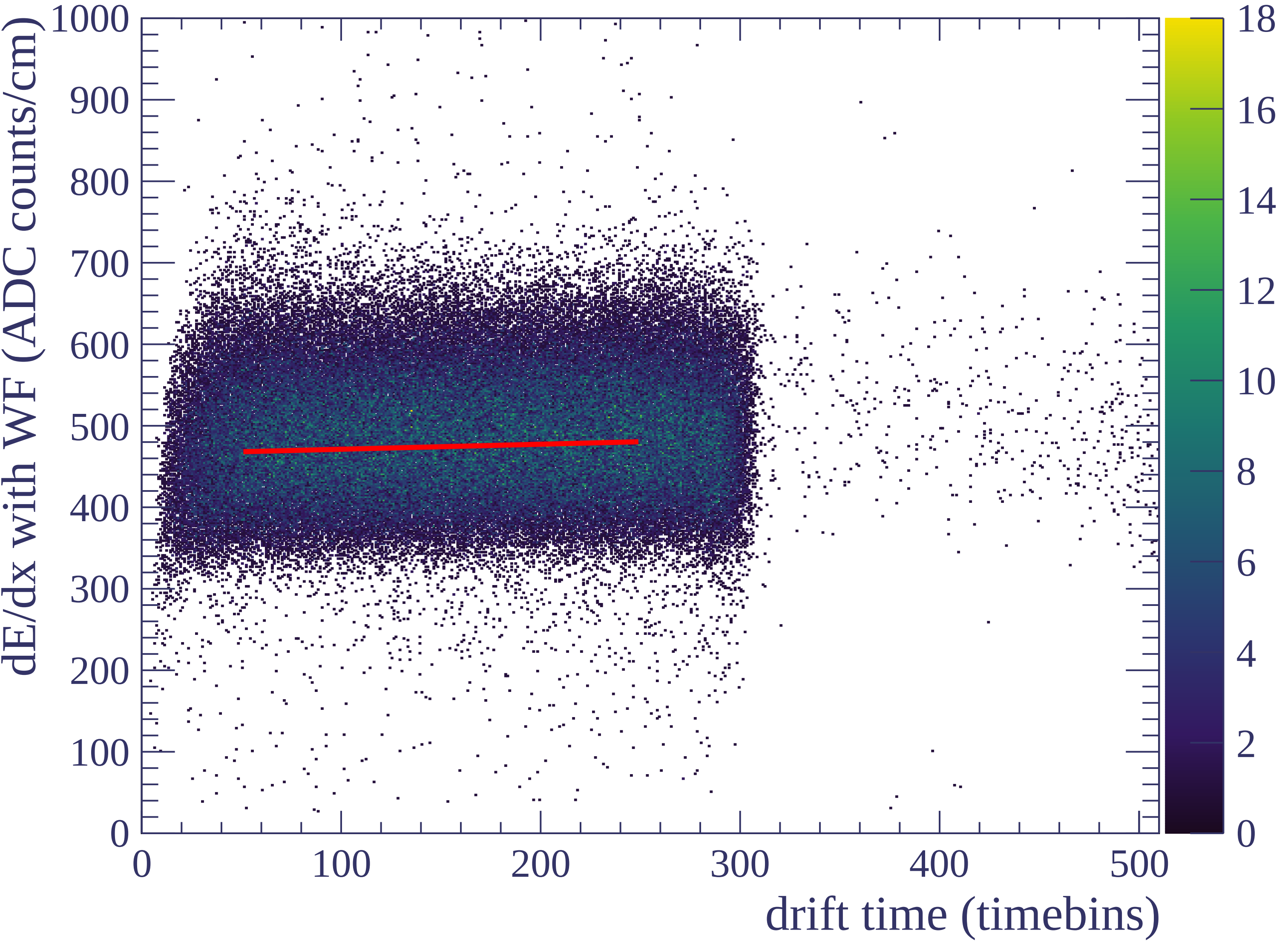

Energy loss in ERAM 47
WF (1280 entries)
σ/μ = 17.03± 0.54 %
μ = 479.0± 2.6
σ = 81.6± 2.1
XP (1280 entries)
σ/μ = 16.65± 0.50 %
μ = 485.2± 2.6
σ = 80.8± 2.0
Energy loss in ERAM 16
WF (721 entries)
σ/μ = 16.58± 0.78 %
μ = 495.9± 3.6
σ = 82.2± 3.3
XP (721 entries)
σ/μ = 16.96± 0.89 %
μ = 501.9± 3.9
σ = 85.1± 3.8
Energy loss in ERAM 14
WF (604 entries)
σ/μ = 18.17± 0.85 %
μ = 535.1± 4.8
σ = 97.2± 3.7
XP (604 entries)
σ/μ = 18.26± 0.93 %
μ = 530.6± 4.7
σ = 96.9± 4.1
Energy loss in ERAM 15
WF (563 entries)
σ/μ = 17.07± 0.80 %
μ = 545.1± 4.5
σ = 93.0± 3.6
XP (563 entries)
σ/μ = 17.08± 0.82 %
μ = 542.9± 4.5
σ = 92.7± 3.7
Energy loss in ERAM 42
WF (9912 entries)
σ/μ = 14.91± 0.17 %
μ = 466.7± 0.8
σ = 69.6± 0.7
XP (9912 entries)
σ/μ = 14.51± 0.16 %
μ = 467.4± 0.8
σ = 67.8± 0.6
Energy loss in ERAM 45
WF (8726 entries)
σ/μ = 14.77± 0.17 %
μ = 483.9± 0.9
σ = 71.5± 0.7
XP (8726 entries)
σ/μ = 14.49± 0.17 %
μ = 479.4± 0.9
σ = 69.5± 0.7
Energy loss in ERAM 37
WF (8413 entries)
σ/μ = 14.61± 0.17 %
μ = 506.6± 0.9
σ = 74.0± 0.7
XP (8413 entries)
σ/μ = 14.05± 0.16 %
μ = 502.6± 0.9
σ = 70.6± 0.7
Energy loss in ERAM 36
WF (7158 entries)
σ/μ = 15.63± 0.22 %
μ = 504.4± 1.1
σ = 78.8± 0.9
XP (7158 entries)
σ/μ = 14.68± 0.17 %
μ = 498.7± 1.0
σ = 73.2± 0.7
Energy loss in ERAM 20
WF (1678 entries)
σ/μ = 15.94± 0.48 %
μ = 480.4± 2.2
σ = 76.6± 1.9
XP (1678 entries)
σ/μ = 14.98± 0.38 %
μ = 488.3± 2.1
σ = 73.1± 1.5
Energy loss in ERAM 38
WF (1116 entries)
σ/μ = 16.05± 0.60 %
μ = 472.6± 2.8
σ = 75.8± 2.4
XP (1116 entries)
σ/μ = 15.89± 0.55 %
μ = 478.9± 2.7
σ = 76.1± 2.2
Energy loss in ERAM 7
WF (963 entries)
σ/μ = 17.29± 0.65 %
μ = 494.8± 3.2
σ = 85.5± 2.7
XP (963 entries)
σ/μ = 16.00± 0.54 %
μ = 495.6± 2.9
σ = 79.3± 2.2
Energy loss in ERAM 44
WF (872 entries)
σ/μ = 15.83± 0.59 %
μ = 487.8± 3.0
σ = 77.2± 2.4
XP (872 entries)
σ/μ = 15.31± 0.57 %
μ = 483.9± 3.0
σ = 74.1± 2.3
Energy loss in ERAM 43
WF (10645 entries)
σ/μ = 15.23± 0.16 %
μ = 491.3± 0.8
σ = 74.8± 0.7
XP (10645 entries)
σ/μ = 14.88± 0.15 %
μ = 490.6± 0.8
σ = 73.0± 0.6
Energy loss in ERAM 39
WF (9332 entries)
σ/μ = 14.54± 0.16 %
μ = 469.2± 0.8
σ = 68.2± 0.6
XP (9332 entries)
σ/μ = 15.10± 0.19 %
μ = 467.8± 0.9
σ = 70.7± 0.8
Energy loss in ERAM 41
WF (8401 entries)
σ/μ = 14.96± 0.19 %
μ = 491.8± 0.9
σ = 73.6± 0.8
XP (8401 entries)
σ/μ = 14.68± 0.19 %
μ = 487.6± 0.9
σ = 71.6± 0.8
Energy loss in ERAM 46
WF (7272 entries)
σ/μ = 15.00± 0.20 %
μ = 476.7± 1.0
σ = 71.5± 0.8
XP (7272 entries)
σ/μ = 14.49± 0.17 %
μ = 472.7± 0.9
σ = 68.5± 0.7
Count
dE/dx (ADC counts/cm)


fph1f_WFXP
Entries 151146
Mean x 493.9
Mean y 490.4
Std Dev x 78.18
Std Dev y 76.48
0 0 36
0 151106 4
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
500
1000
1500
2000
2500

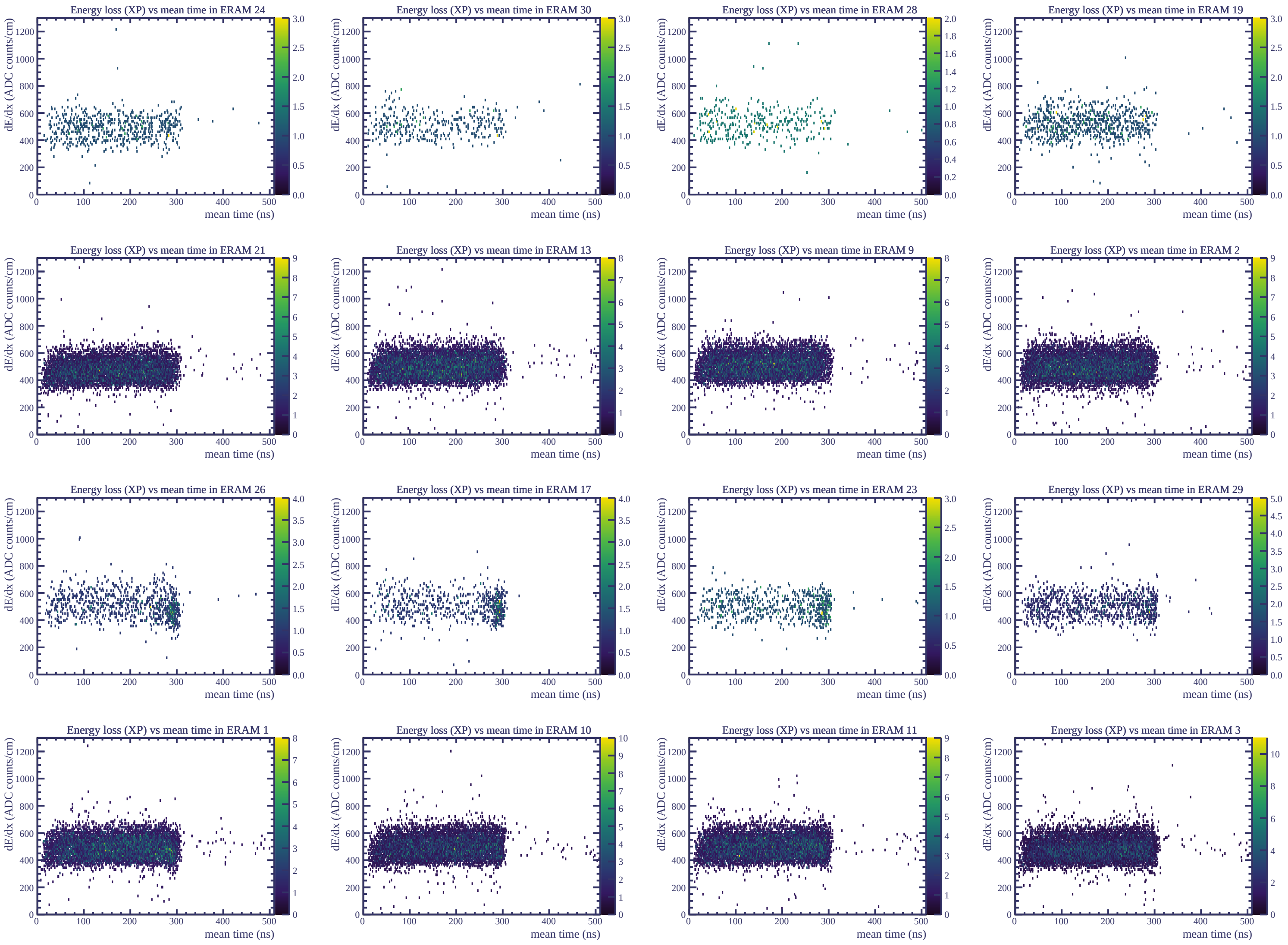

Waveforms sum
Crossed pads
Resolution (%)
Drift time (timebins)
0
5
10
15
20
25
30
0
100
200
300
400
500



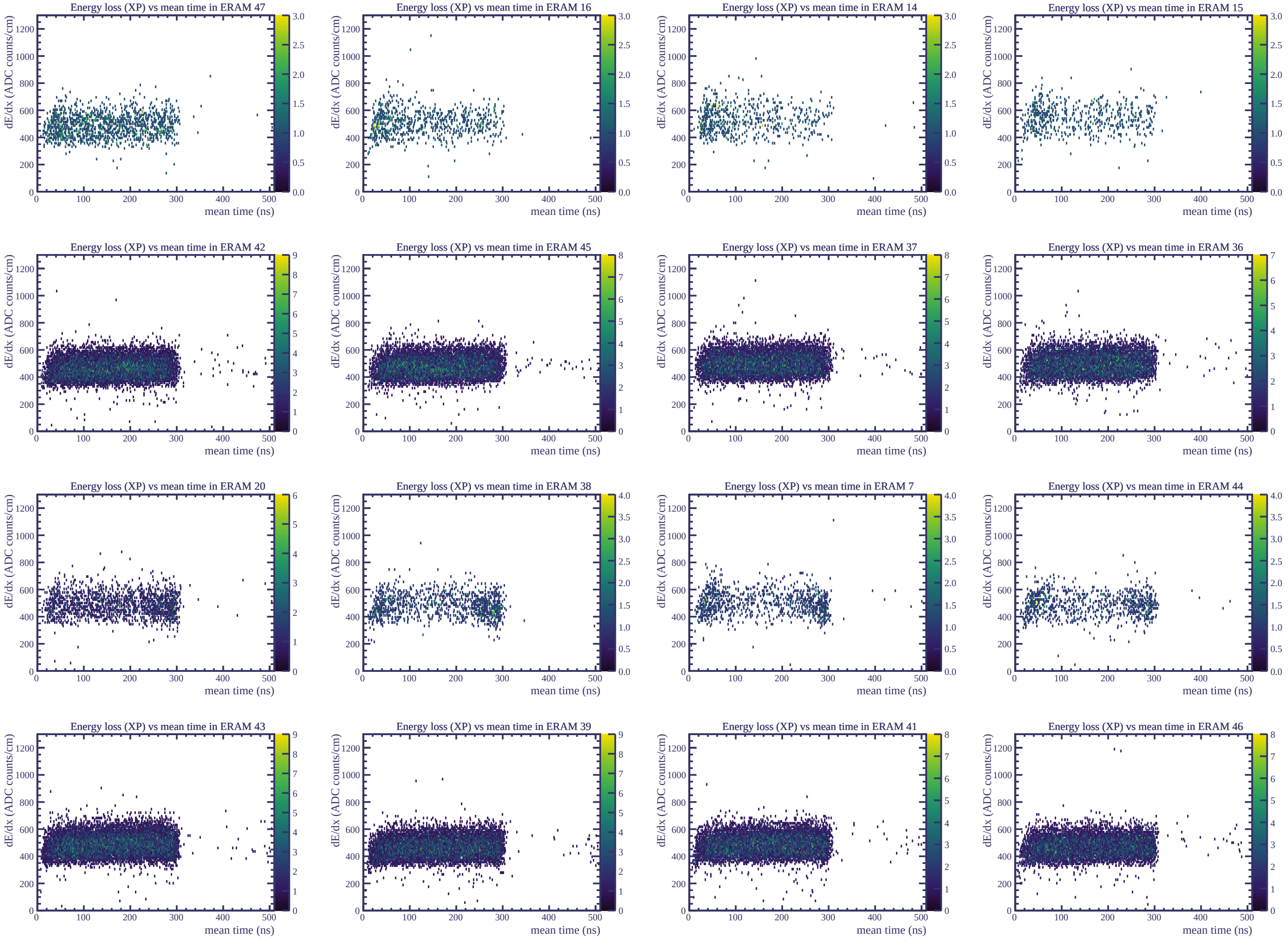
Mean (ADC counts/cm)
800
700
600
500
400
300
200
0
100
200
300
400
500
Drift time (timebins)
Waveforms sum
Crossed pads

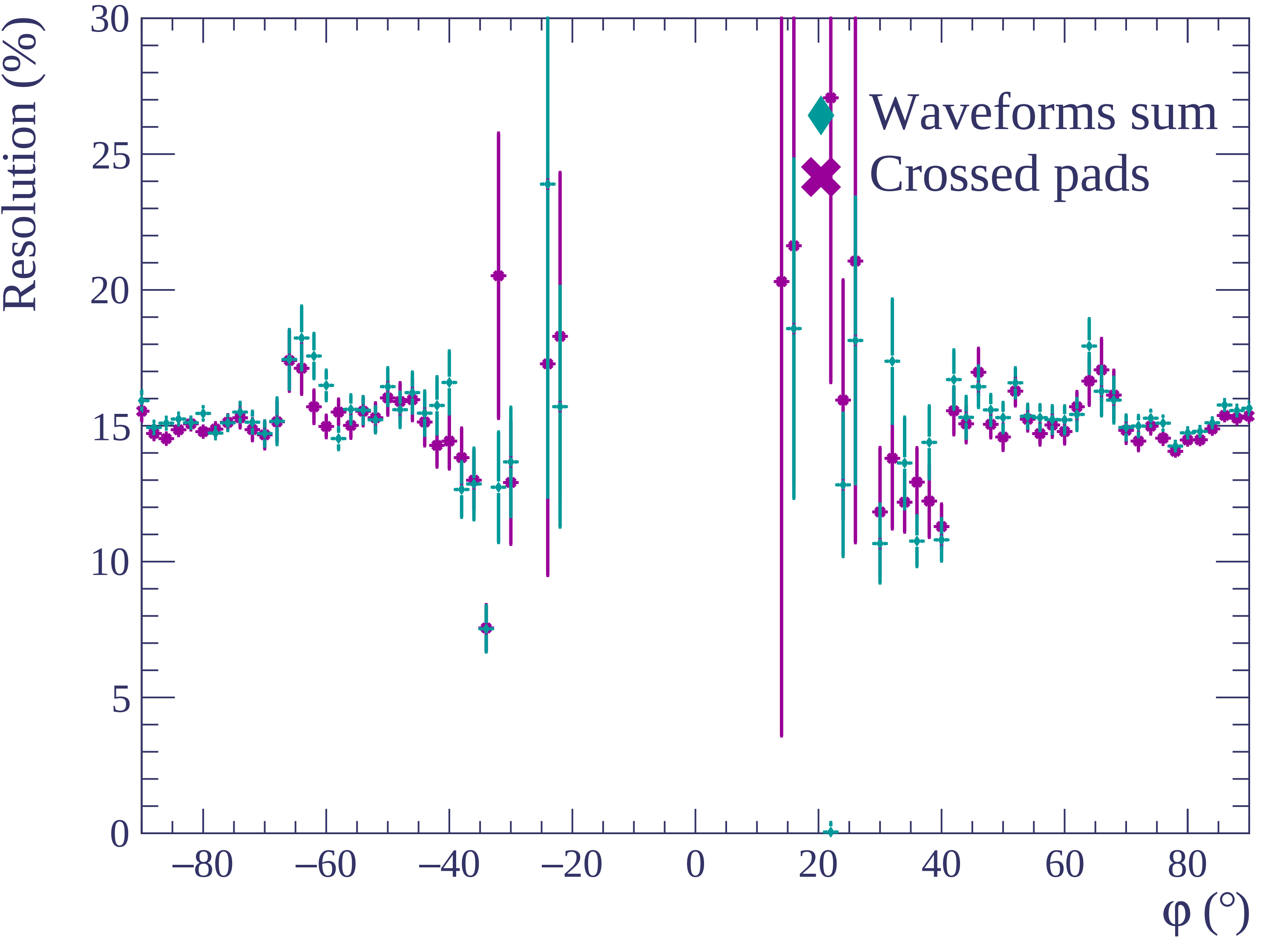


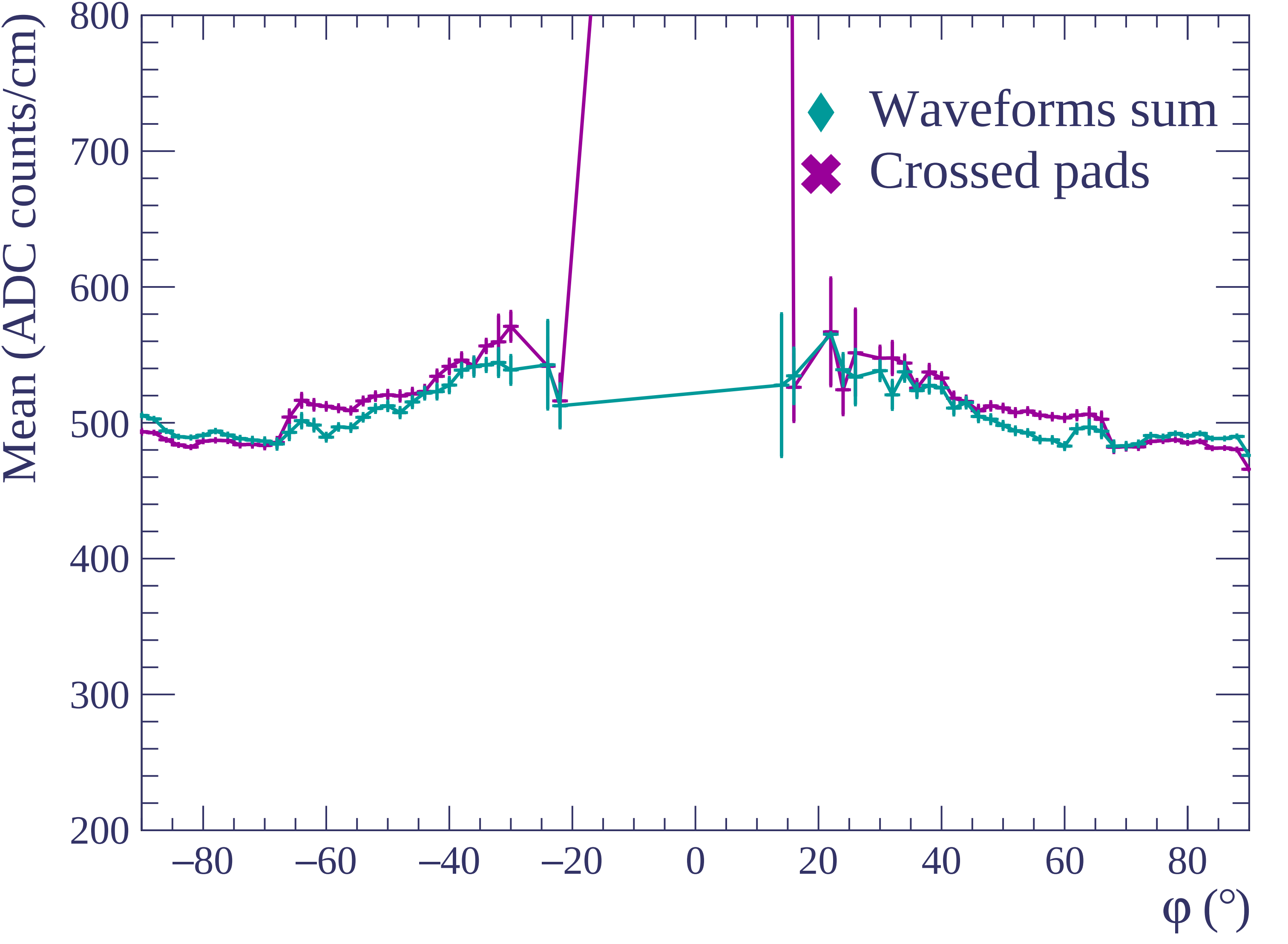
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16
18



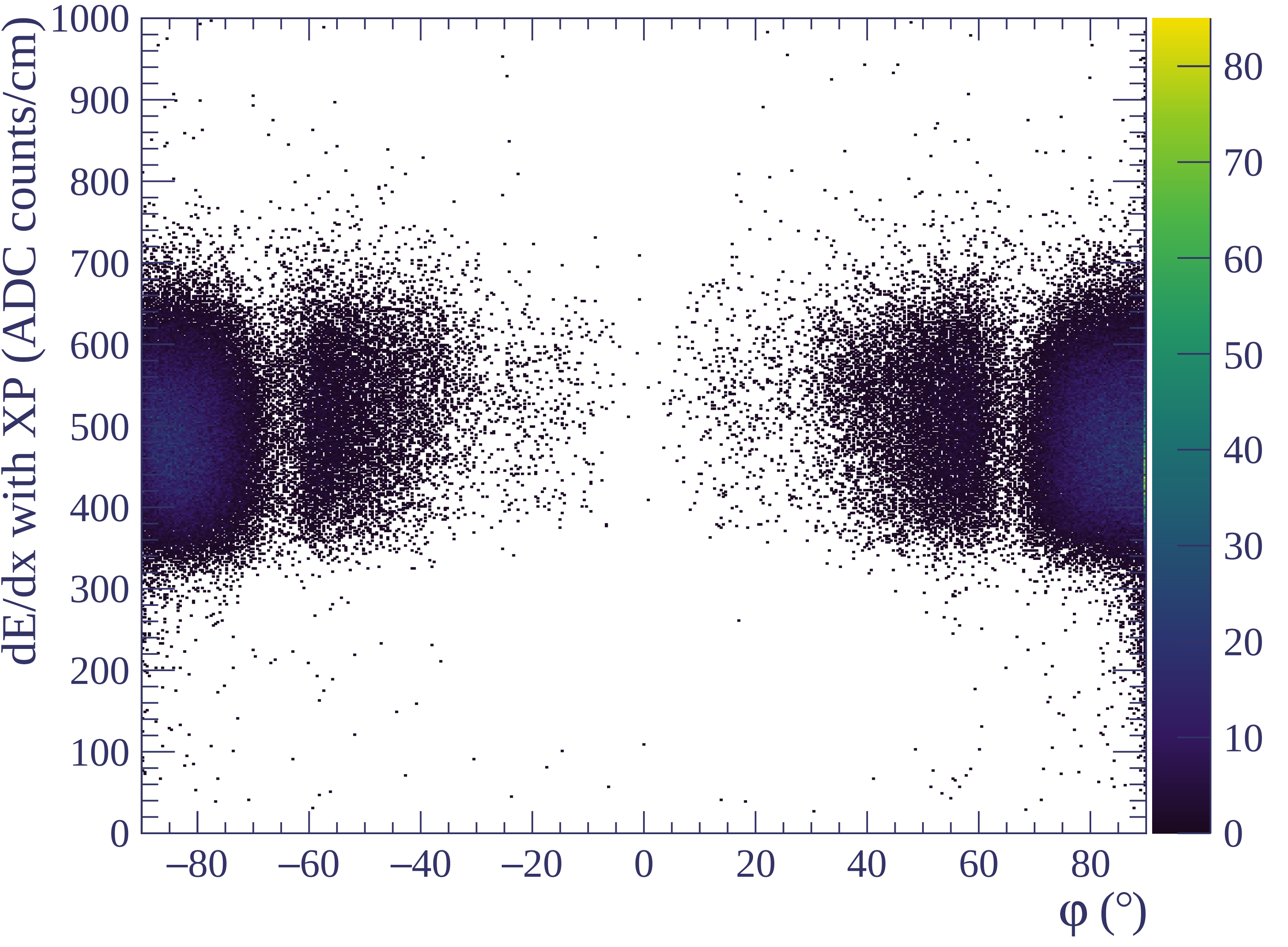
Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)


Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

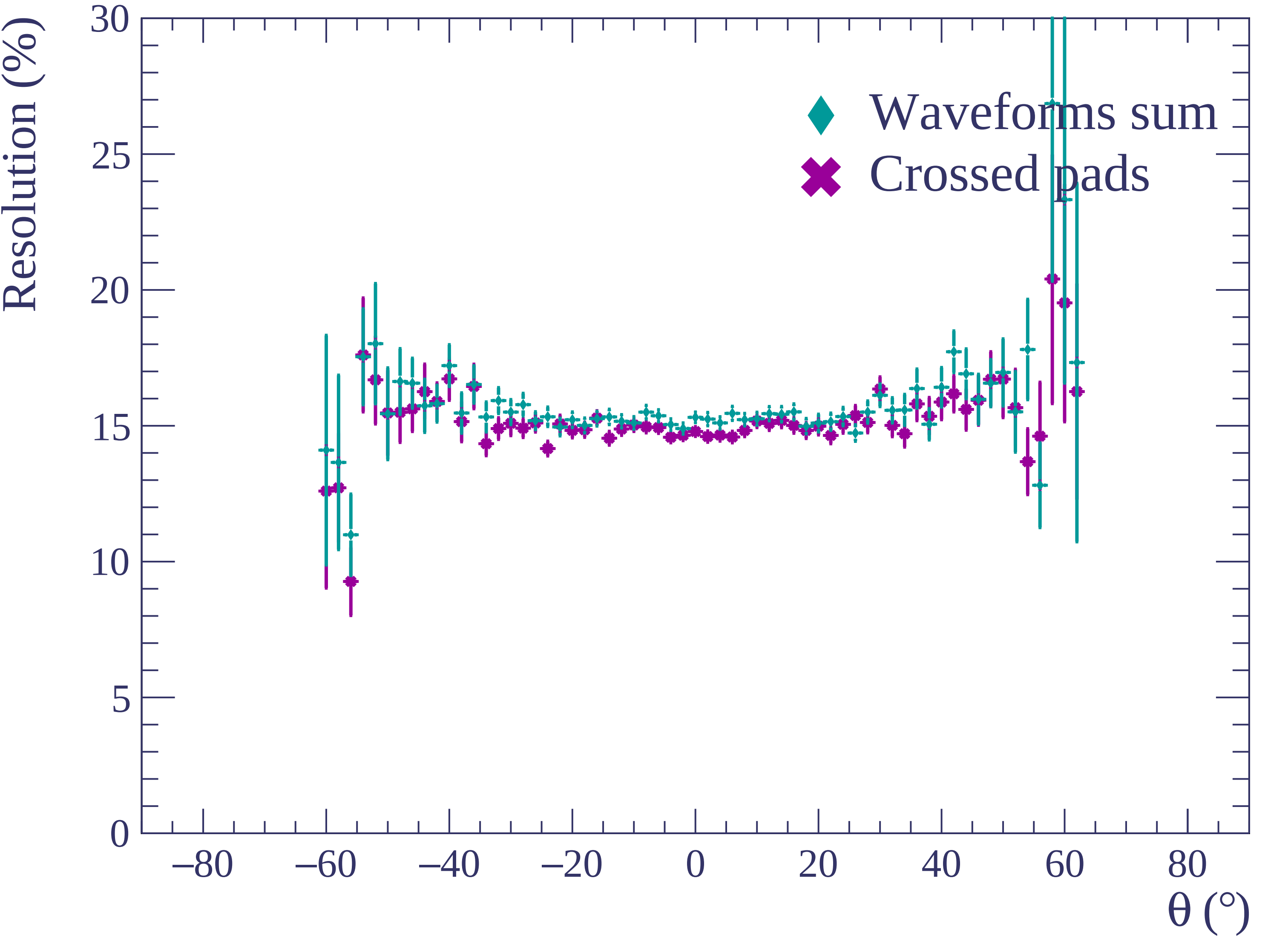

Resolution (%)
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads



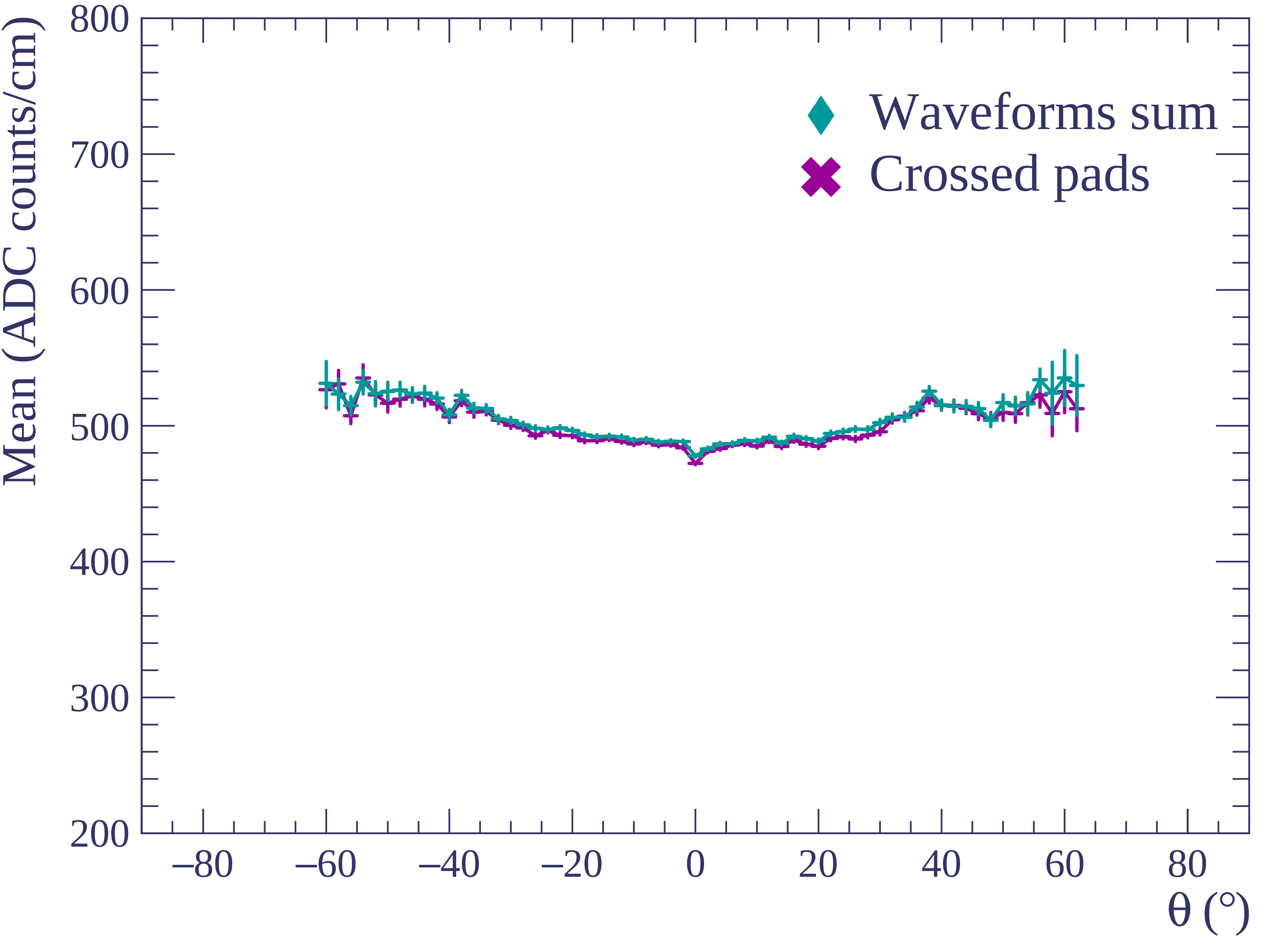
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads



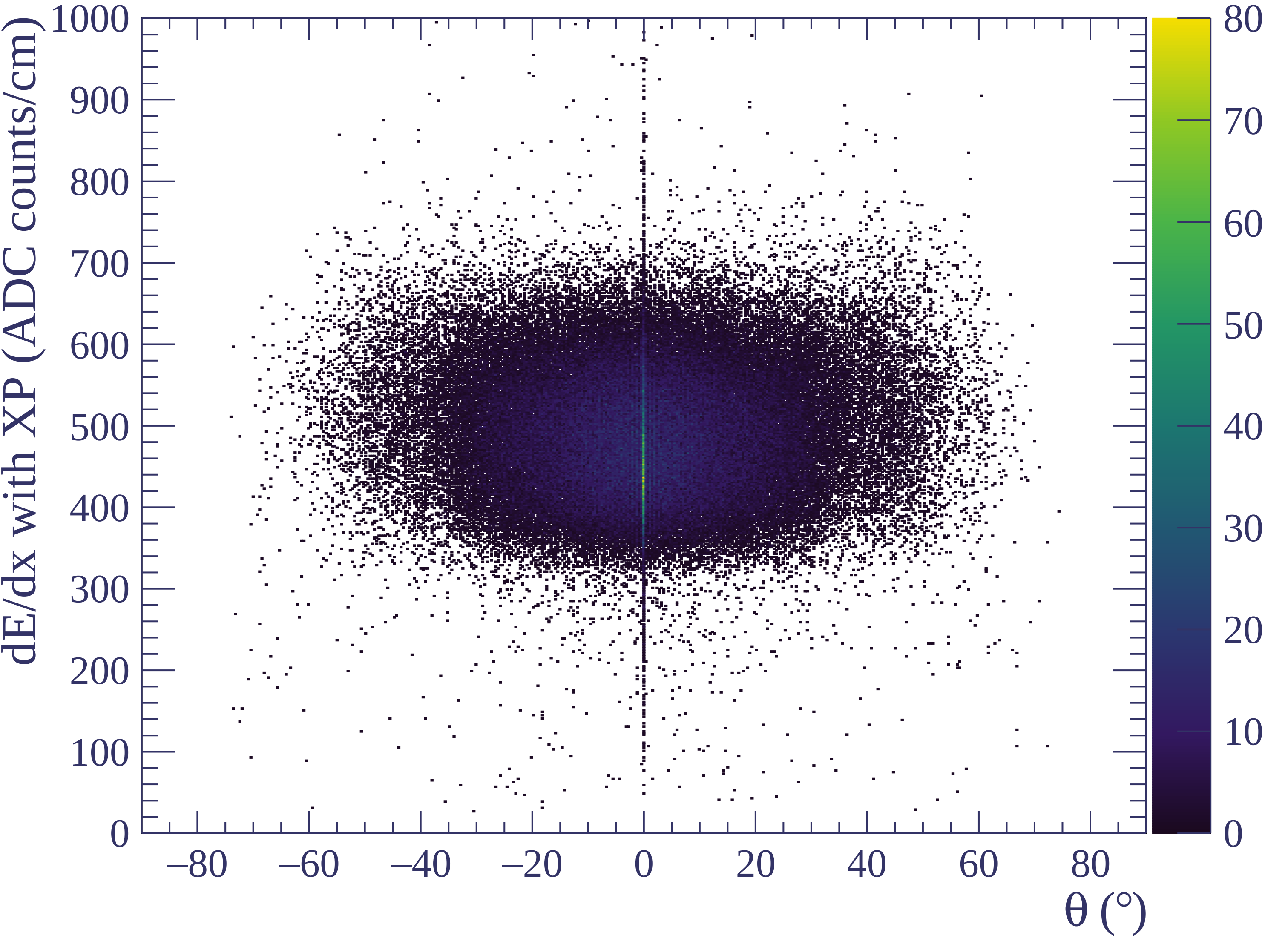
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
80
70
60
50
40
30
20
10
0



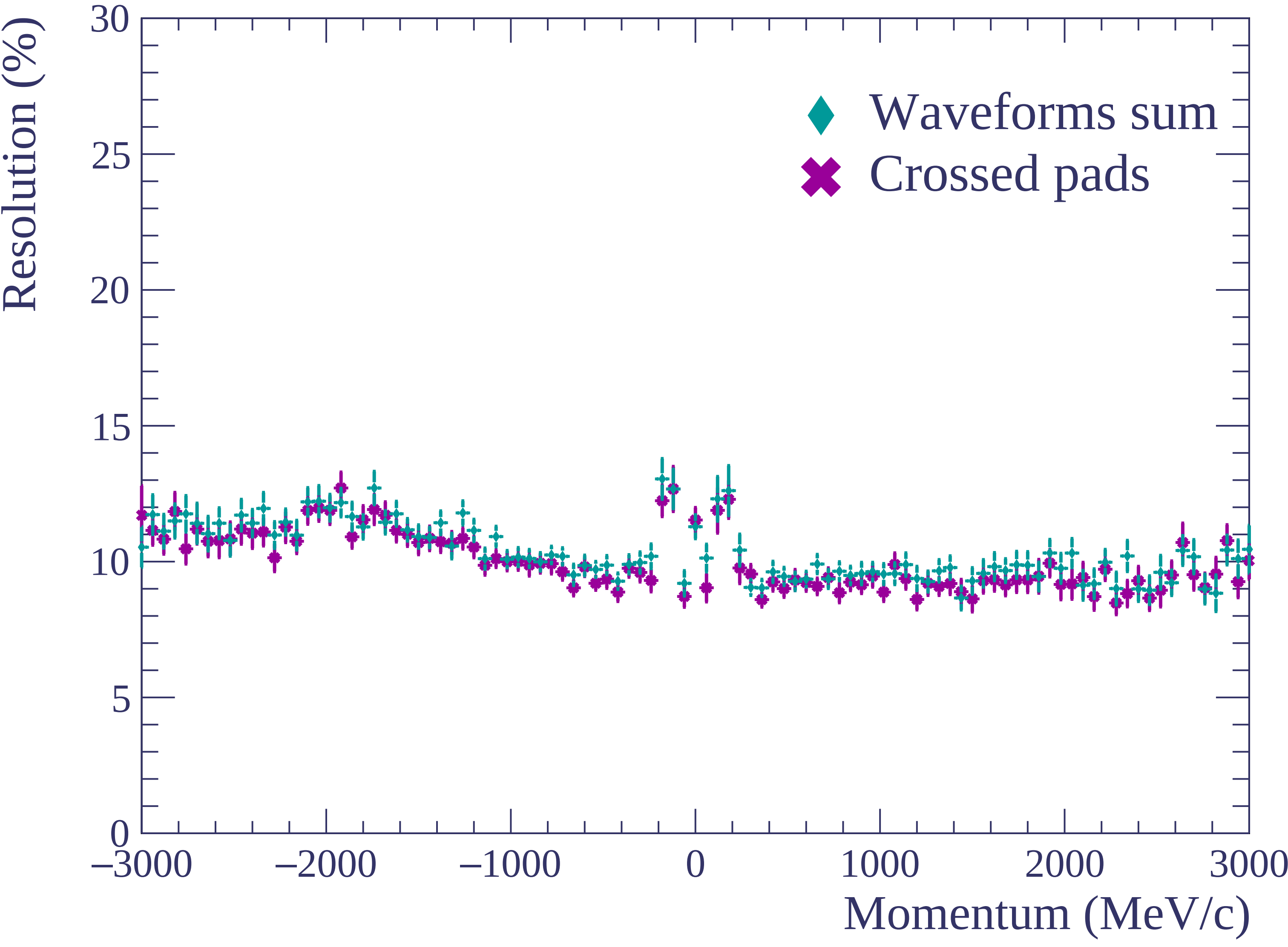
Resolution (%)
θ (°)
Waveforms sum
Crossed pads
0
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80

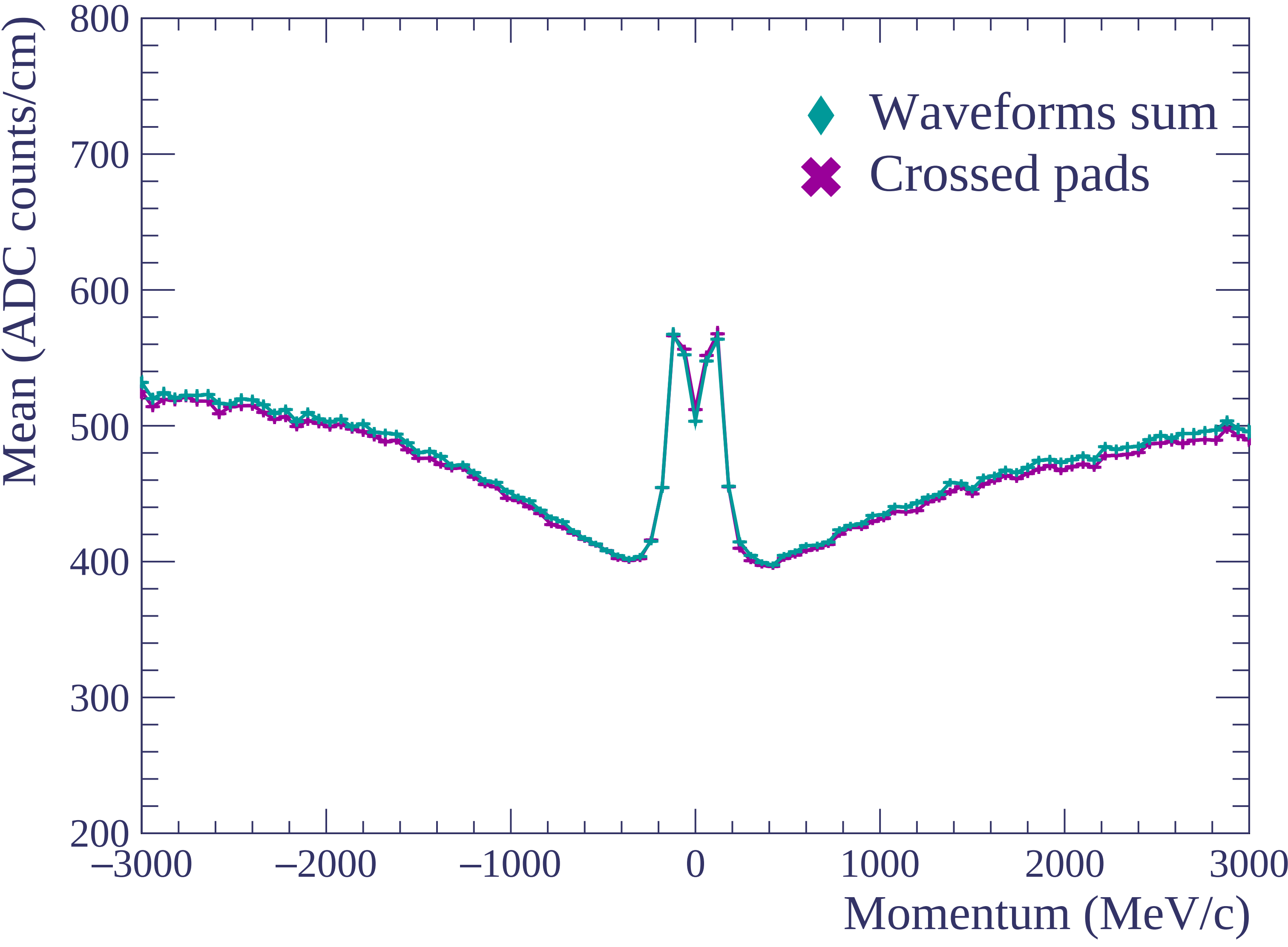


Waveforms sum
Crossed pads
Mean (ADC counts/cm)
θ (°)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80

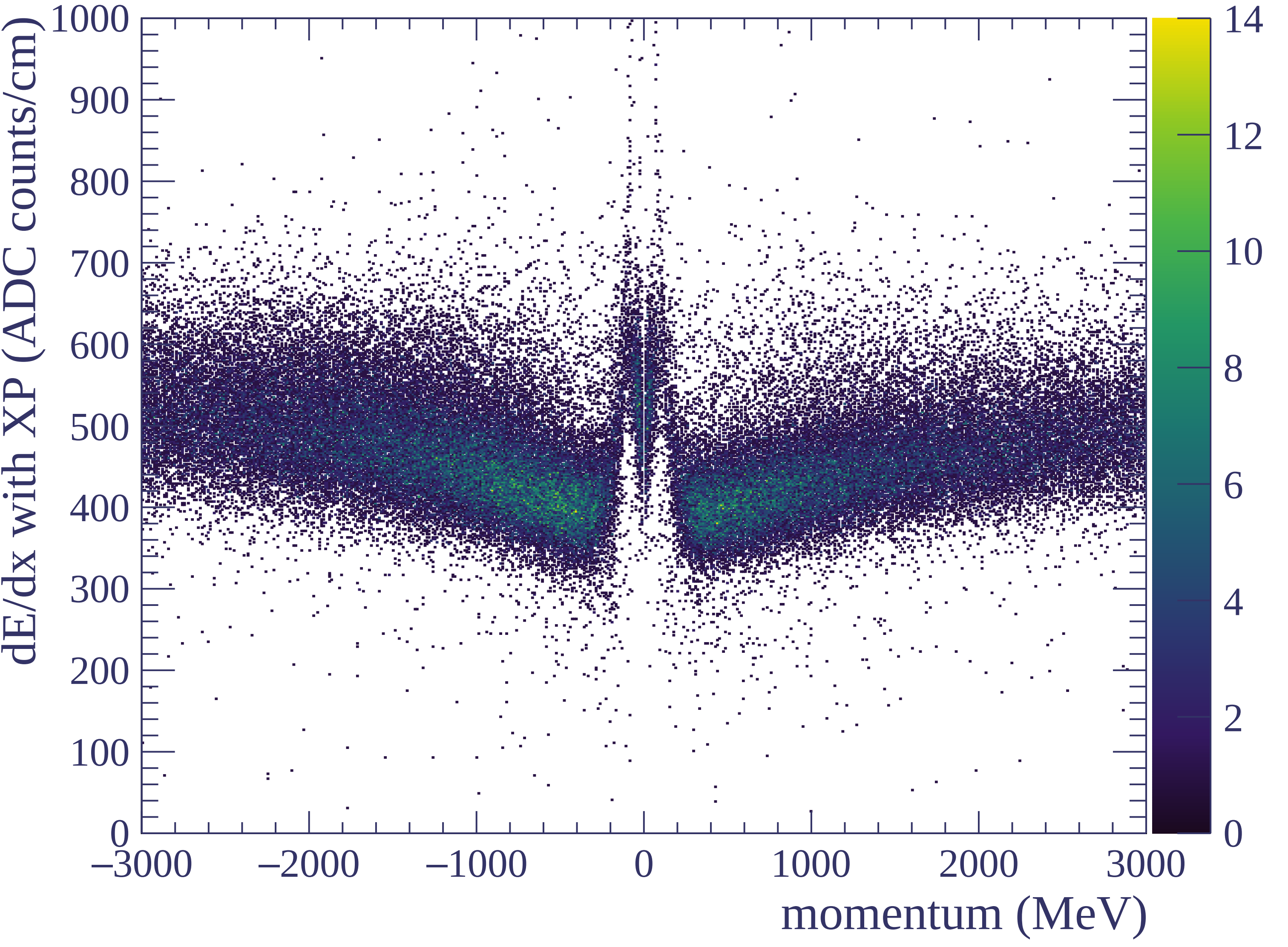


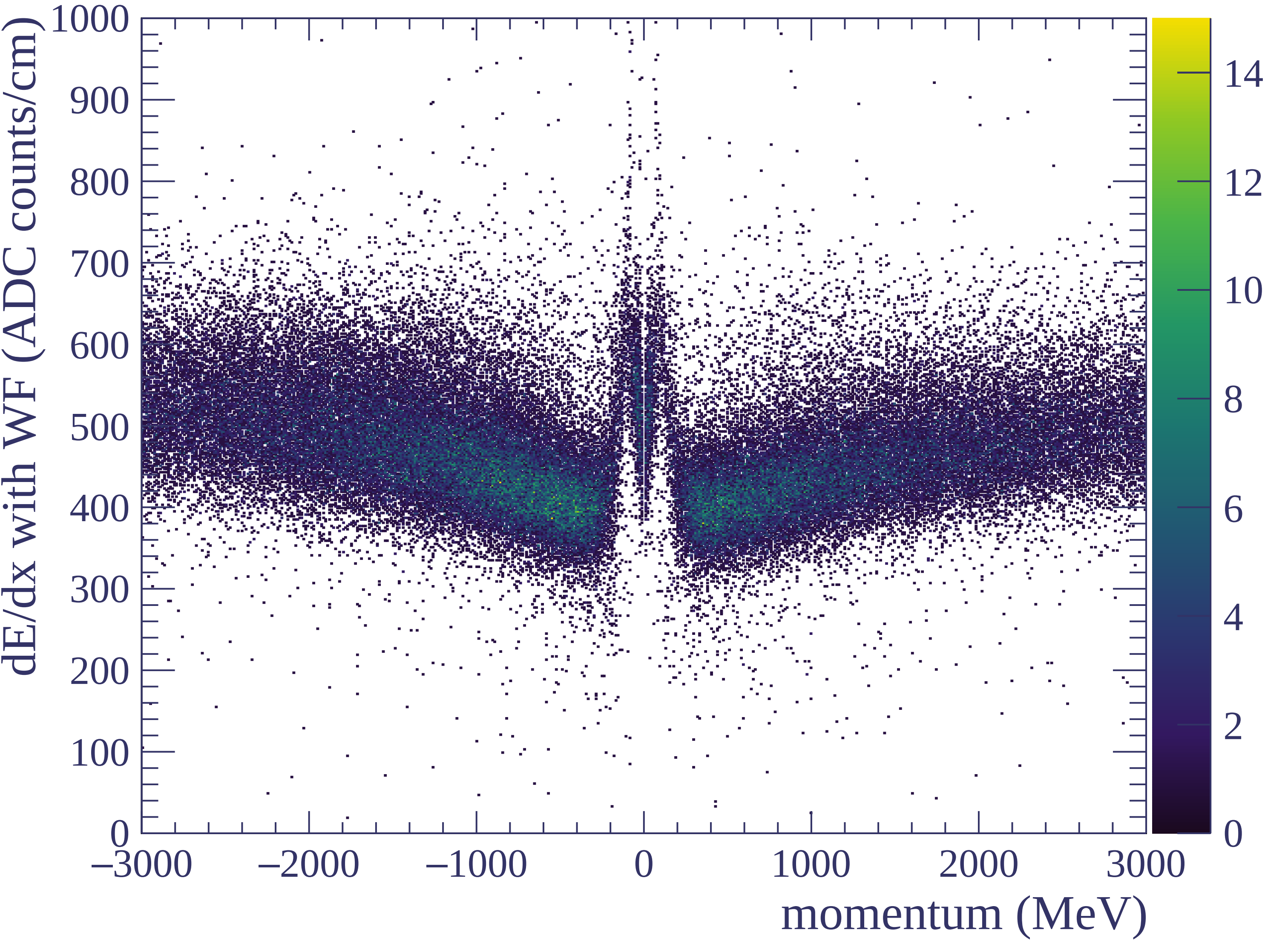
dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
80
70
60
50
40
30
20
10
0


Waveforms sum
Crossed pads
Resolution (%)
Momentum (MeV/c)
0
5
10
15
20
25
30
−3000
−2000
−1000
0
1000
2000
3000

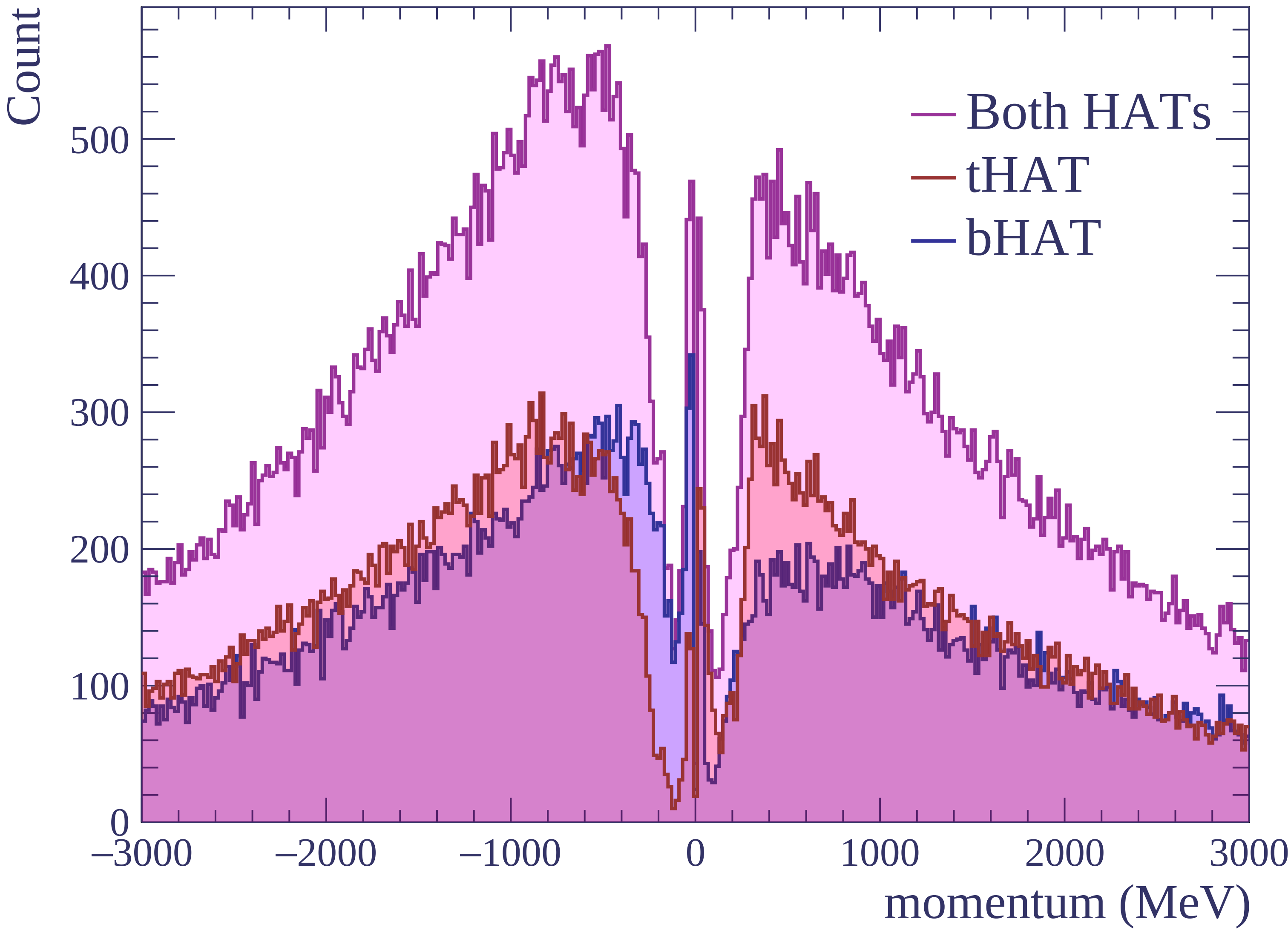


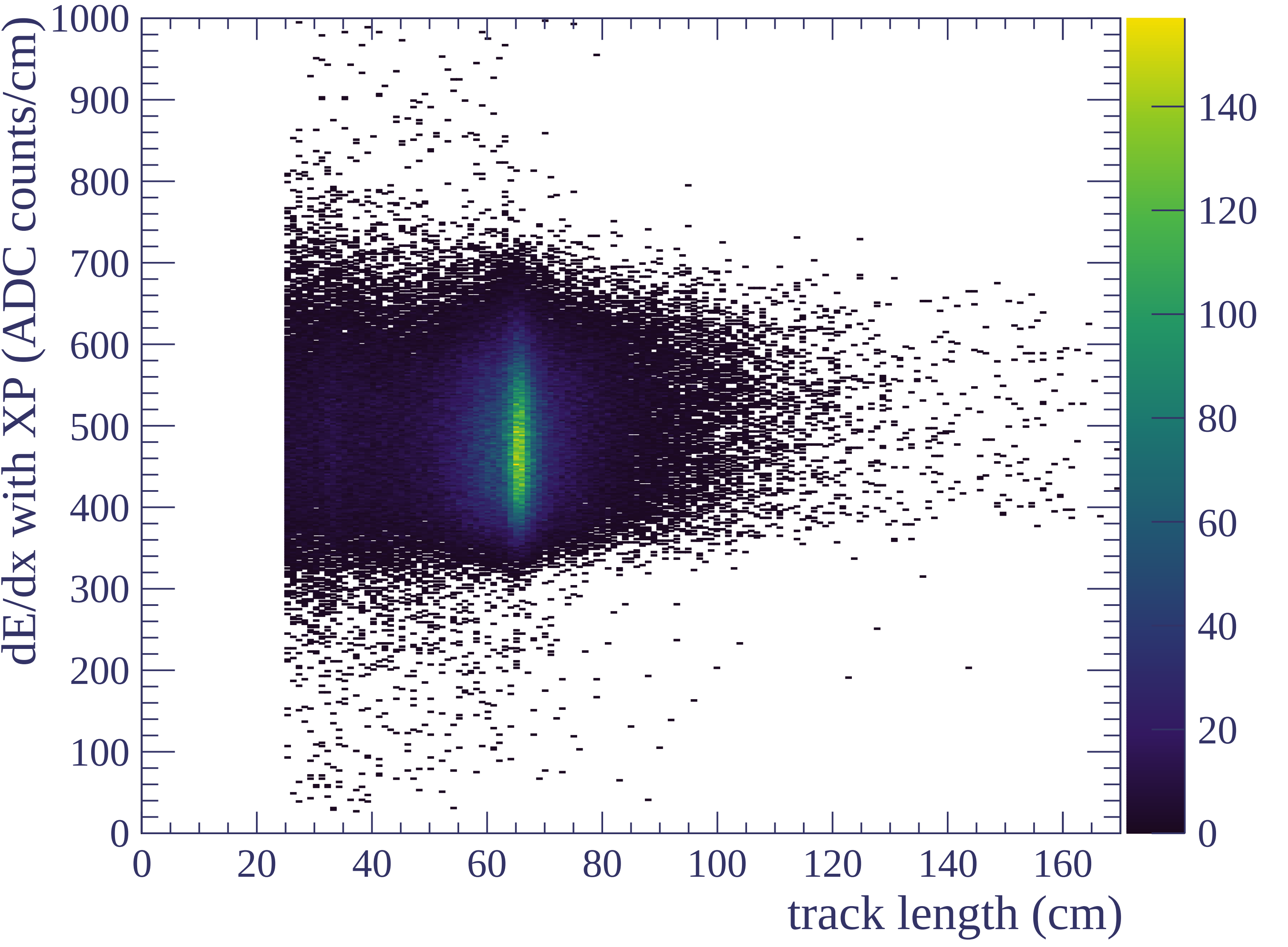
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
14
12
10
8
6
4
2
0

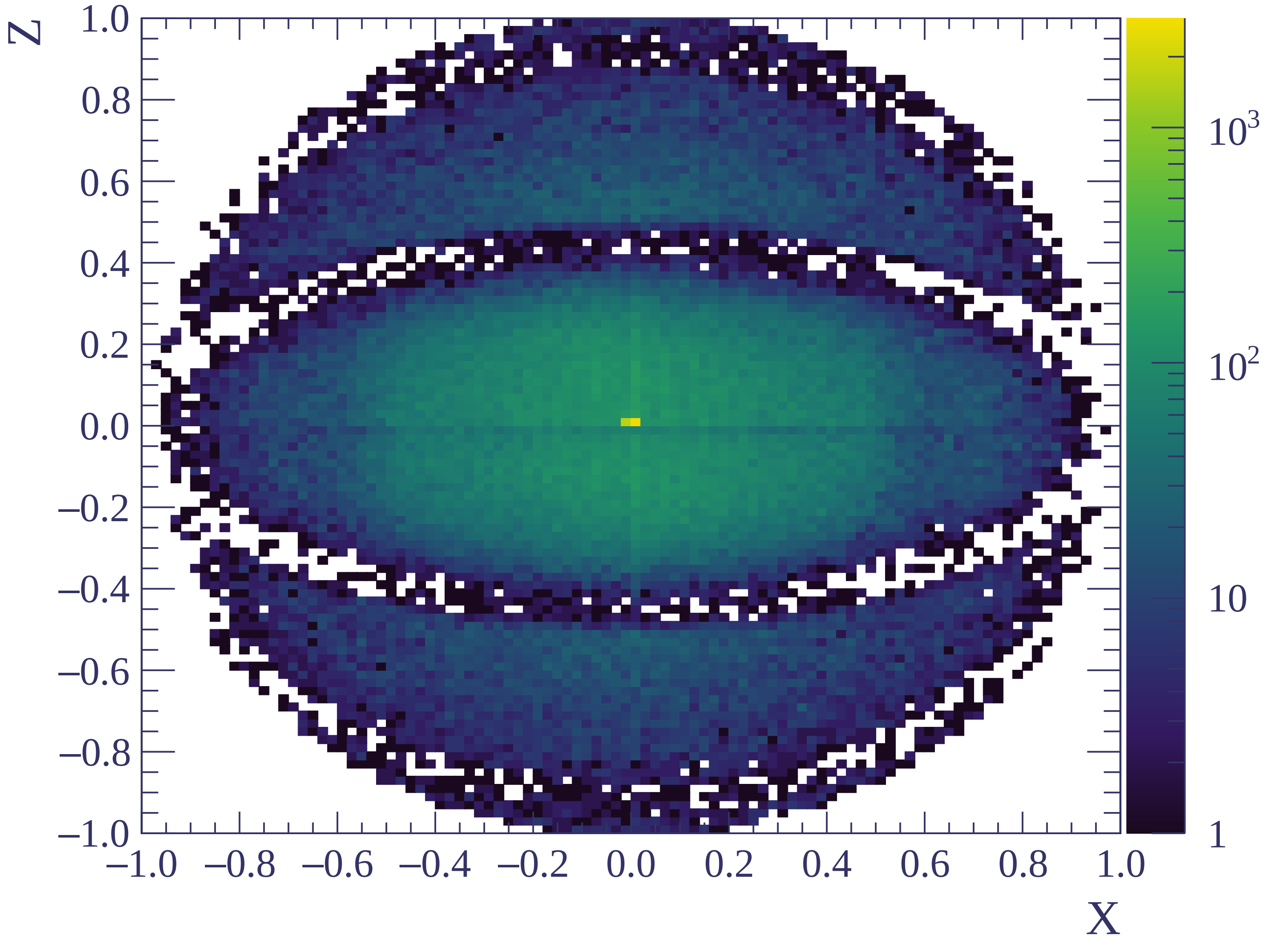


dE/dx with WF (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
14
12
10
8
6
4
2
0

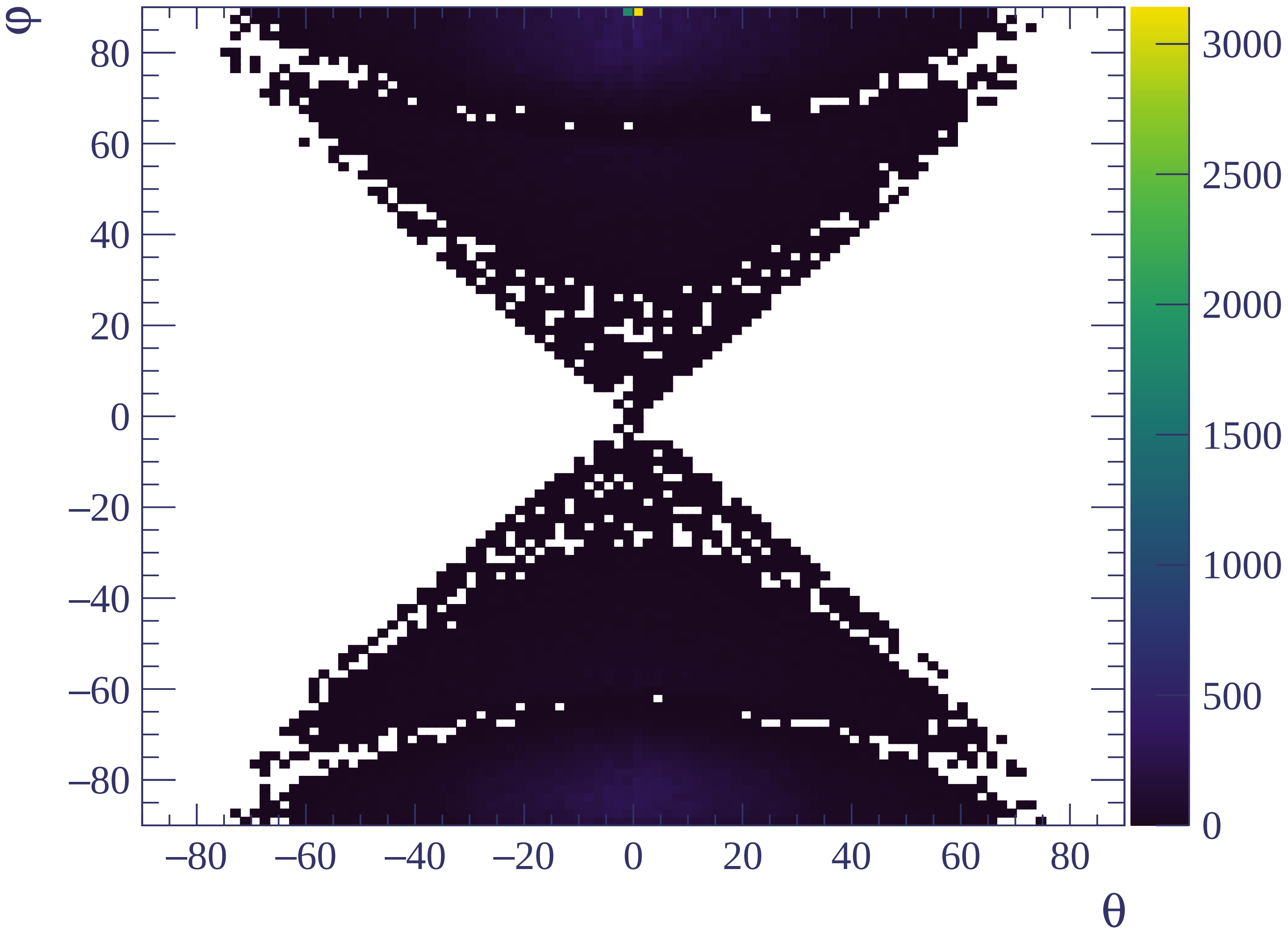

Count
momentum (MeV)
Both HATs
tHAT
bHAT
0
100
200
300
400
500
−3000
−2000
−1000
0
1000
2000
3000

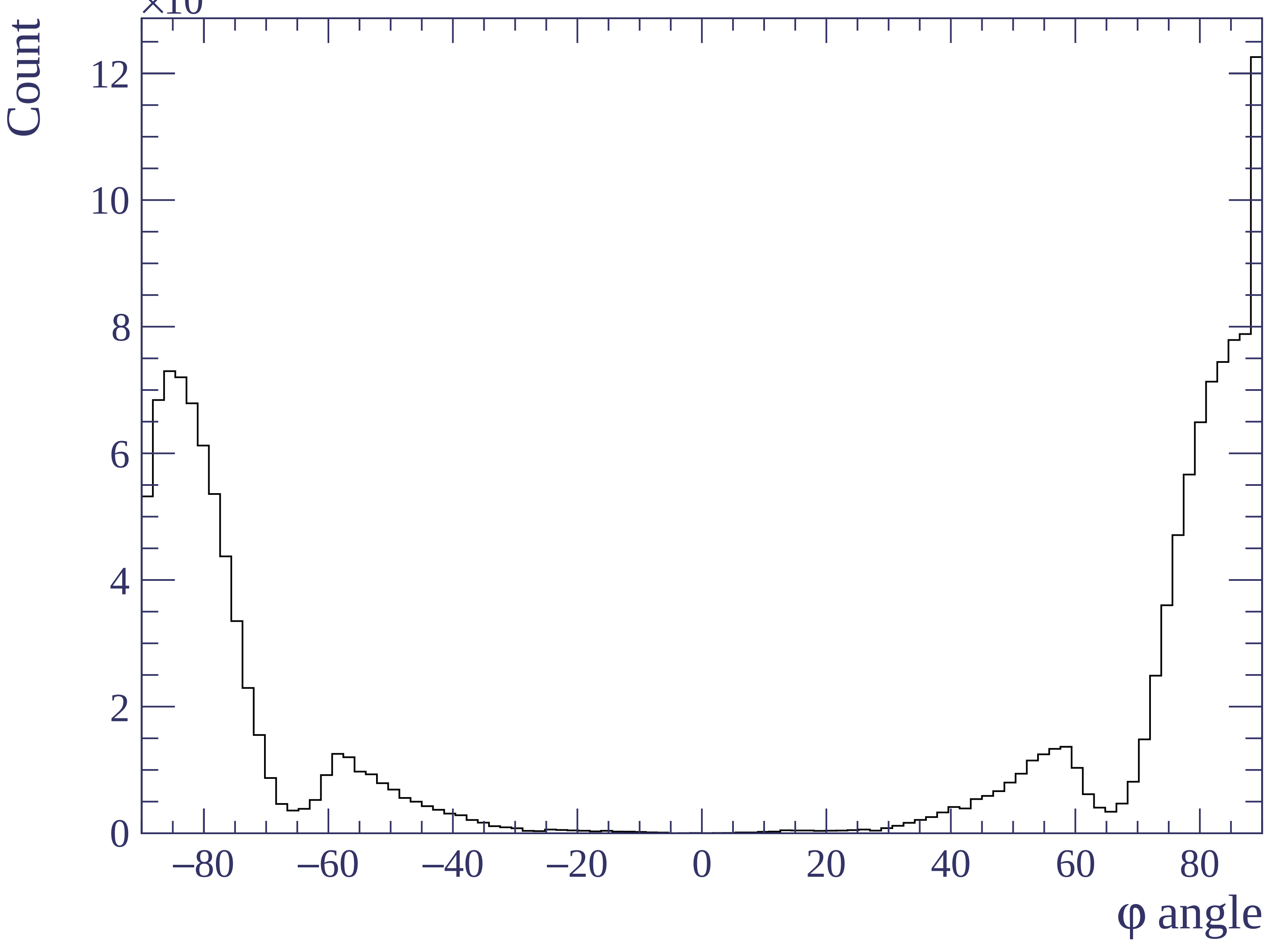


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)
140
120
100
80
60
40
20
0

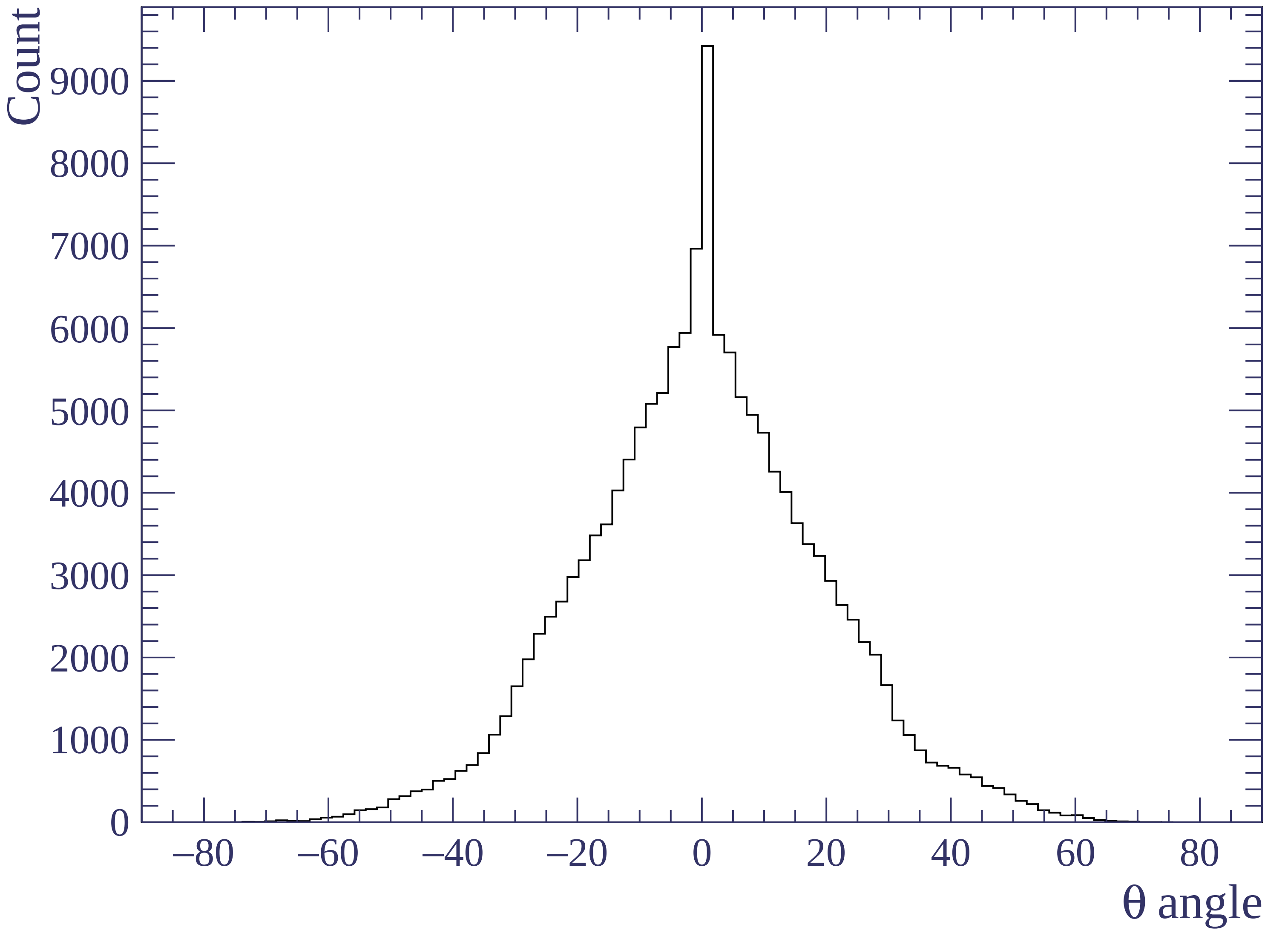


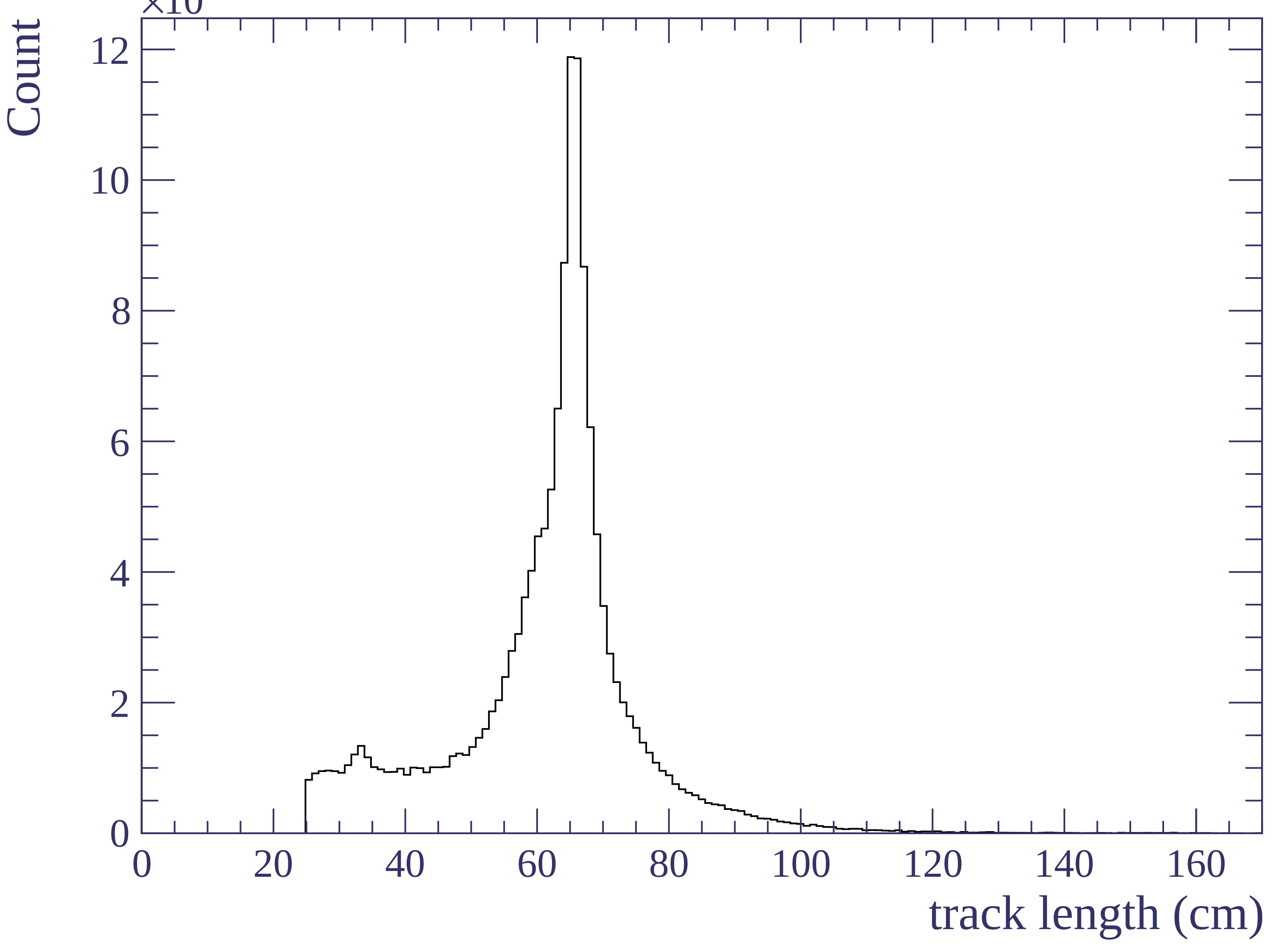
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
10^3
10^2
10
1

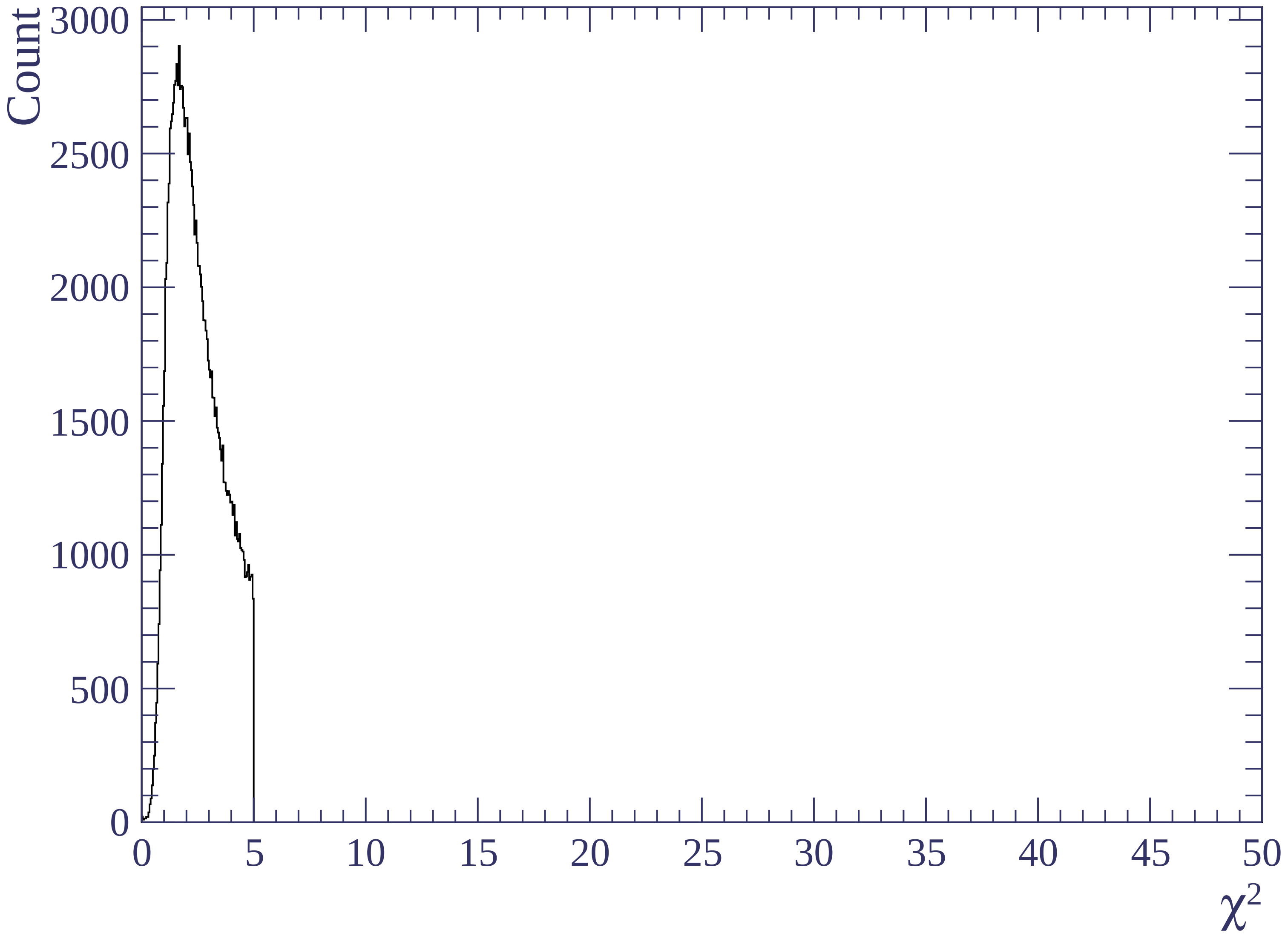

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
3000
2500
2000
1500
1000
500
0



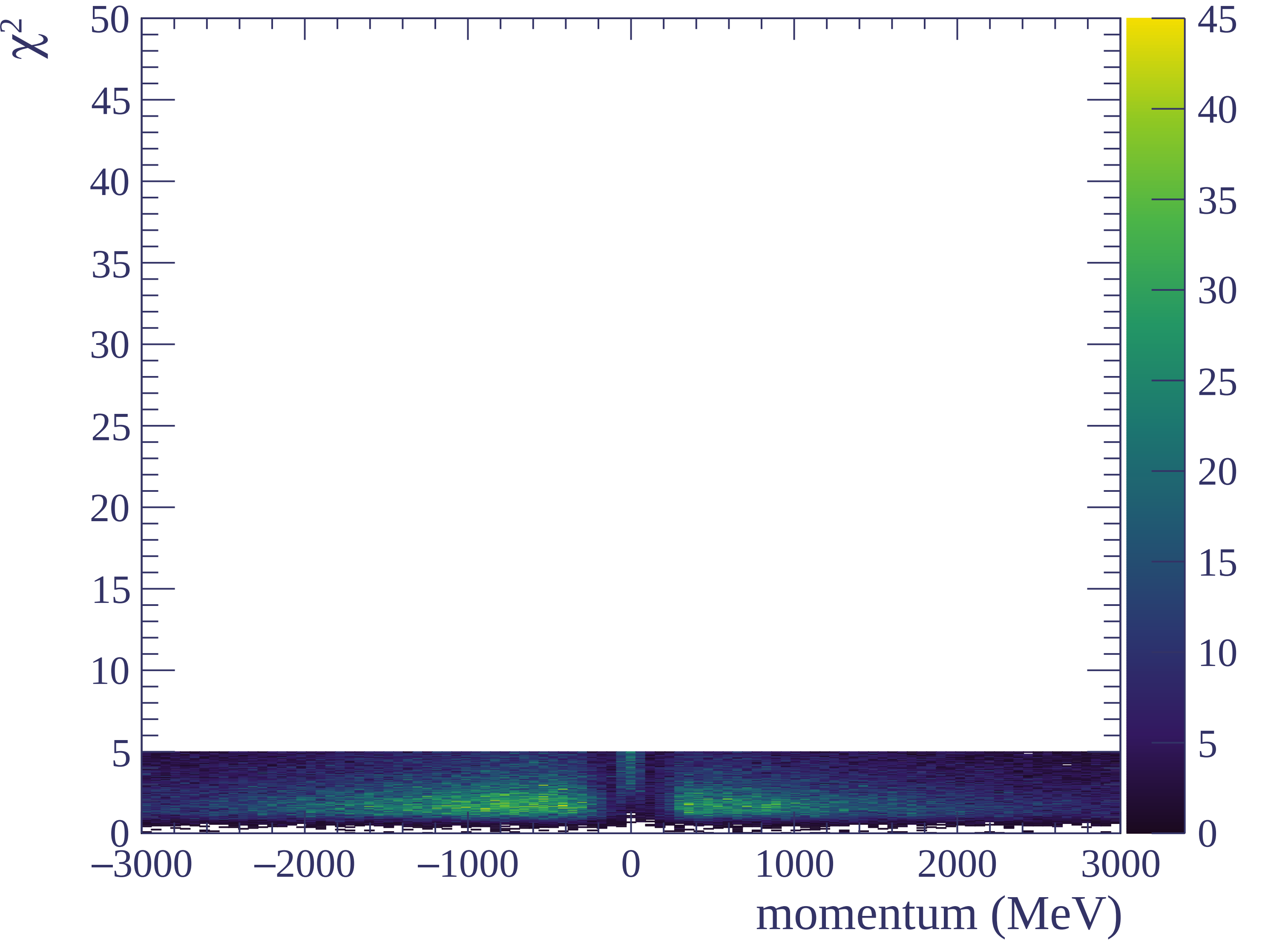
×10
Count
12
10
8
6
4
2
0
−80
−60
−40
−20
0
20
40
60
80
φ angle


Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

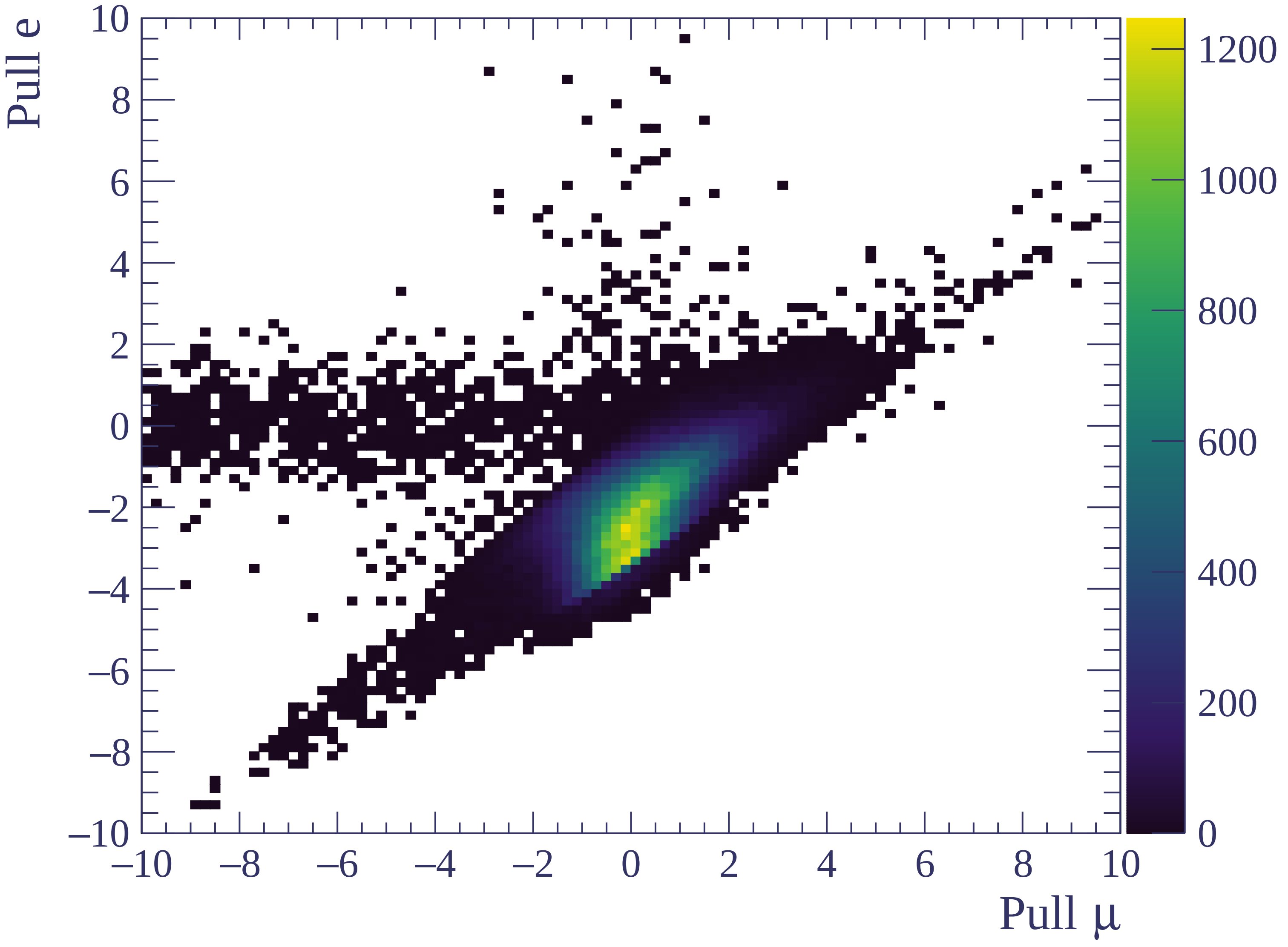

×10
Count
0
2
4
6
8
10
12
0
20
40
60
80
100
120
140
160
track length (cm)

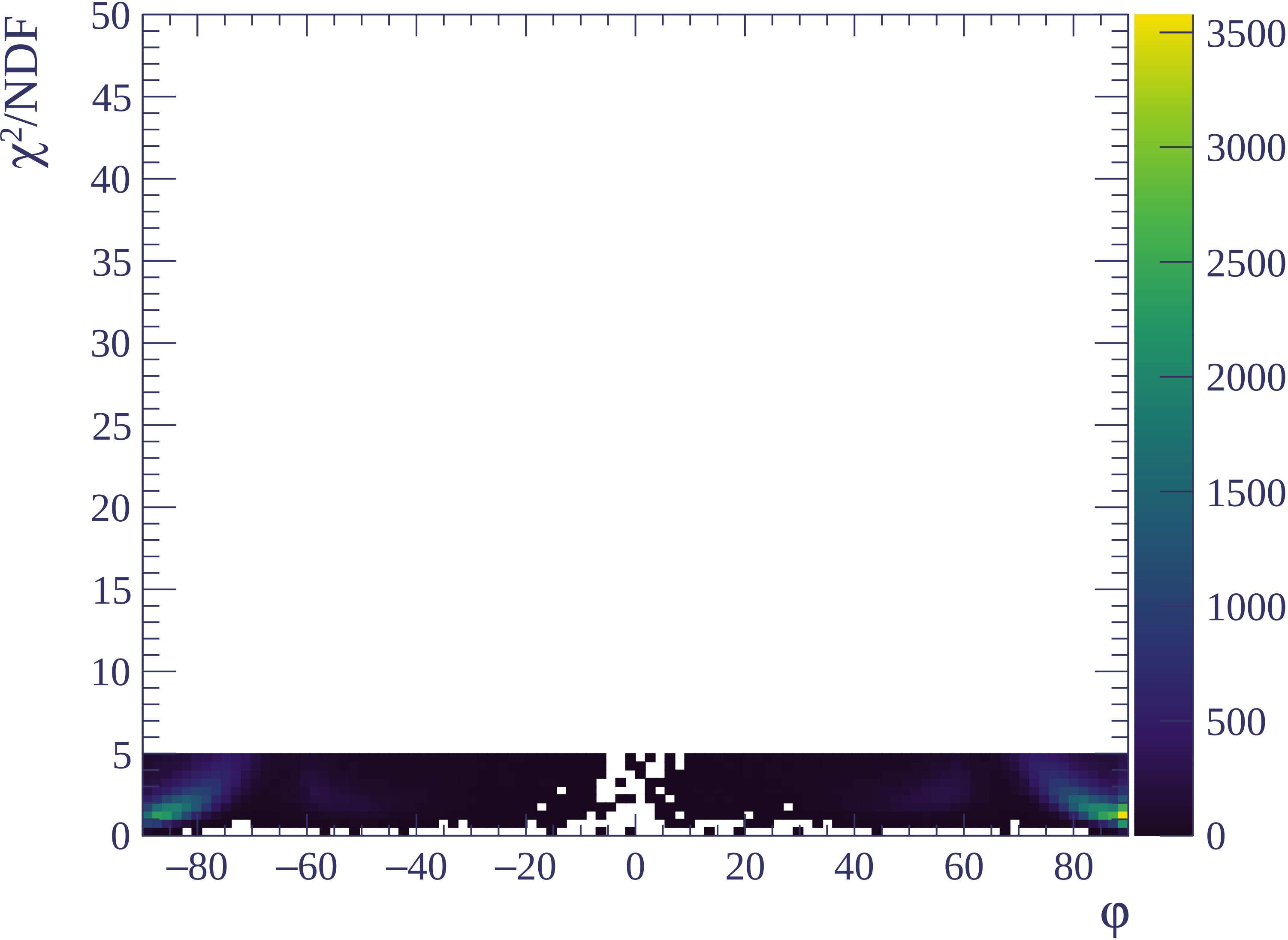


Count
3000
2500
2000
1500
1000
500
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

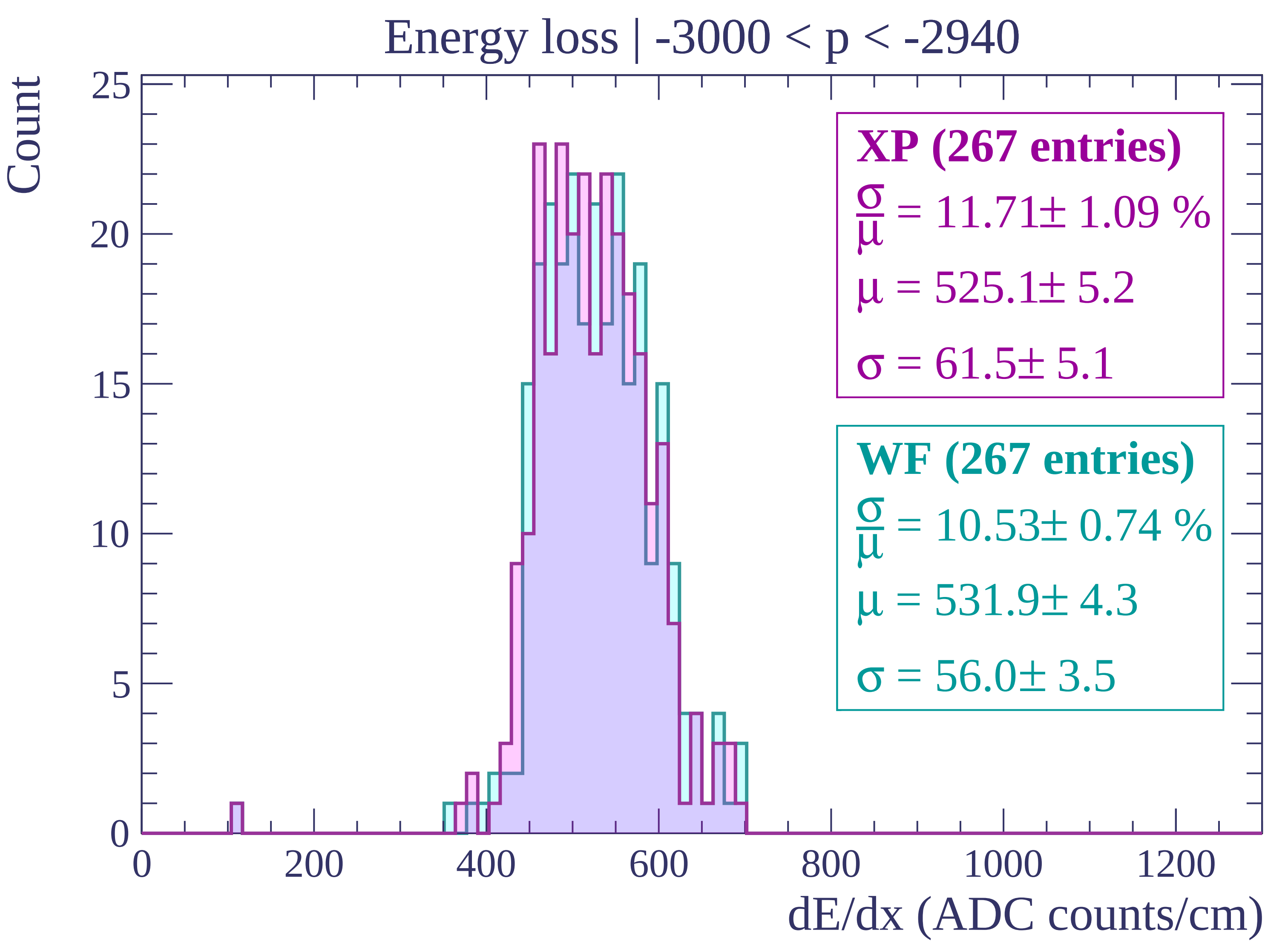


χ^2
50
45
40
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
45
40
35
30
25
20
15
10
5
0

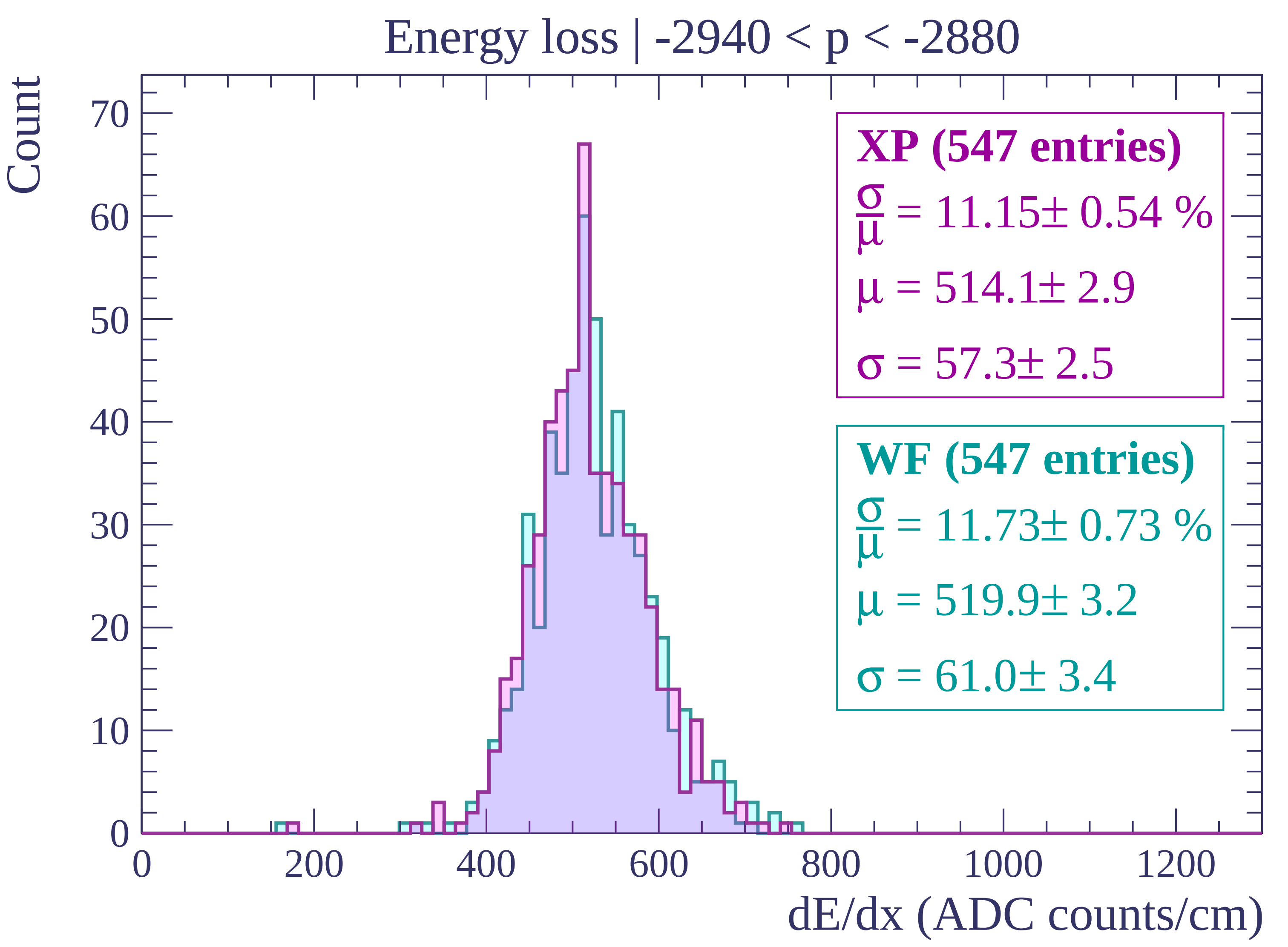

Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
1200
1000
800
600
400
200



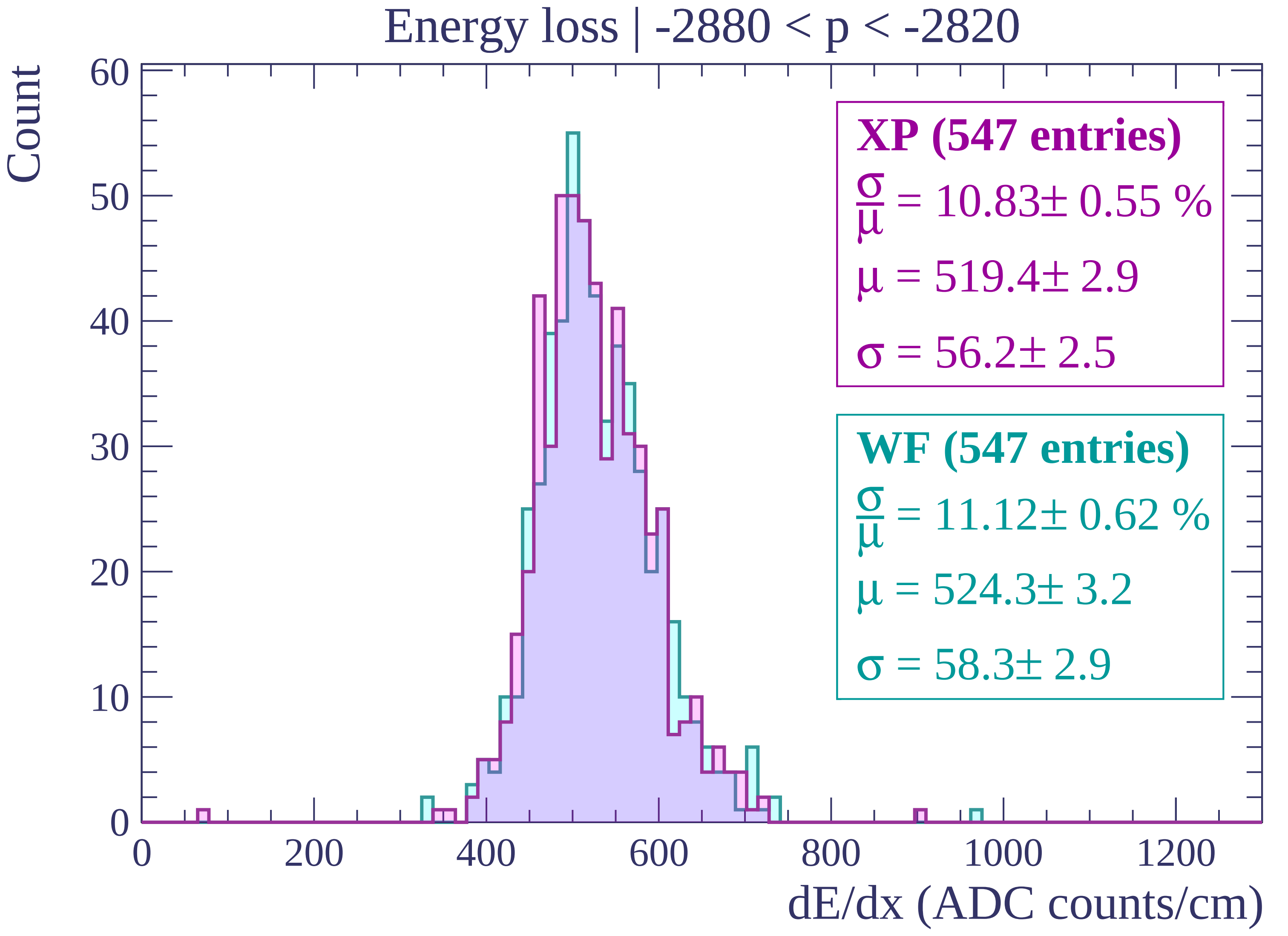
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
3500
3000
2500
2000
1500
1000
500
0



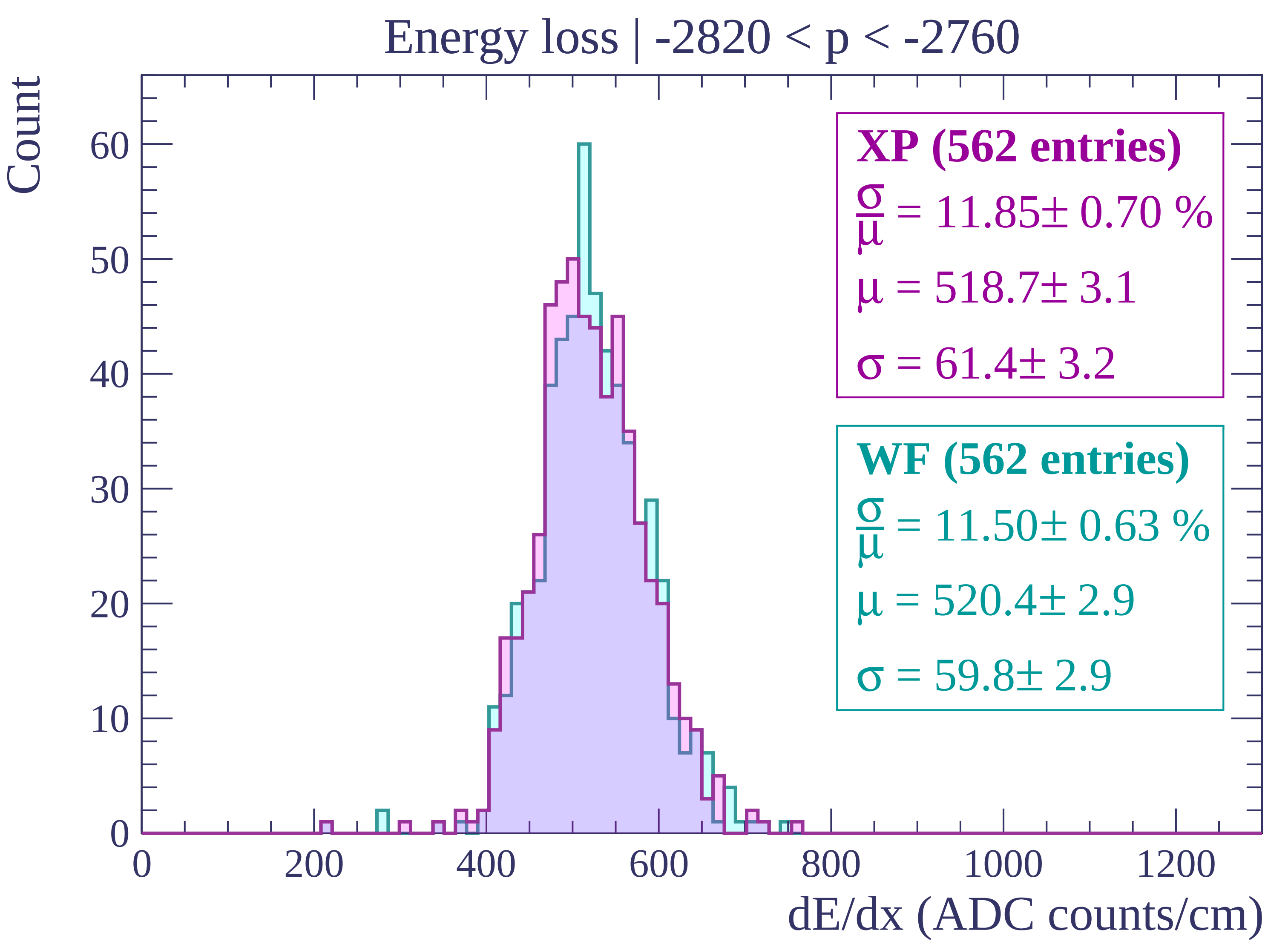
Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
XP (267 entries)
σ/μ = 11.71± 1.09 %
μ = 525.1± 5.2
σ = 61.5± 5.1
WF (267 entries)
σ/μ = 10.53± 0.74 %
μ = 531.9± 4.3
σ = 56.0± 3.5



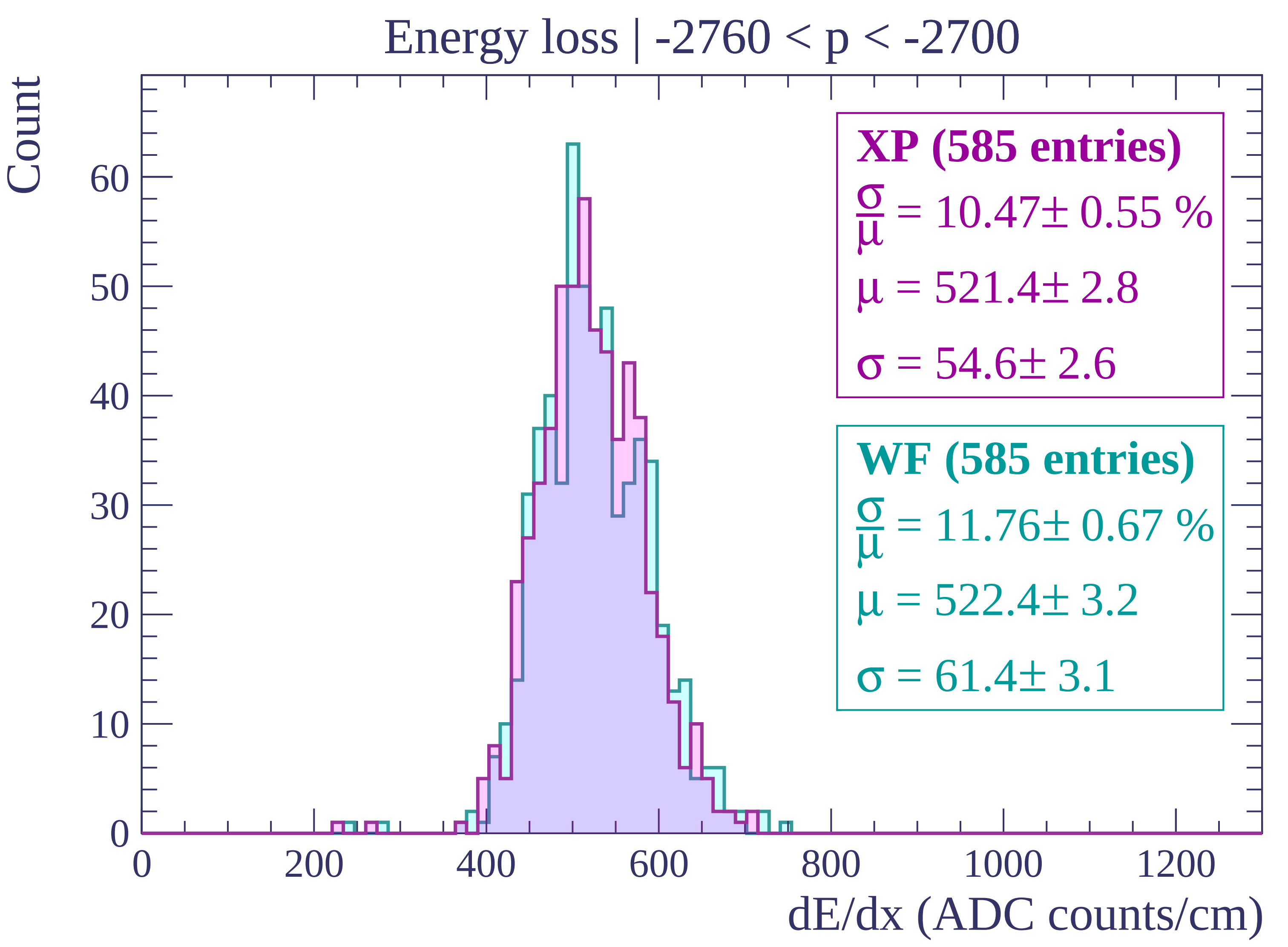
Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP (547 entries)
σ/μ = 11.15± 0.54 %
μ = 514.1± 2.9
σ = 57.3± 2.5
WF (547 entries)
σ/μ = 11.73± 0.73 %
μ = 519.9± 3.2
σ = 61.0± 3.4



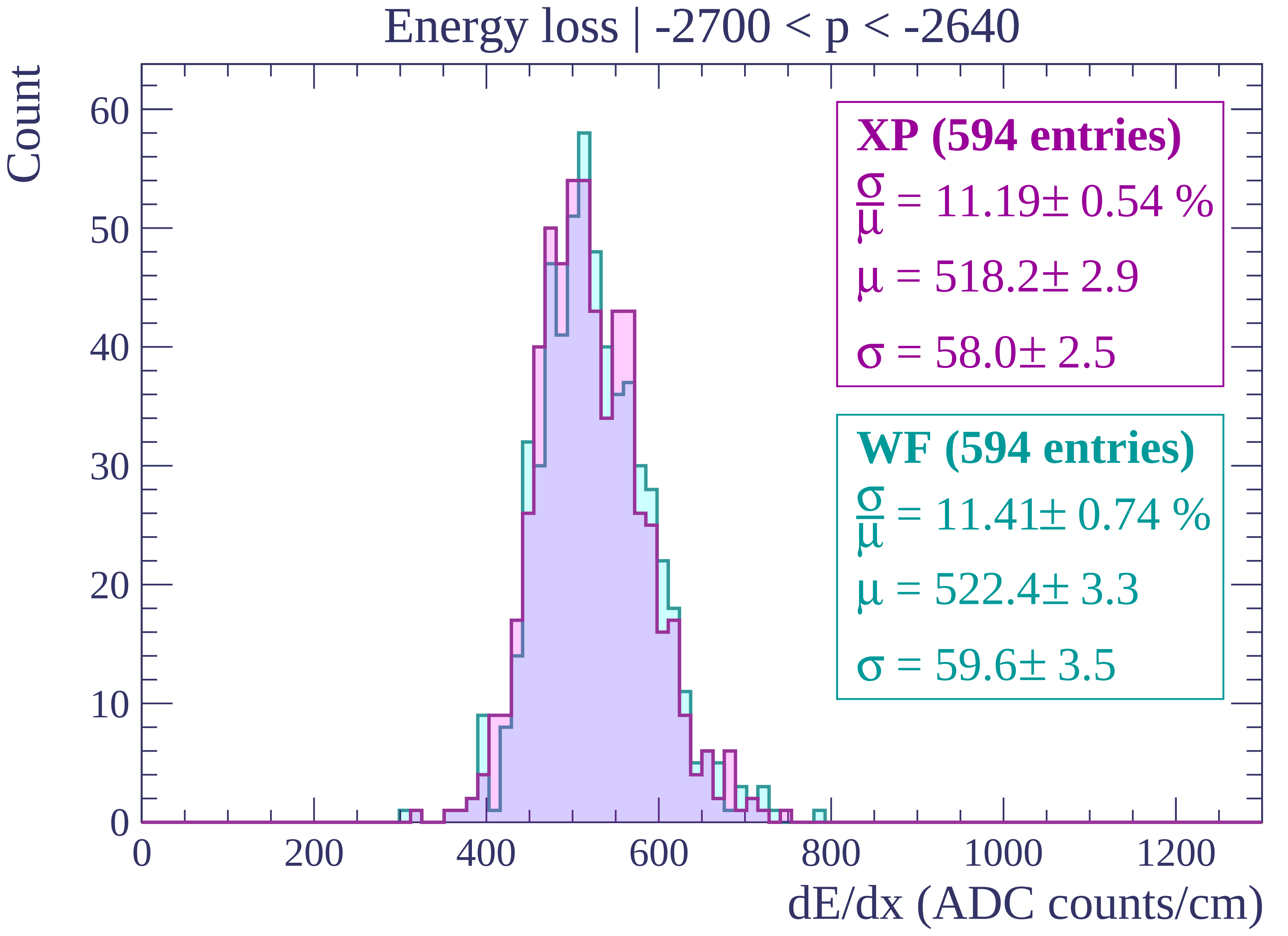
Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (547 entries)
σ/μ = 10.83± 0.55 %
μ = 519.4± 2.9
σ = 56.2± 2.5
WF (547 entries)
σ/μ = 11.12± 0.62 %
μ = 524.3± 3.2
σ = 58.3± 2.9



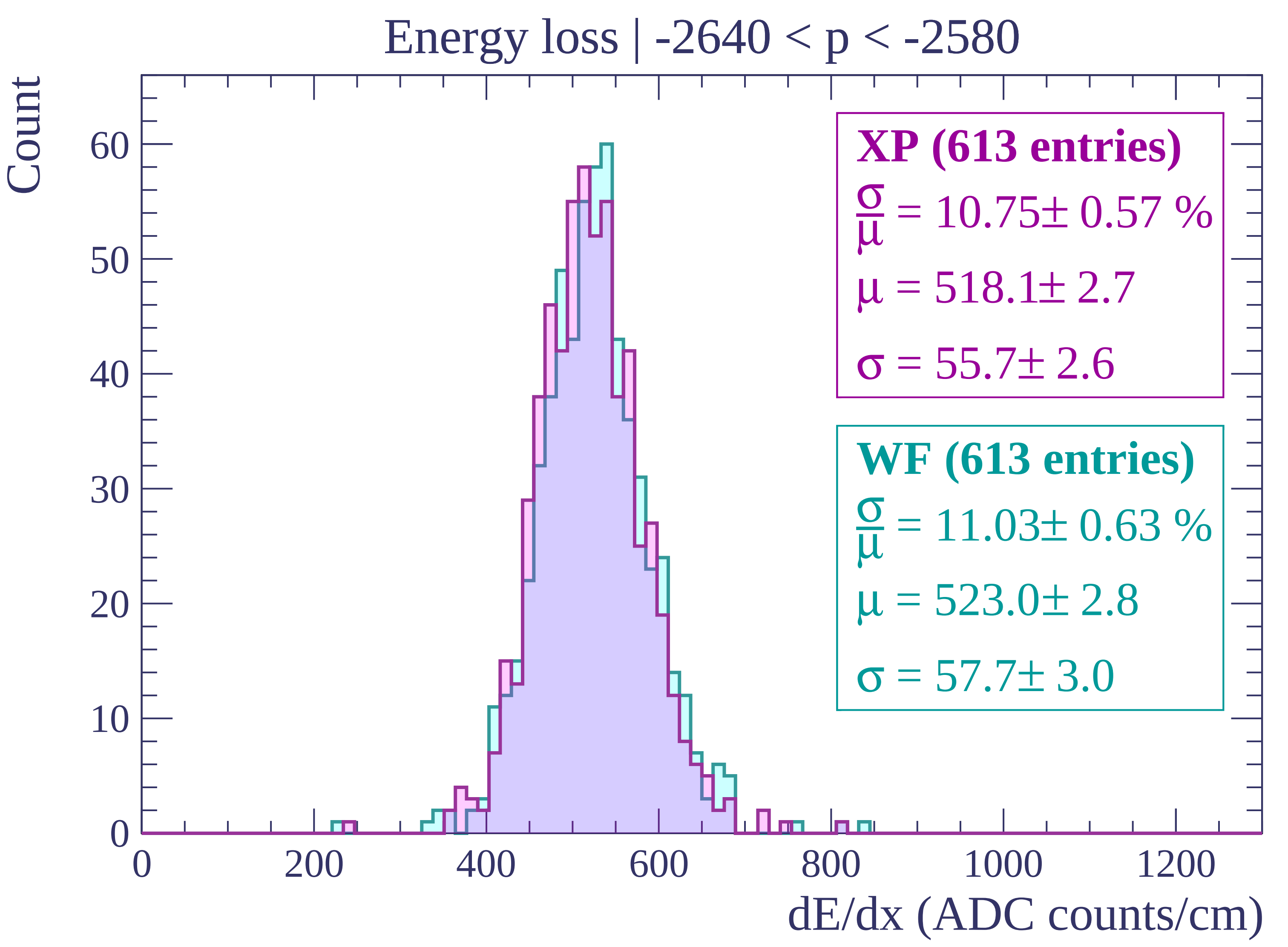
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (562 entries)
σ/μ = 11.85± 0.70 %
μ = 518.7± 3.1
σ = 61.4± 3.2
WF (562 entries)
σ/μ = 11.50± 0.63 %
μ = 520.4± 2.9
σ = 59.8± 2.9



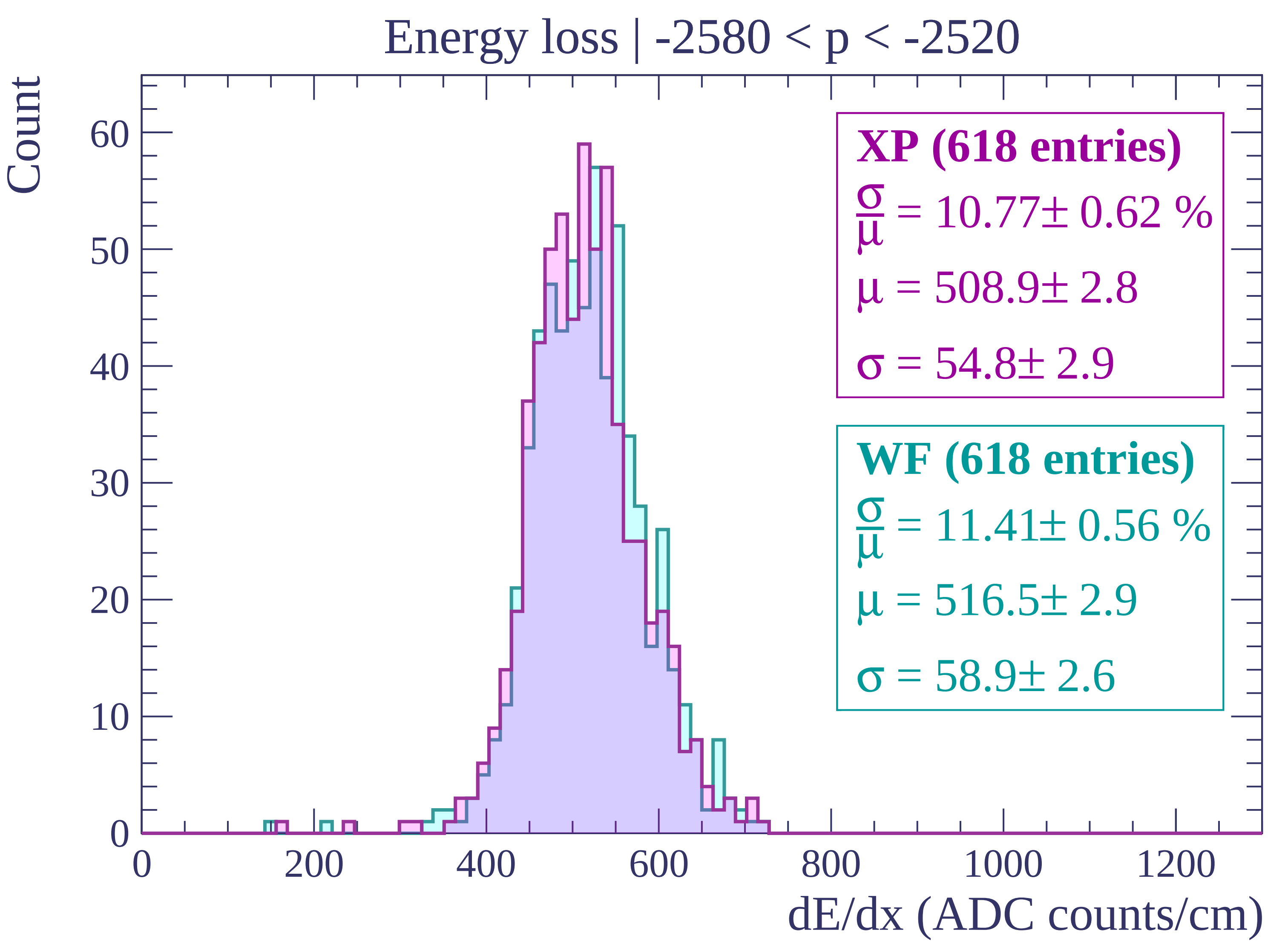
Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (585 entries)
σ/μ = 10.47± 0.55 %
μ = 521.4± 2.8
σ = 54.6± 2.6
WF (585 entries)
σ/μ = 11.76± 0.67 %
μ = 522.4± 3.2
σ = 61.4± 3.1

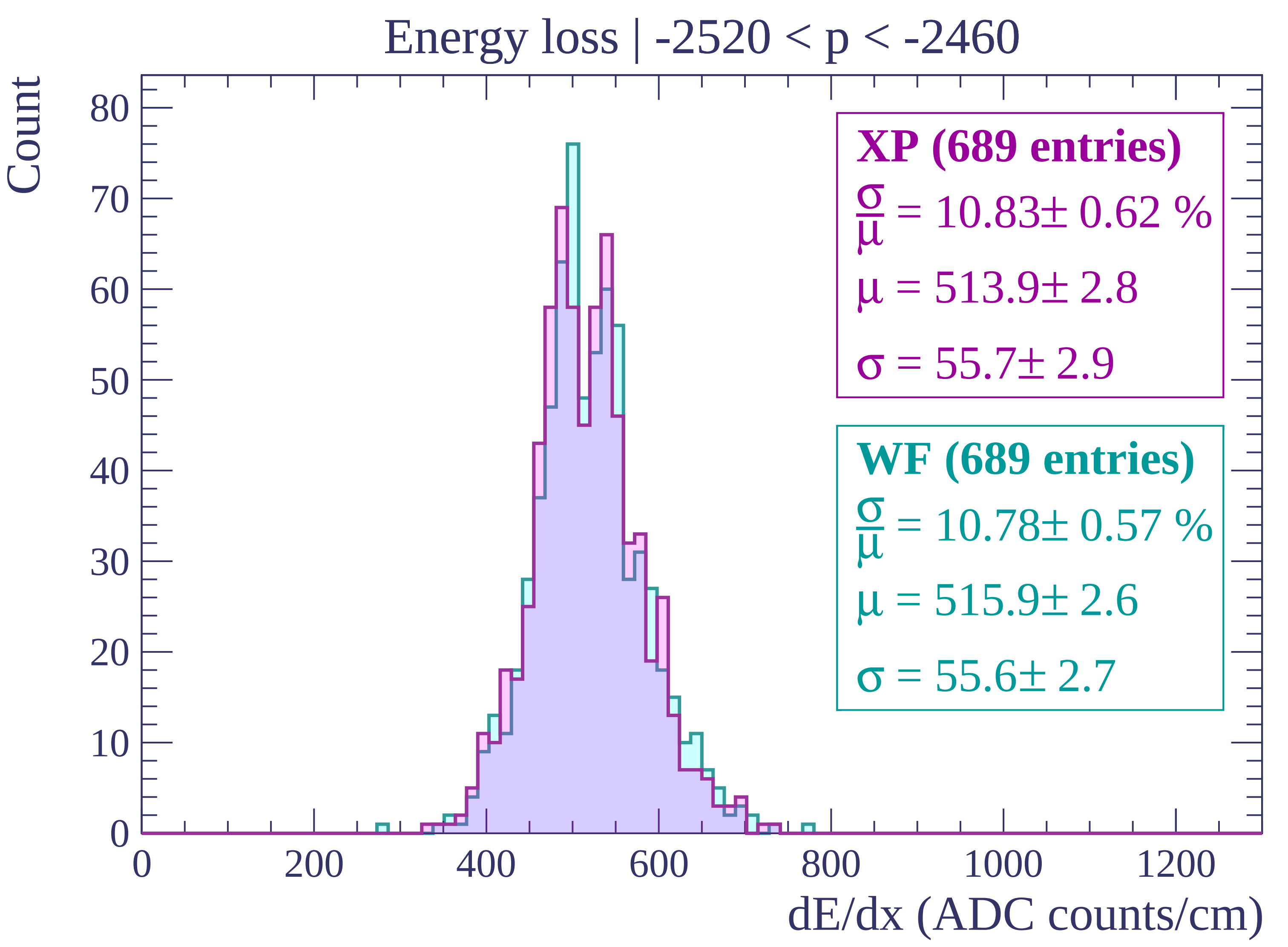


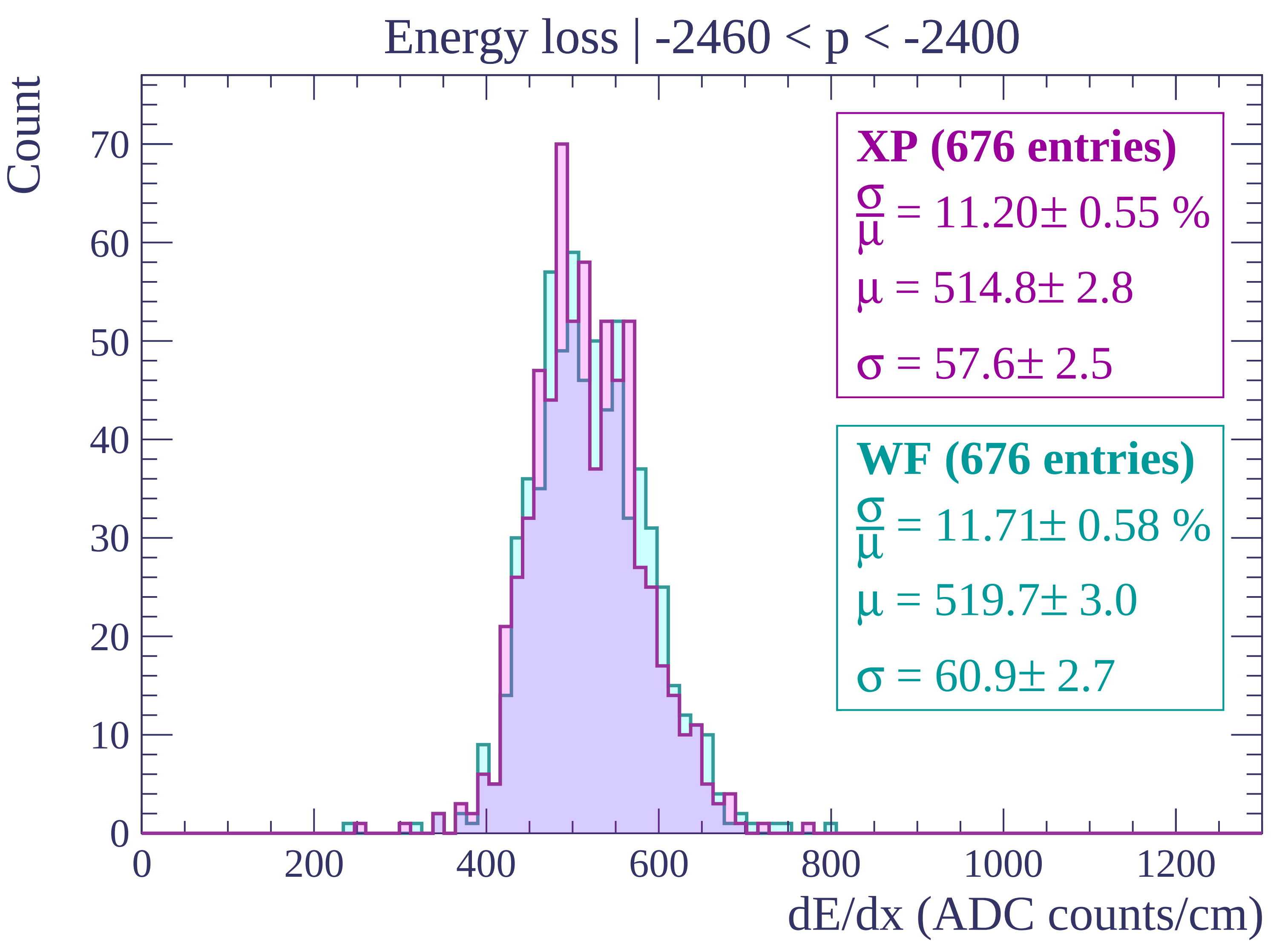
Energy loss | -2700 < p < -2640
Count
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (594 entries)
σ/μ = 11.19± 0.54 %
μ = 518.2± 2.9
σ = 58.0± 2.5
WF (594 entries)
σ/μ = 11.41± 0.74 %
μ = 522.4± 3.3
σ = 59.6± 3.5



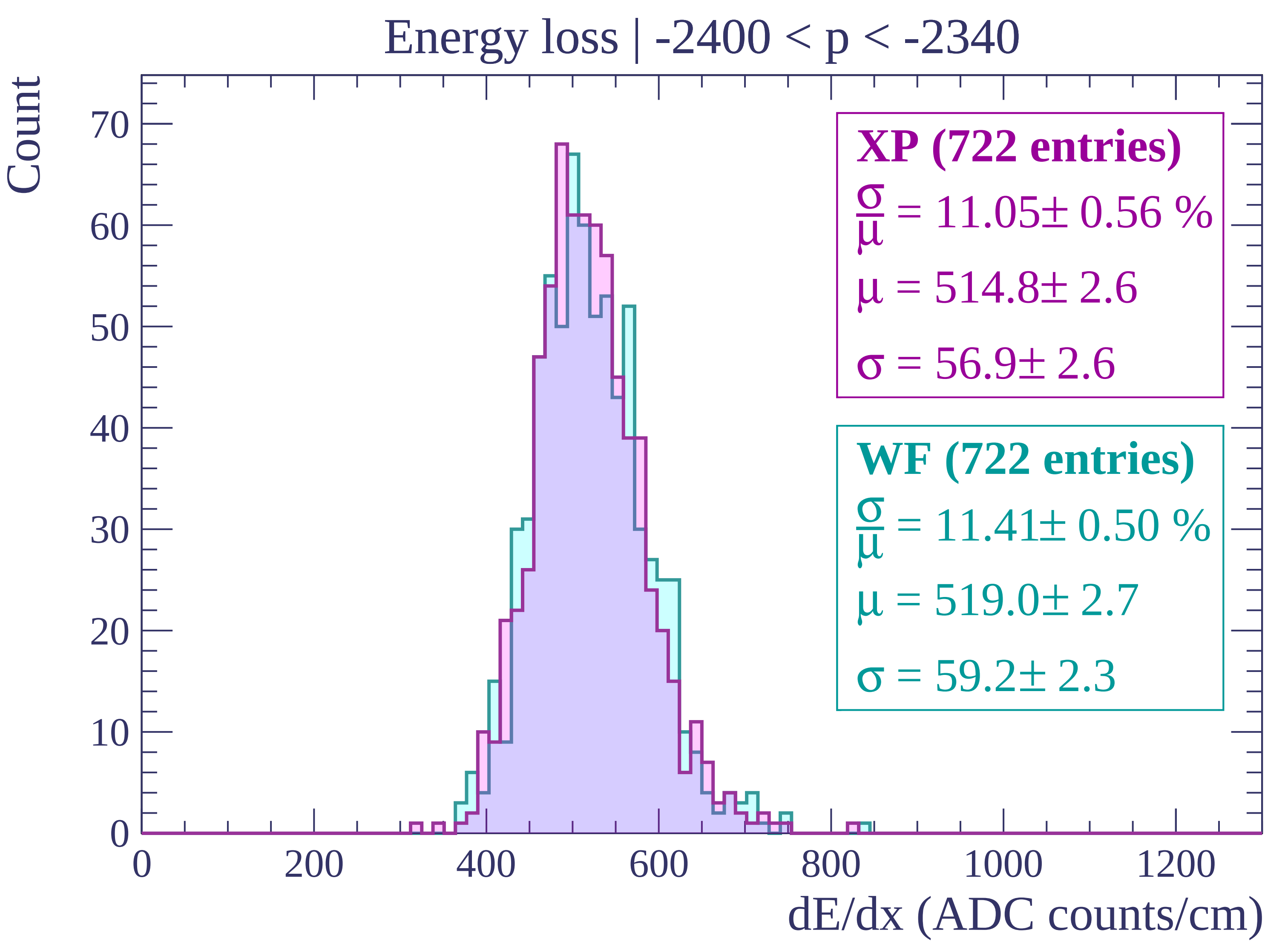
Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP (613 entries)
$\frac{\sigma}{\mu}$ = 10.75± 0.57 %
μ = 518.1± 2.7
σ = 55.7± 2.6
WF (613 entries)
$\frac{\sigma}{\mu}$ = 11.03± 0.63 %
μ = 523.0± 2.8
σ = 57.7± 3.0


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (618 entries)
σ/μ = 10.77± 0.62 %
μ = 508.9± 2.8
σ = 54.8± 2.9
WF (618 entries)
σ/μ = 11.41± 0.56 %
μ = 516.5± 2.9
σ = 58.9± 2.6

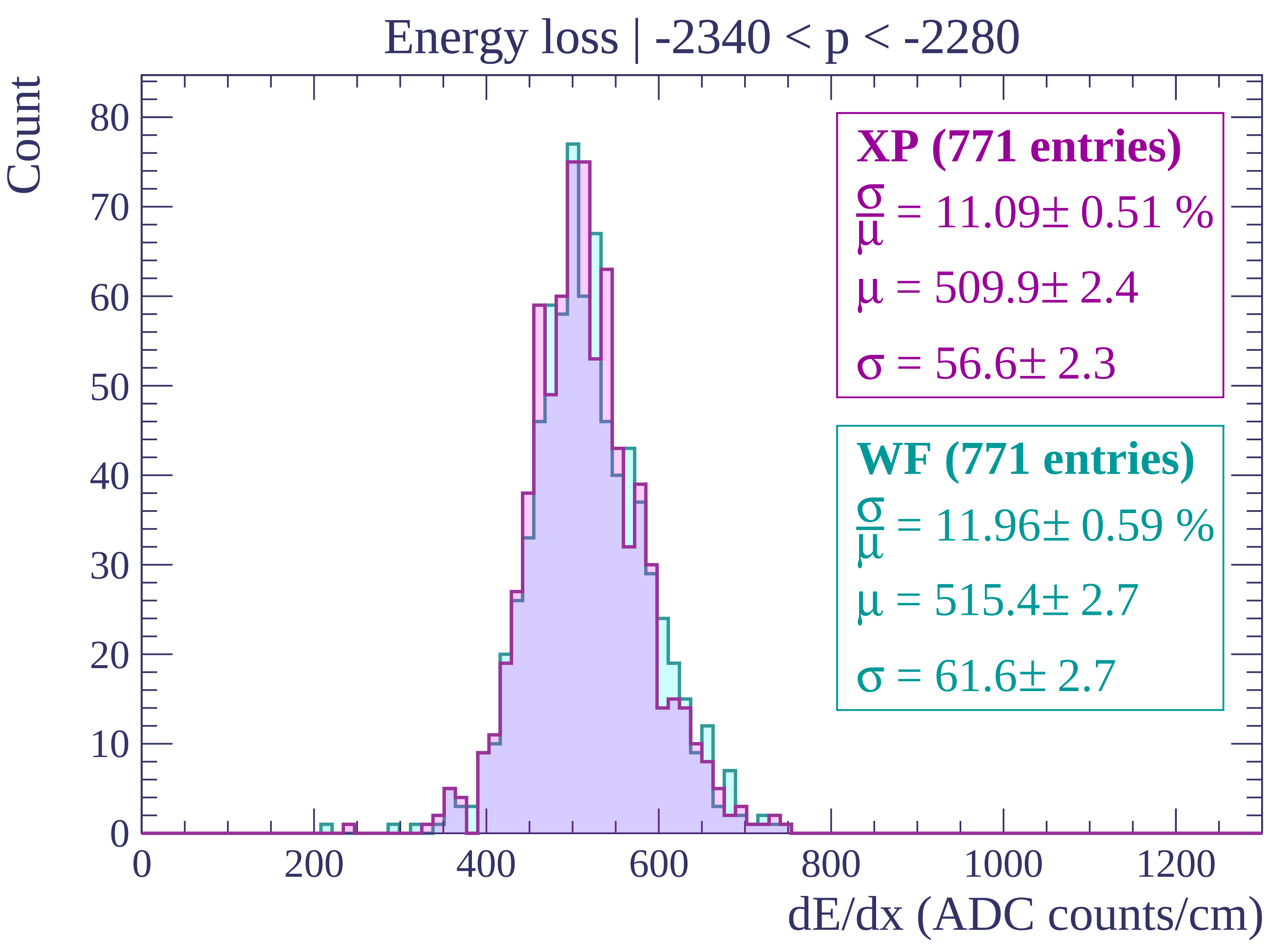

Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP (689 entries)
σ/μ = 10.83± 0.62 %
μ = 513.9± 2.8
σ = 55.7± 2.9
WF (689 entries)
σ/μ = 10.78± 0.57 %
μ = 515.9± 2.6
σ = 55.6± 2.7



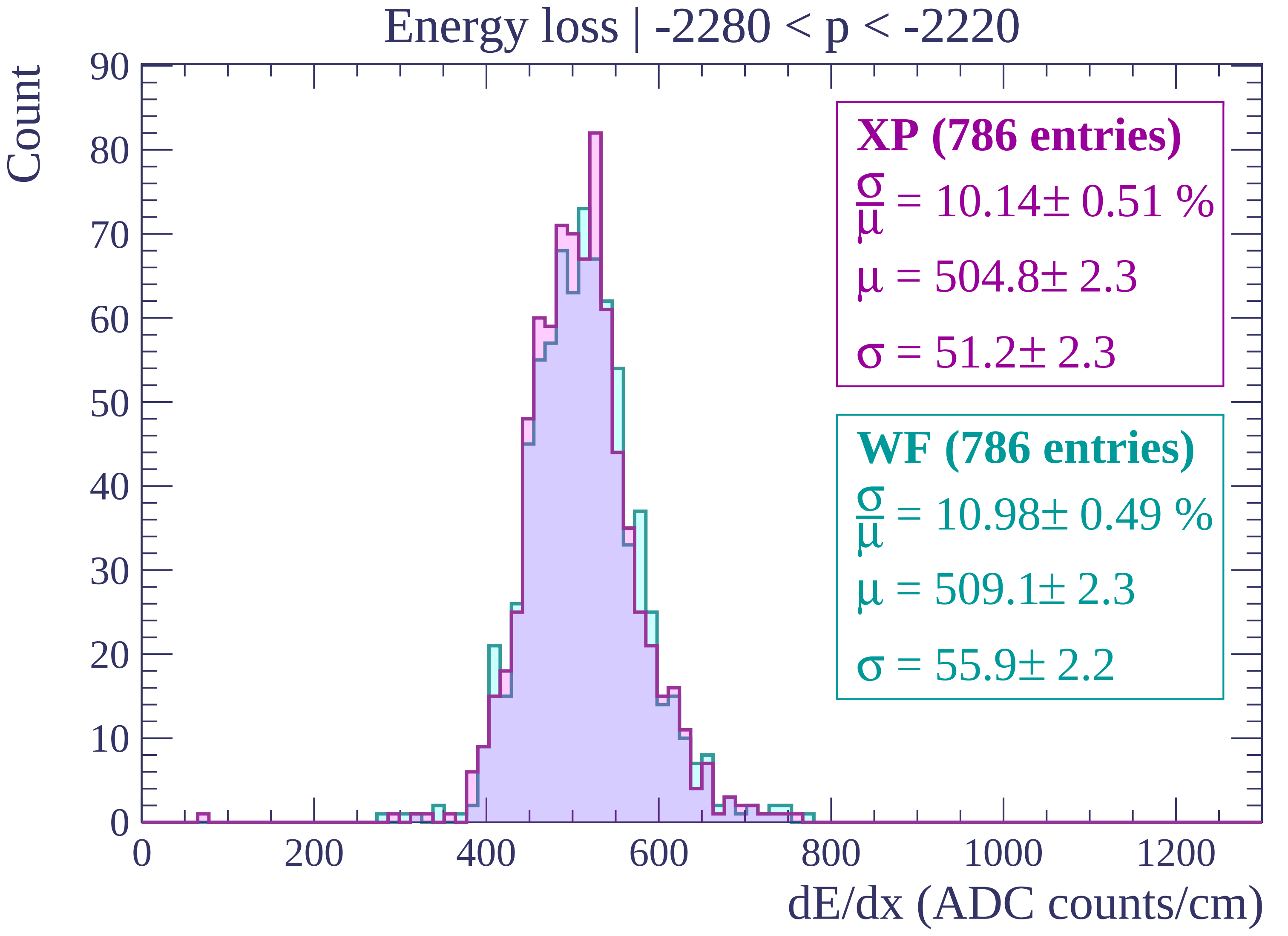
Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (676 entries)
σ/μ = 11.20± 0.55 %
μ = 514.8± 2.8
σ = 57.6± 2.5
WF (676 entries)
σ/μ = 11.71± 0.58 %
μ = 519.7± 3.0
σ = 60.9± 2.7



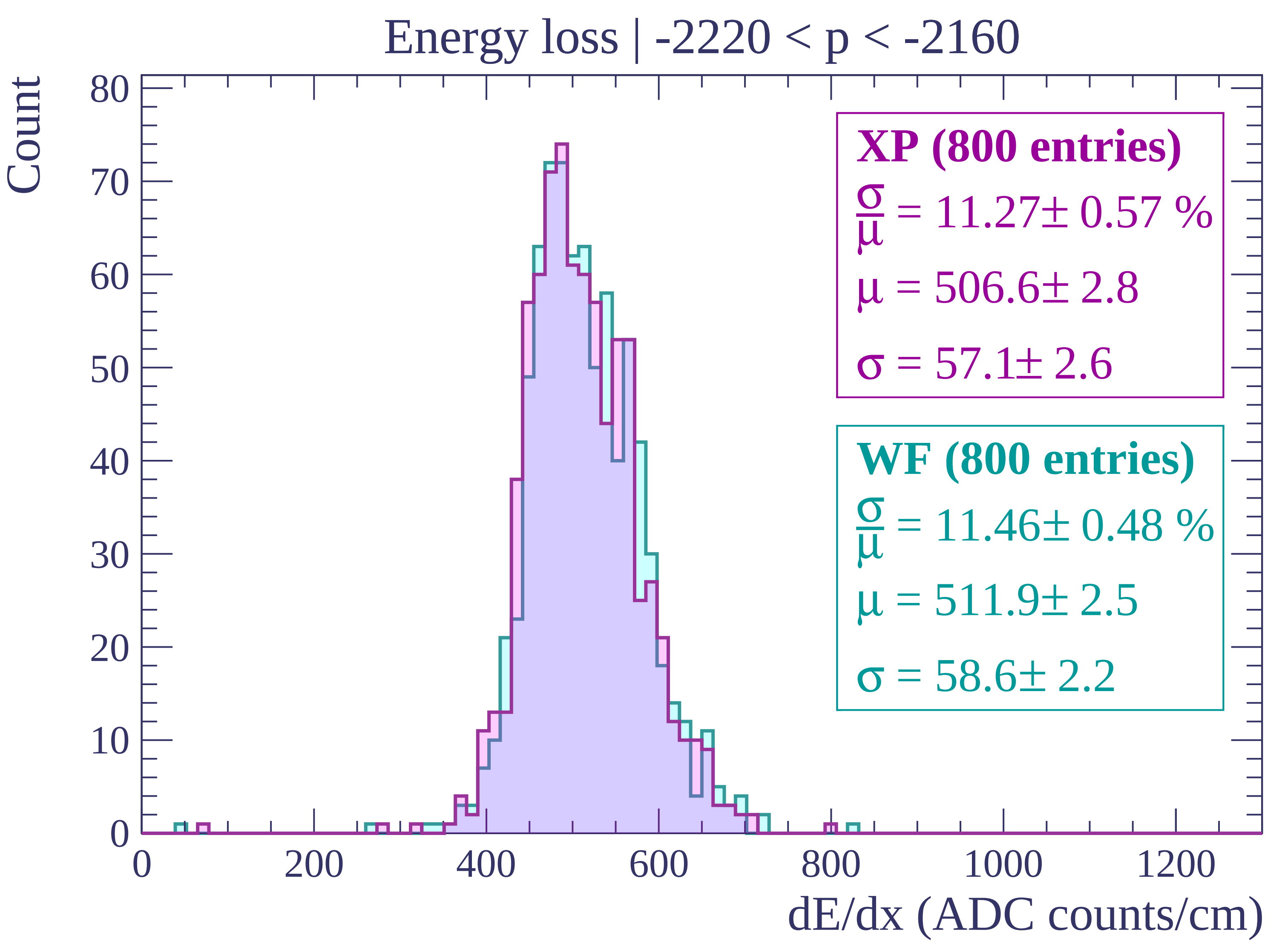
Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (722 entries)
σ/μ = 11.05± 0.56 %
μ = 514.8± 2.6
σ = 56.9± 2.6
WF (722 entries)
σ/μ = 11.41± 0.50 %
μ = 519.0± 2.7
σ = 59.2± 2.3

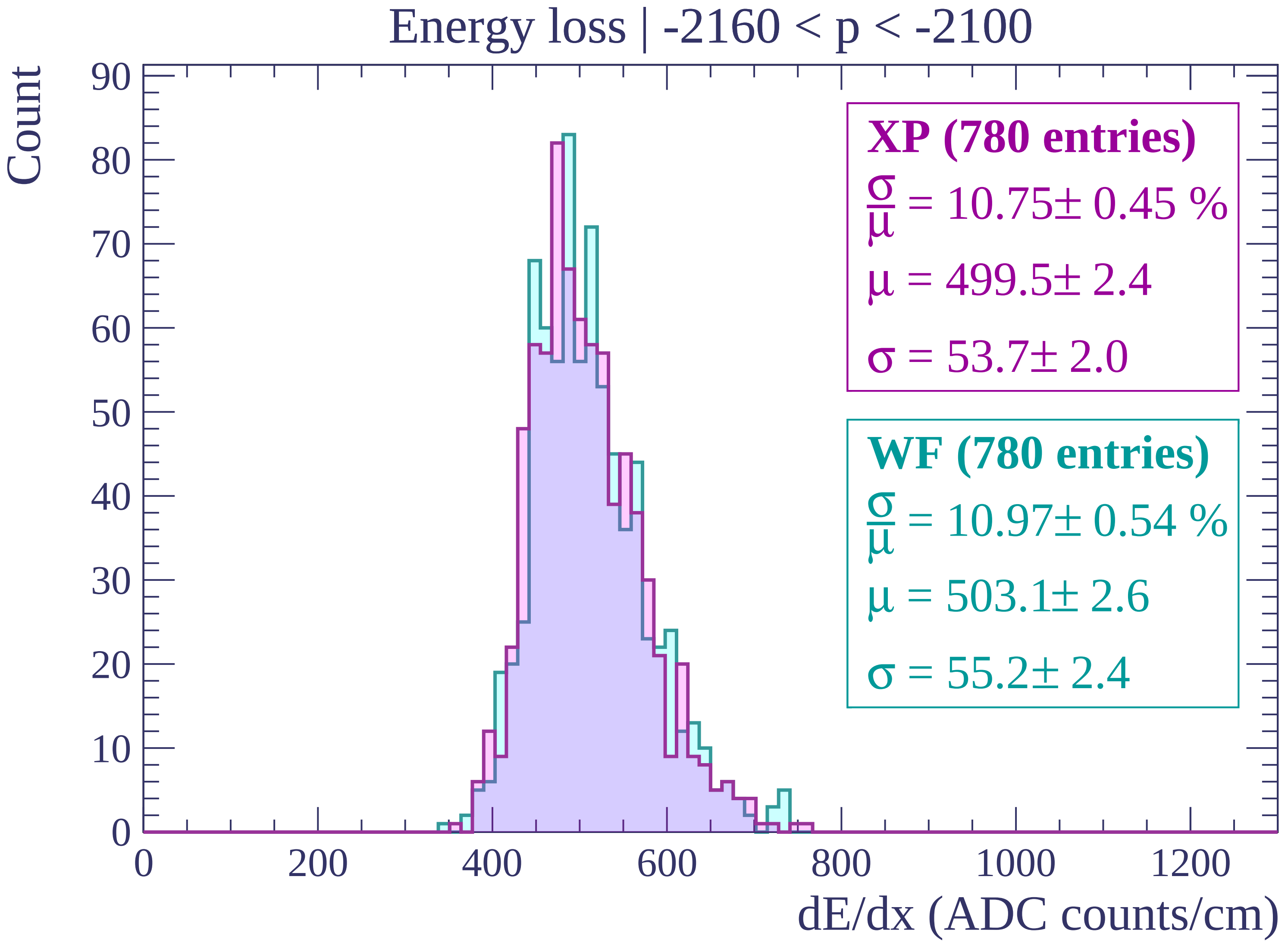


Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (771 entries)
σ/μ = 11.09± 0.51 %
μ = 509.9± 2.4
σ = 56.6± 2.3
WF (771 entries)
σ/μ = 11.96± 0.59 %
μ = 515.4± 2.7
σ = 61.6± 2.7

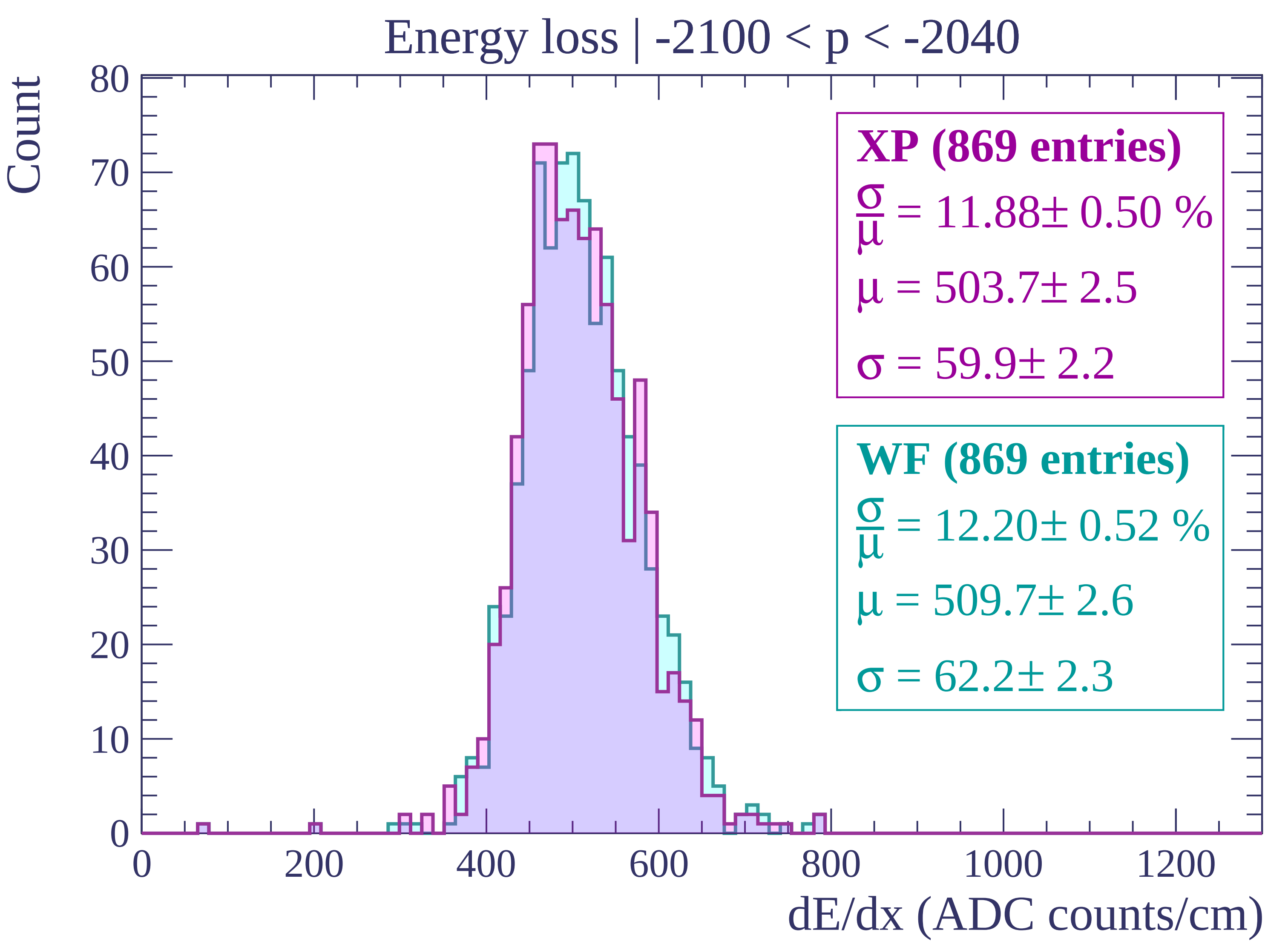


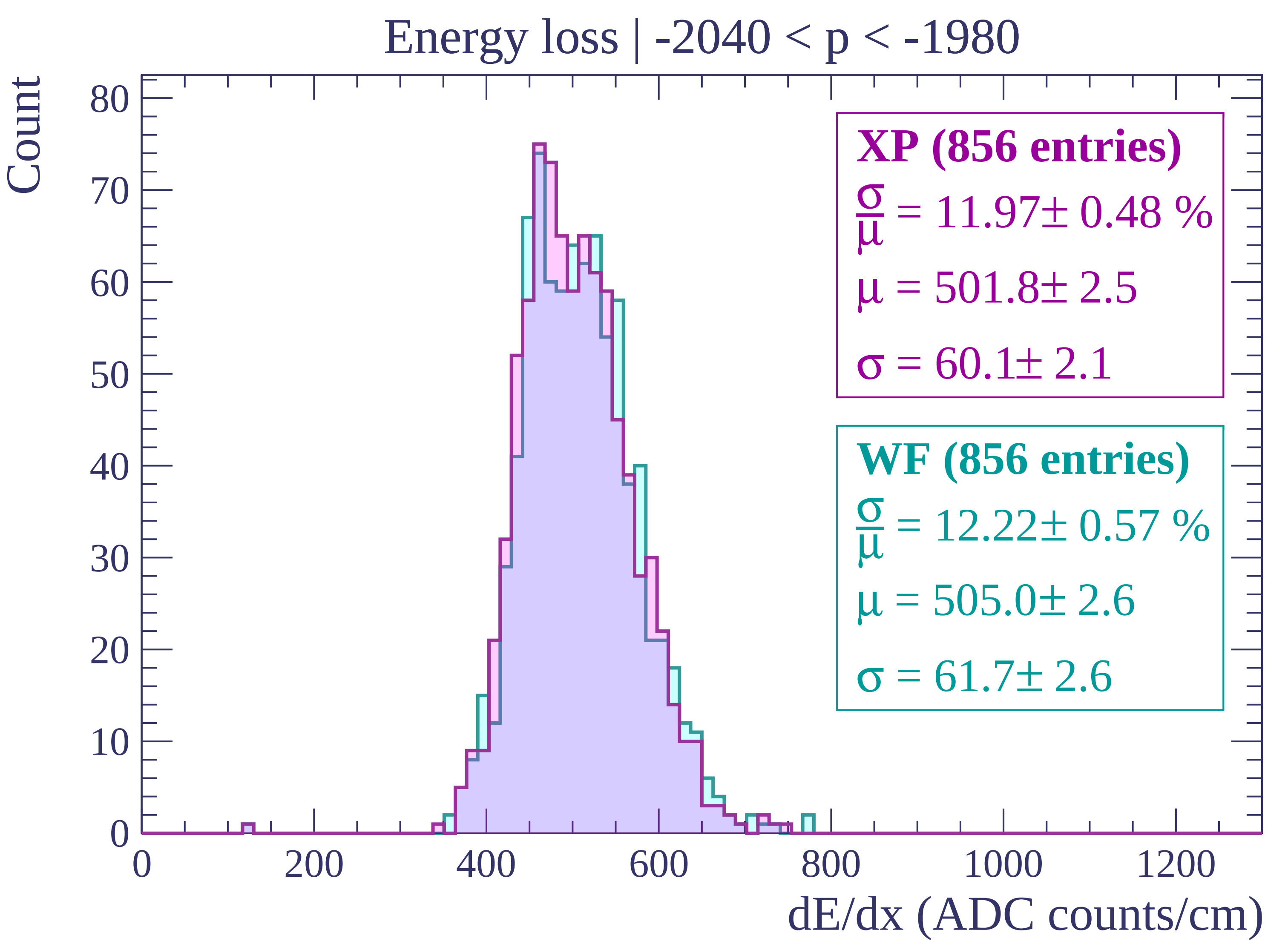
Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (786 entries)
σ/μ = 10.14± 0.51 %
μ = 504.8± 2.3
σ = 51.2± 2.3
WF (786 entries)
σ/μ = 10.98± 0.49 %
μ = 509.1± 2.3
σ = 55.9± 2.2


Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
XP (800 entries)
σ/μ = 11.27± 0.57 %
μ = 506.6± 2.8
σ = 57.1± 2.6
WF (800 entries)
σ/μ = 11.46± 0.48 %
μ = 511.9± 2.5
σ = 58.6± 2.2

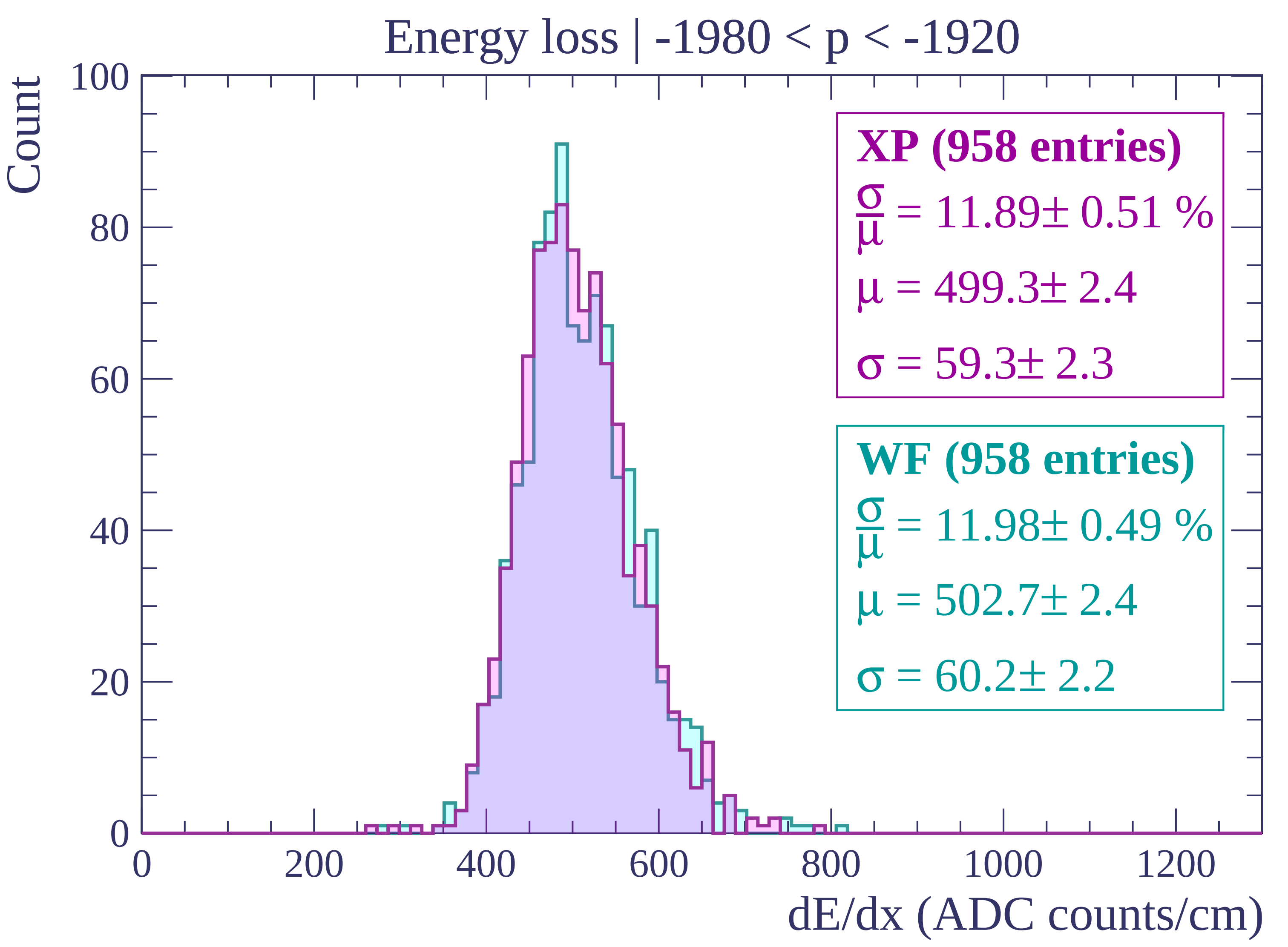


Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (780 entries)
σ/μ = 10.75± 0.45 %
μ = 499.5± 2.4
σ = 53.7± 2.0
WF (780 entries)
σ/μ = 10.97± 0.54 %
μ = 503.1± 2.6
σ = 55.2± 2.4

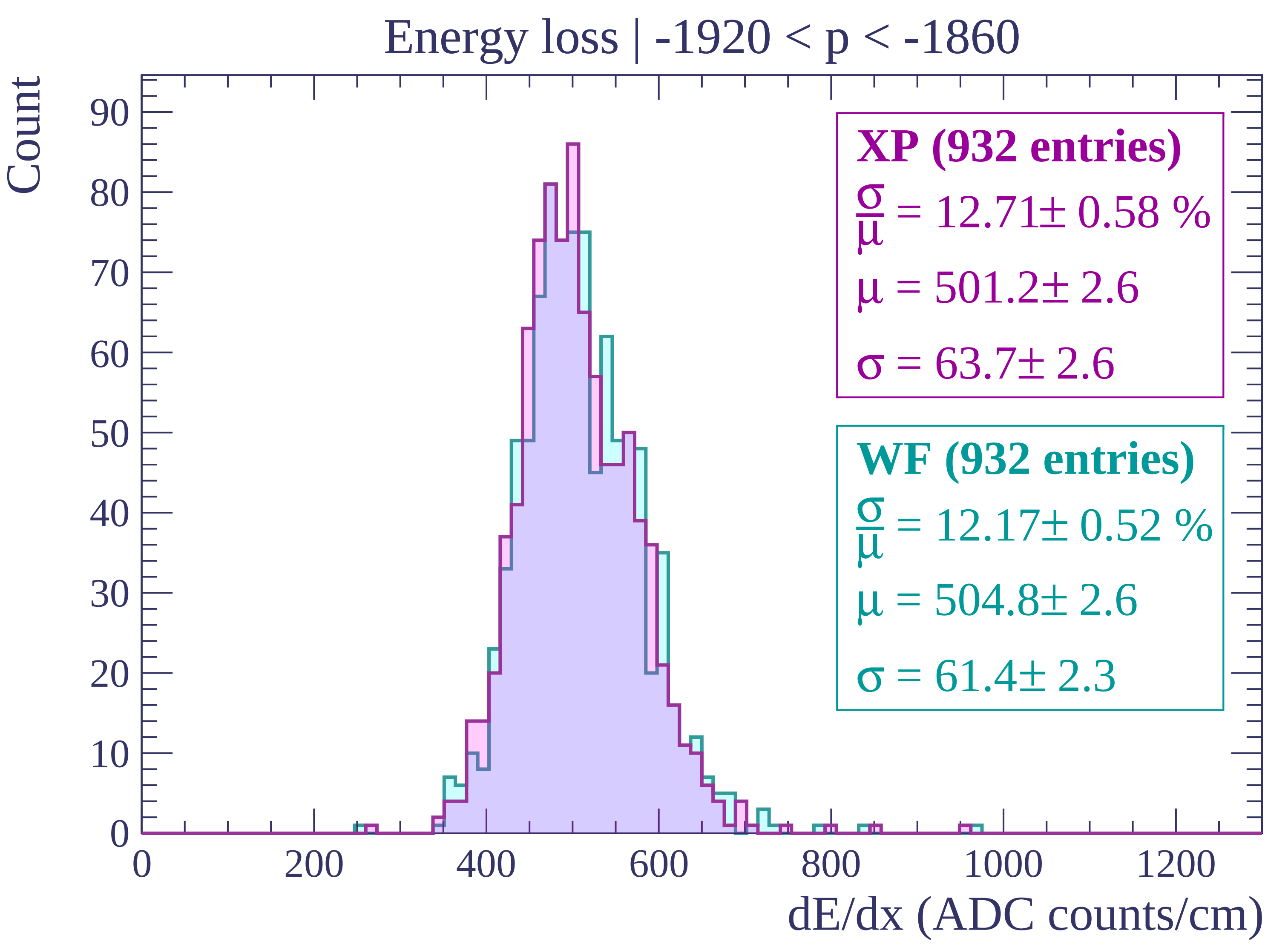

Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (869 entries)
σ/μ = 11.88± 0.50 %
μ = 503.7± 2.5
σ = 59.9± 2.2
WF (869 entries)
σ/μ = 12.20± 0.52 %
μ = 509.7± 2.6
σ = 62.2± 2.3

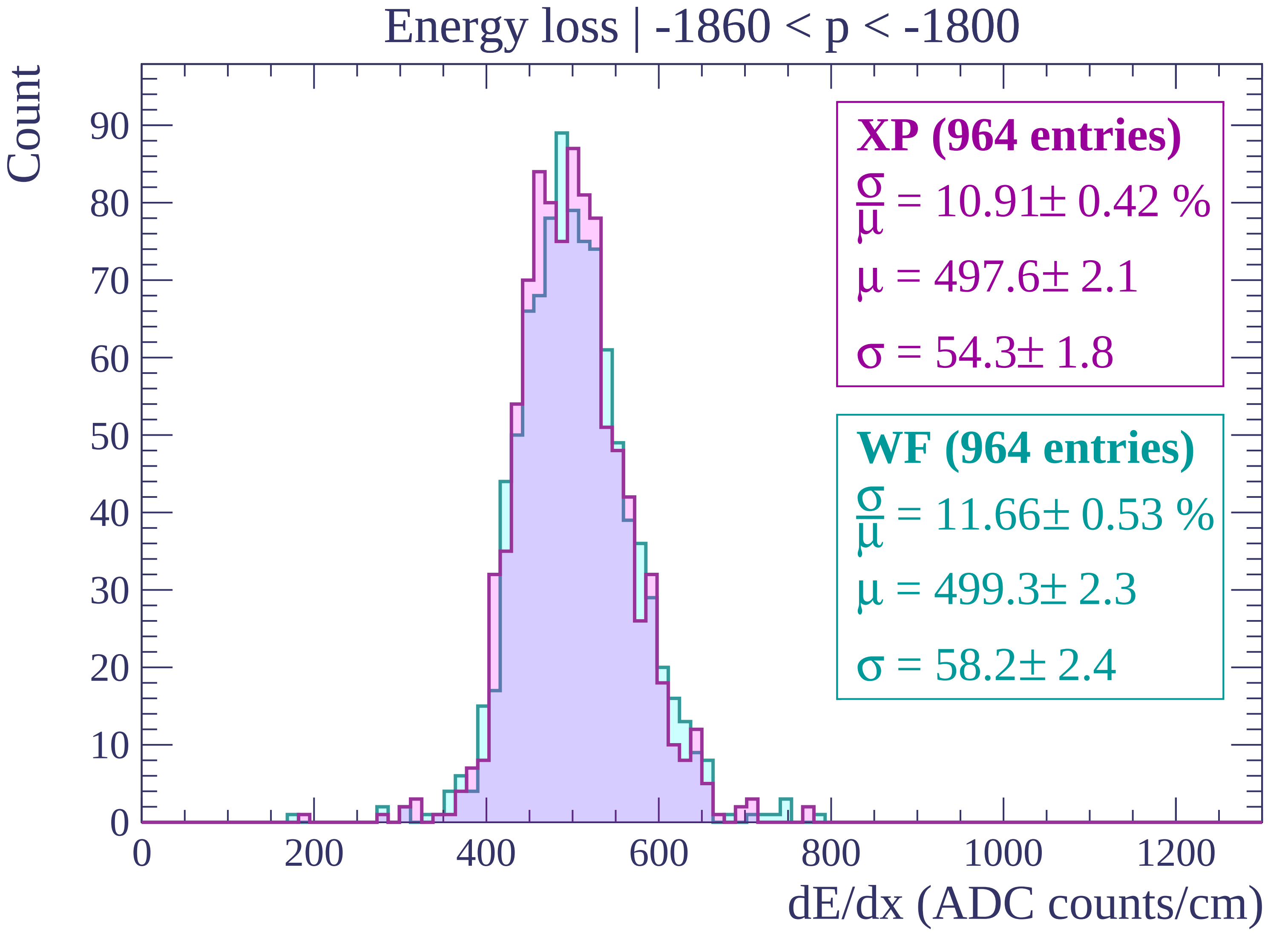


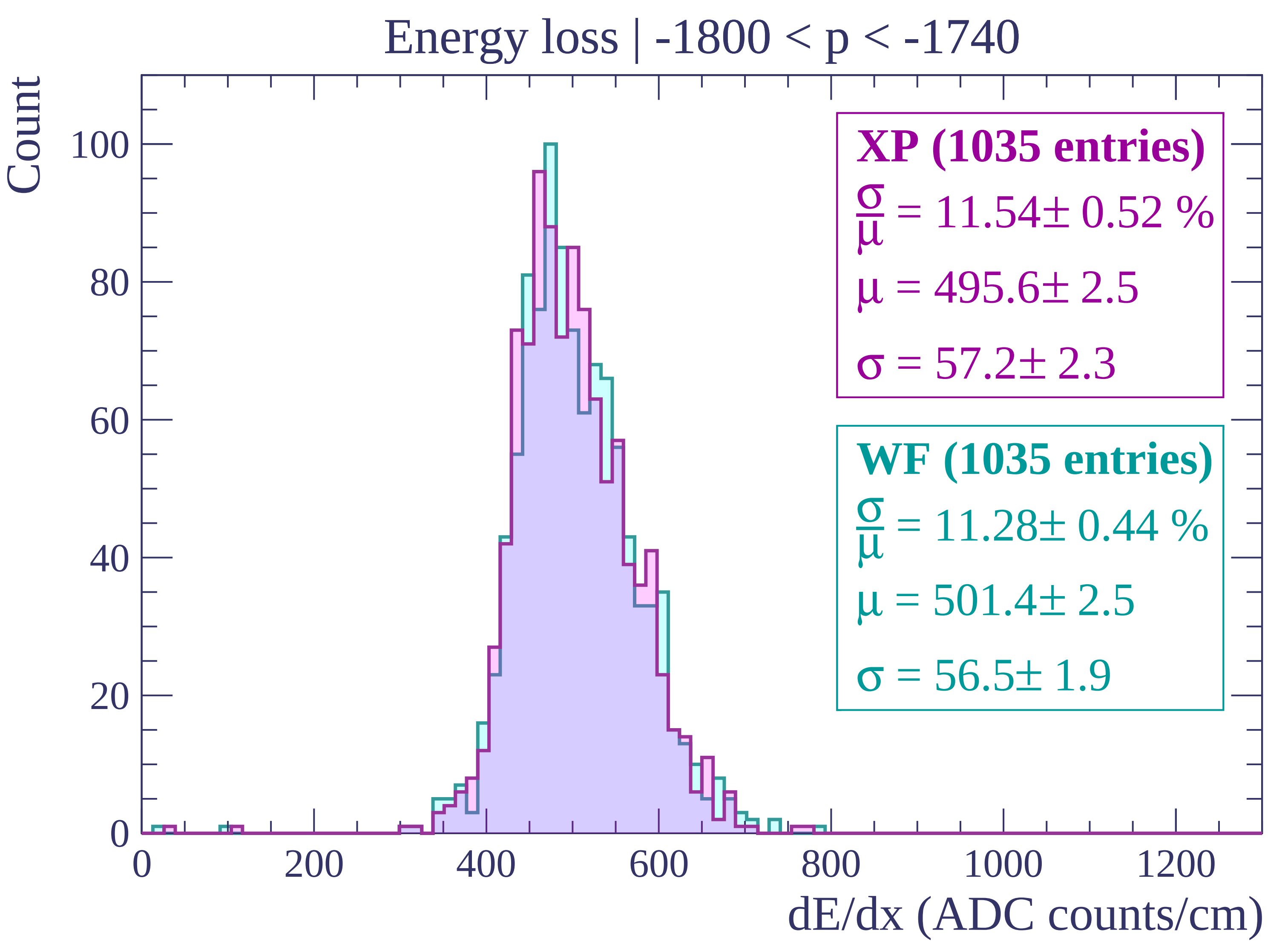
Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
XP (856 entries)
σ/μ = 11.97± 0.48 %
μ = 501.8± 2.5
σ = 60.1± 2.1
WF (856 entries)
σ/μ = 12.22± 0.57 %
μ = 505.0± 2.6
σ = 61.7± 2.6

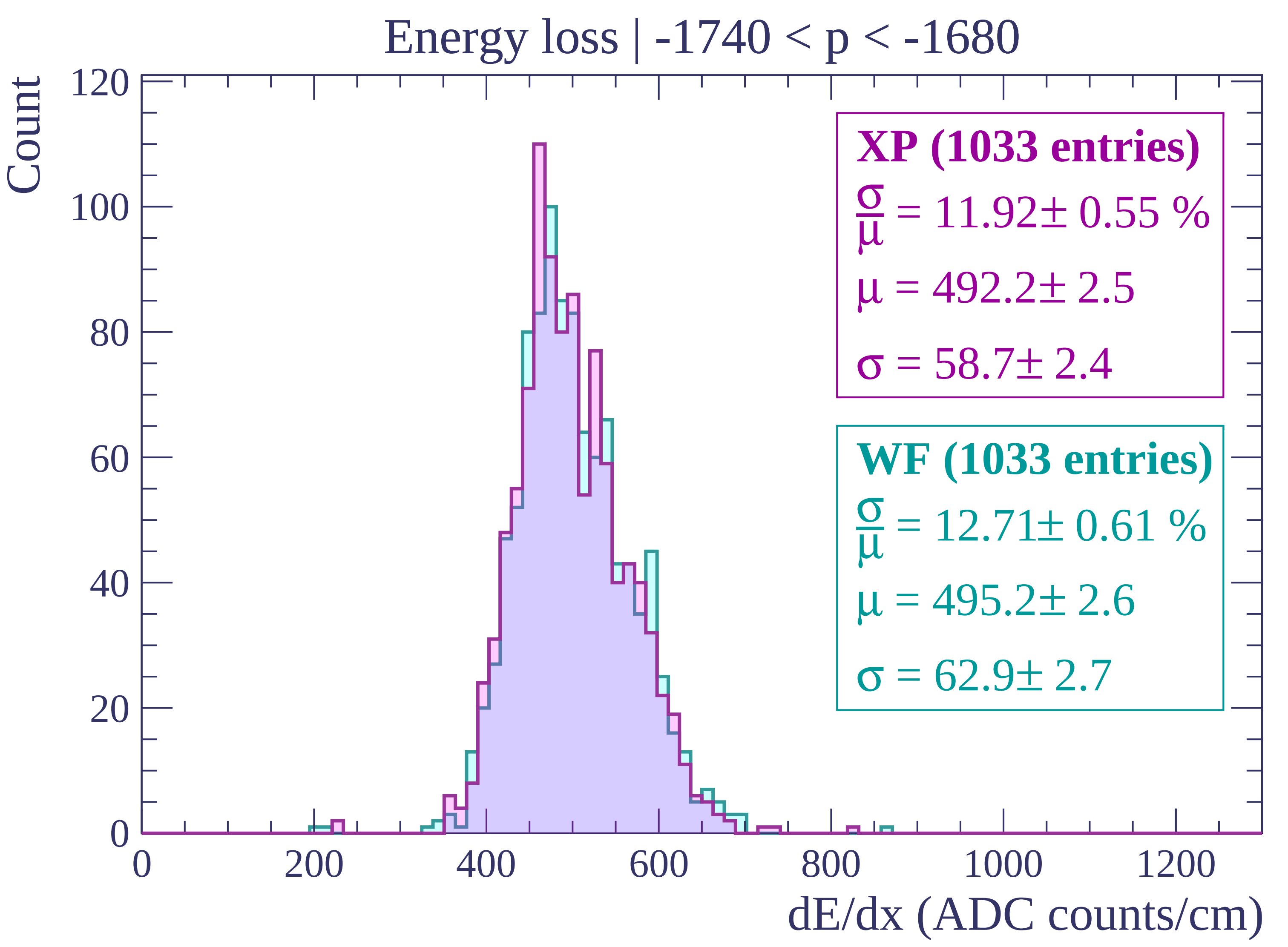


Energy loss | -1980 < p < -1920
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (958 entries)
σ/μ = 11.89± 0.51 %
μ = 499.3± 2.4
σ = 59.3± 2.3
WF (958 entries)
σ/μ = 11.98± 0.49 %
μ = 502.7± 2.4
σ = 60.2± 2.2

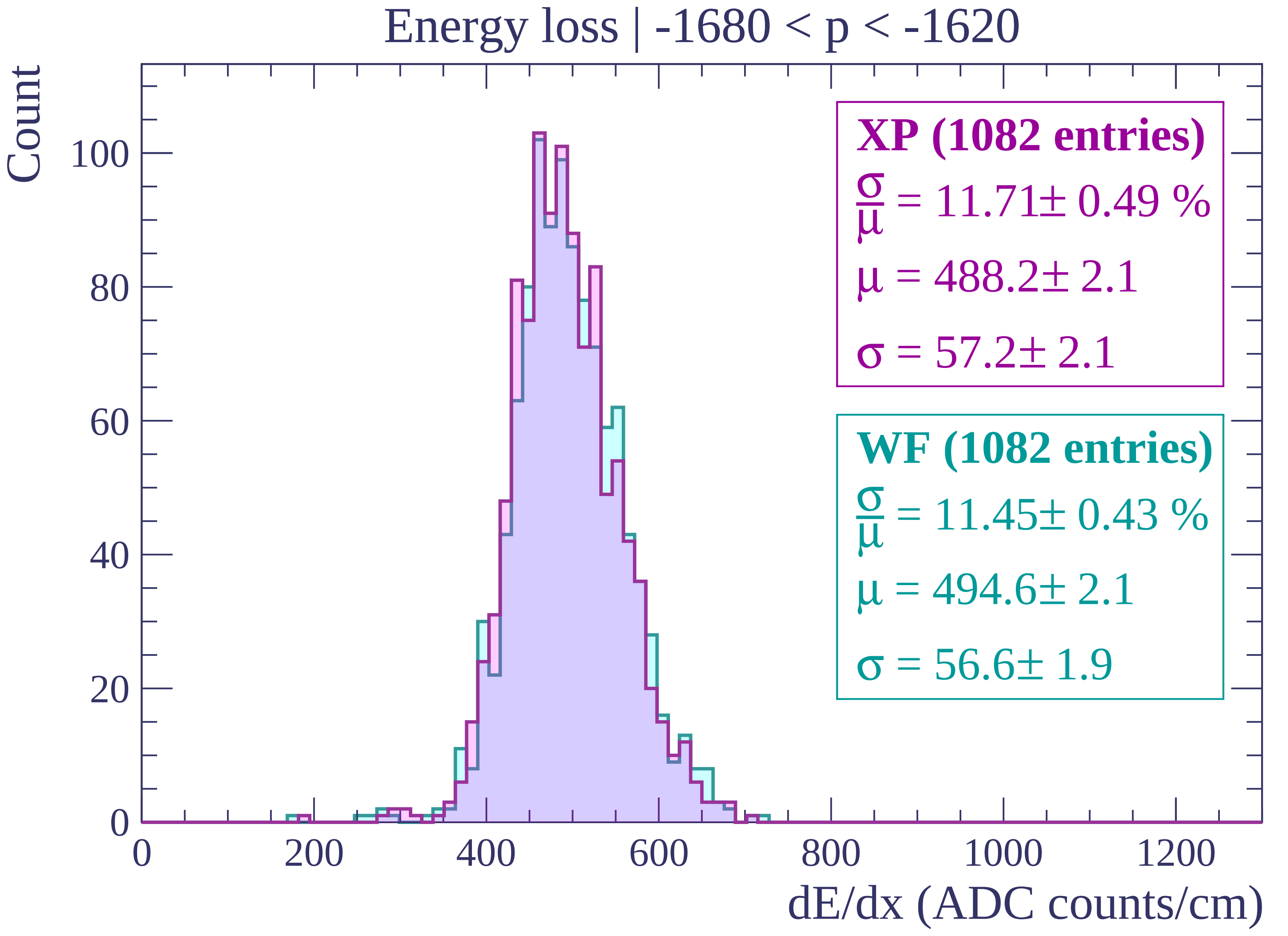

Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
XP (932 entries)
σ/μ = 12.71 ± 0.58 %
μ = 501.2 ± 2.6
σ = 63.7 ± 2.6
WF (932 entries)
σ/μ = 12.17 ± 0.52 %
μ = 504.8 ± 2.6
σ = 61.4 ± 2.3
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200


Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (964 entries)
σ/μ = 10.91± 0.42 %
μ = 497.6± 2.1
σ = 54.3± 1.8
WF (964 entries)
σ/μ = 11.66± 0.53 %
μ = 499.3± 2.3
σ = 58.2± 2.4

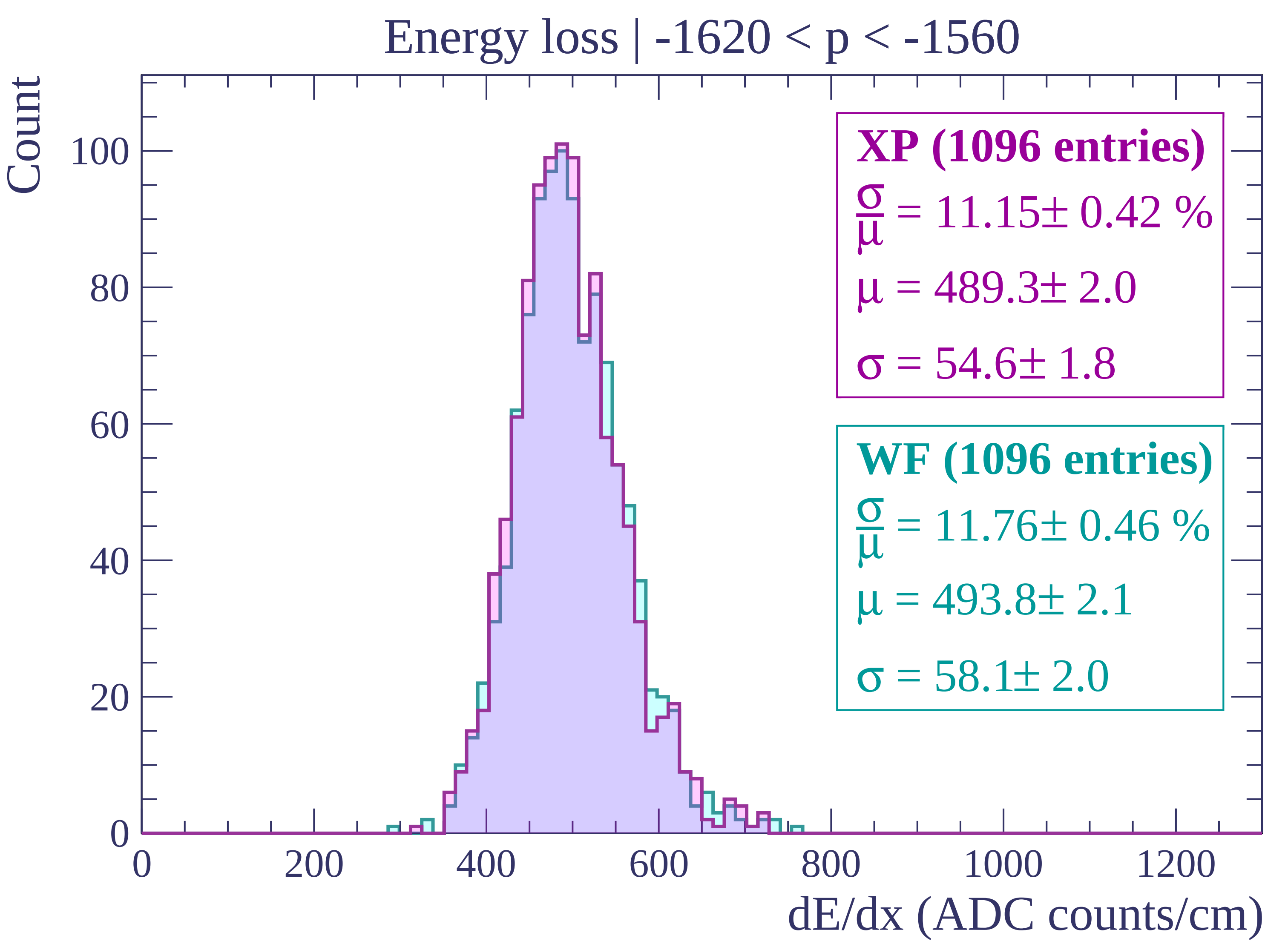

Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1035 entries)
σ/μ = 11.54± 0.52 %
μ = 495.6± 2.5
σ = 57.2± 2.3
WF (1035 entries)
σ/μ = 11.28± 0.44 %
μ = 501.4± 2.5
σ = 56.5± 1.9



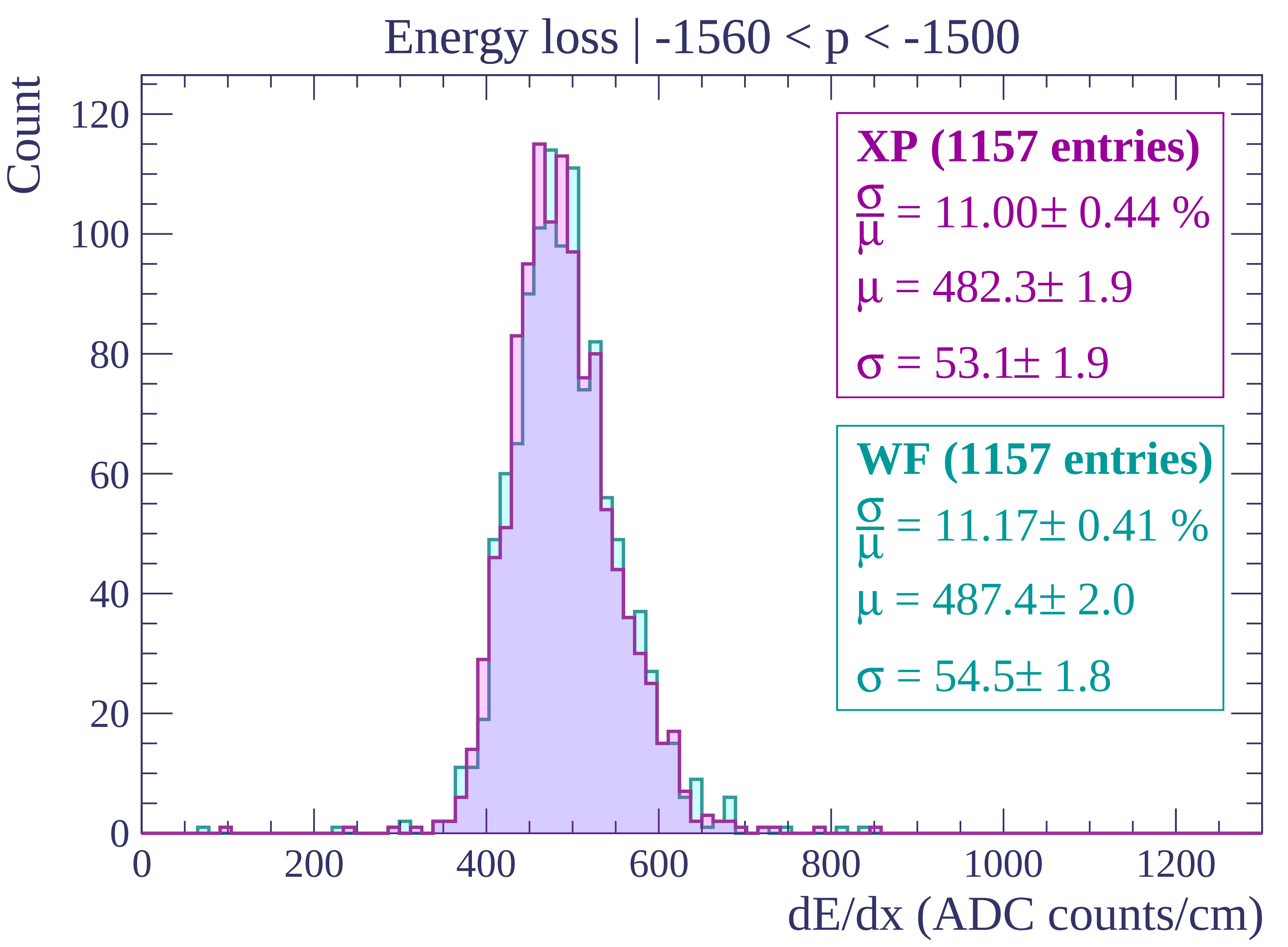
Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
XP (1033 entries)
$\frac{\sigma}{\mu}$ = 11.92± 0.55 %
μ = 492.2± 2.5
σ = 58.7± 2.4
WF (1033 entries)
$\frac{\sigma}{\mu}$ = 12.71± 0.61 %
μ = 495.2± 2.6
σ = 62.9± 2.7



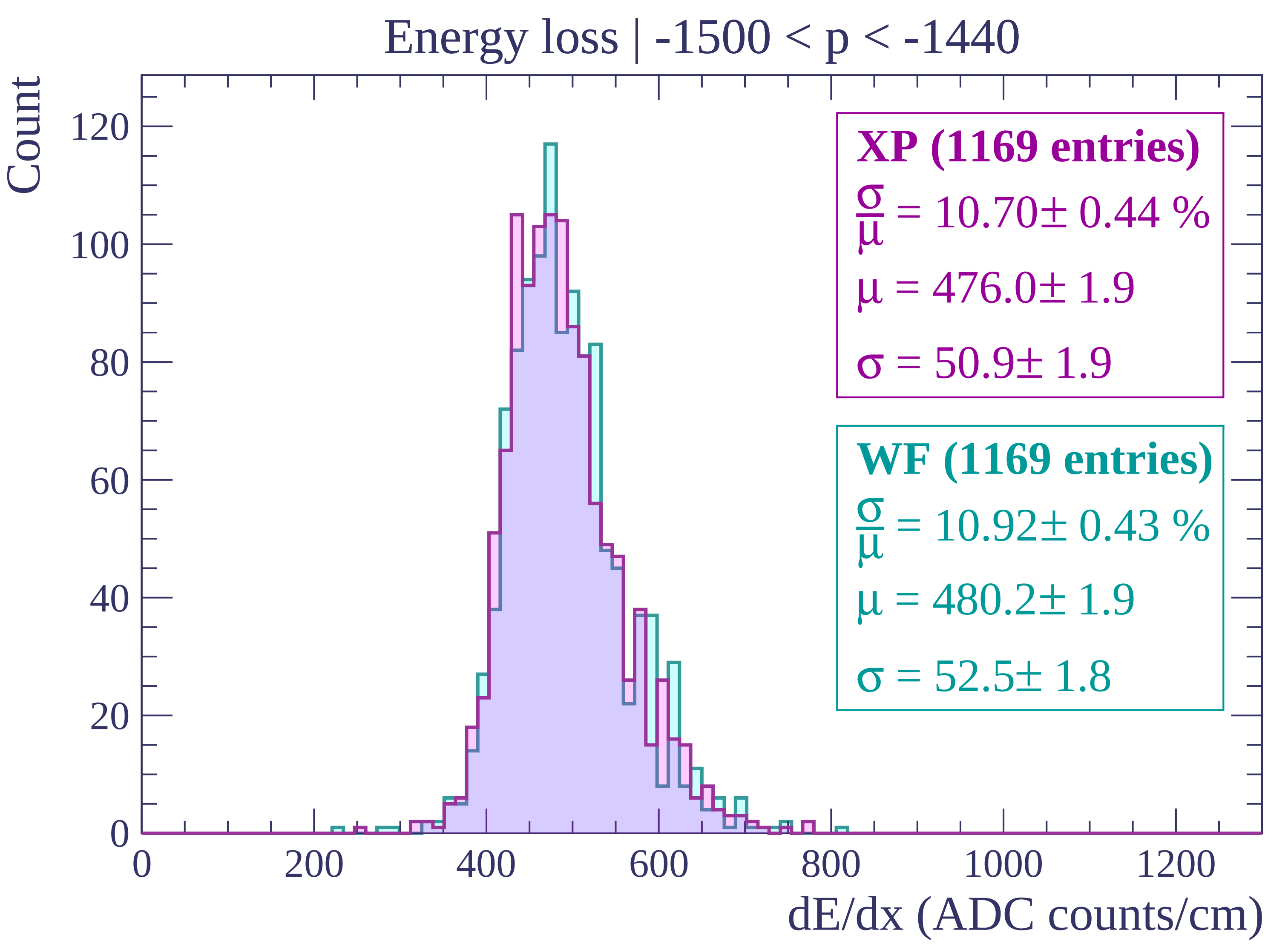
Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1082 entries)
$\frac{\sigma}{\mu}$ = 11.71± 0.49 %
μ = 488.2± 2.1
σ = 57.2± 2.1
WF (1082 entries)
$\frac{\sigma}{\mu}$ = 11.45± 0.43 %
μ = 494.6± 2.1
σ = 56.6± 1.9



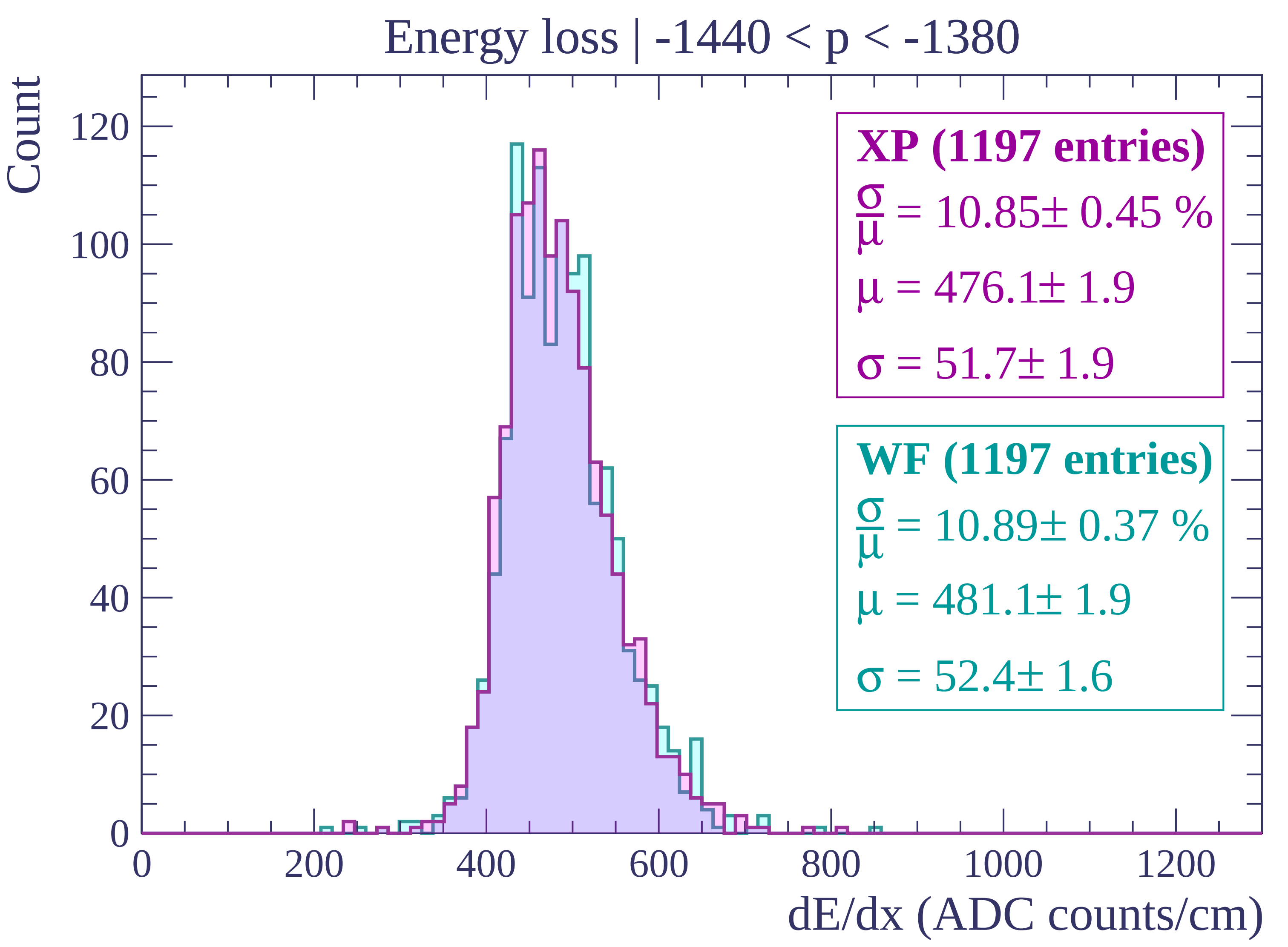
Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1096 entries)
$\frac{\sigma}{\mu}$ = 11.15± 0.42 %
μ = 489.3± 2.0
σ = 54.6± 1.8
WF (1096 entries)
$\frac{\sigma}{\mu}$ = 11.76± 0.46 %
μ = 493.8± 2.1
σ = 58.1± 2.0

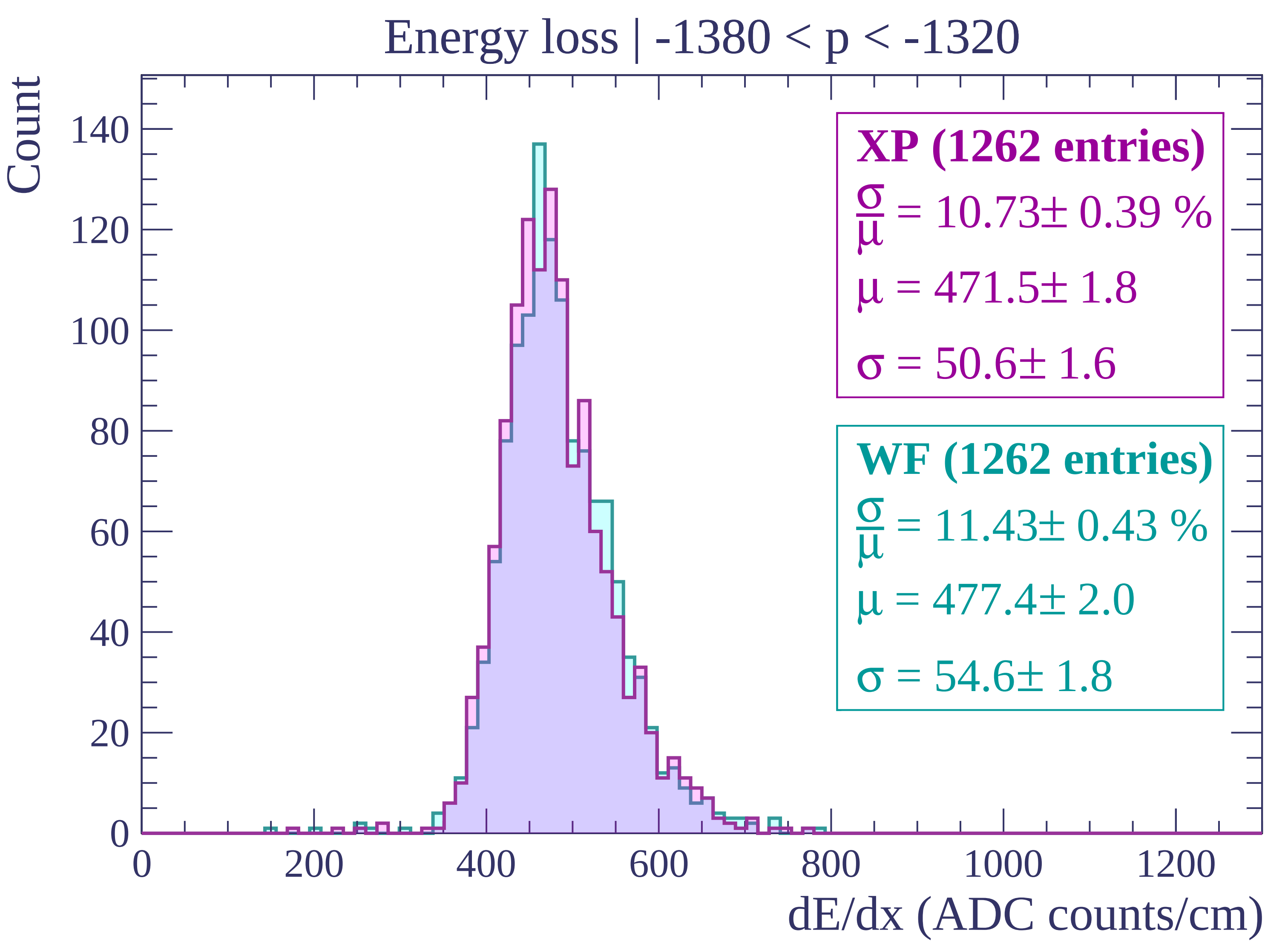


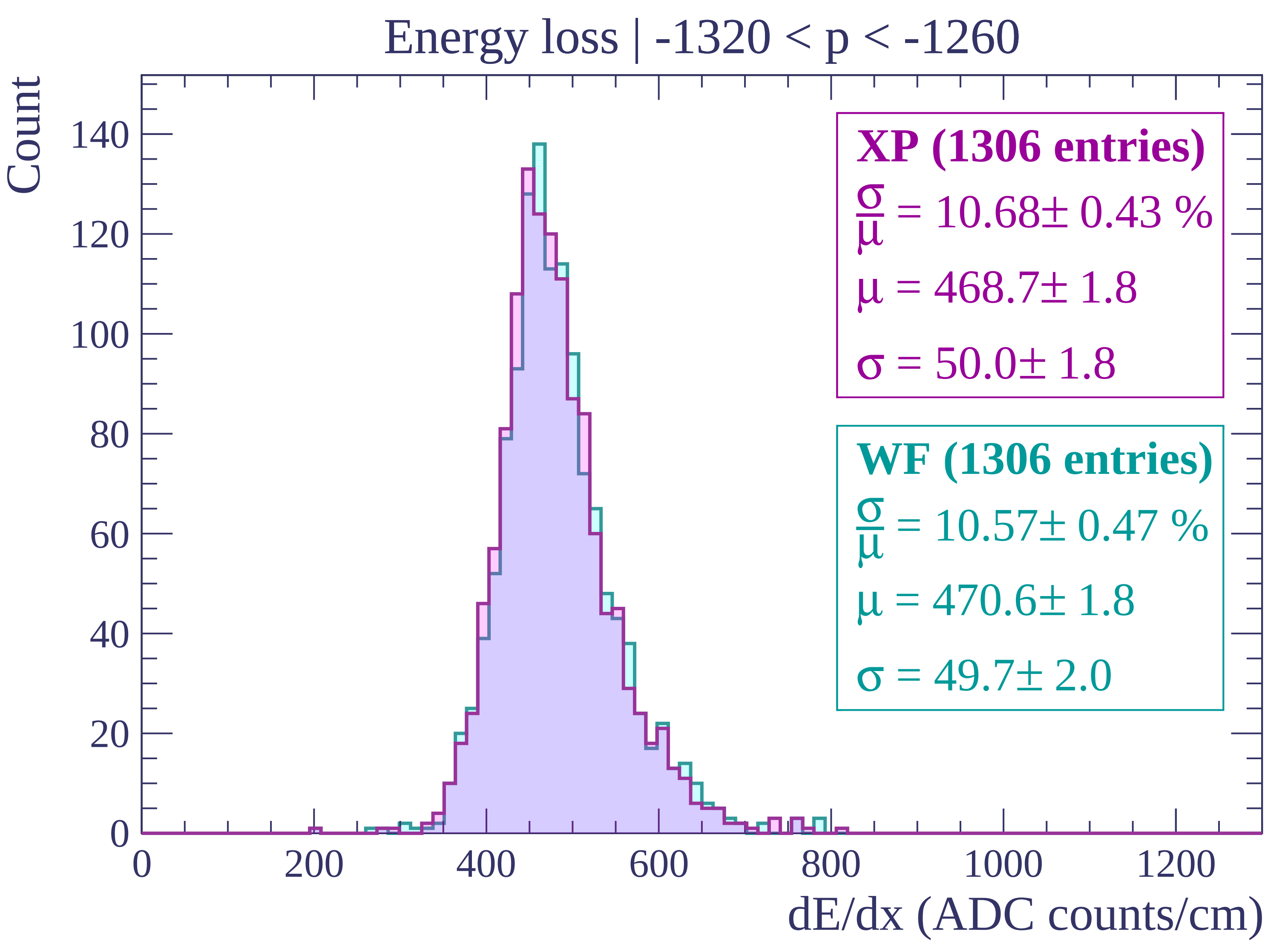
Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP (1157 entries)
σ/μ = 11.00± 0.44 %
μ = 482.3± 1.9
σ = 53.1± 1.9
WF (1157 entries)
σ/μ = 11.17± 0.41 %
μ = 487.4± 2.0
σ = 54.5± 1.8



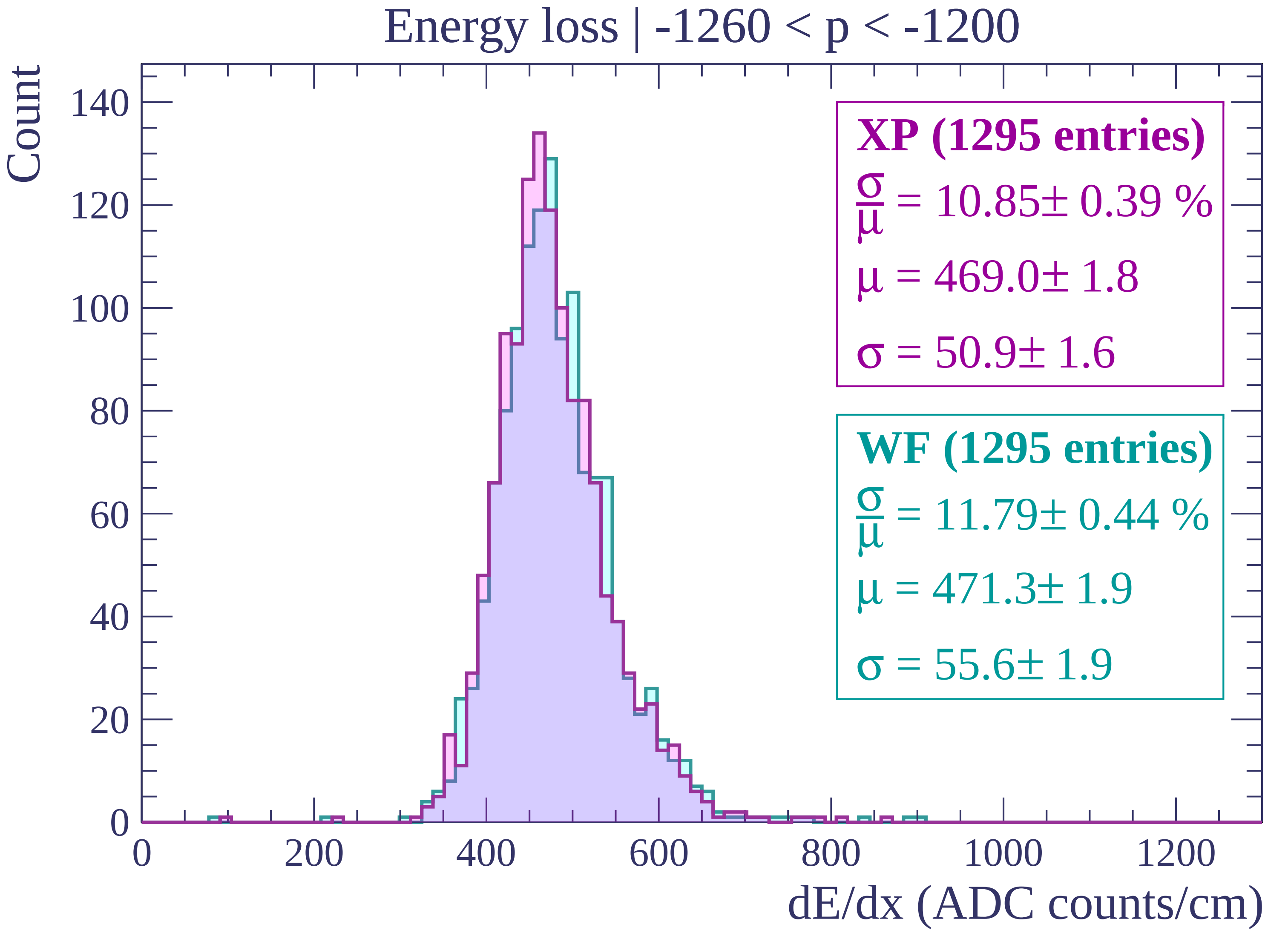
Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP (1169 entries)
σ/μ = 10.70± 0.44 %
μ = 476.0± 1.9
σ = 50.9± 1.9
WF (1169 entries)
σ/μ = 10.92± 0.43 %
μ = 480.2± 1.9
σ = 52.5± 1.8

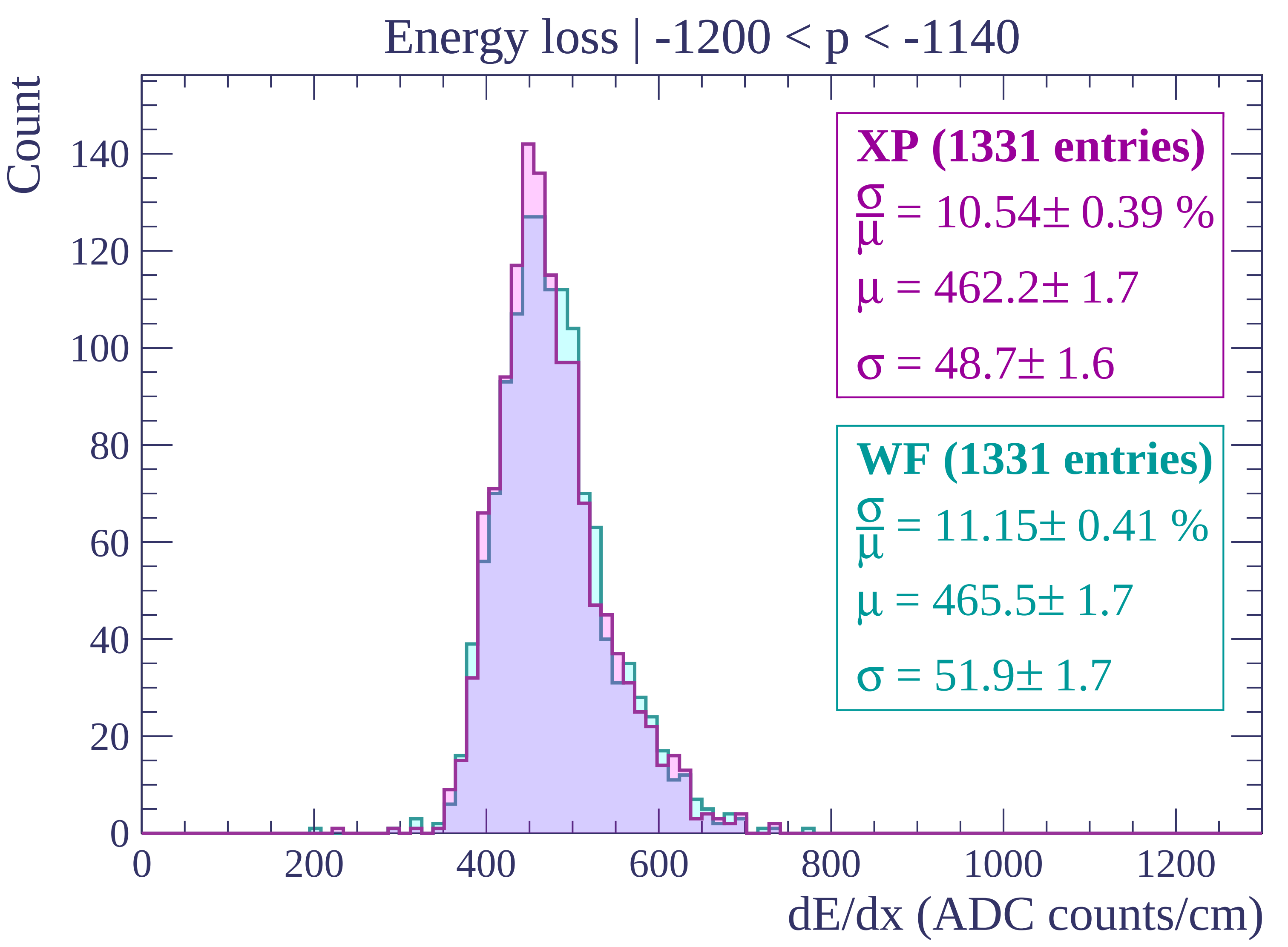

Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1197 entries)
σ/μ = 10.85± 0.45 %
μ = 476.1± 1.9
σ = 51.7± 1.9
WF (1197 entries)
σ/μ = 10.89± 0.37 %
μ = 481.1± 1.9
σ = 52.4± 1.6

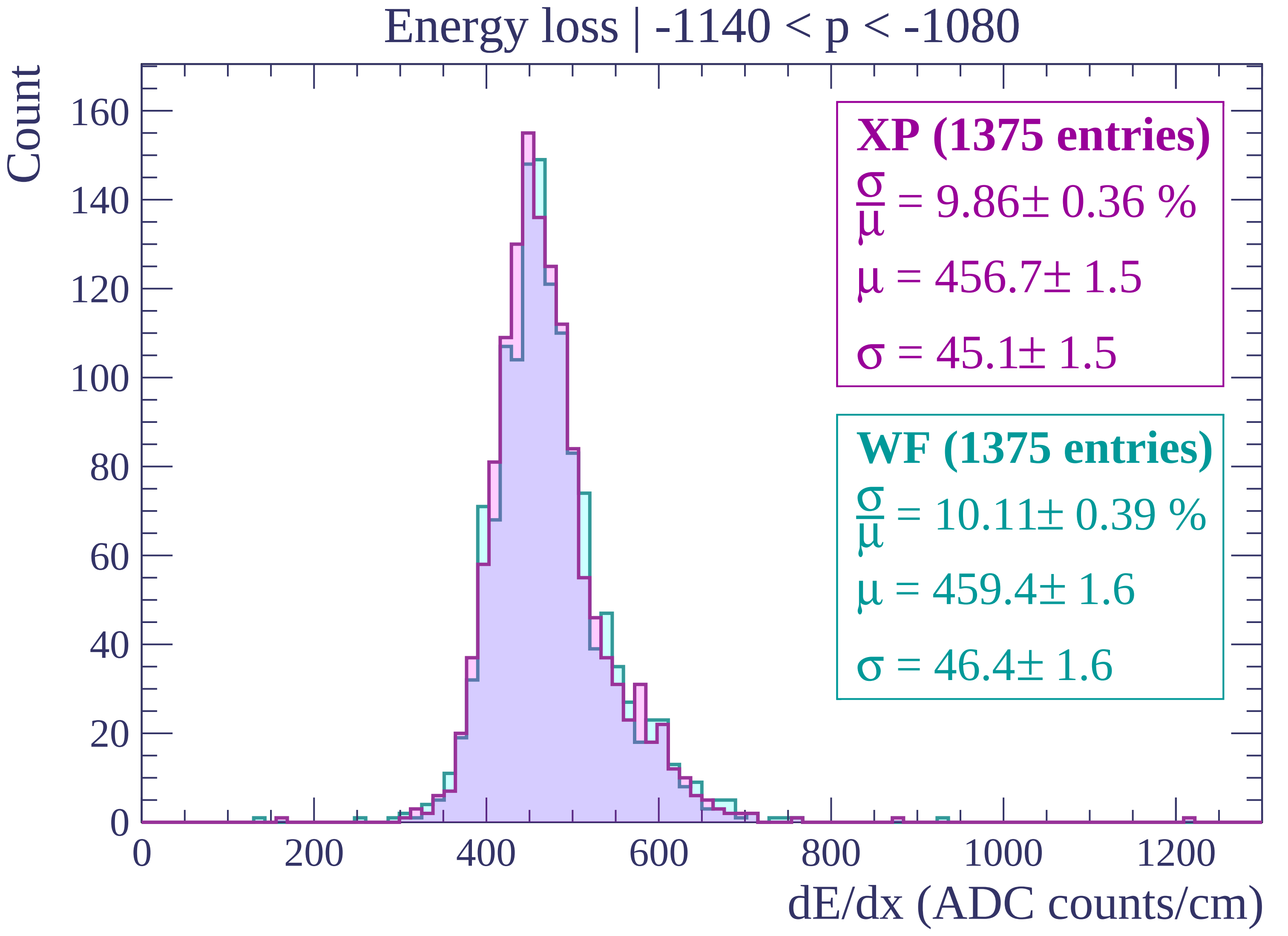

Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1262 entries)
σ/μ = 10.73± 0.39 %
μ = 471.5± 1.8
σ = 50.6± 1.6
WF (1262 entries)
σ/μ = 11.43± 0.43 %
μ = 477.4± 2.0
σ = 54.6± 1.8



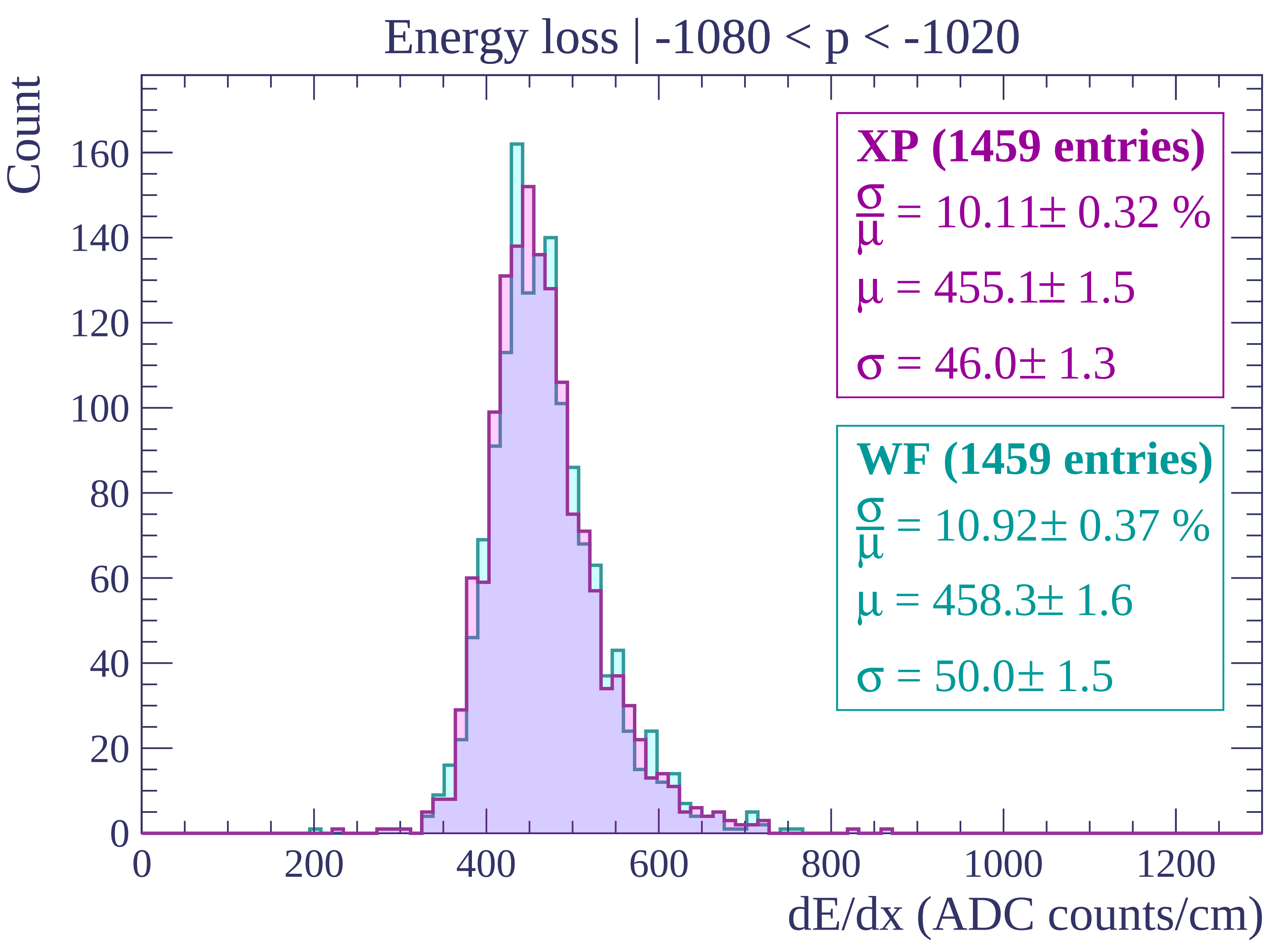
Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1306 entries)
$\frac{\sigma}{\mu}$ = 10.68± 0.43 %
μ = 468.7± 1.8
σ = 50.0± 1.8
WF (1306 entries)
$\frac{\sigma}{\mu}$ = 10.57± 0.47 %
μ = 470.6± 1.8
σ = 49.7± 2.0

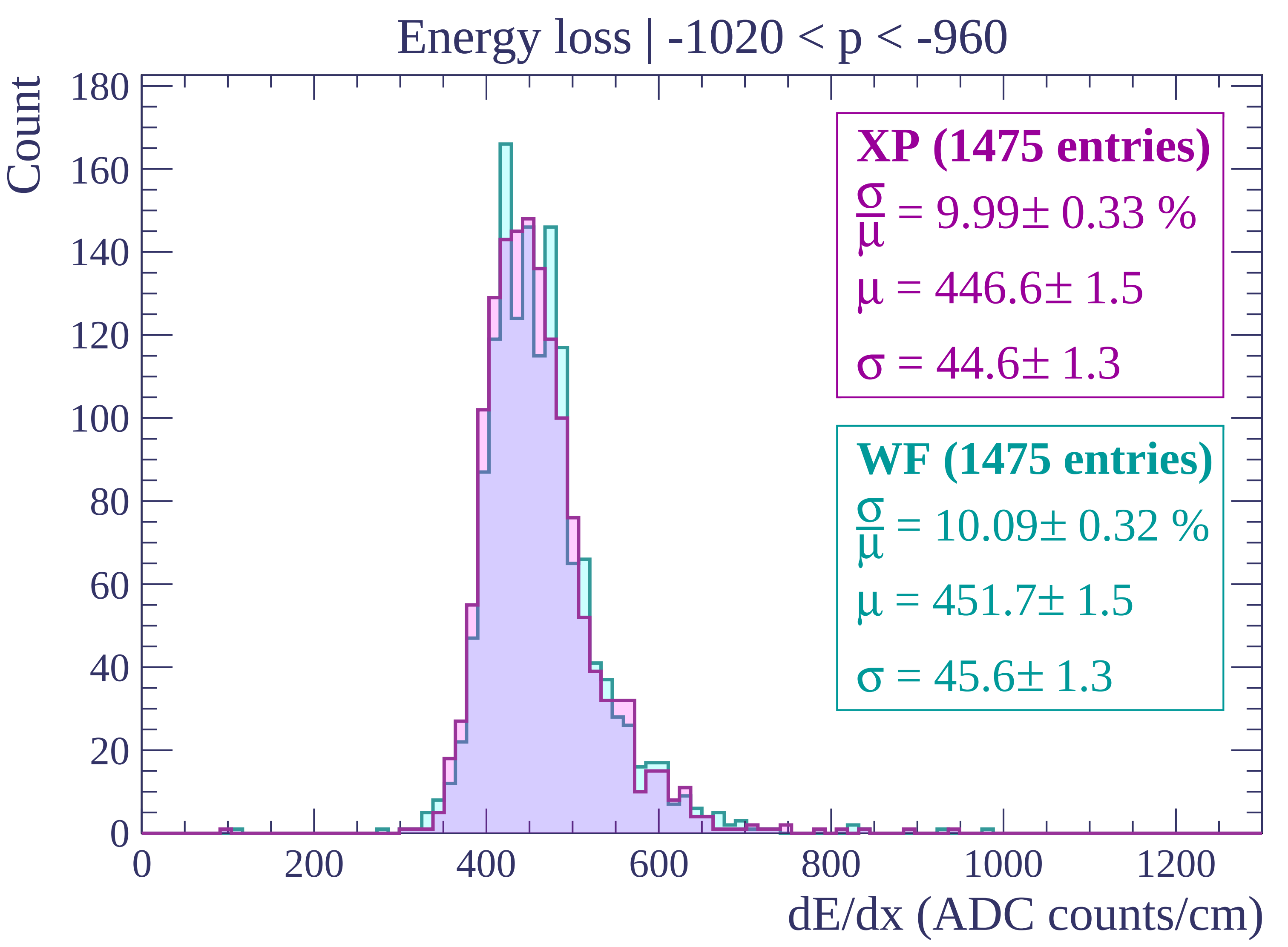

Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP (1295 entries)
σ/μ = 10.85± 0.39 %
μ = 469.0± 1.8
σ = 50.9± 1.6
WF (1295 entries)
σ/μ = 11.79± 0.44 %
μ = 471.3± 1.9
σ = 55.6± 1.9

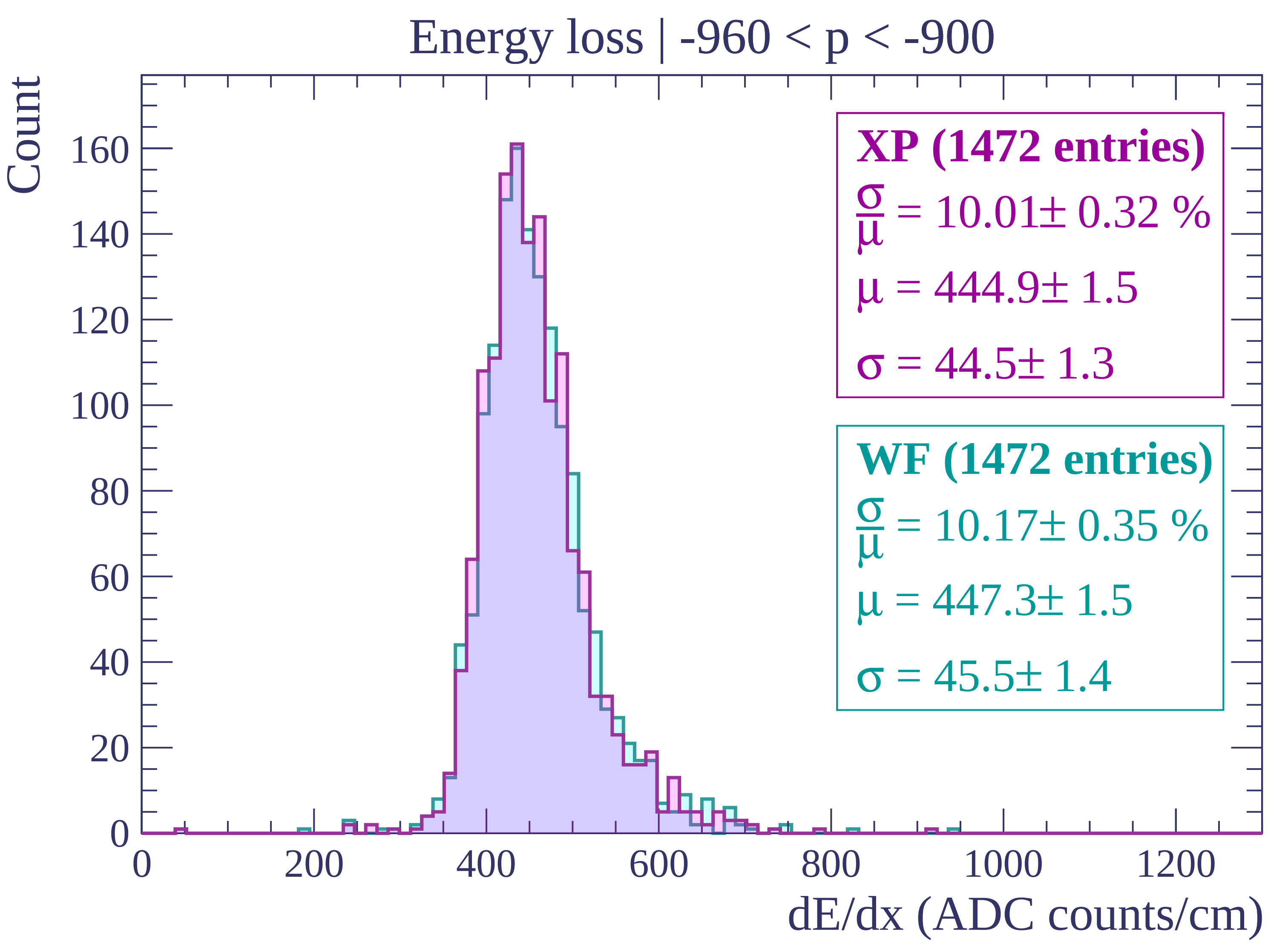

Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1331 entries)
σ/μ = 10.54± 0.39 %
μ = 462.2± 1.7
σ = 48.7± 1.6
WF (1331 entries)
σ/μ = 11.15± 0.41 %
μ = 465.5± 1.7
σ = 51.9± 1.7



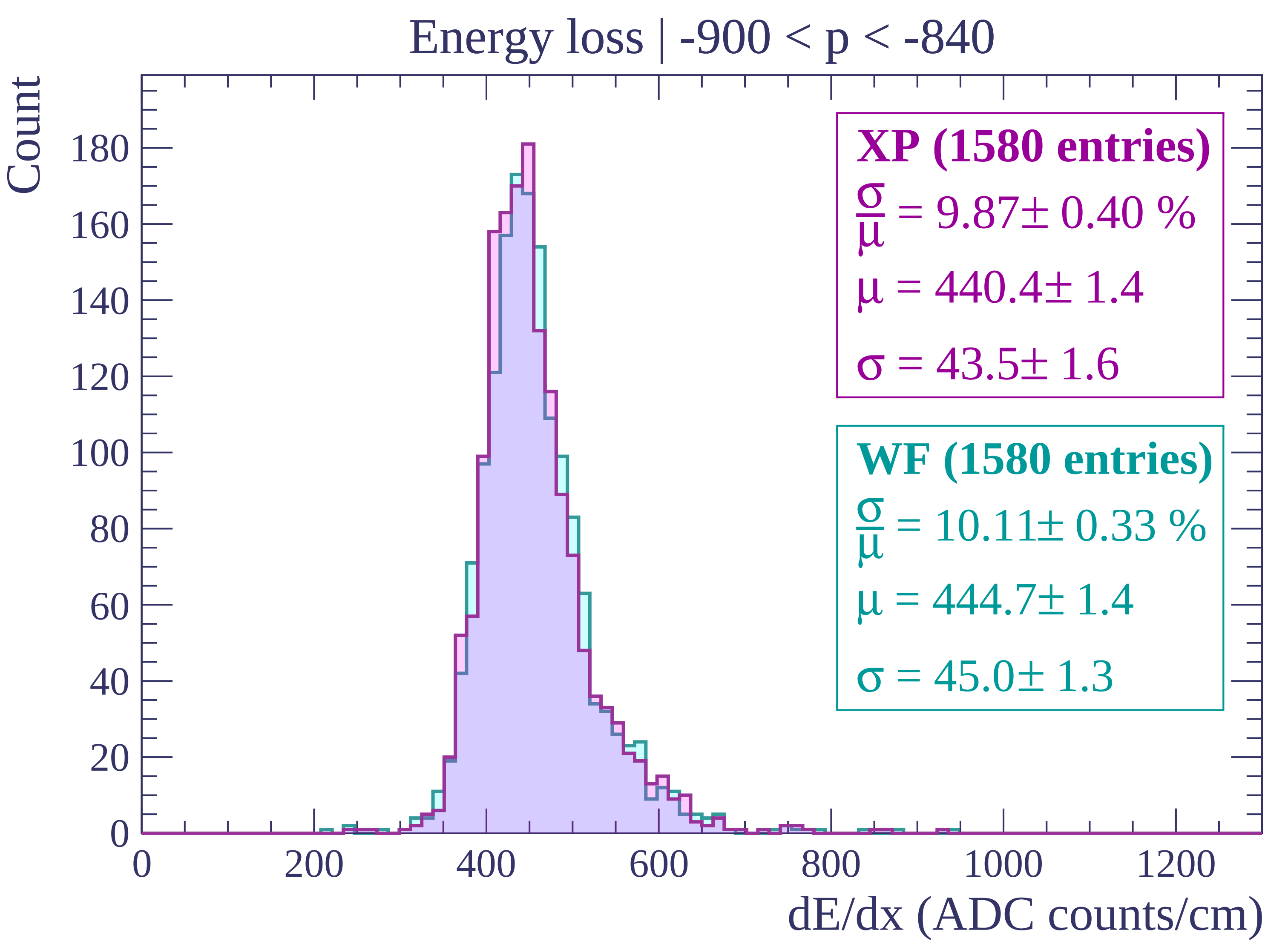
Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP (1375 entries)
σ/μ = 9.86± 0.36 %
μ = 456.7± 1.5
σ = 45.1± 1.5
WF (1375 entries)
σ/μ = 10.11± 0.39 %
μ = 459.4± 1.6
σ = 46.4± 1.6


Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
XP (1459 entries)
σ/μ = 10.11± 0.32 %
μ = 455.1± 1.5
σ = 46.0± 1.3
WF (1459 entries)
σ/μ = 10.92± 0.37 %
μ = 458.3± 1.6
σ = 50.0± 1.5
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200

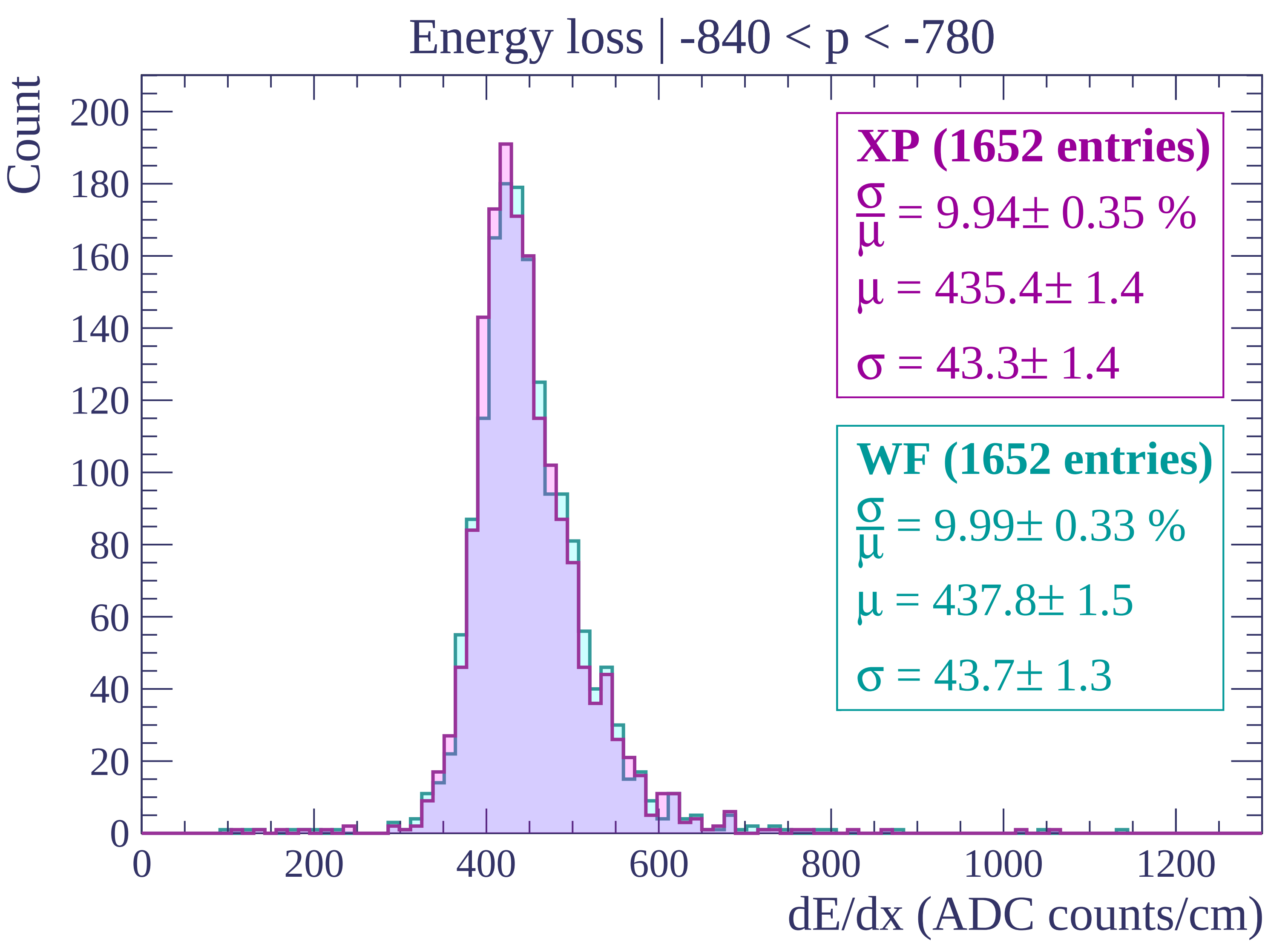

Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1475 entries)
$\frac{\sigma}{\mu}$ = 9.99± 0.33 %
μ = 446.6± 1.5
σ = 44.6± 1.3
WF (1475 entries)
$\frac{\sigma}{\mu}$ = 10.09± 0.32 %
μ = 451.7± 1.5
σ = 45.6± 1.3



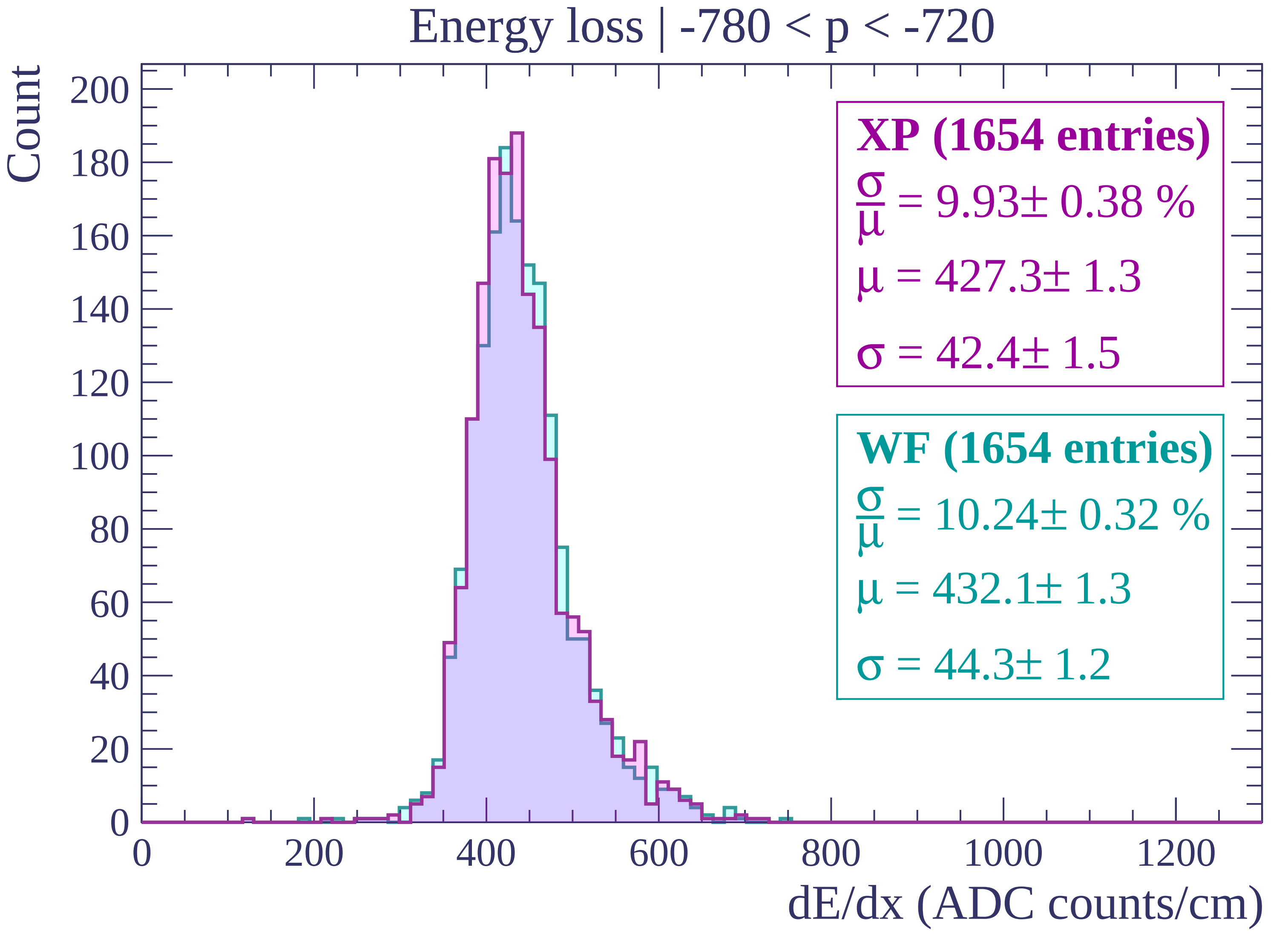
Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
XP (1472 entries)
σ/μ = 10.01± 0.32 %
μ = 444.9± 1.5
σ = 44.5± 1.3
WF (1472 entries)
σ/μ = 10.17± 0.35 %
μ = 447.3± 1.5
σ = 45.5± 1.4
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



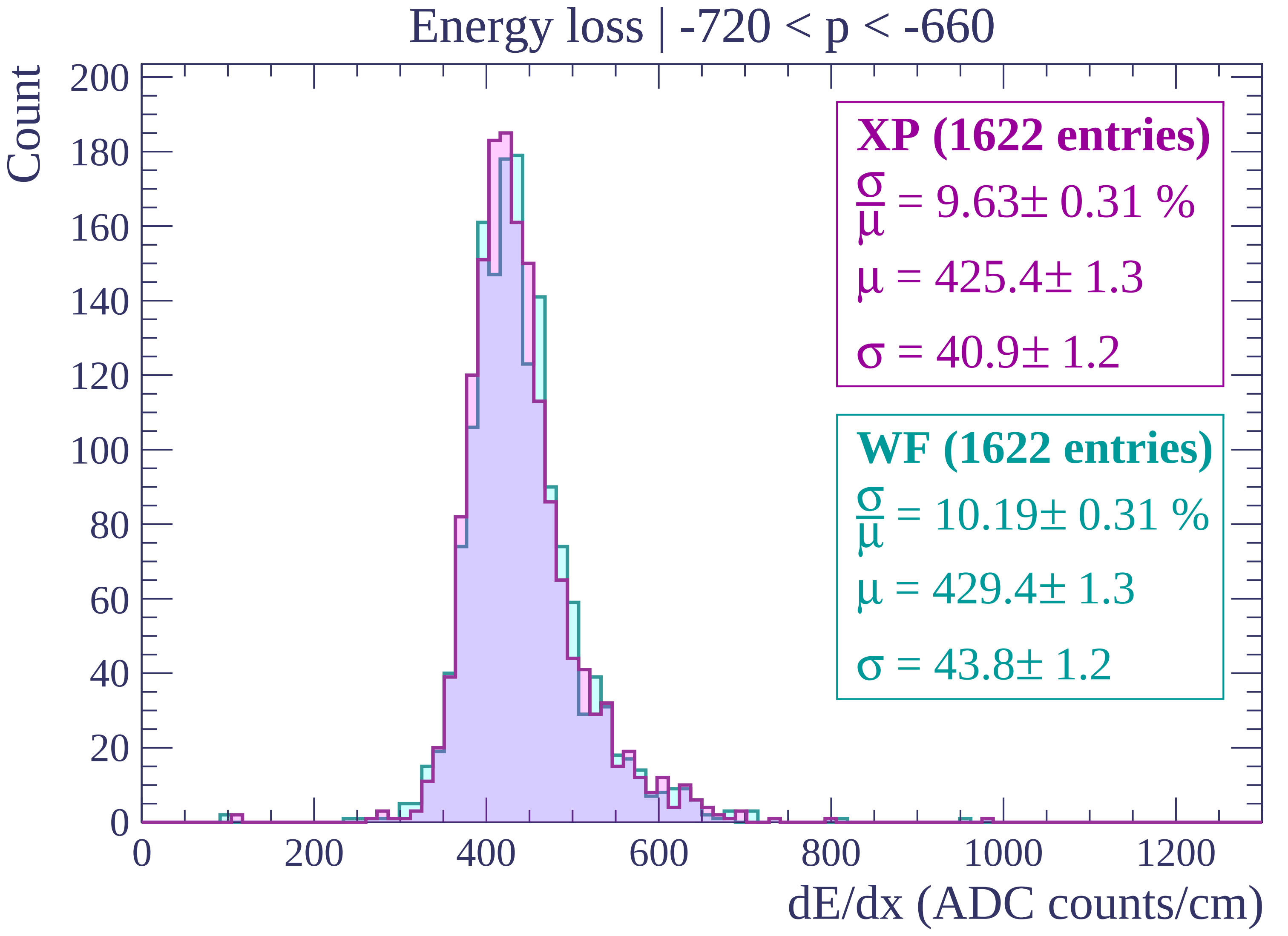
Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1580 entries)
σ/μ = 9.87± 0.40 %
μ = 440.4± 1.4
σ = 43.5± 1.6
WF (1580 entries)
σ/μ = 10.11± 0.33 %
μ = 444.7± 1.4
σ = 45.0± 1.3



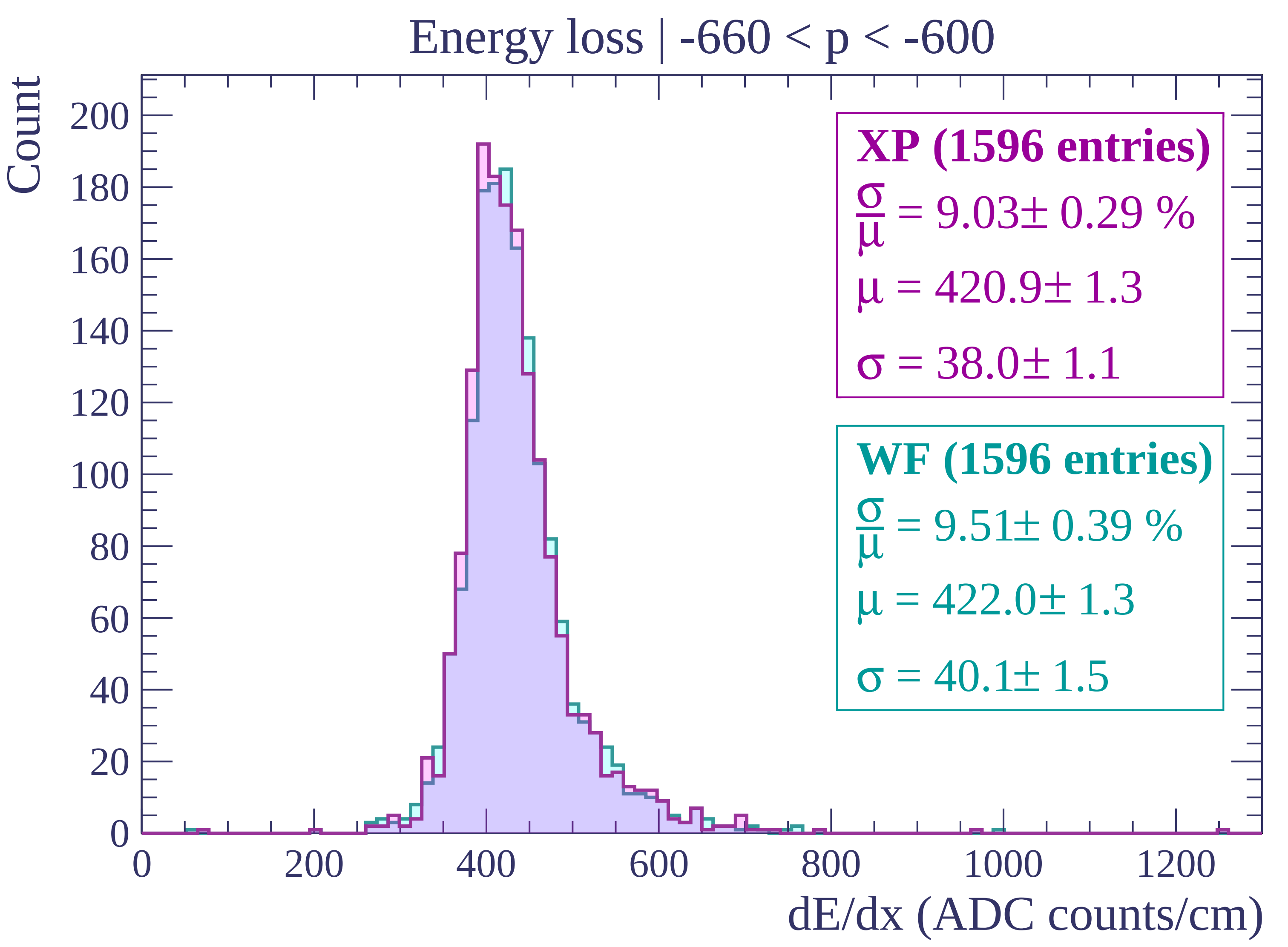
Energy loss | -840 < p < -780
Count
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1652 entries)
$\frac{\sigma}{\mu}$ = 9.94± 0.35 %
μ = 435.4± 1.4
σ = 43.3± 1.4
WF (1652 entries)
$\frac{\sigma}{\mu}$ = 9.99± 0.33 %
μ = 437.8± 1.5
σ = 43.7± 1.3

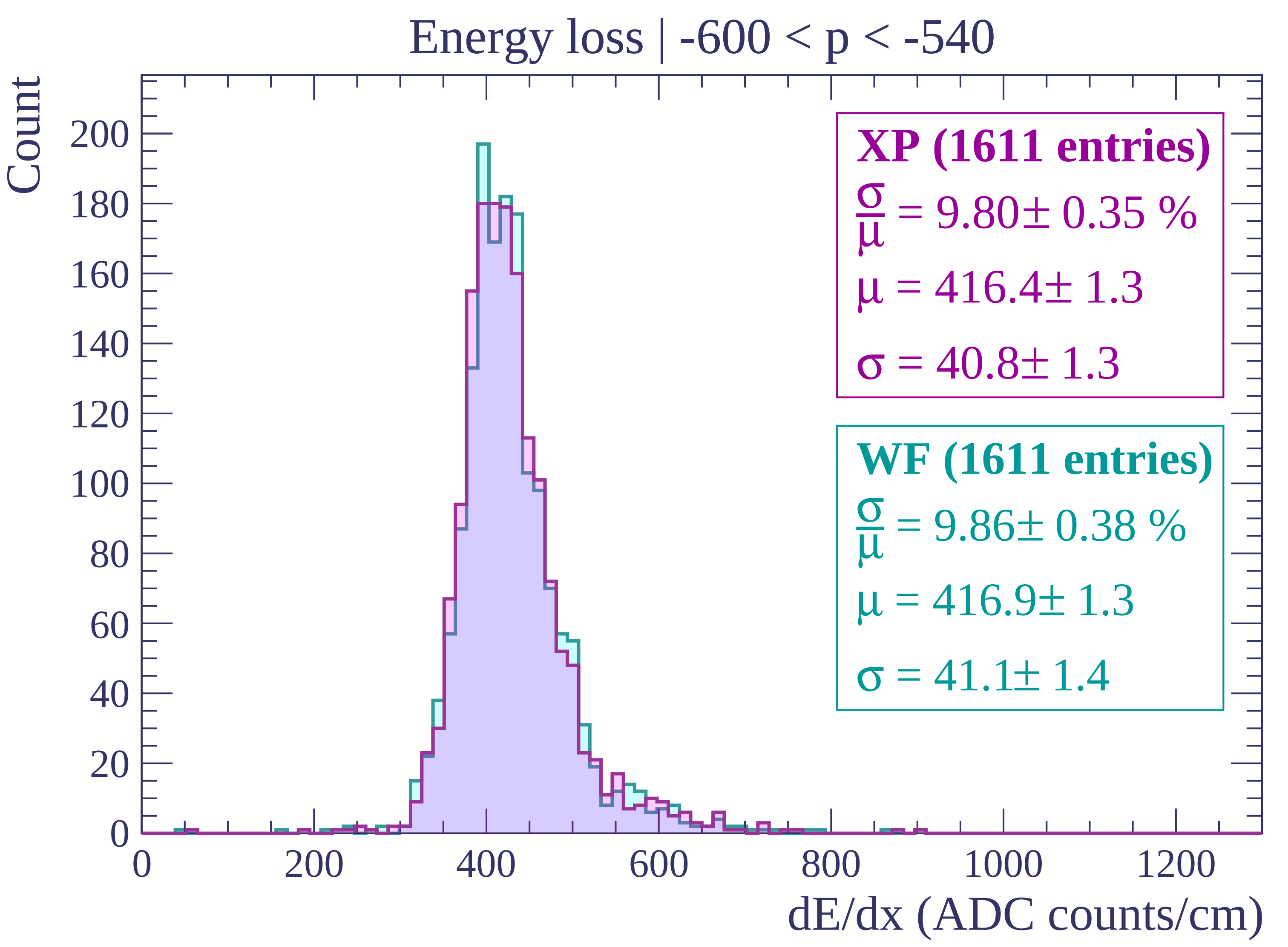


Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP (1654 entries)
σ/μ = 9.93± 0.38 %
μ = 427.3± 1.3
σ = 42.4± 1.5
WF (1654 entries)
σ/μ = 10.24± 0.32 %
μ = 432.1± 1.3
σ = 44.3± 1.2

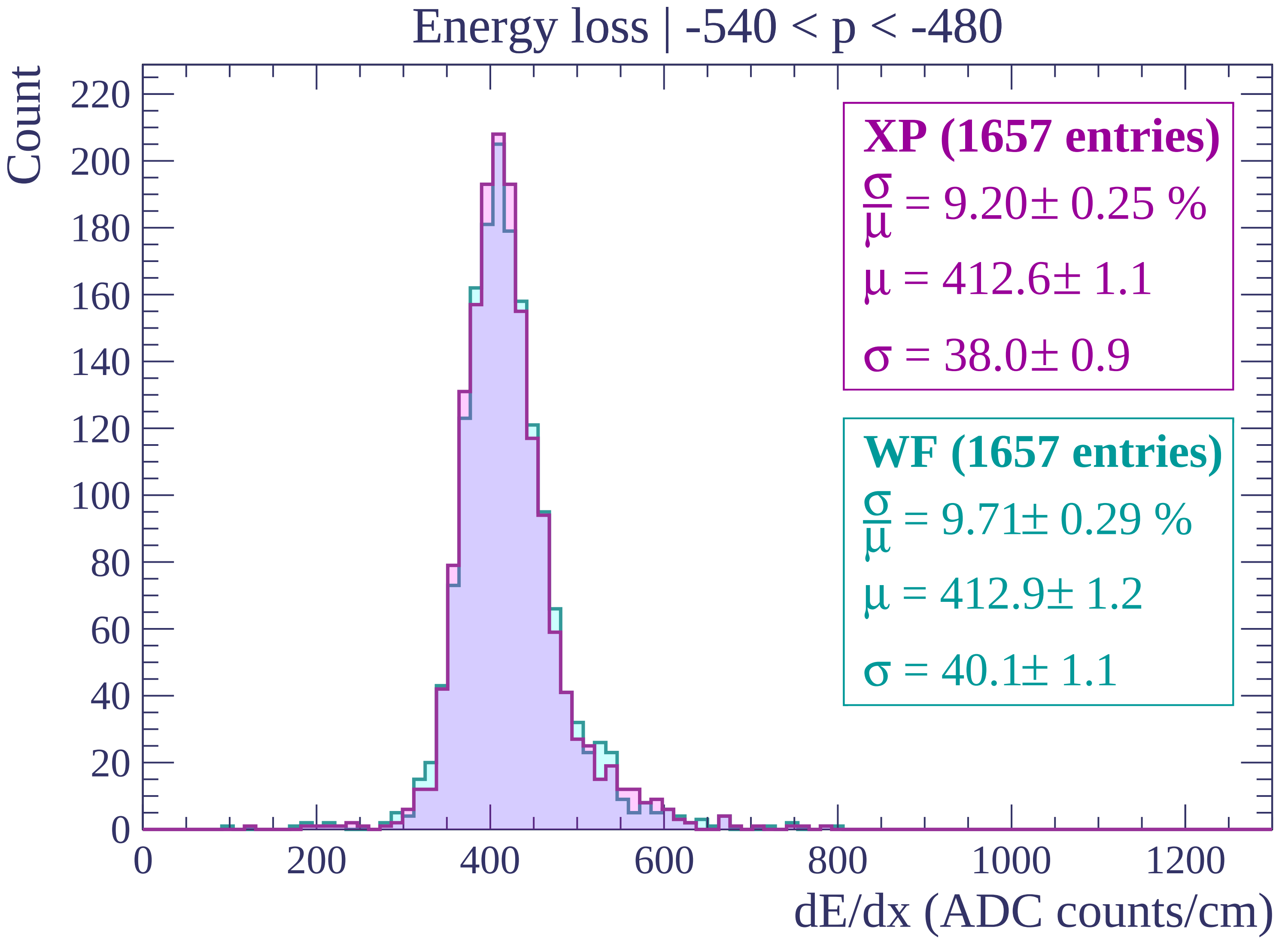


Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (1622 entries)
σ/μ = 9.63± 0.31 %
μ = 425.4± 1.3
σ = 40.9± 1.2
WF (1622 entries)
σ/μ = 10.19± 0.31 %
μ = 429.4± 1.3
σ = 43.8± 1.2

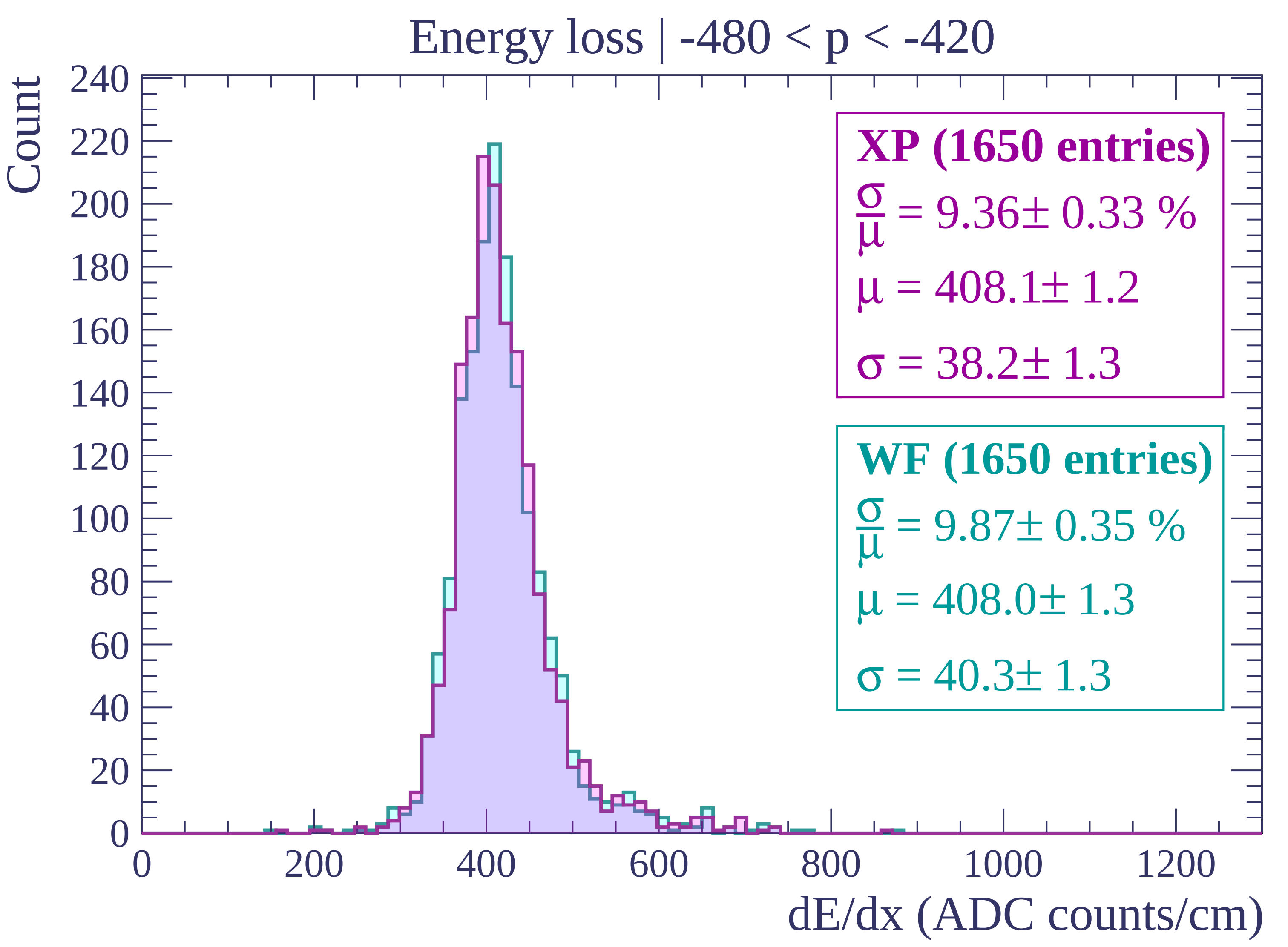

Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
XP (1596 entries)
σ/μ = 9.03± 0.29 %
μ = 420.9± 1.3
σ = 38.0± 1.1
WF (1596 entries)
σ/μ = 9.51± 0.39 %
μ = 422.0± 1.3
σ = 40.1± 1.5
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200

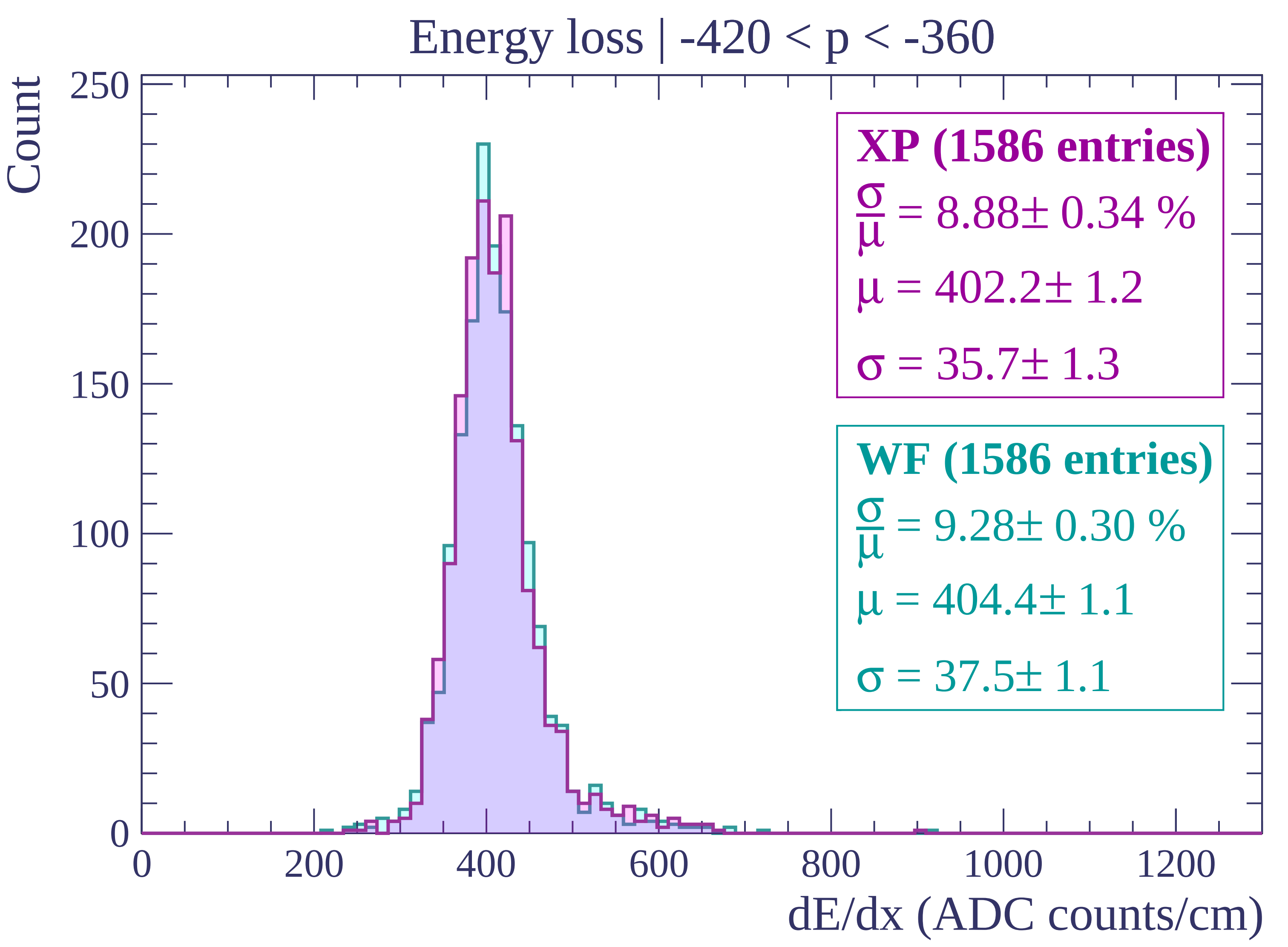


Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
XP (1611 entries)
σ/μ = 9.80± 0.35 %
μ = 416.4± 1.3
σ = 40.8± 1.3
WF (1611 entries)
σ/μ = 9.86± 0.38 %
μ = 416.9± 1.3
σ = 41.1± 1.4

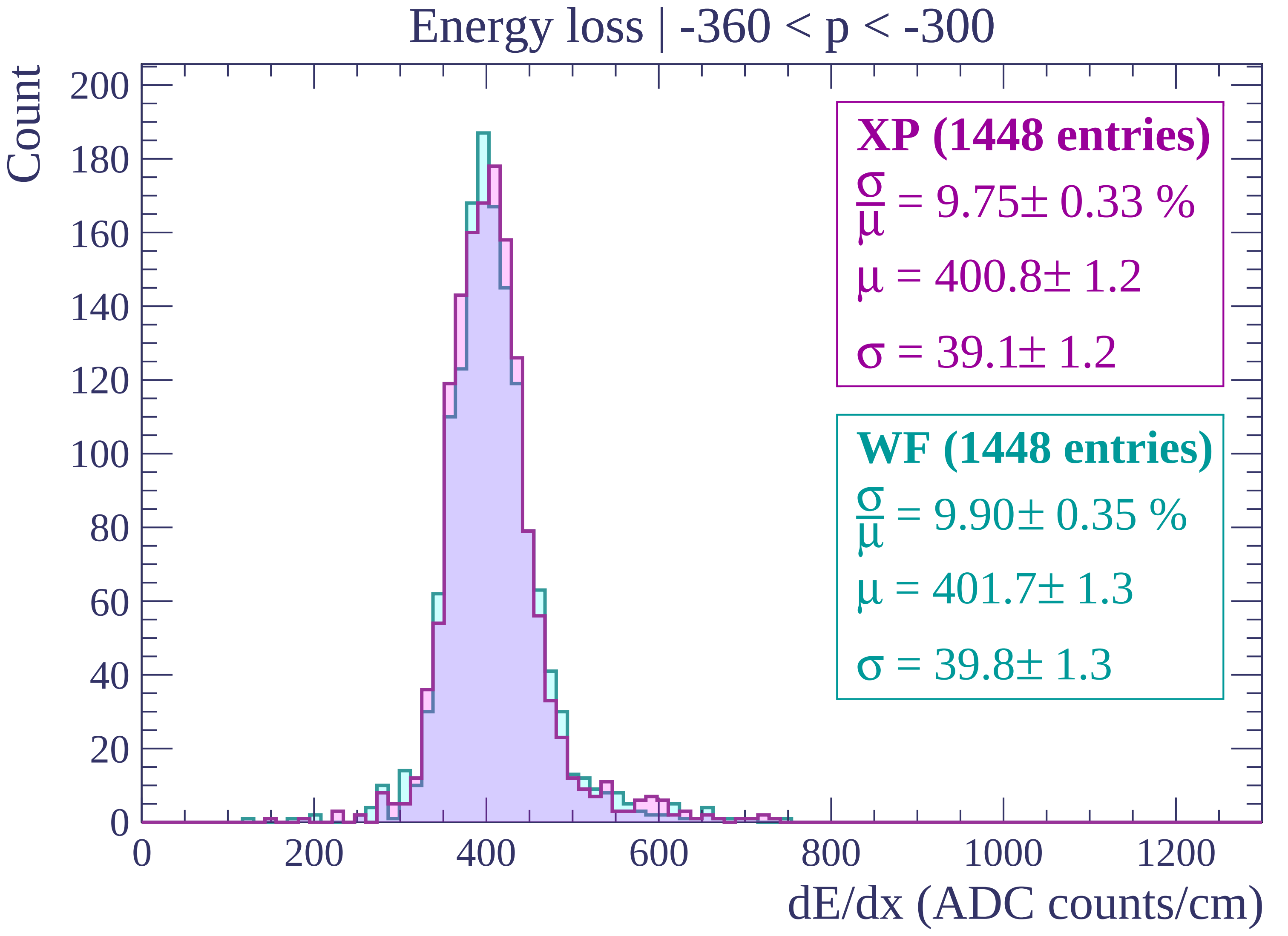


Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1657 entries)
σ/μ = 9.20± 0.25 %
μ = 412.6± 1.1
σ = 38.0± 0.9
WF (1657 entries)
σ/μ = 9.71± 0.29 %
μ = 412.9± 1.2
σ = 40.1± 1.1

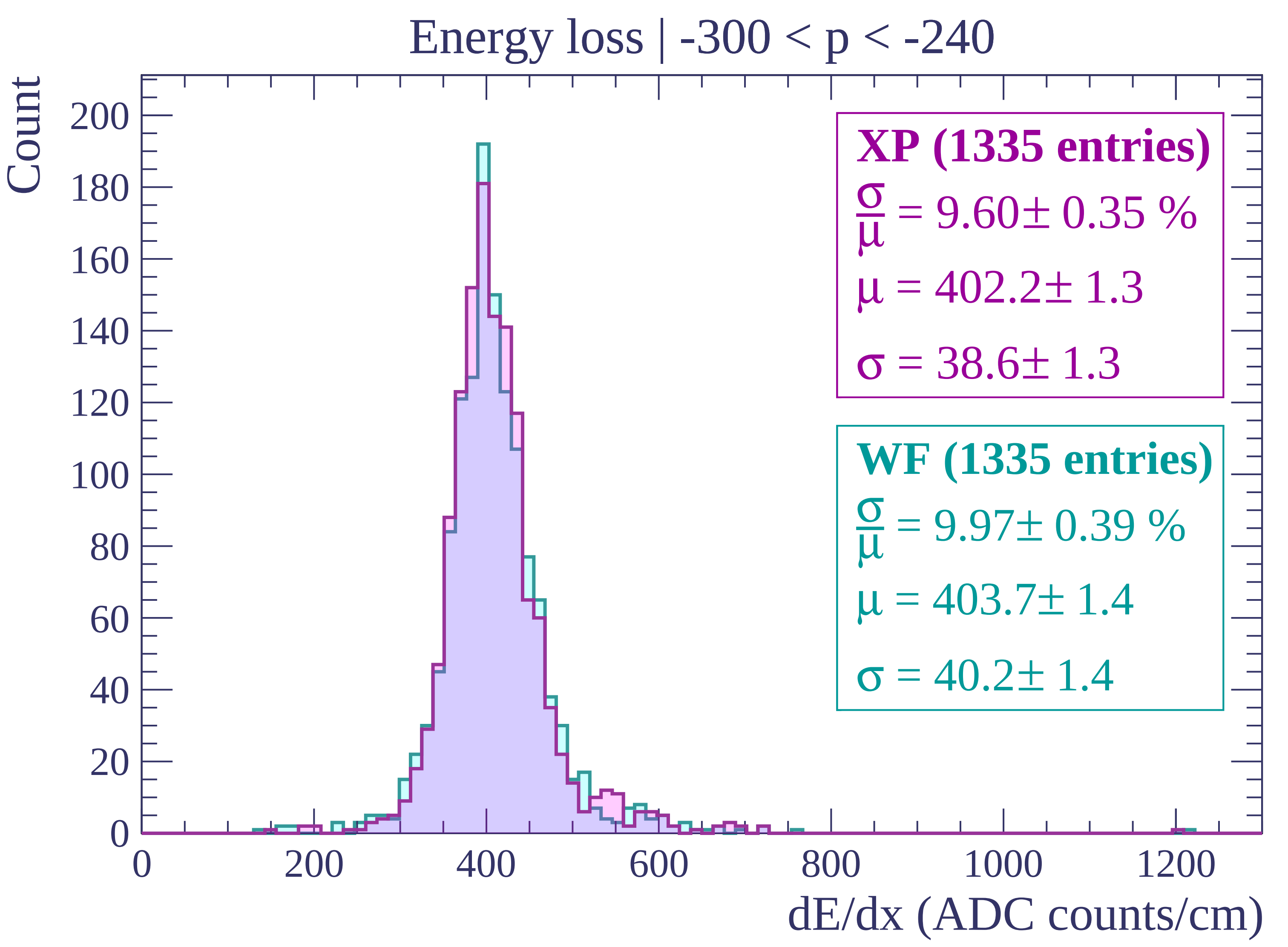

Energy loss | -480 < p < -420
Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1650 entries)
σ/μ = 9.36± 0.33 %
μ = 408.1± 1.2
σ = 38.2± 1.3
WF (1650 entries)
σ/μ = 9.87± 0.35 %
μ = 408.0± 1.3
σ = 40.3± 1.3

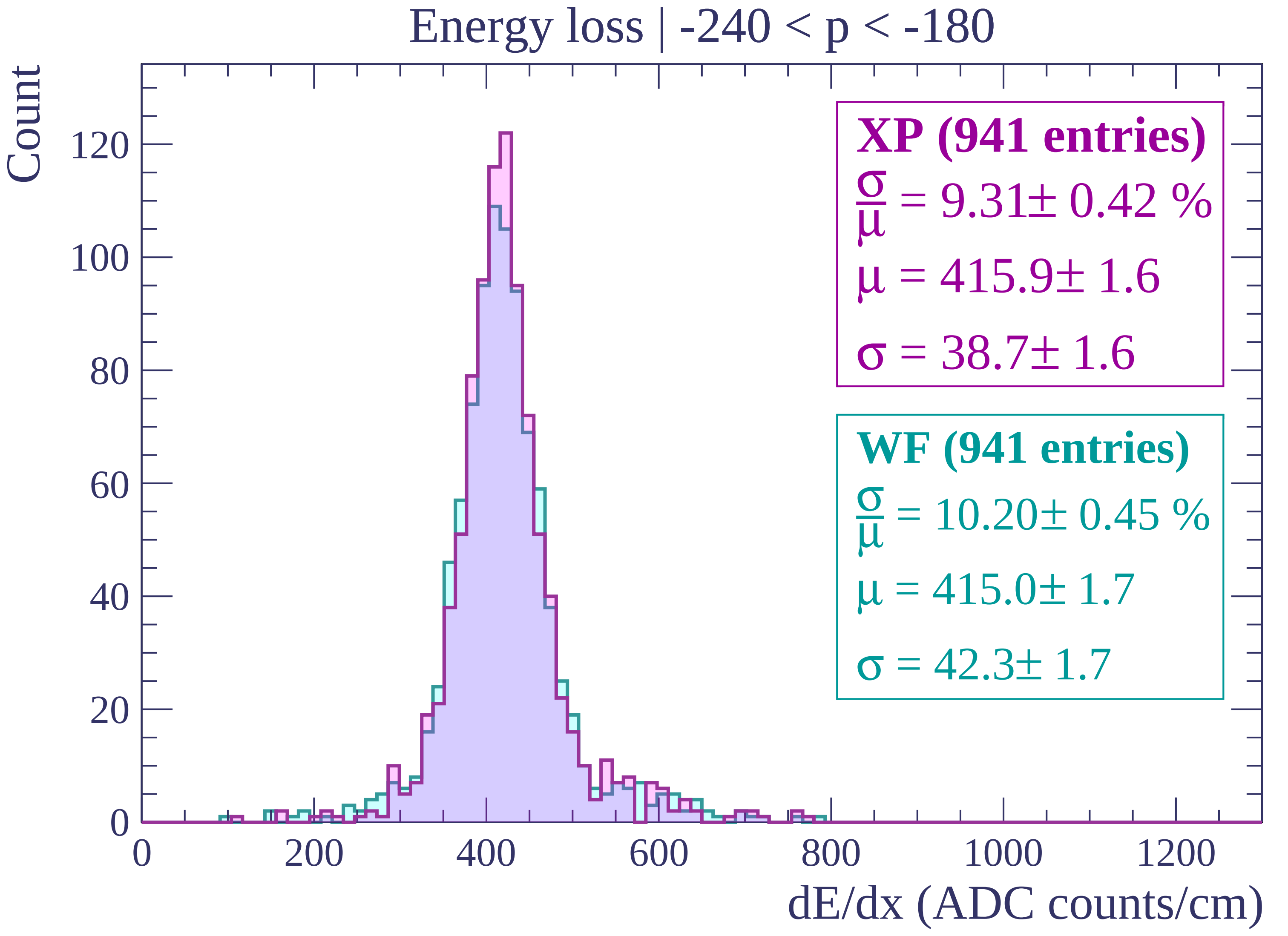


Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (1586 entries)
$\frac{\sigma}{\mu}$ = 8.88± 0.34 %
μ = 402.2± 1.2
σ = 35.7± 1.3
WF (1586 entries)
$\frac{\sigma}{\mu}$ = 9.28± 0.30 %
μ = 404.4± 1.1
σ = 37.5± 1.1

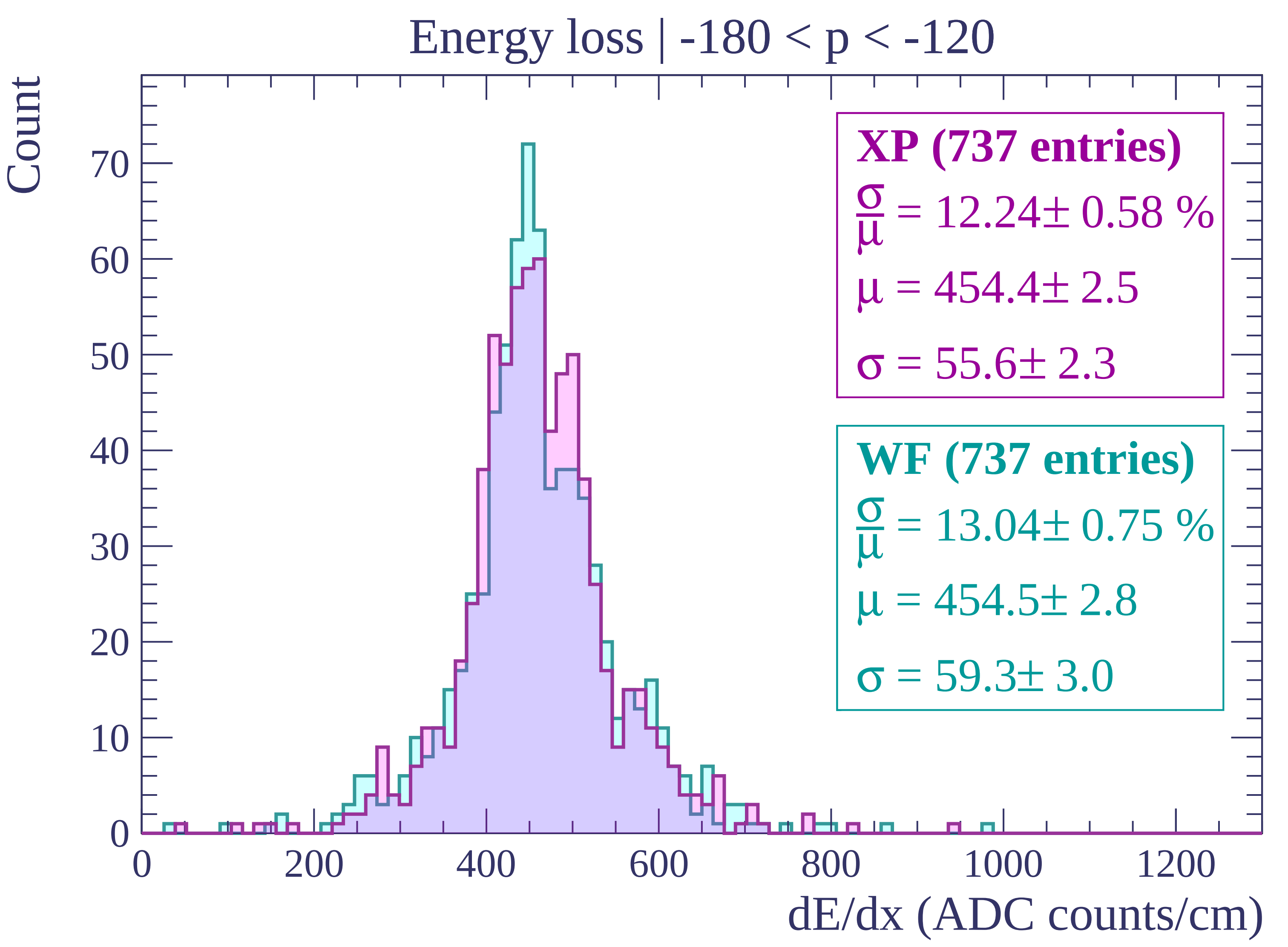


Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (1448 entries)
σ/μ = 9.75± 0.33 %
μ = 400.8± 1.2
σ = 39.1± 1.2
WF (1448 entries)
σ/μ = 9.90± 0.35 %
μ = 401.7± 1.3
σ = 39.8± 1.3

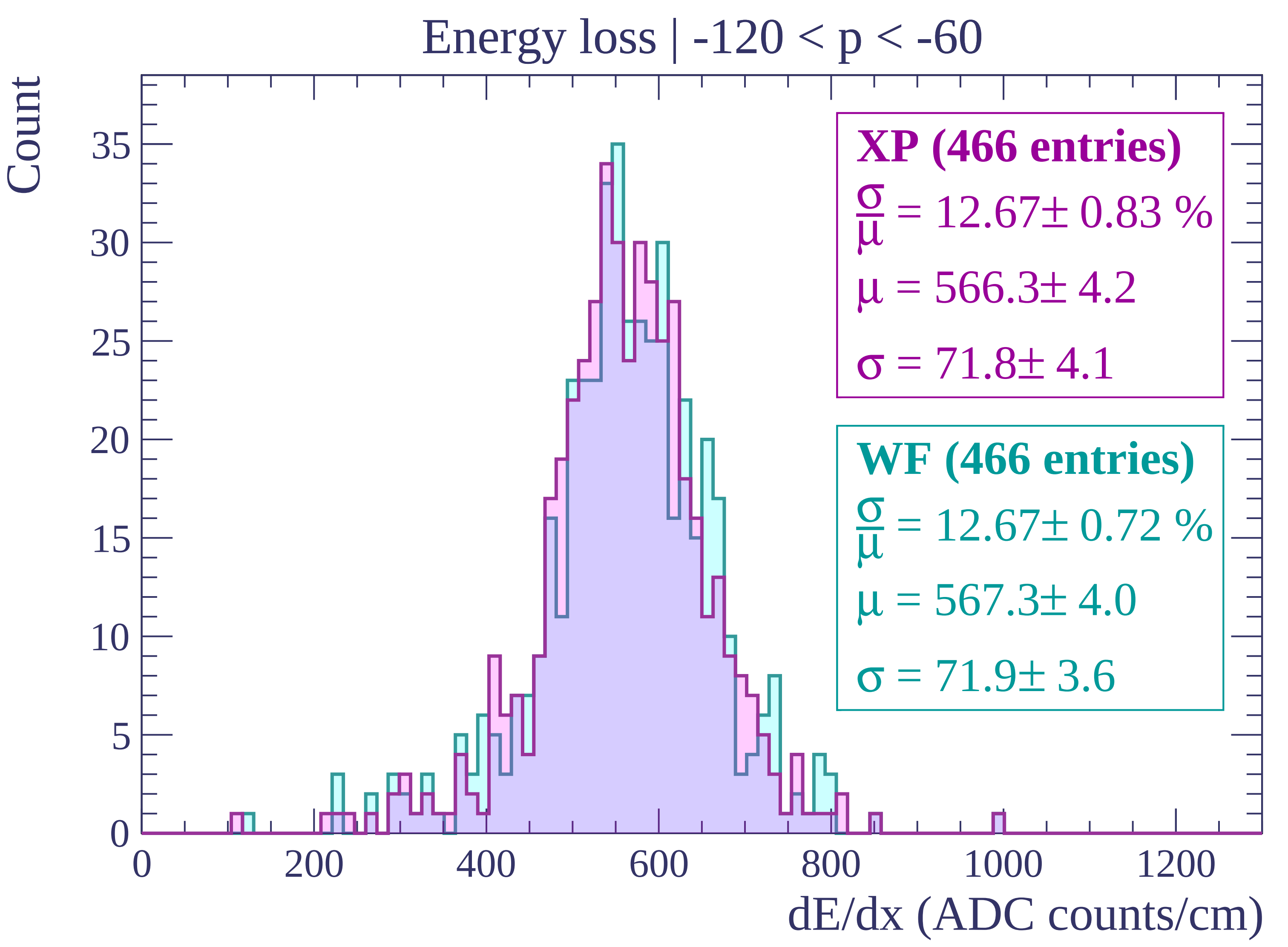

Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
XP (1335 entries)
$\frac{\sigma}{\mu}$ = 9.60± 0.35 %
μ = 402.2± 1.3
σ = 38.6± 1.3
WF (1335 entries)
$\frac{\sigma}{\mu}$ = 9.97± 0.39 %
μ = 403.7± 1.4
σ = 40.2± 1.4
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200



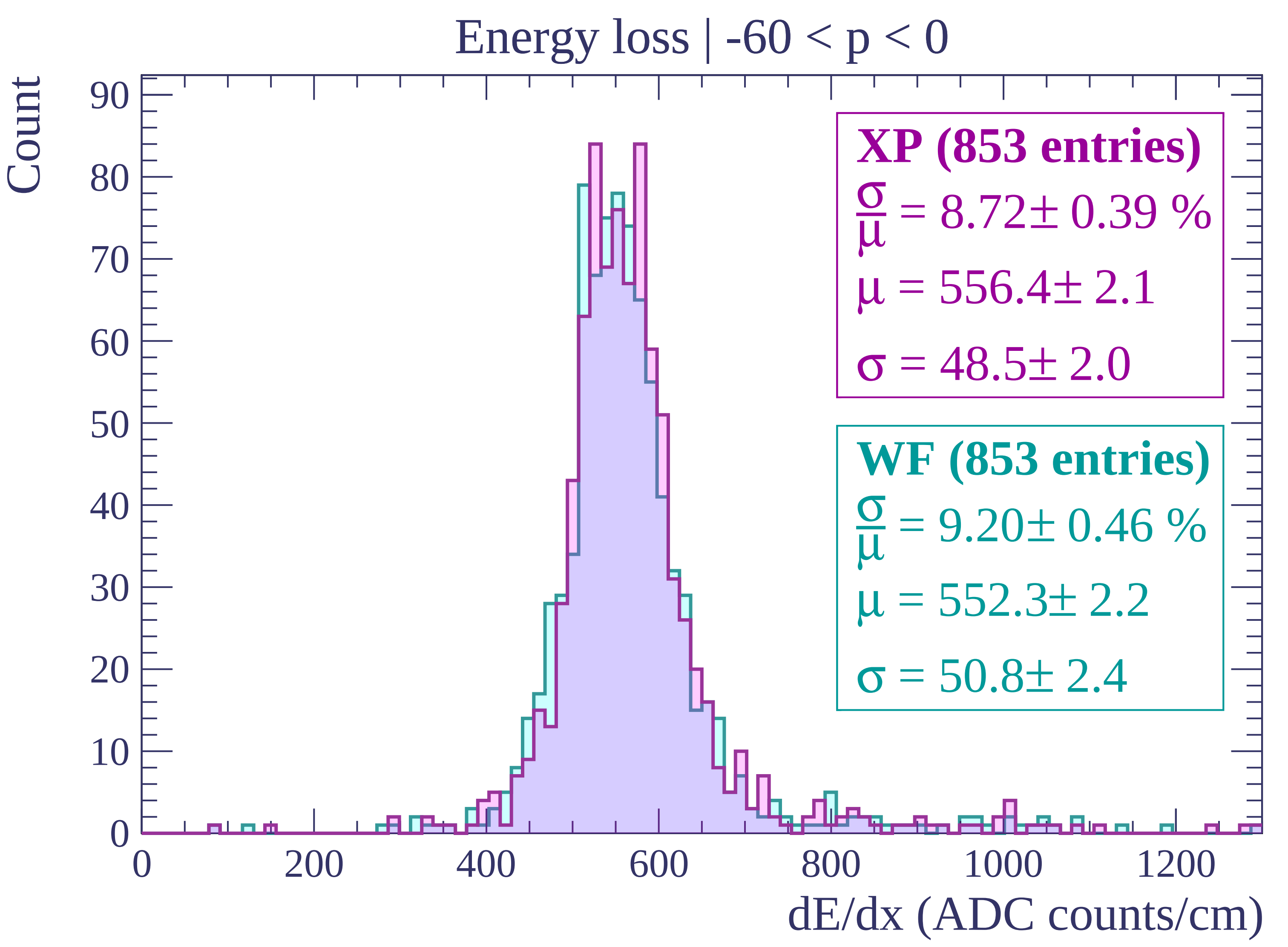
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (941 entries)
$\frac{\sigma}{\mu}$ = 9.31± 0.42 %
μ = 415.9± 1.6
σ = 38.7± 1.6
WF (941 entries)
$\frac{\sigma}{\mu}$ = 10.20± 0.45 %
μ = 415.0± 1.7
σ = 42.3± 1.7



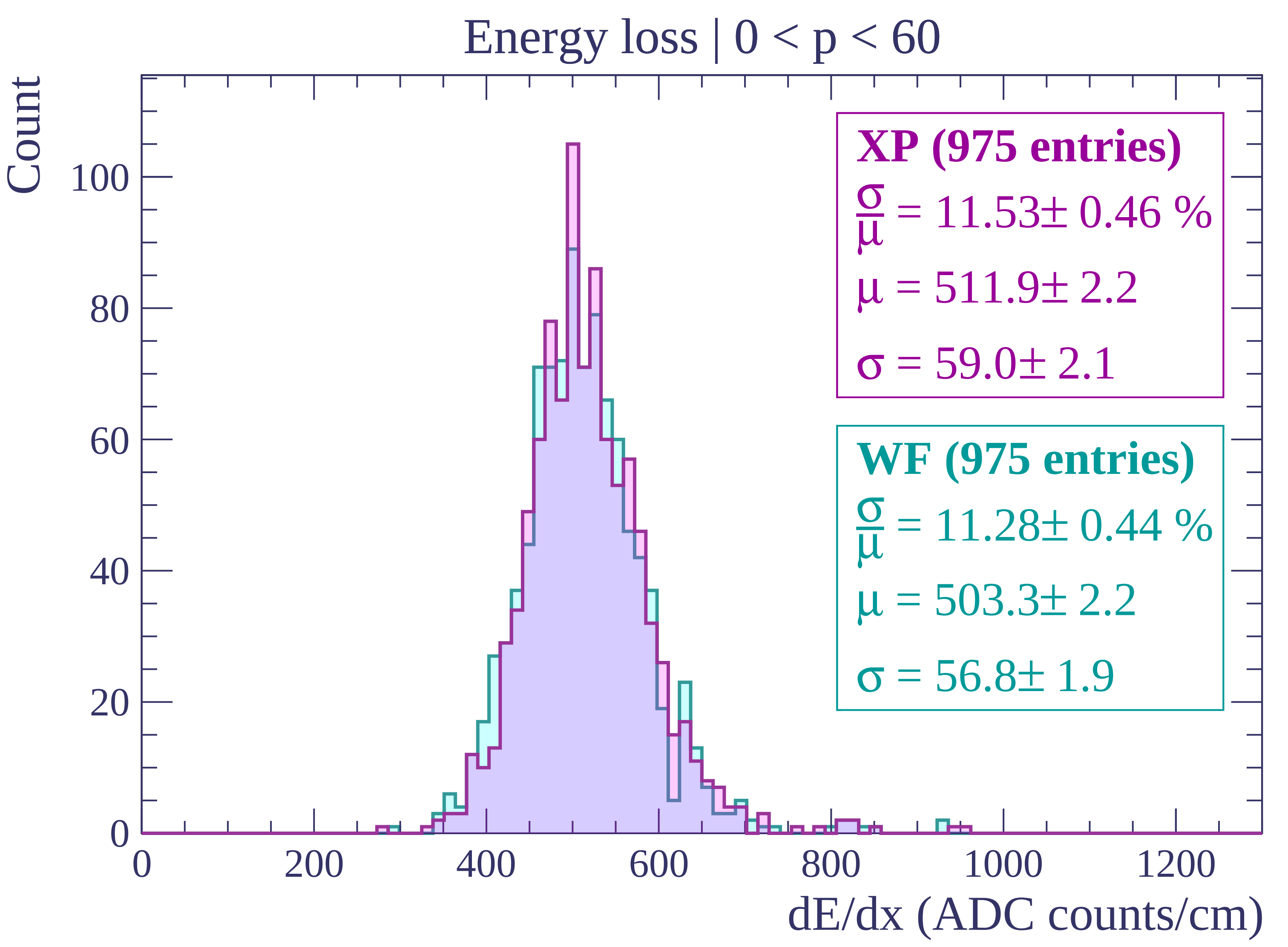
Energy loss | -180 < p < -120
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (737 entries)
σ/μ = 12.24± 0.58 %
μ = 454.4± 2.5
σ = 55.6± 2.3
WF (737 entries)
σ/μ = 13.04± 0.75 %
μ = 454.5± 2.8
σ = 59.3± 3.0



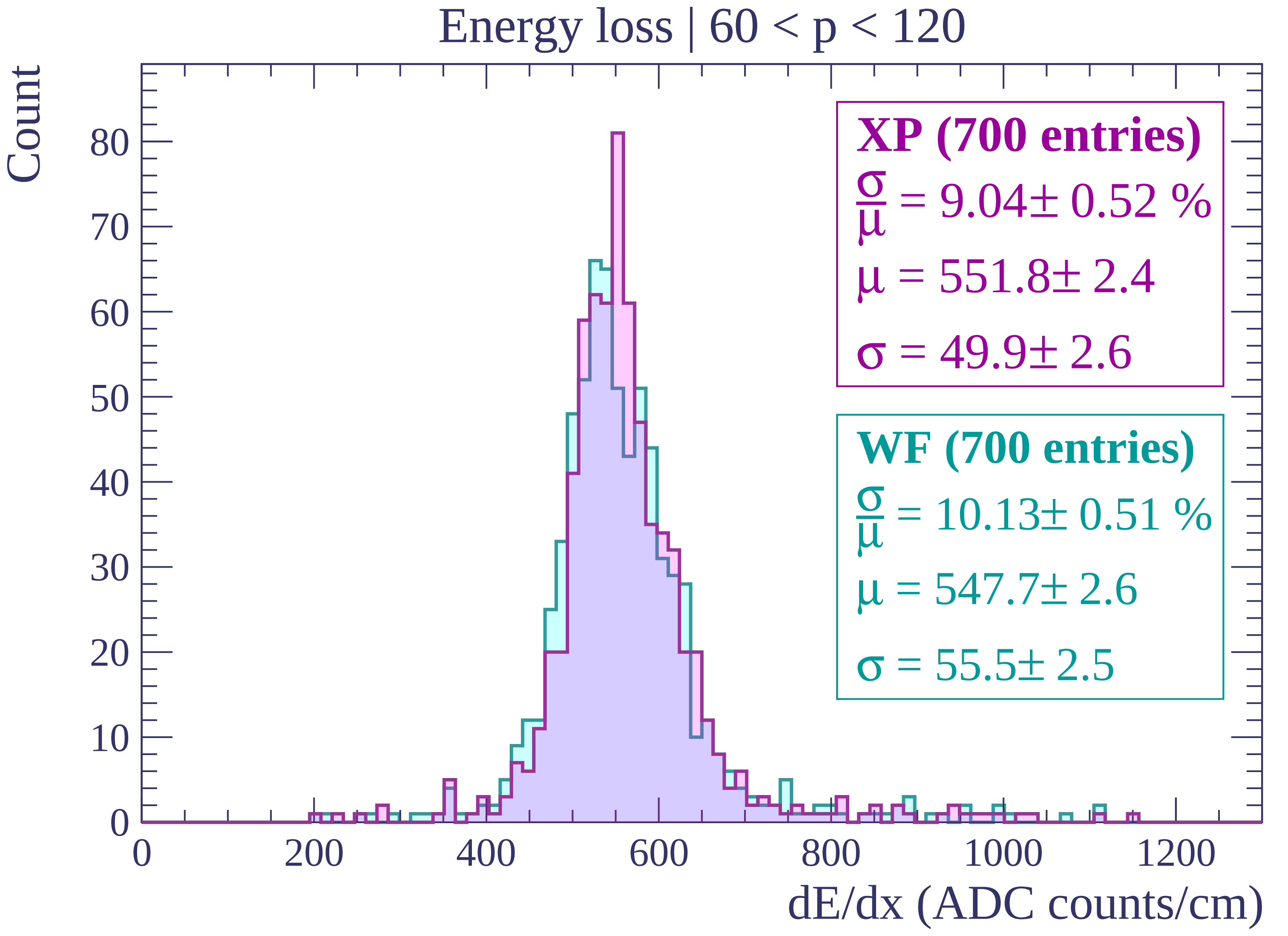
Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP (466 entries)
σ/μ = 12.67± 0.83 %
μ = 566.3± 4.2
σ = 71.8± 4.1
WF (466 entries)
σ/μ = 12.67± 0.72 %
μ = 567.3± 4.0
σ = 71.9± 3.6

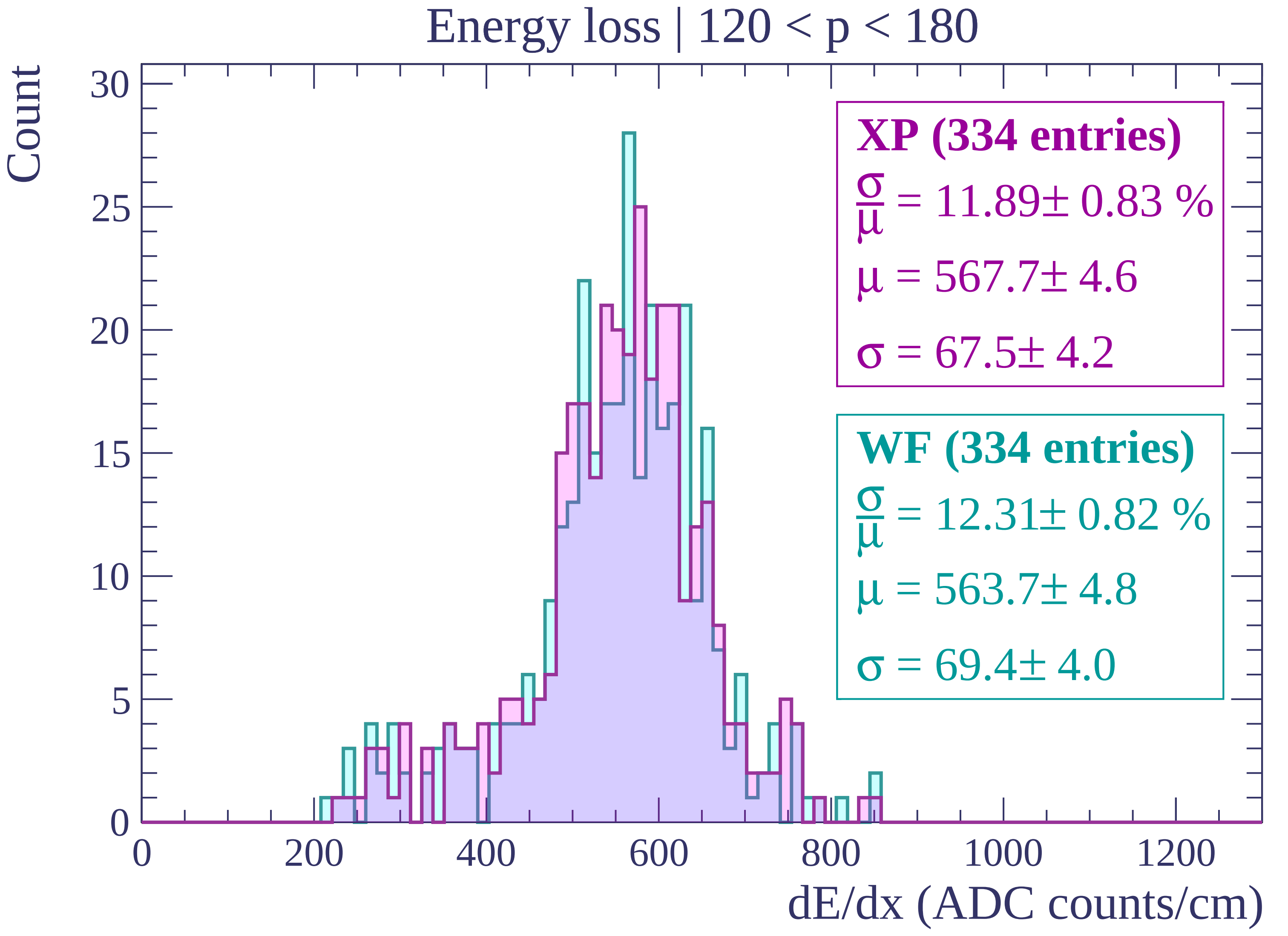


Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (853 entries)
σ/μ = 8.72± 0.39 %
μ = 556.4± 2.1
σ = 48.5± 2.0
WF (853 entries)
σ/μ = 9.20± 0.46 %
μ = 552.3± 2.2
σ = 50.8± 2.4

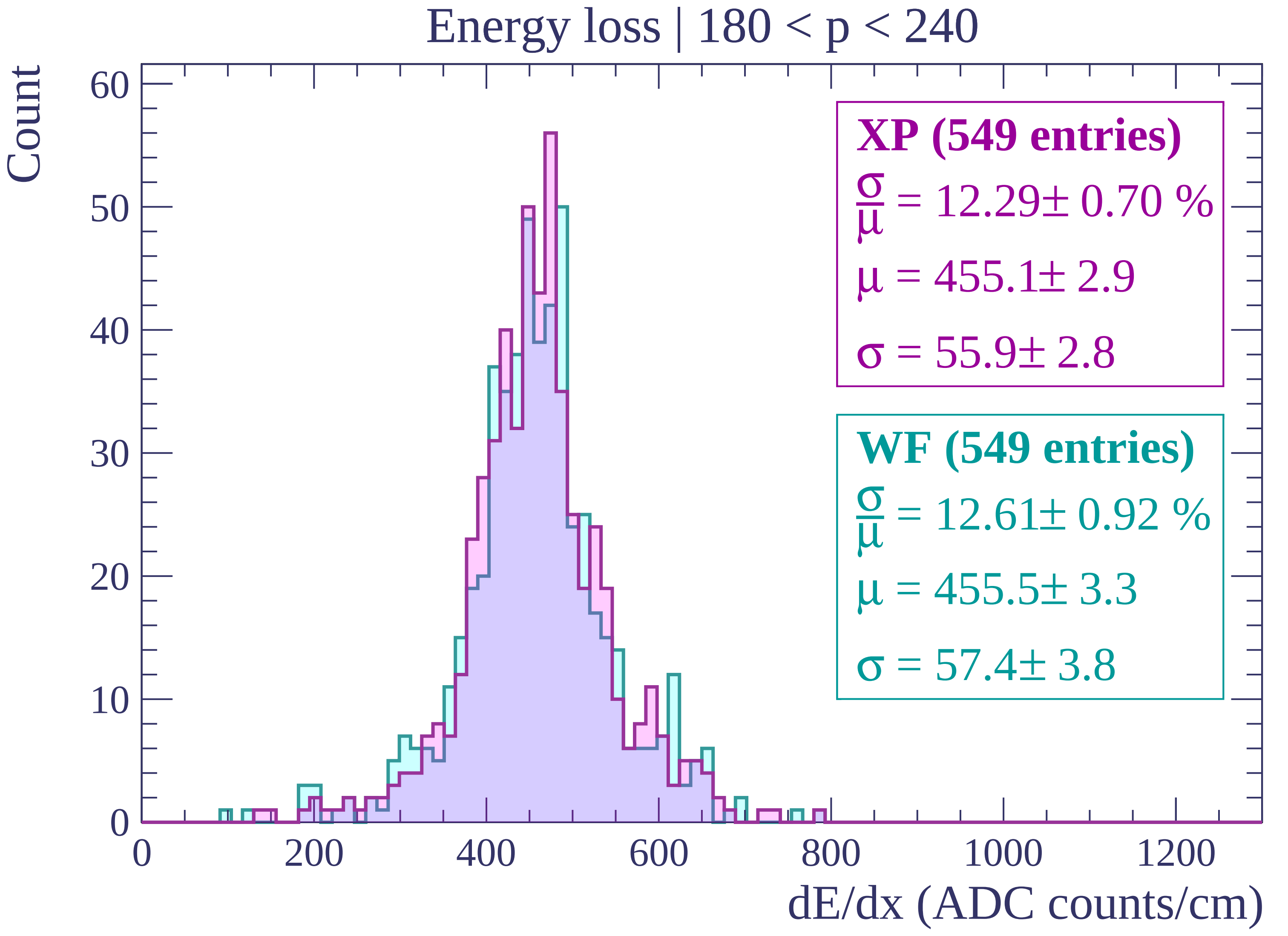


Energy loss | 0 < p < 60
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (975 entries)
$\frac{\sigma}{\mu}$ = 11.53± 0.46 %
μ = 511.9± 2.2
σ = 59.0± 2.1
WF (975 entries)
$\frac{\sigma}{\mu}$ = 11.28± 0.44 %
μ = 503.3± 2.2
σ = 56.8± 1.9

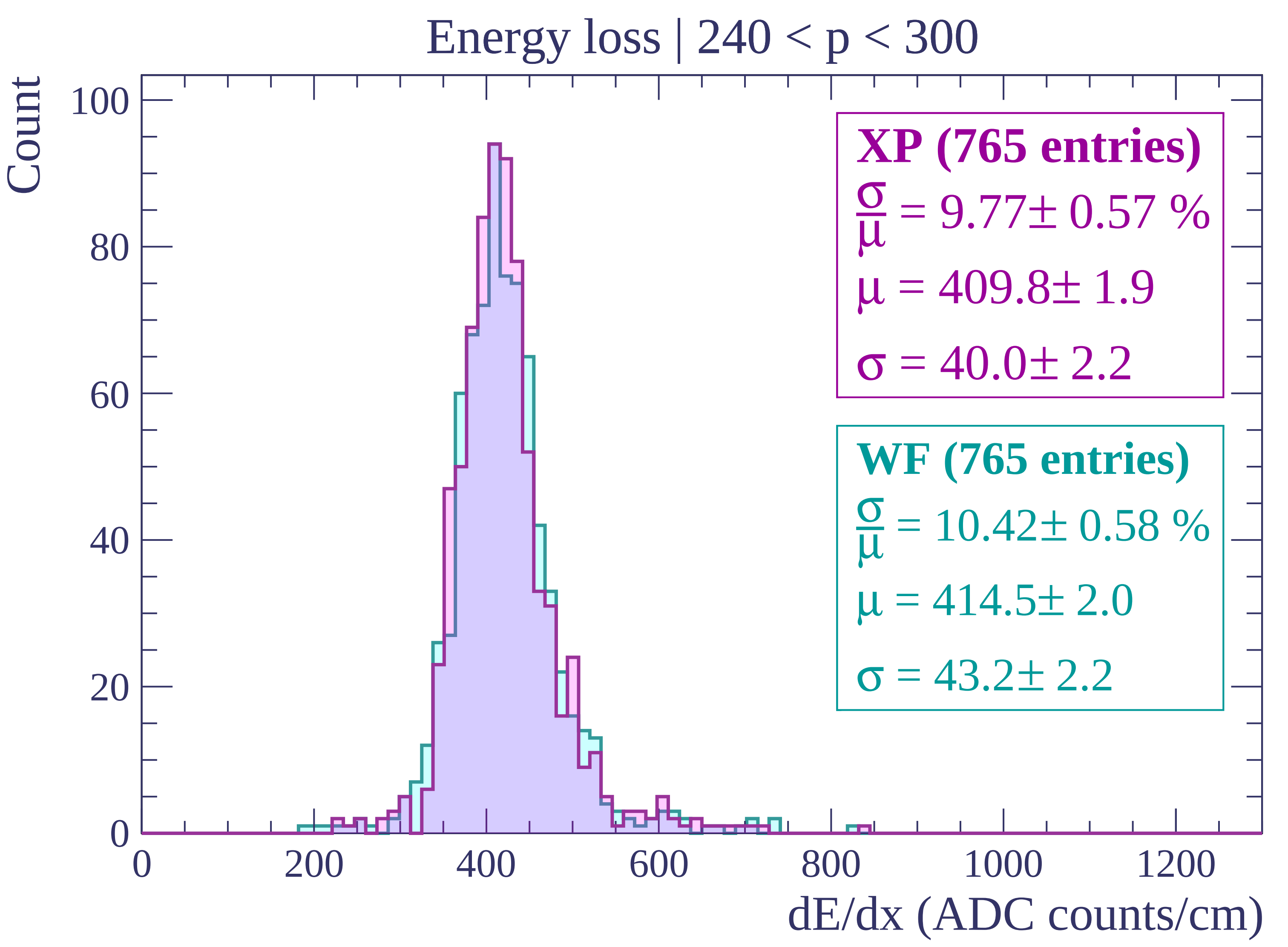

Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
XP (700 entries)
$\frac{\sigma}{\mu}$ = 9.04± 0.52 %
μ = 551.8± 2.4
σ = 49.9± 2.6
WF (700 entries)
$\frac{\sigma}{\mu}$ = 10.13± 0.51 %
μ = 547.7± 2.6
σ = 55.5± 2.5
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



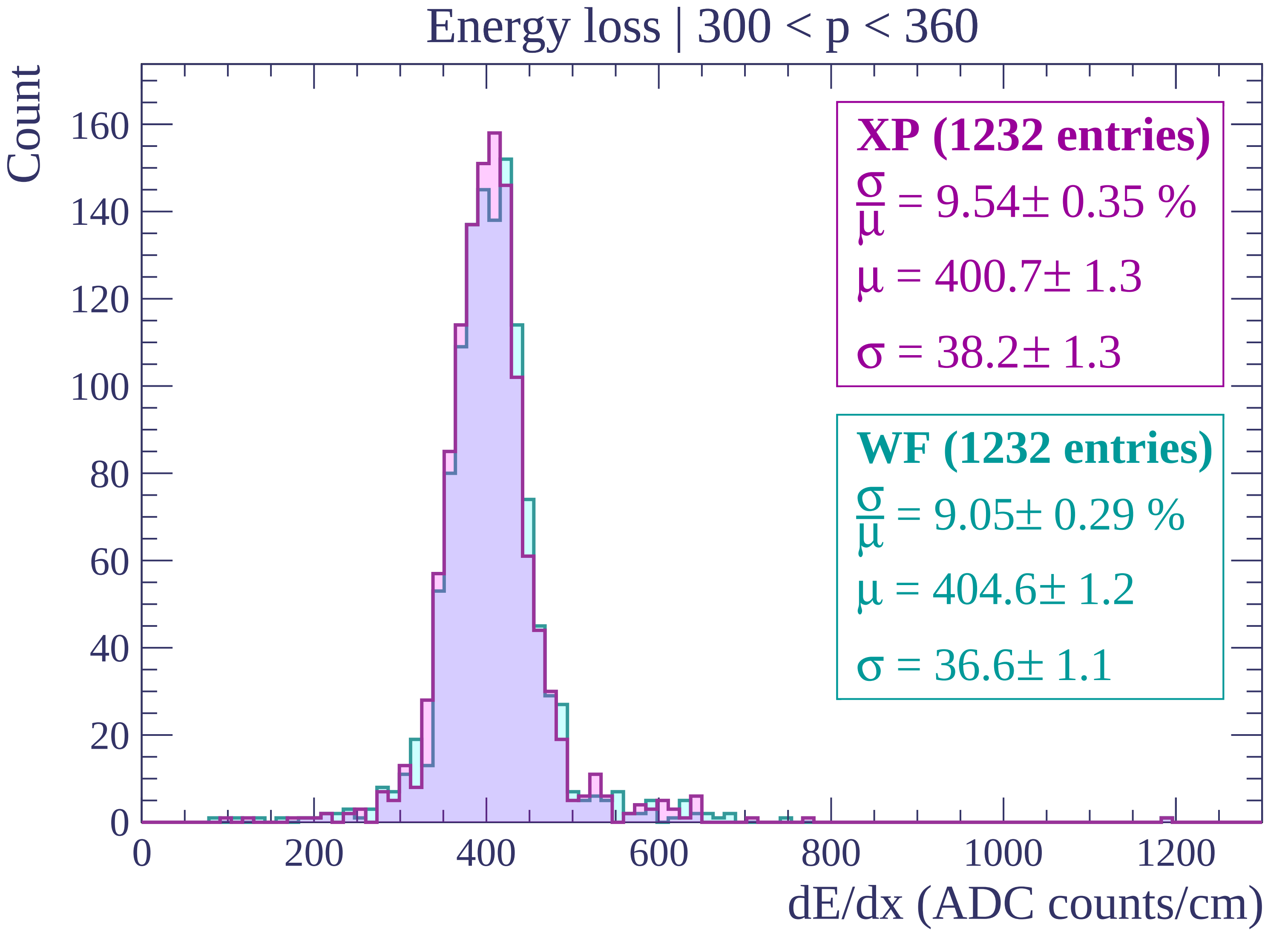
Energy loss | 120 < p < 180
Count
XP (334 entries)
σ/μ = 11.89± 0.83 %
μ = 567.7± 4.6
σ = 67.5± 4.2
WF (334 entries)
σ/μ = 12.31± 0.82 %
μ = 563.7± 4.8
σ = 69.4± 4.0
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

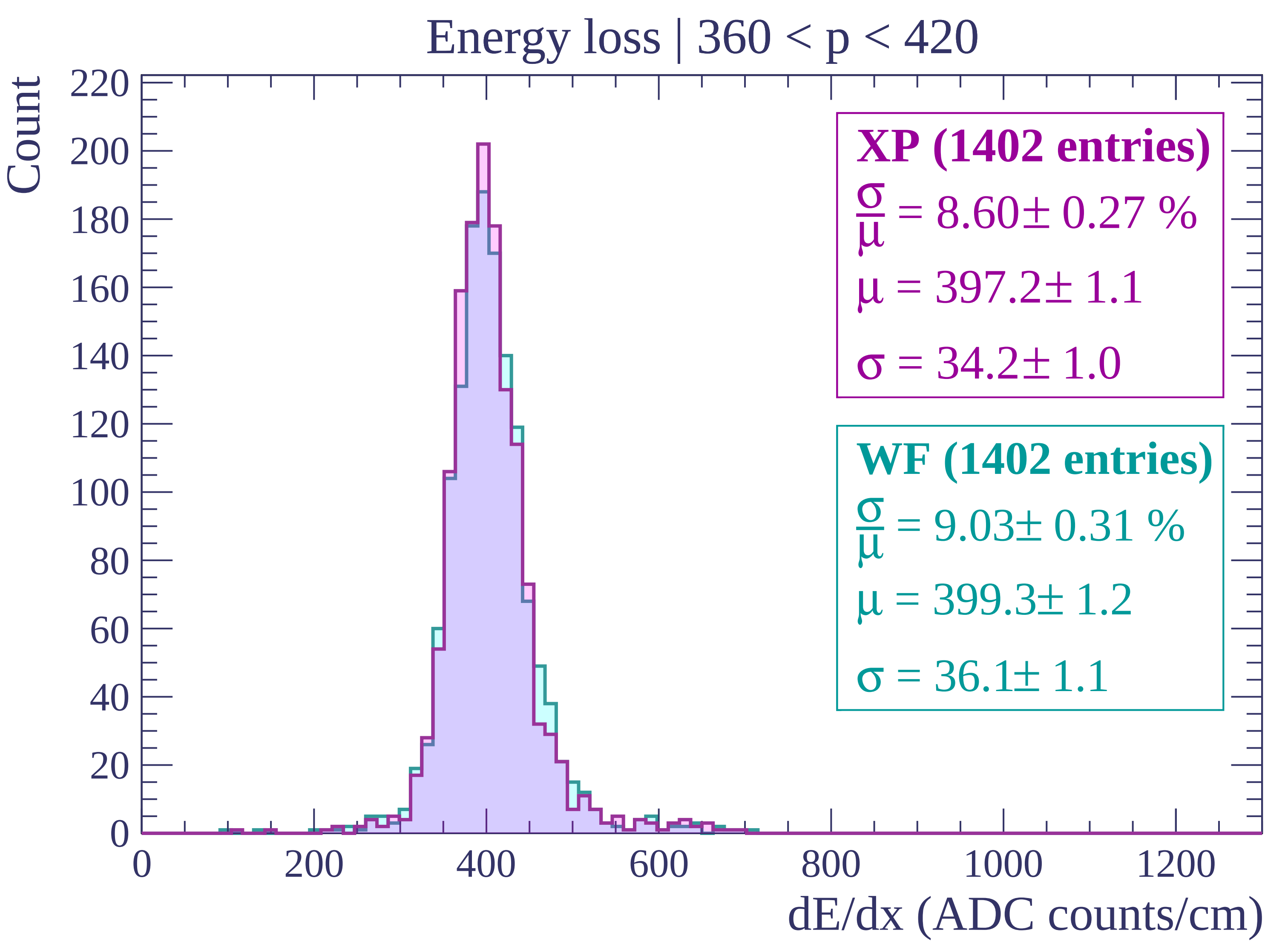


Energy loss | 180 < p < 240
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (549 entries)
$\frac{\sigma}{\mu}$ = 12.29± 0.70 %
μ = 455.1± 2.9
σ = 55.9± 2.8
WF (549 entries)
$\frac{\sigma}{\mu}$ = 12.61± 0.92 %
μ = 455.5± 3.3
σ = 57.4± 3.8

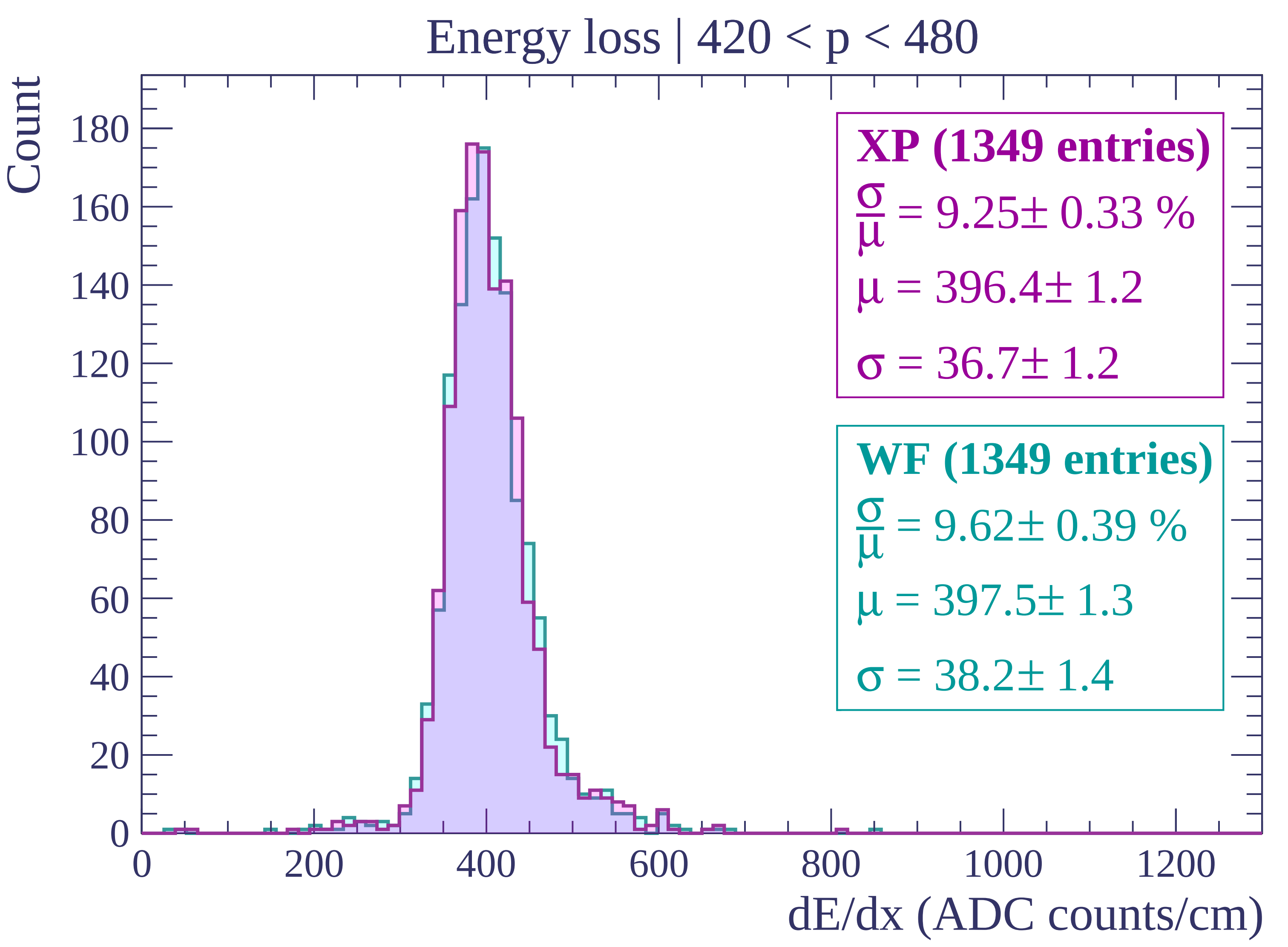


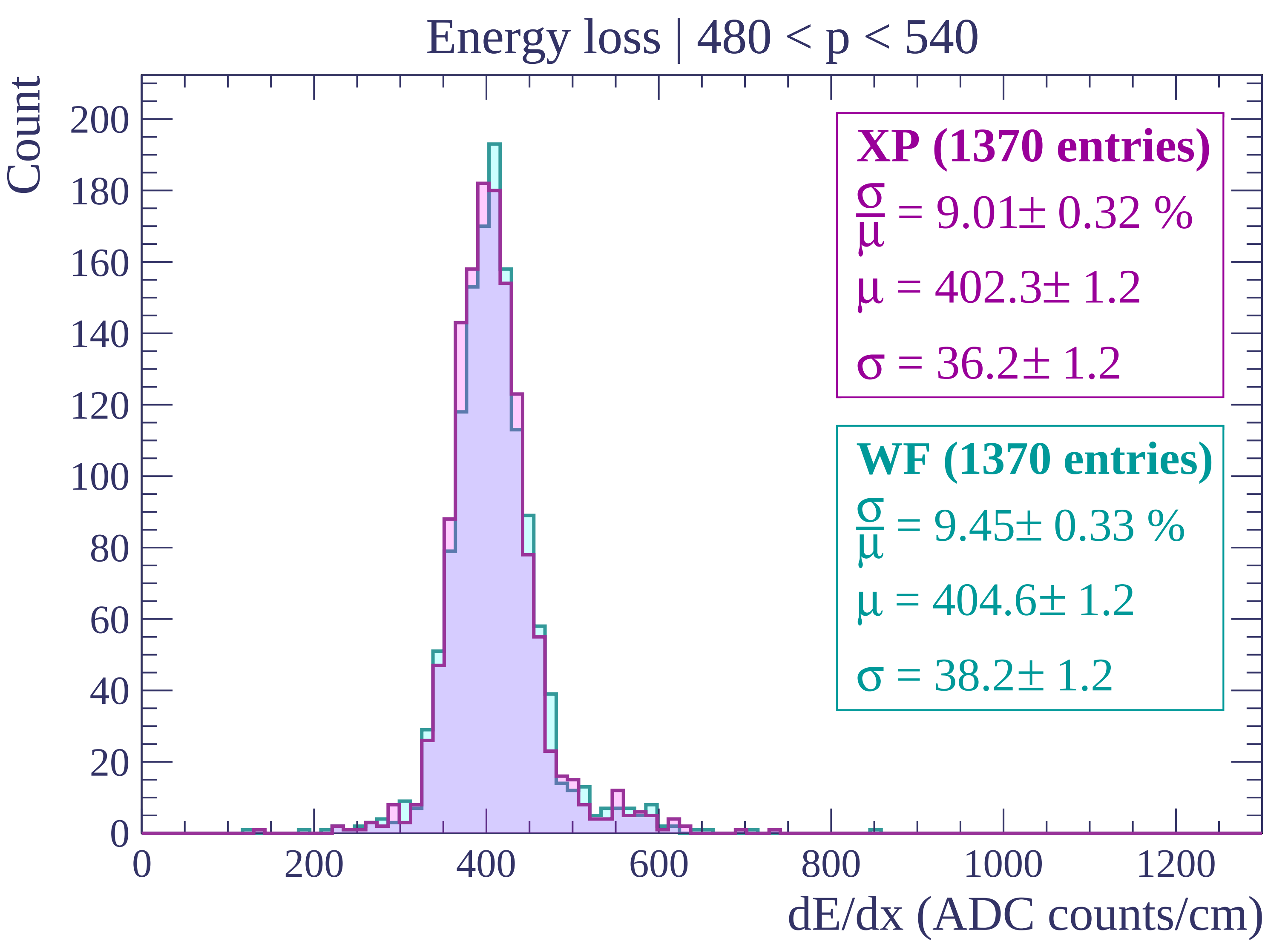
Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (765 entries)
$\frac{\sigma}{\mu}$ = 9.77± 0.57 %
μ = 409.8± 1.9
σ = 40.0± 2.2
WF (765 entries)
$\frac{\sigma}{\mu}$ = 10.42± 0.58 %
μ = 414.5± 2.0
σ = 43.2± 2.2


Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP (1232 entries)
σ/μ = 9.54± 0.35 %
μ = 400.7± 1.3
σ = 38.2± 1.3
WF (1232 entries)
σ/μ = 9.05± 0.29 %
μ = 404.6± 1.2
σ = 36.6± 1.1

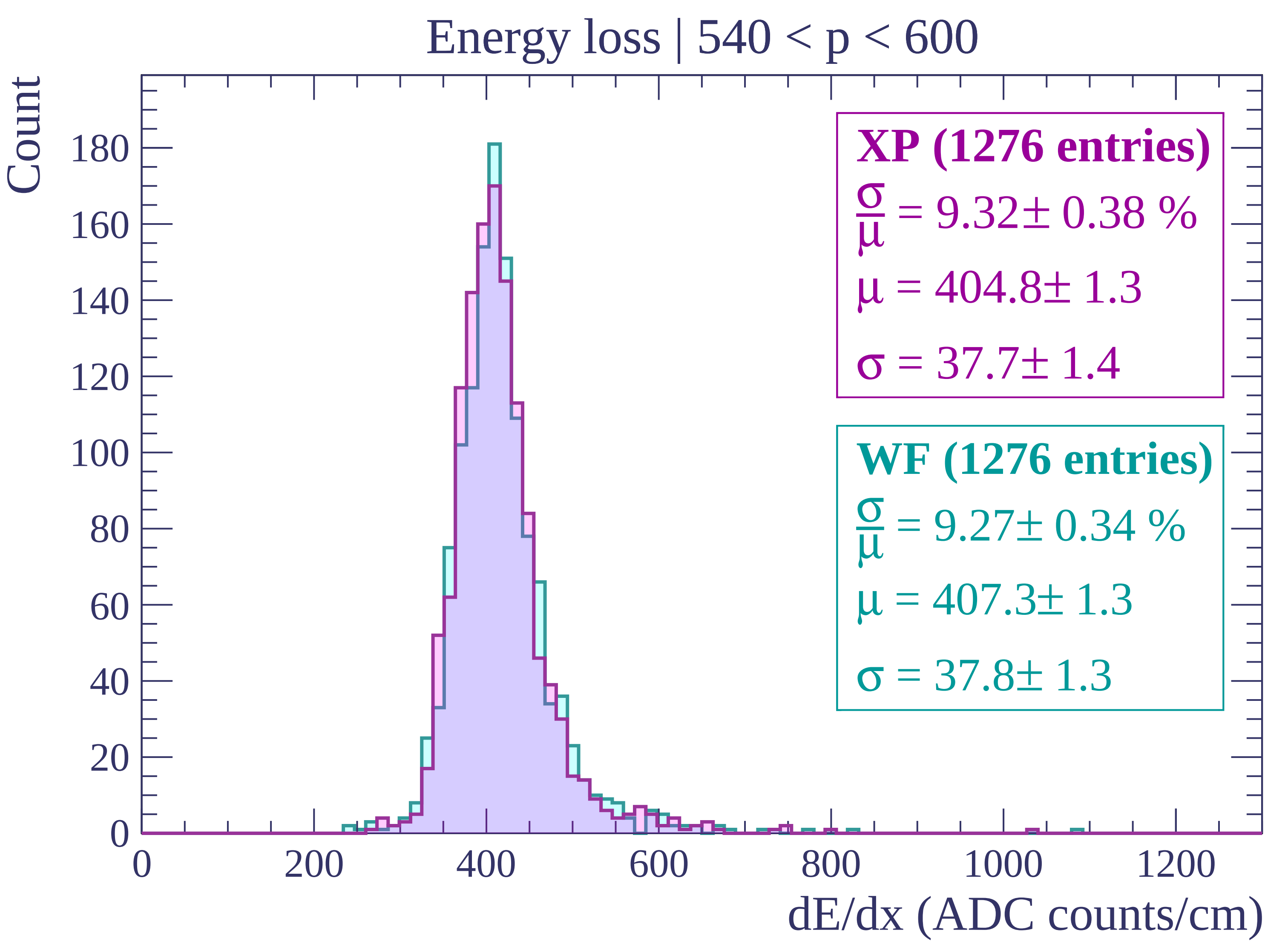


Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
XP (1402 entries)
σ/μ = 8.60± 0.27 %
μ = 397.2± 1.1
σ = 34.2± 1.0
WF (1402 entries)
σ/μ = 9.03± 0.31 %
μ = 399.3± 1.2
σ = 36.1± 1.1
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200

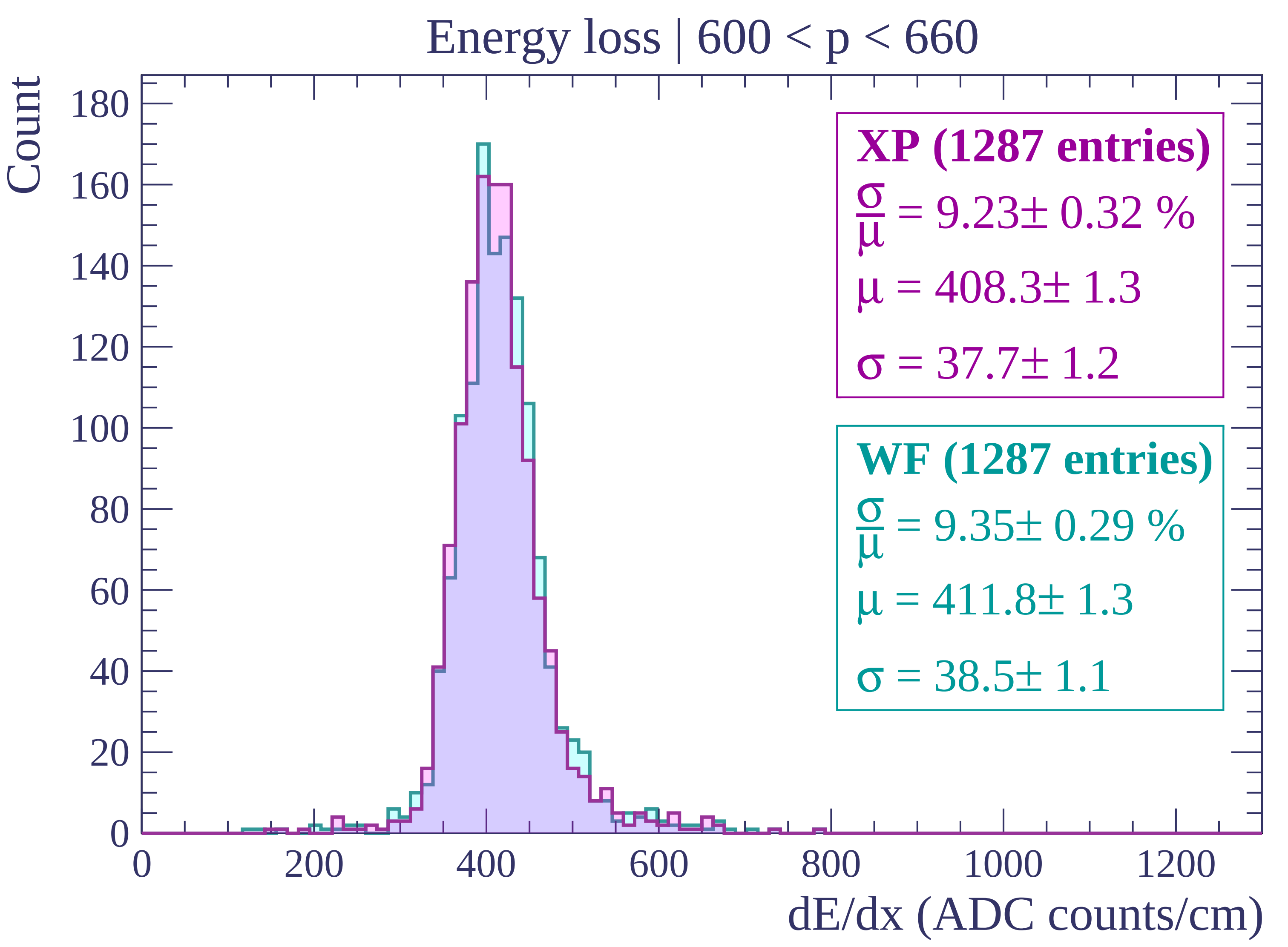

Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1349 entries)
σ/μ = 9.25± 0.33 %
μ = 396.4± 1.2
σ = 36.7± 1.2
WF (1349 entries)
σ/μ = 9.62± 0.39 %
μ = 397.5± 1.3
σ = 38.2± 1.4

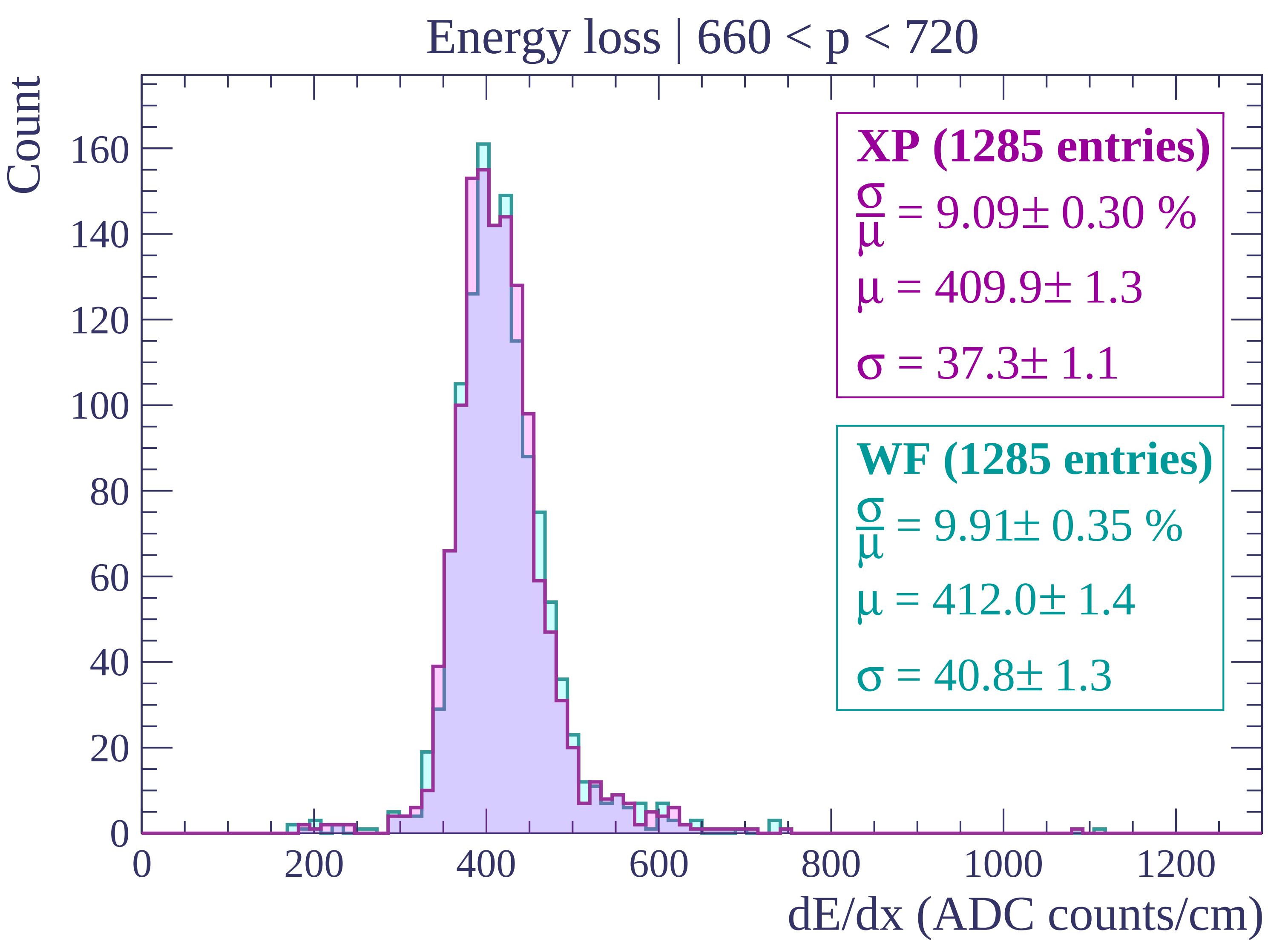


Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (1370 entries)
σ/μ = 9.01± 0.32 %
μ = 402.3± 1.2
σ = 36.2± 1.2
WF (1370 entries)
σ/μ = 9.45± 0.33 %
μ = 404.6± 1.2
σ = 38.2± 1.2

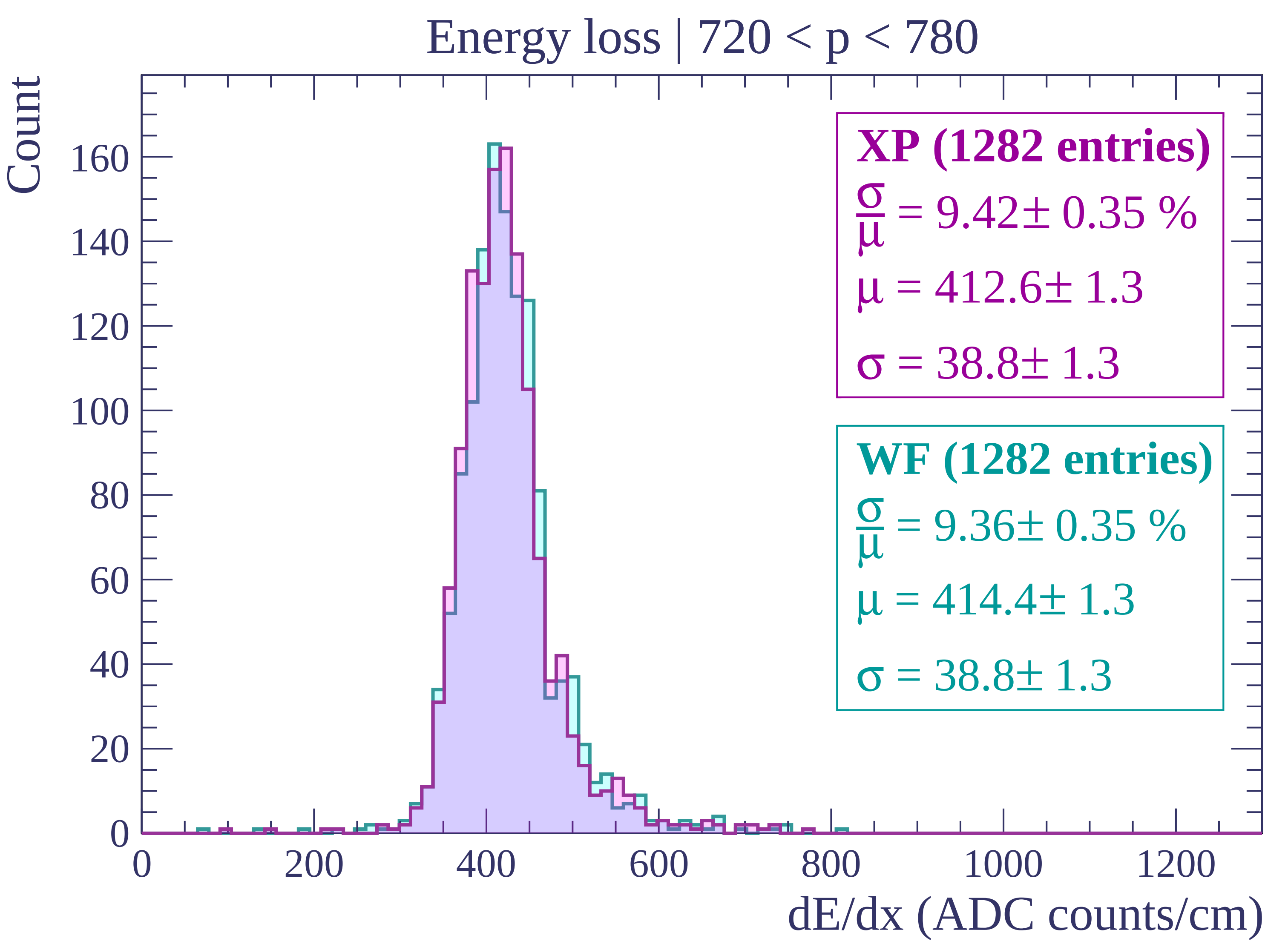


Energy loss | 540 < p < 600
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1276 entries)
$\frac{\sigma}{\mu}$ = 9.32± 0.38 %
μ = 404.8± 1.3
σ = 37.7± 1.4
WF (1276 entries)
$\frac{\sigma}{\mu}$ = 9.27± 0.34 %
μ = 407.3± 1.3
σ = 37.8± 1.3

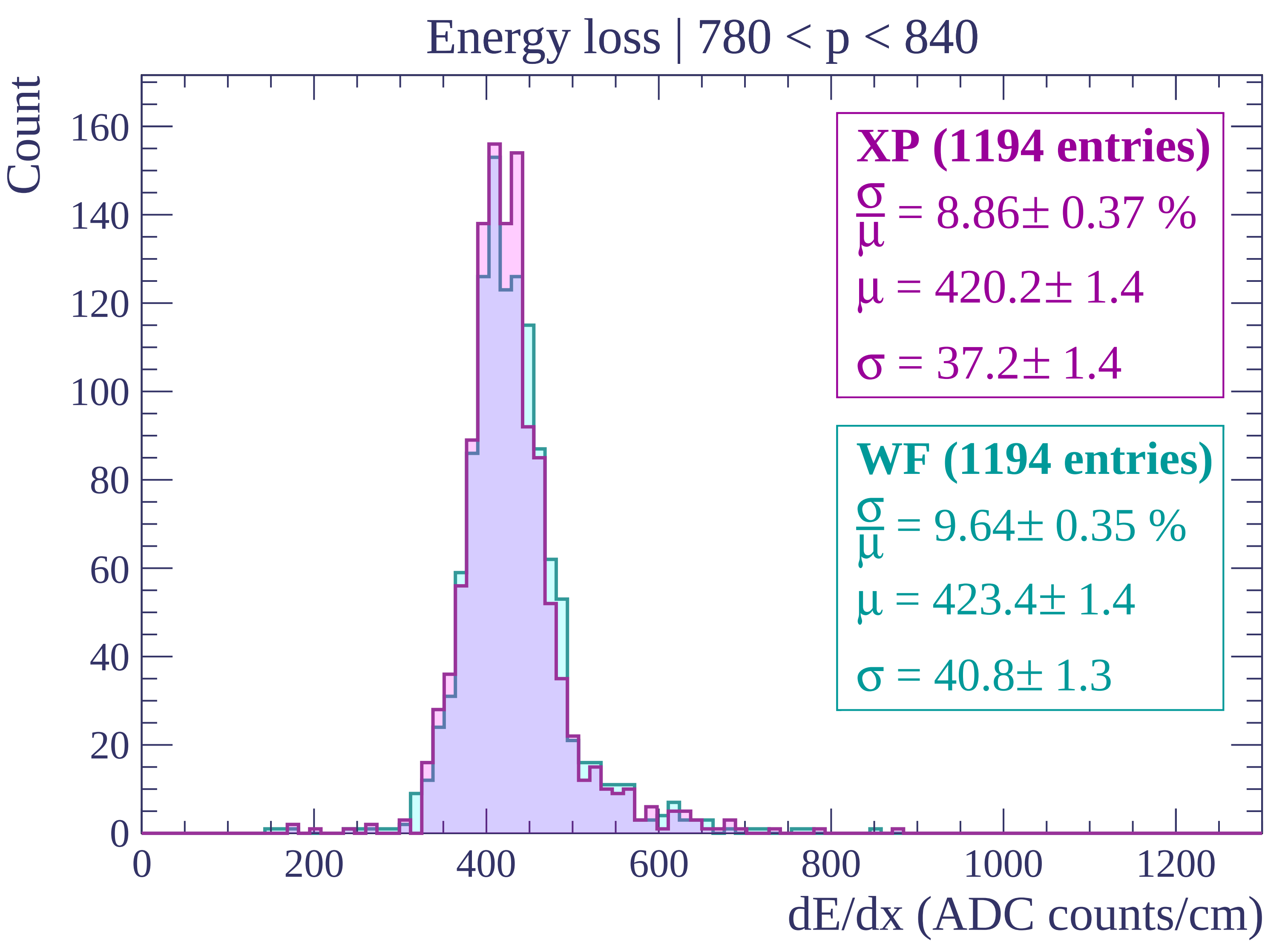

Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP (1287 entries)
σ/μ = 9.23± 0.32 %
μ = 408.3± 1.3
σ = 37.7± 1.2
WF (1287 entries)
σ/μ = 9.35± 0.29 %
μ = 411.8± 1.3
σ = 38.5± 1.1

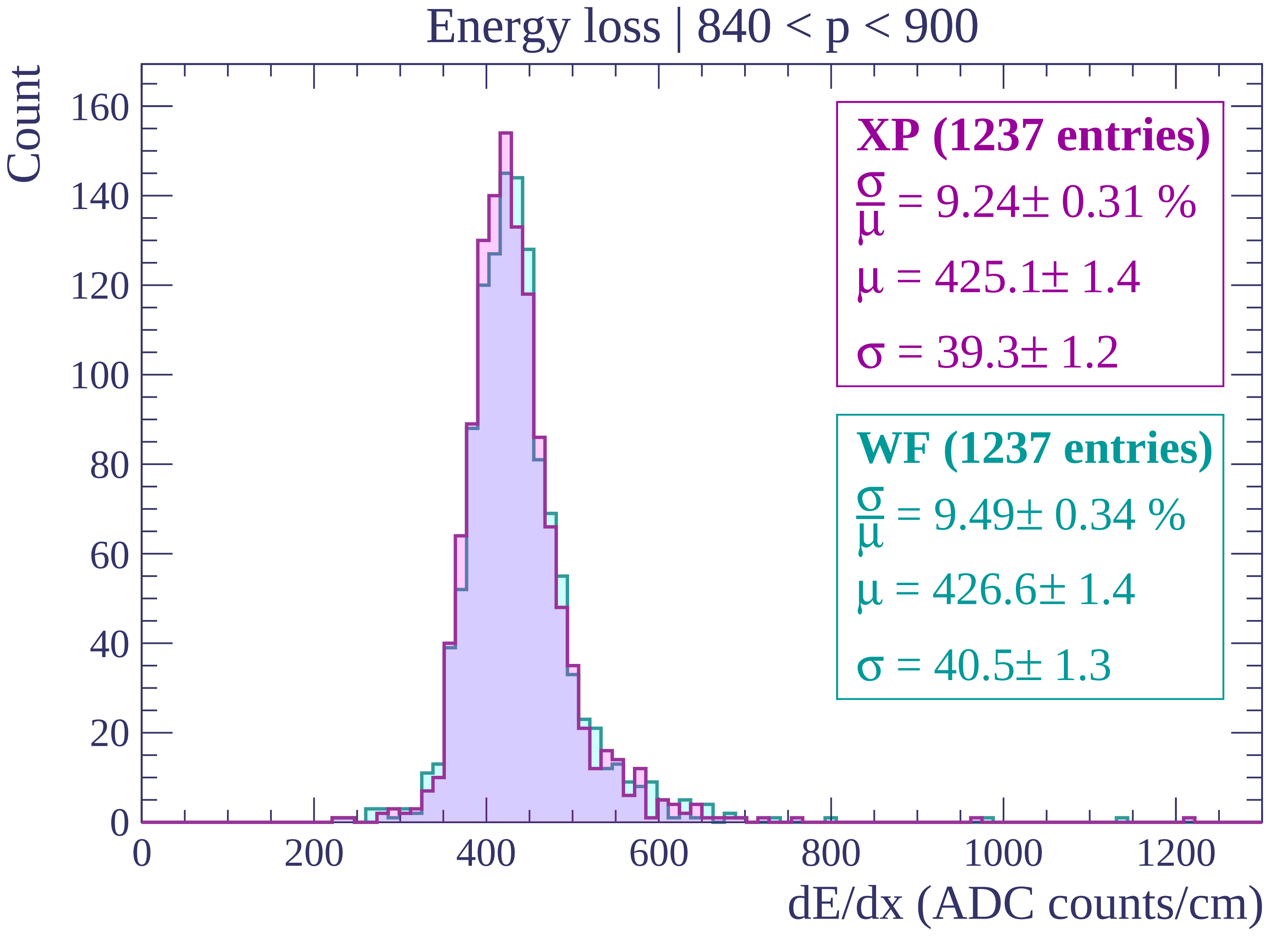


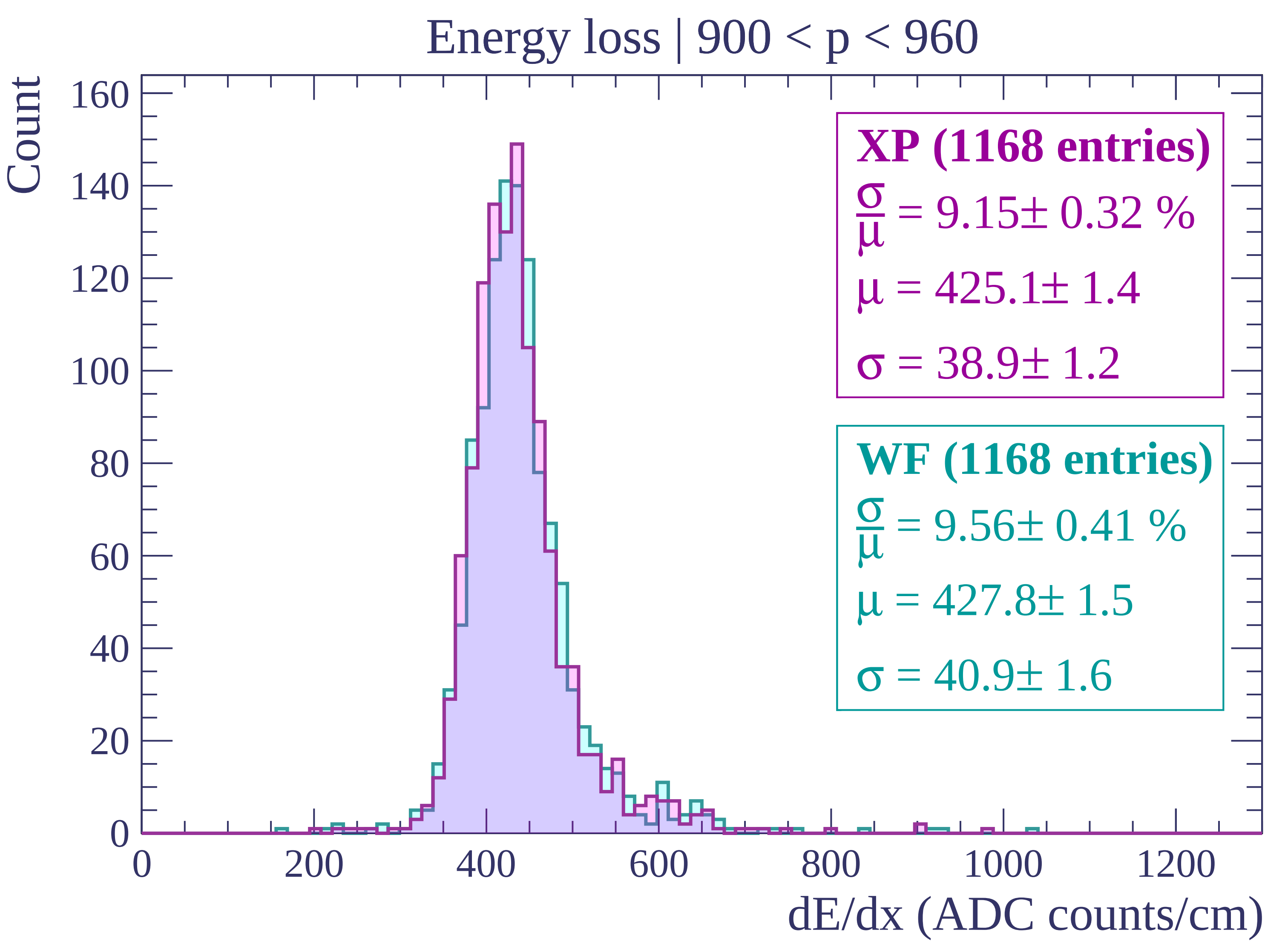
Energy loss | 660 < p < 720
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1285 entries)
$\frac{\sigma}{\mu}$ = 9.09± 0.30 %
μ = 409.9± 1.3
σ = 37.3± 1.1
WF (1285 entries)
$\frac{\sigma}{\mu}$ = 9.91± 0.35 %
μ = 412.0± 1.4
σ = 40.8± 1.3



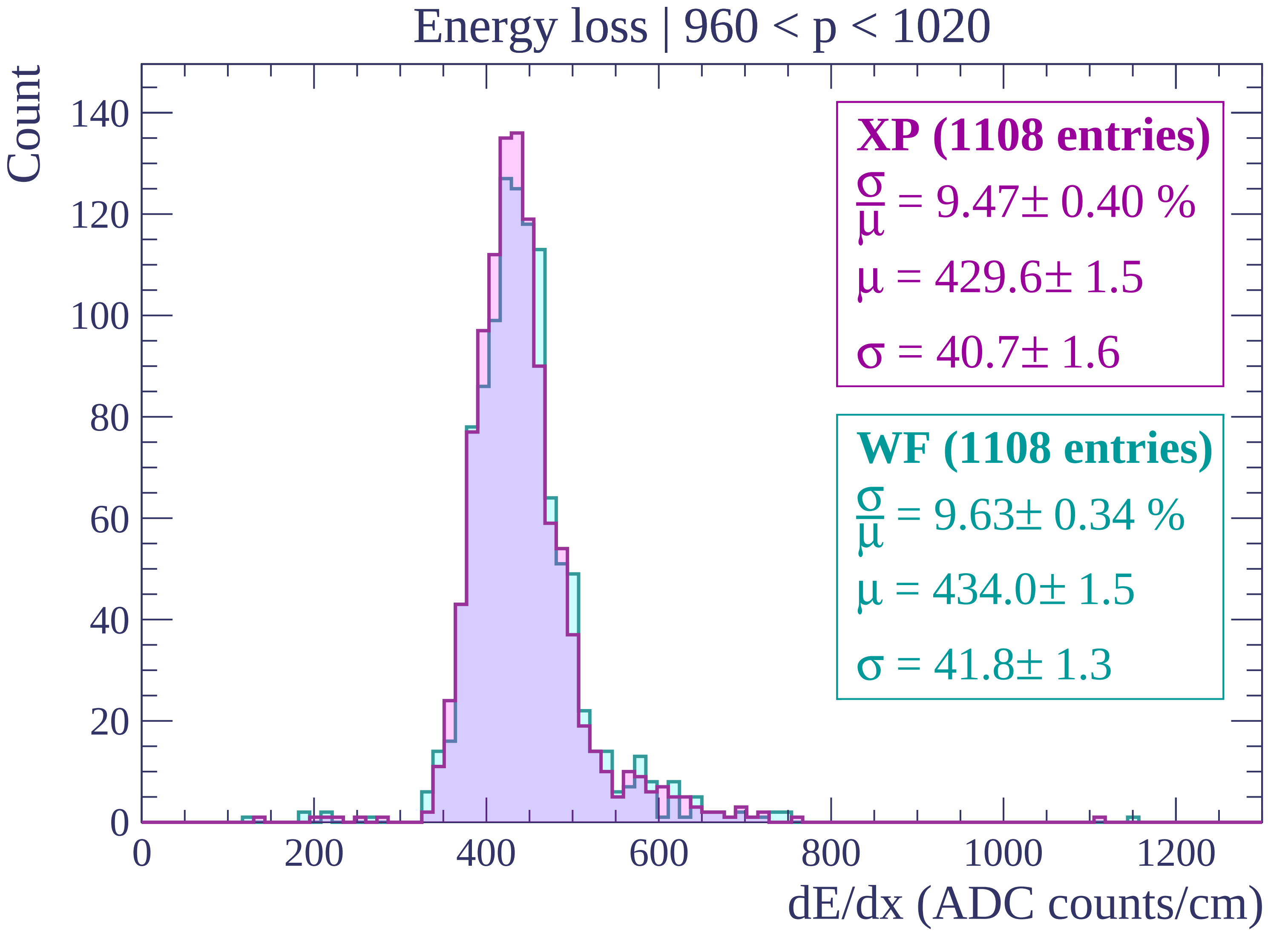
Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP (1282 entries)
σ/μ = 9.42± 0.35 %
μ = 412.6± 1.3
σ = 38.8± 1.3
WF (1282 entries)
σ/μ = 9.36± 0.35 %
μ = 414.4± 1.3
σ = 38.8± 1.3


Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
XP (1194 entries)
$\frac{\sigma}{\mu}$ = 8.86± 0.37 %
μ = 420.2± 1.4
σ = 37.2± 1.4
WF (1194 entries)
$\frac{\sigma}{\mu}$ = 9.64± 0.35 %
μ = 423.4± 1.4
σ = 40.8± 1.3
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200

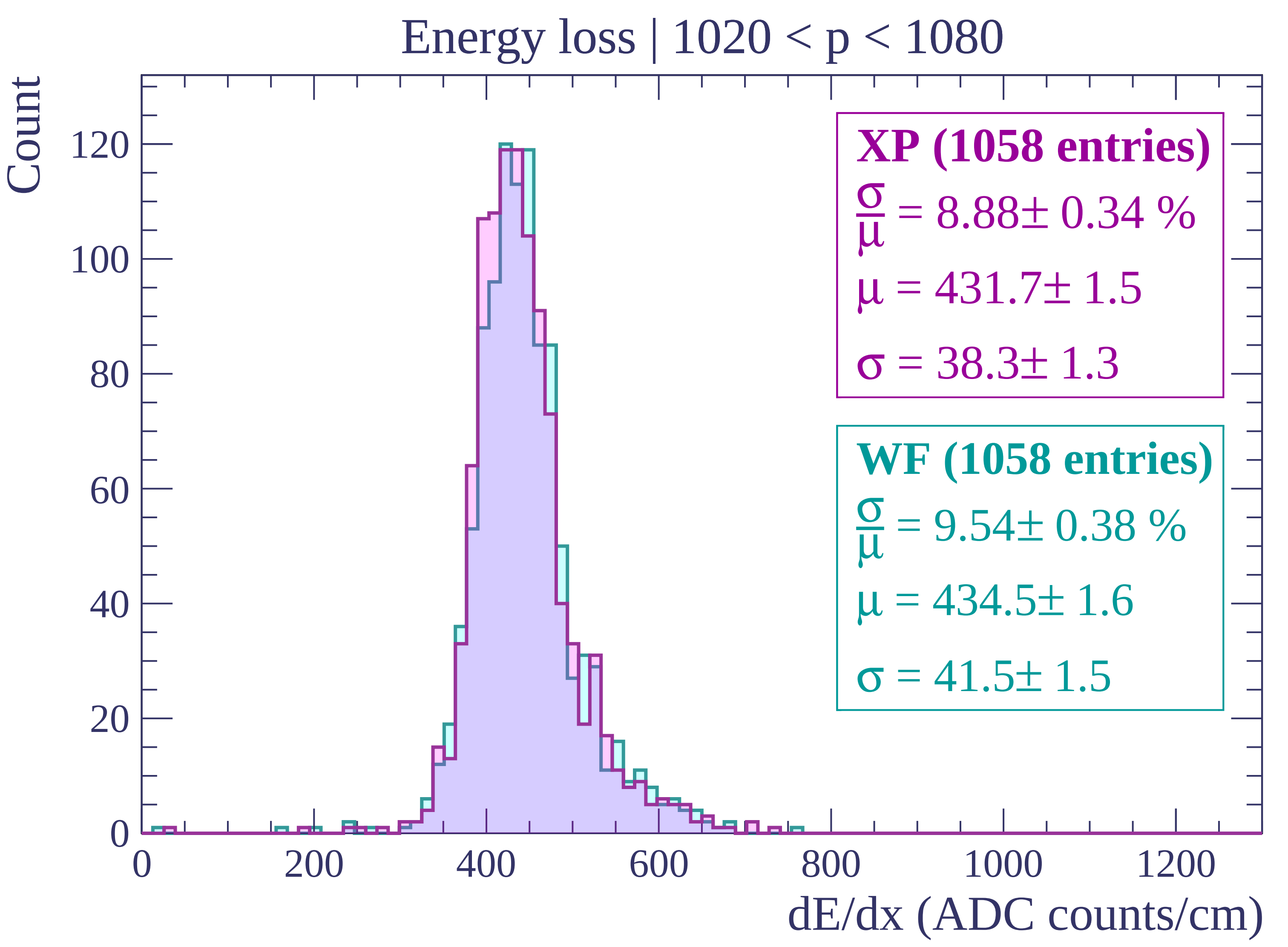

Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
XP (1237 entries)
σ/μ = 9.24± 0.31 %
μ = 425.1± 1.4
σ = 39.3± 1.2
WF (1237 entries)
σ/μ = 9.49± 0.34 %
μ = 426.6± 1.4
σ = 40.5± 1.3



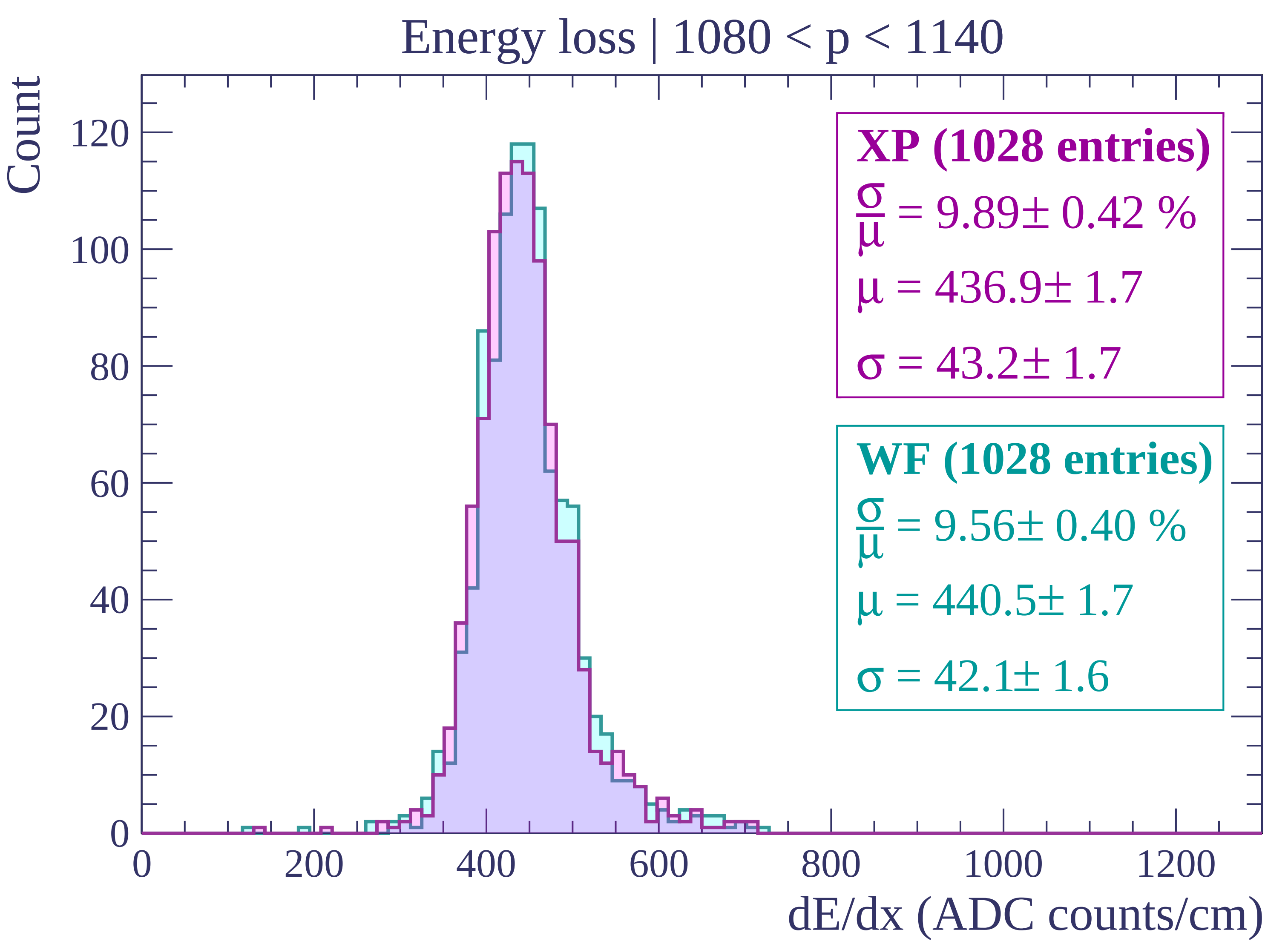
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1168 entries)
σ/μ = 9.15± 0.32 %
μ = 425.1± 1.4
σ = 38.9± 1.2
WF (1168 entries)
σ/μ = 9.56± 0.41 %
μ = 427.8± 1.5
σ = 40.9± 1.6



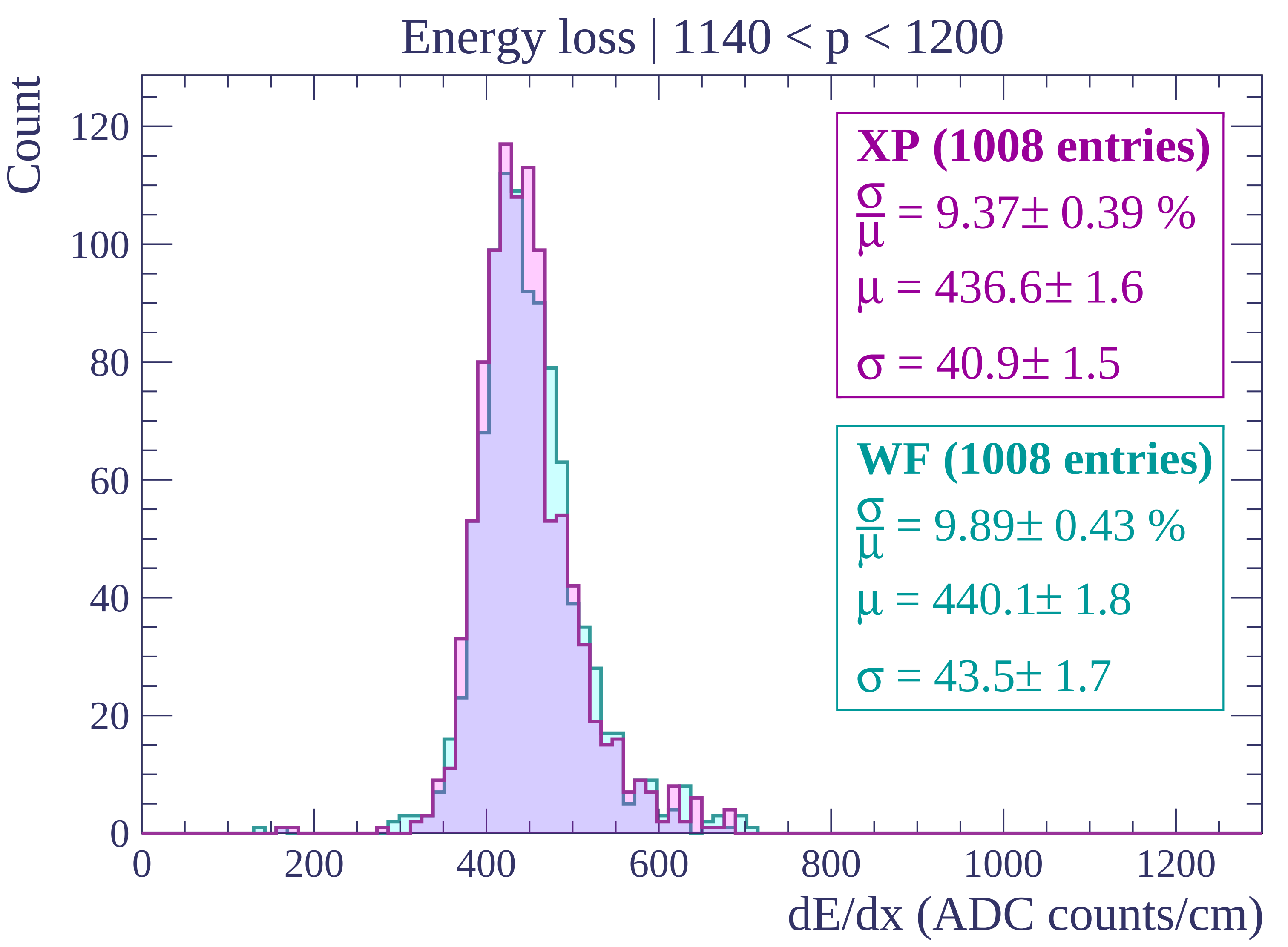
Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1108 entries)
$\frac{\sigma}{\mu}$ = 9.47± 0.40 %
μ = 429.6± 1.5
σ = 40.7± 1.6
WF (1108 entries)
$\frac{\sigma}{\mu}$ = 9.63± 0.34 %
μ = 434.0± 1.5
σ = 41.8± 1.3

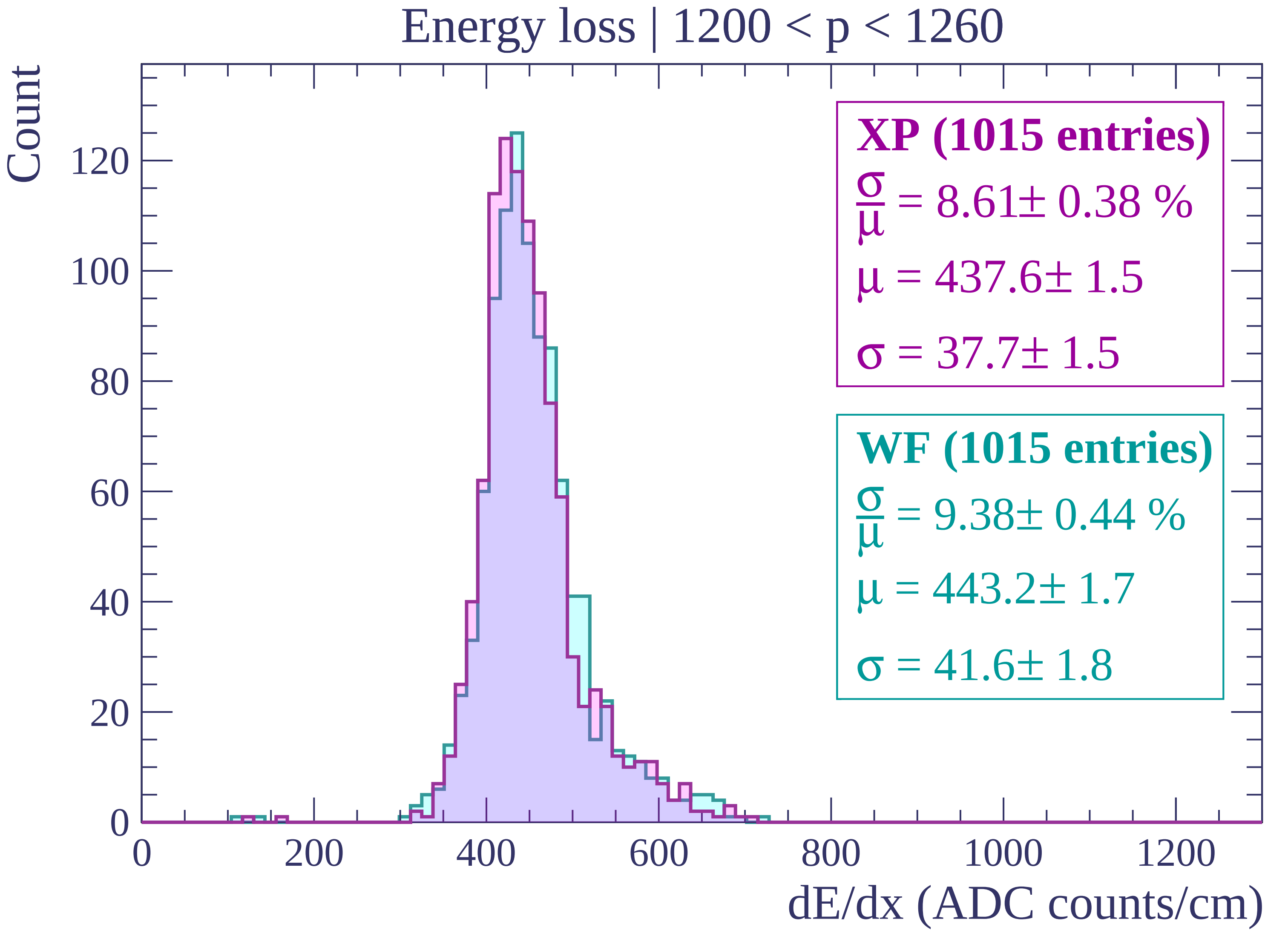


Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1058 entries)
$\frac{\sigma}{\mu}$ = 8.88± 0.34 %
μ = 431.7± 1.5
σ = 38.3± 1.3
WF (1058 entries)
$\frac{\sigma}{\mu}$ = 9.54± 0.38 %
μ = 434.5± 1.6
σ = 41.5± 1.5

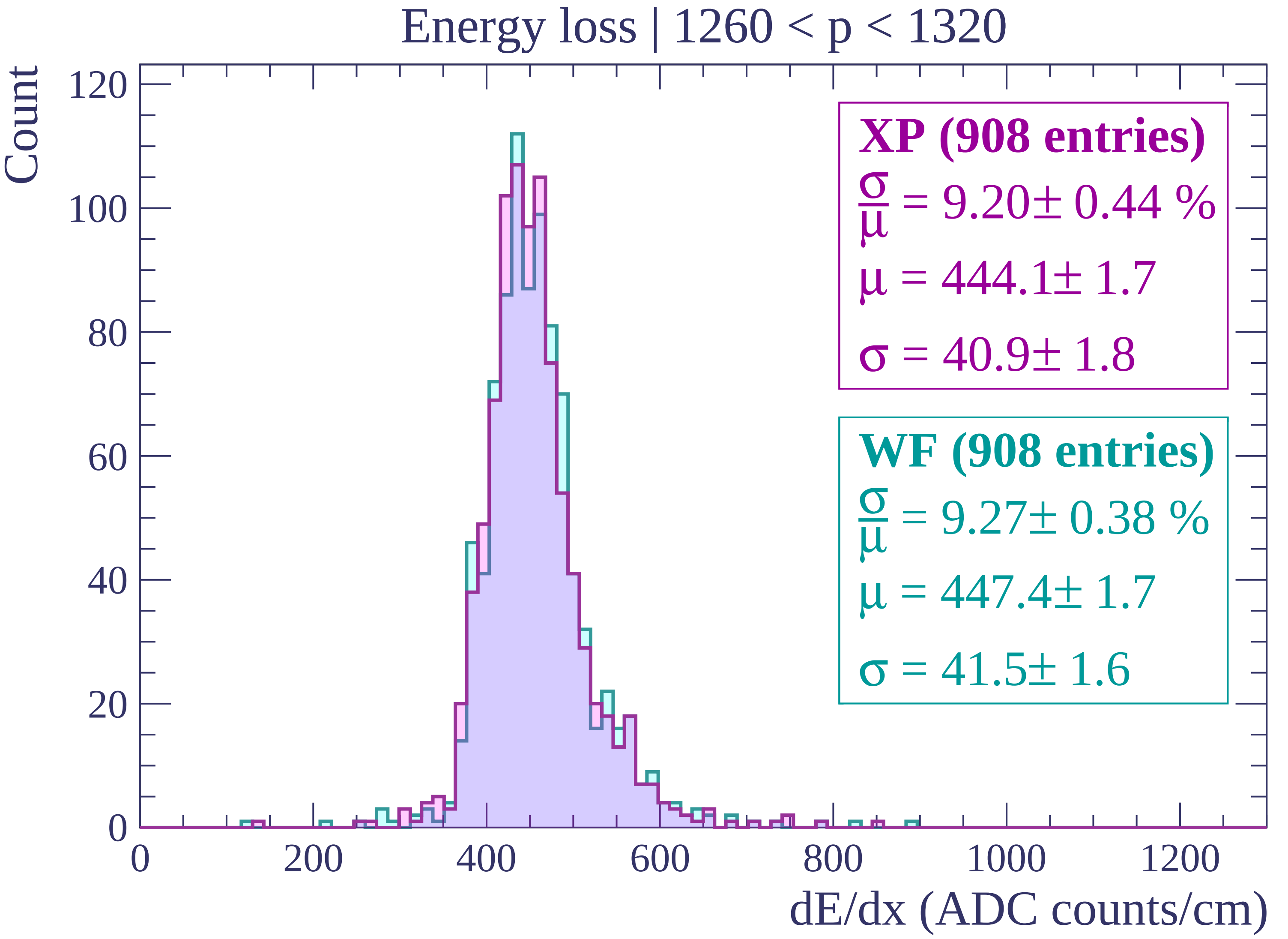


Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP (1028 entries)
σ/μ = 9.89± 0.42 %
μ = 436.9± 1.7
σ = 43.2± 1.7
WF (1028 entries)
σ/μ = 9.56± 0.40 %
μ = 440.5± 1.7
σ = 42.1± 1.6

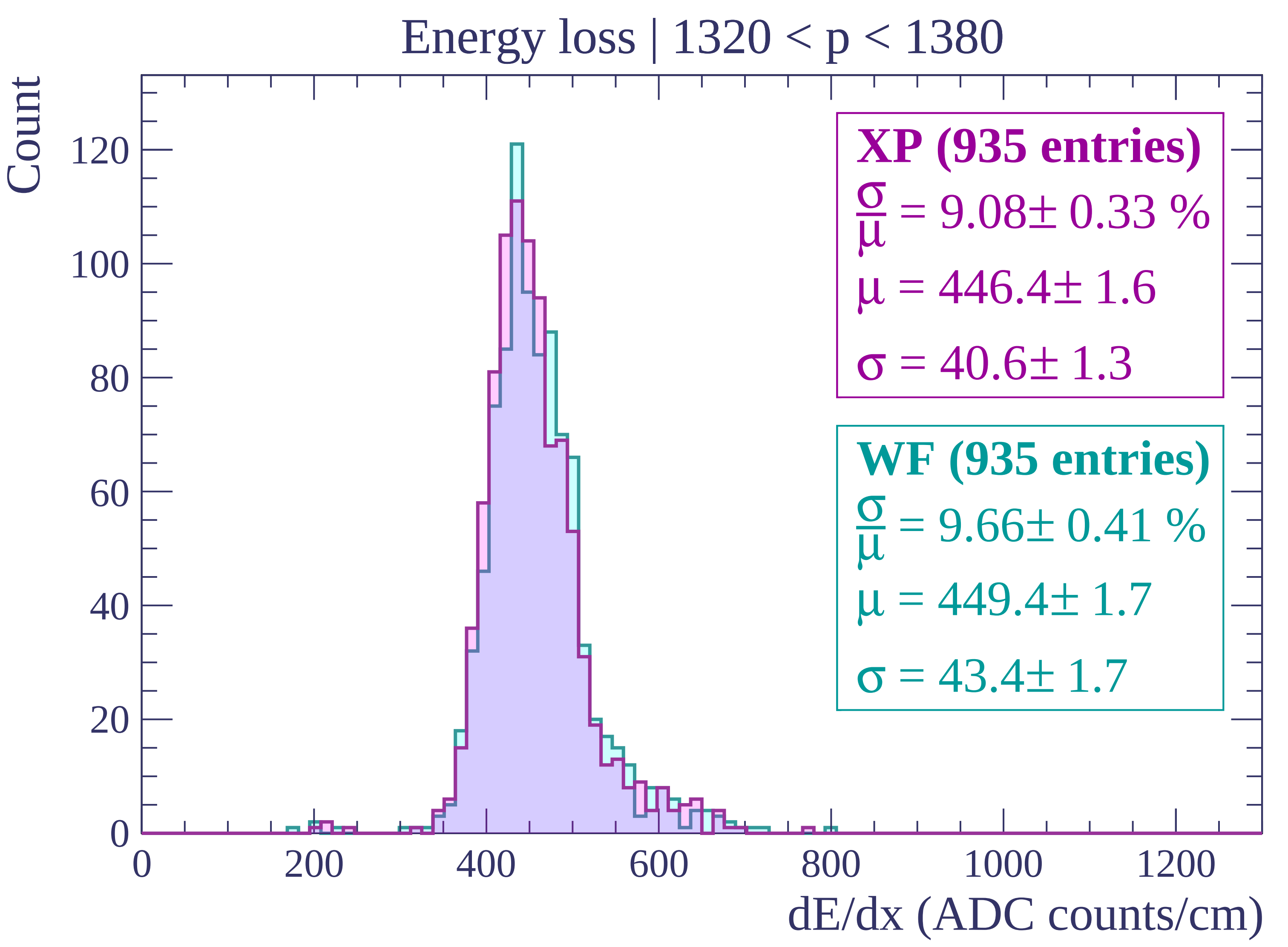

Energy loss | 1140 < p < 1200
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1008 entries)
$\frac{\sigma}{\mu}$ = 9.37± 0.39 %
μ = 436.6± 1.6
σ = 40.9± 1.5
WF (1008 entries)
$\frac{\sigma}{\mu}$ = 9.89± 0.43 %
μ = 440.1± 1.8
σ = 43.5± 1.7



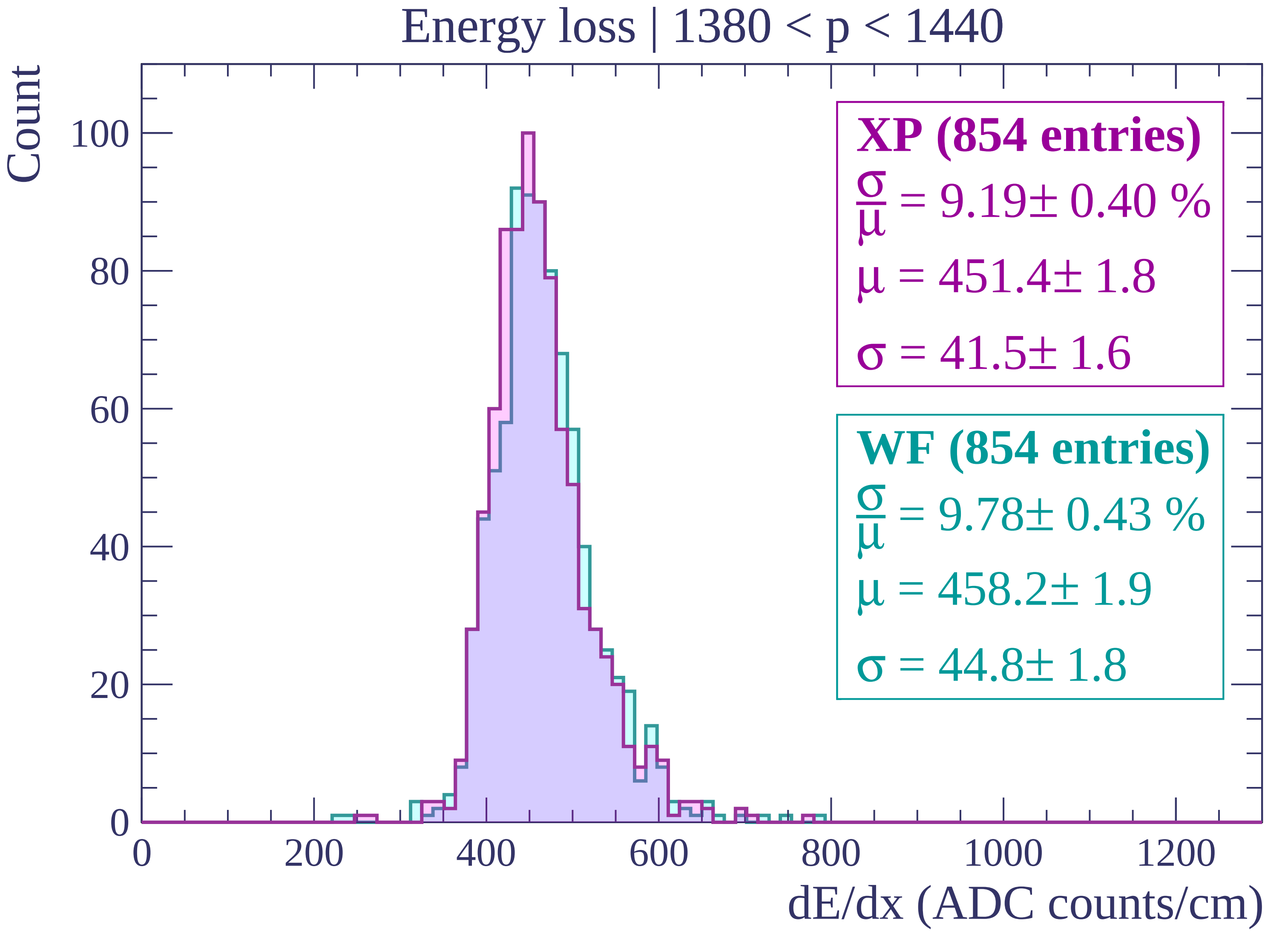
Energy loss | 1200 < p < 1260
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1015 entries)
σ/μ = 8.61± 0.38 %
μ = 437.6± 1.5
σ = 37.7± 1.5
WF (1015 entries)
σ/μ = 9.38± 0.44 %
μ = 443.2± 1.7
σ = 41.6± 1.8

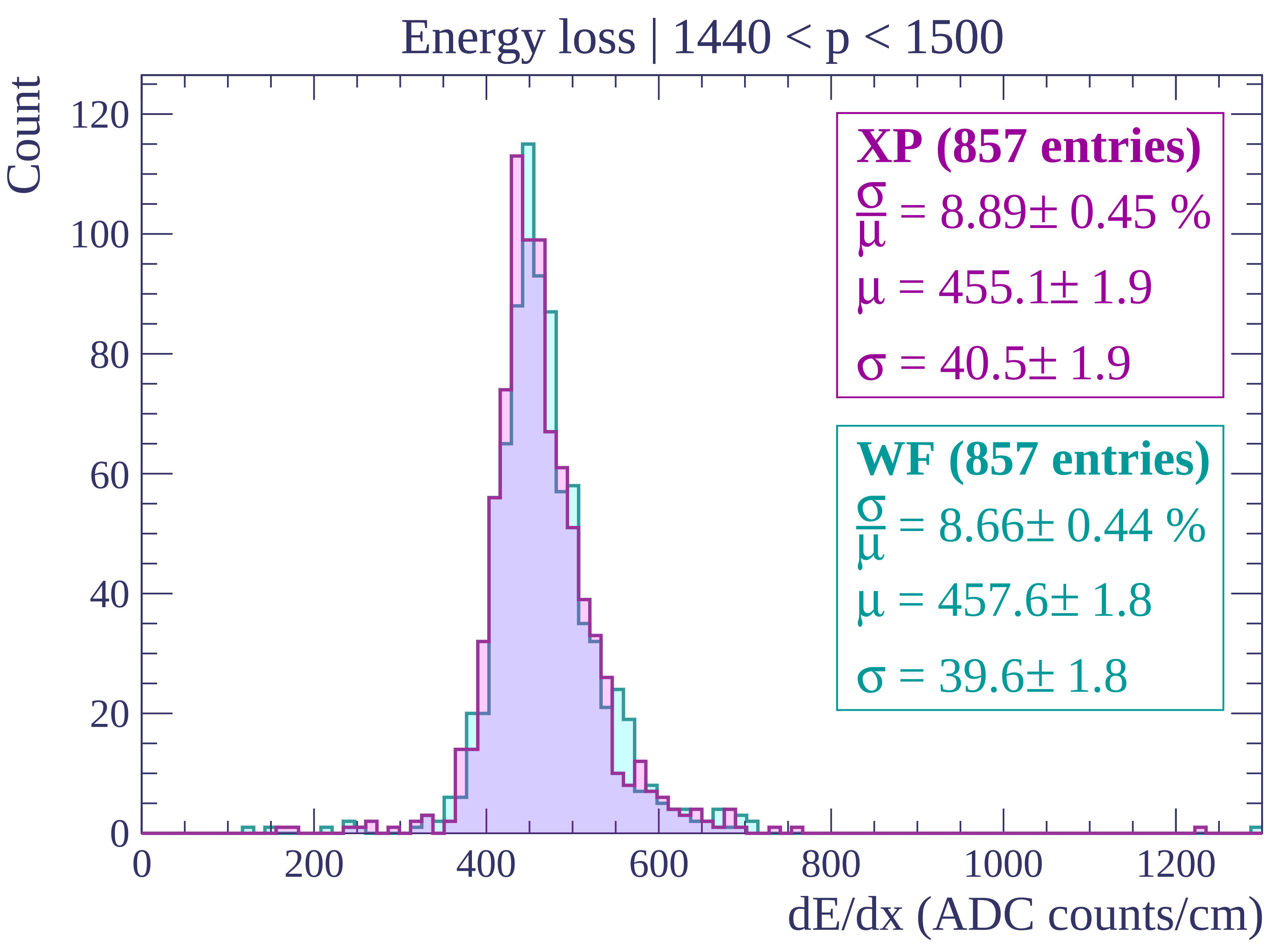


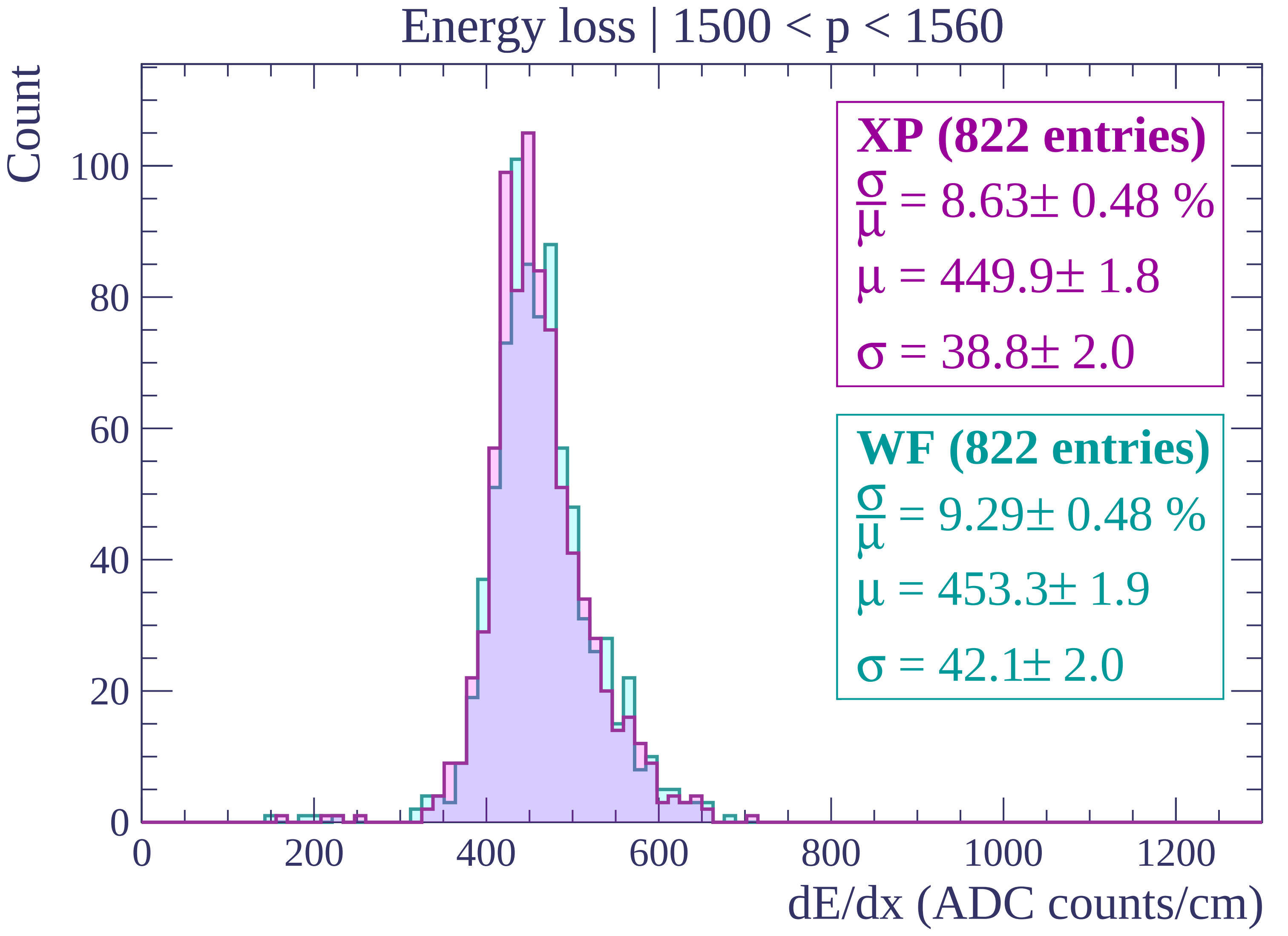
Energy loss | 1260 < p < 1320
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (908 entries)
σ/μ = 9.20± 0.44 %
μ = 444.1± 1.7
σ = 40.9± 1.8
WF (908 entries)
σ/μ = 9.27± 0.38 %
μ = 447.4± 1.7
σ = 41.5± 1.6



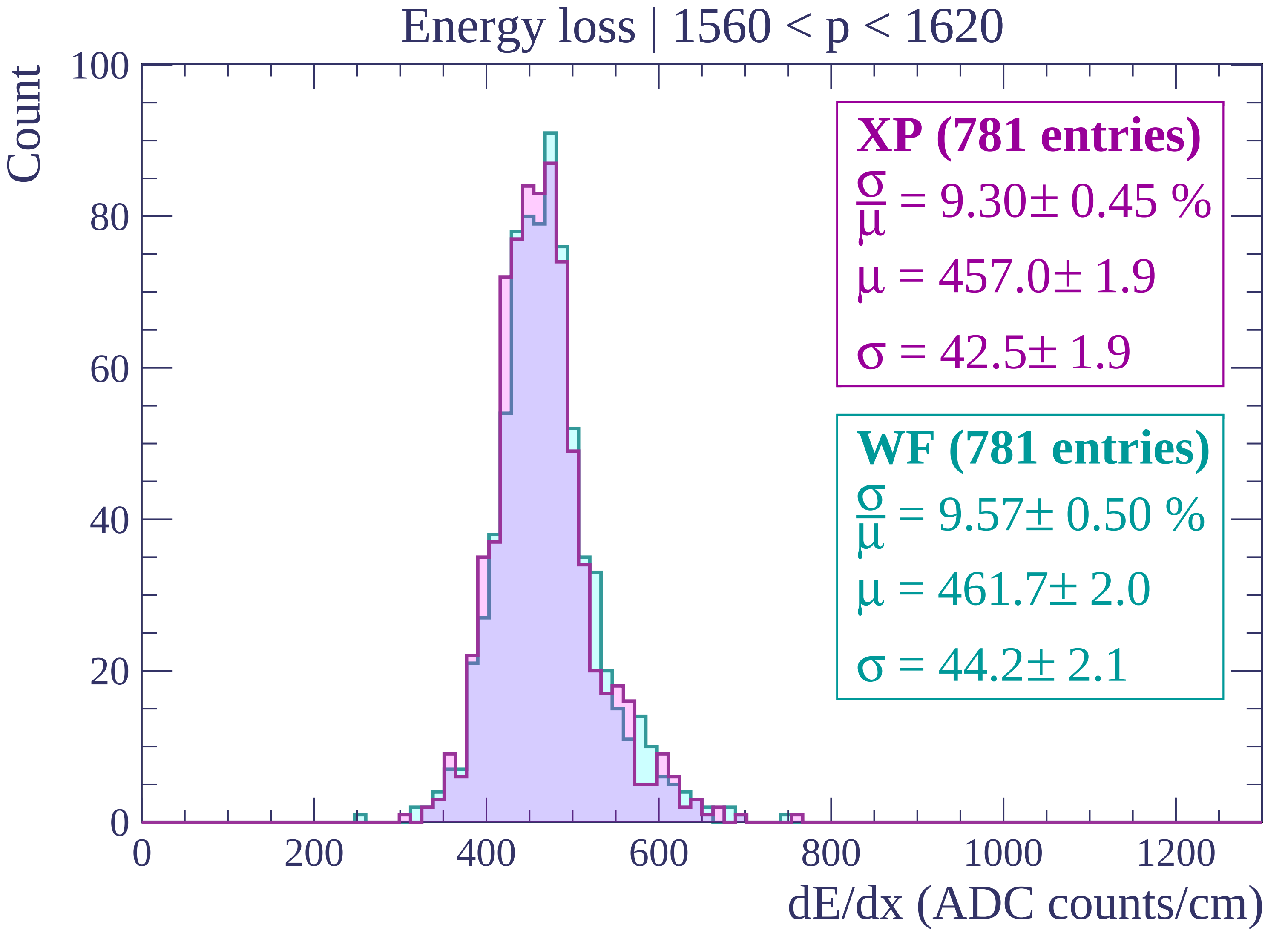
Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (935 entries)
σ/μ = 9.08± 0.33 %
μ = 446.4± 1.6
σ = 40.6± 1.3
WF (935 entries)
σ/μ = 9.66± 0.41 %
μ = 449.4± 1.7
σ = 43.4± 1.7

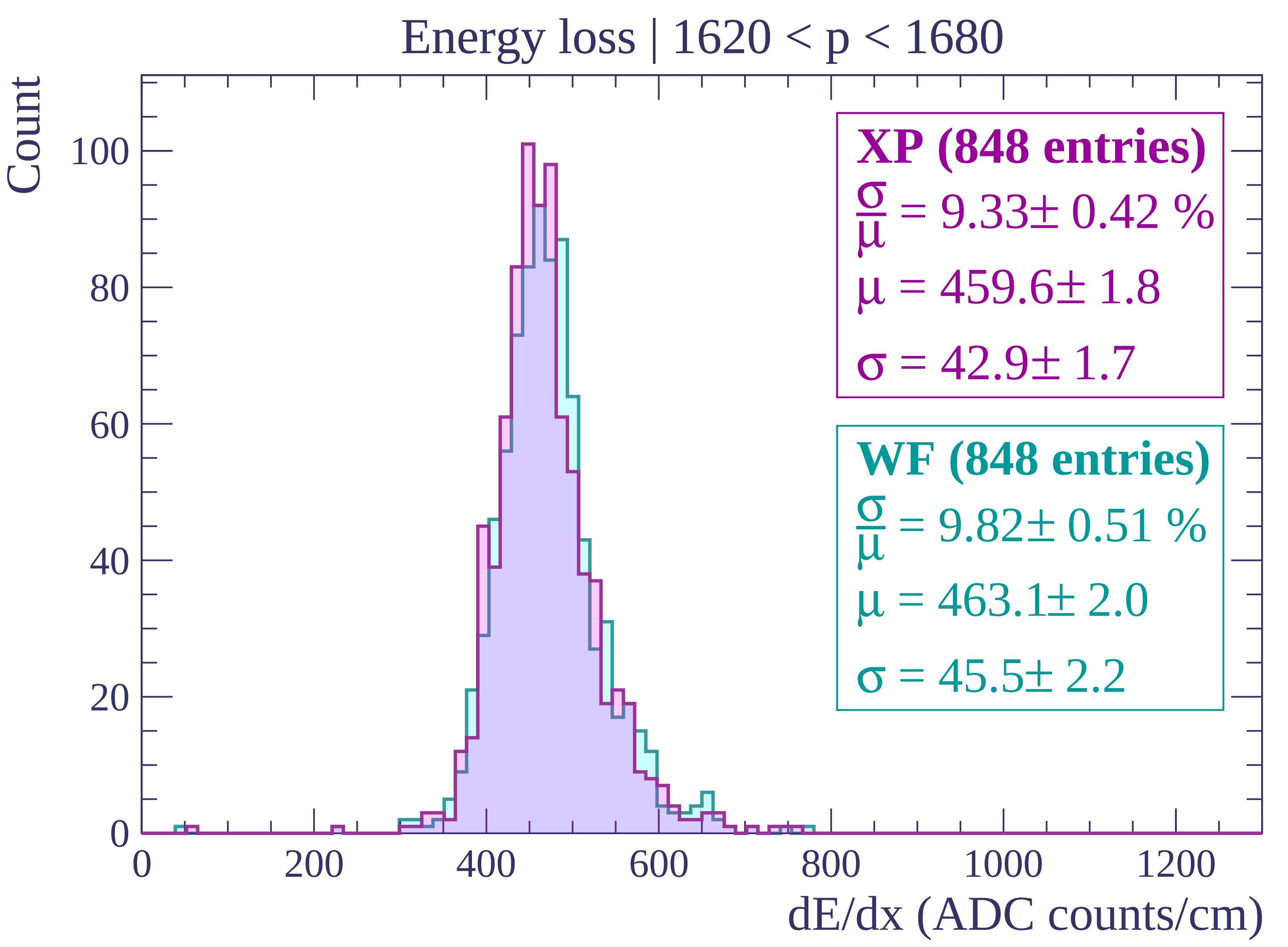

Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (854 entries)
$\frac{\sigma}{\mu}$ = 9.19± 0.40 %
μ = 451.4± 1.8
σ = 41.5± 1.6
WF (854 entries)
$\frac{\sigma}{\mu}$ = 9.78± 0.43 %
μ = 458.2± 1.9
σ = 44.8± 1.8


Energy loss | 1440 < p < 1500
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (857 entries)
$\frac{\sigma}{\mu}$ = 8.89± 0.45 %
μ = 455.1± 1.9
σ = 40.5± 1.9
WF (857 entries)
$\frac{\sigma}{\mu}$ = 8.66± 0.44 %
μ = 457.6± 1.8
σ = 39.6± 1.8

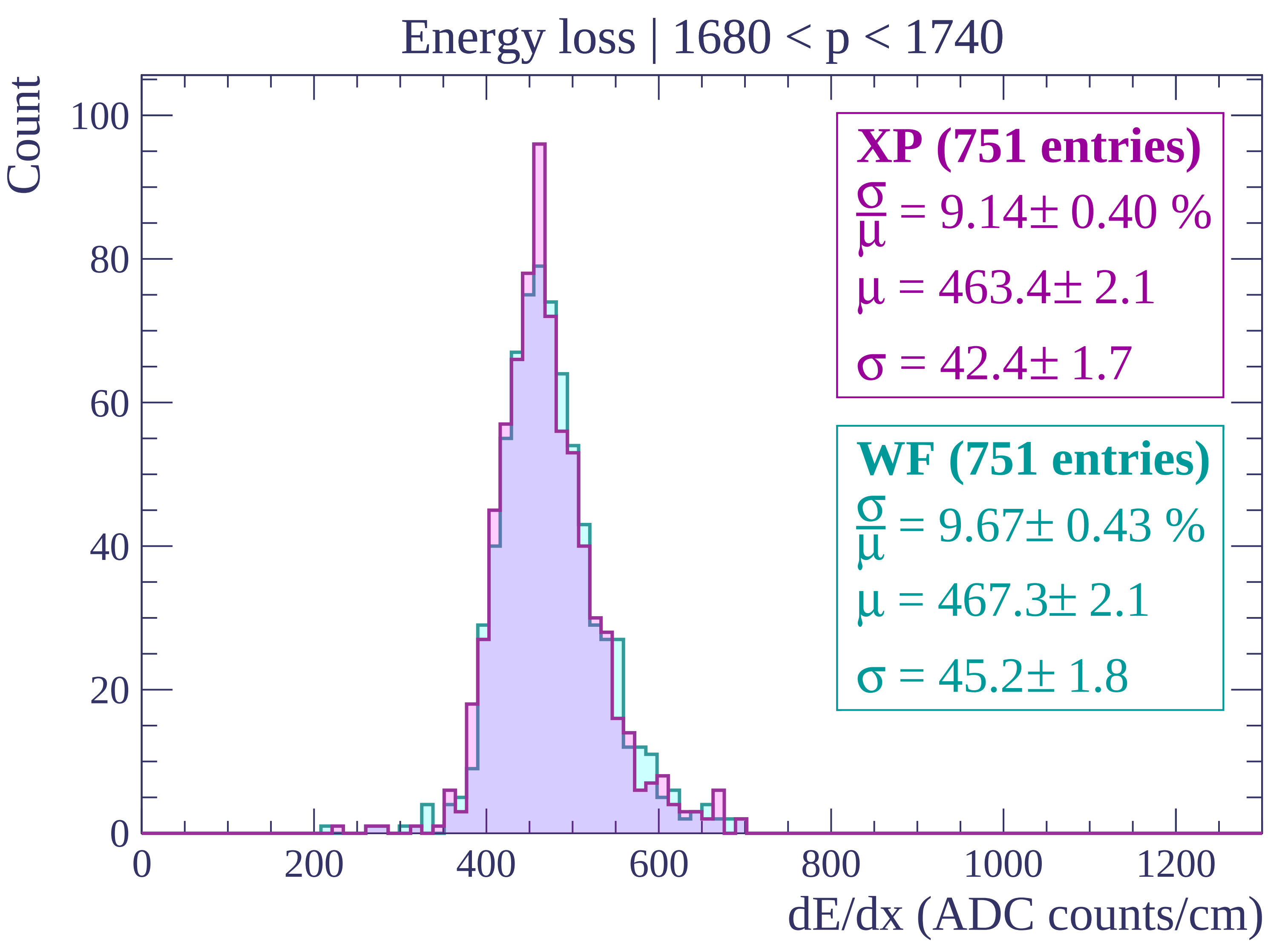


Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (822 entries)
σ/μ = 8.63± 0.48 %
μ = 449.9± 1.8
σ = 38.8± 2.0
WF (822 entries)
σ/μ = 9.29± 0.48 %
μ = 453.3± 1.9
σ = 42.1± 2.0

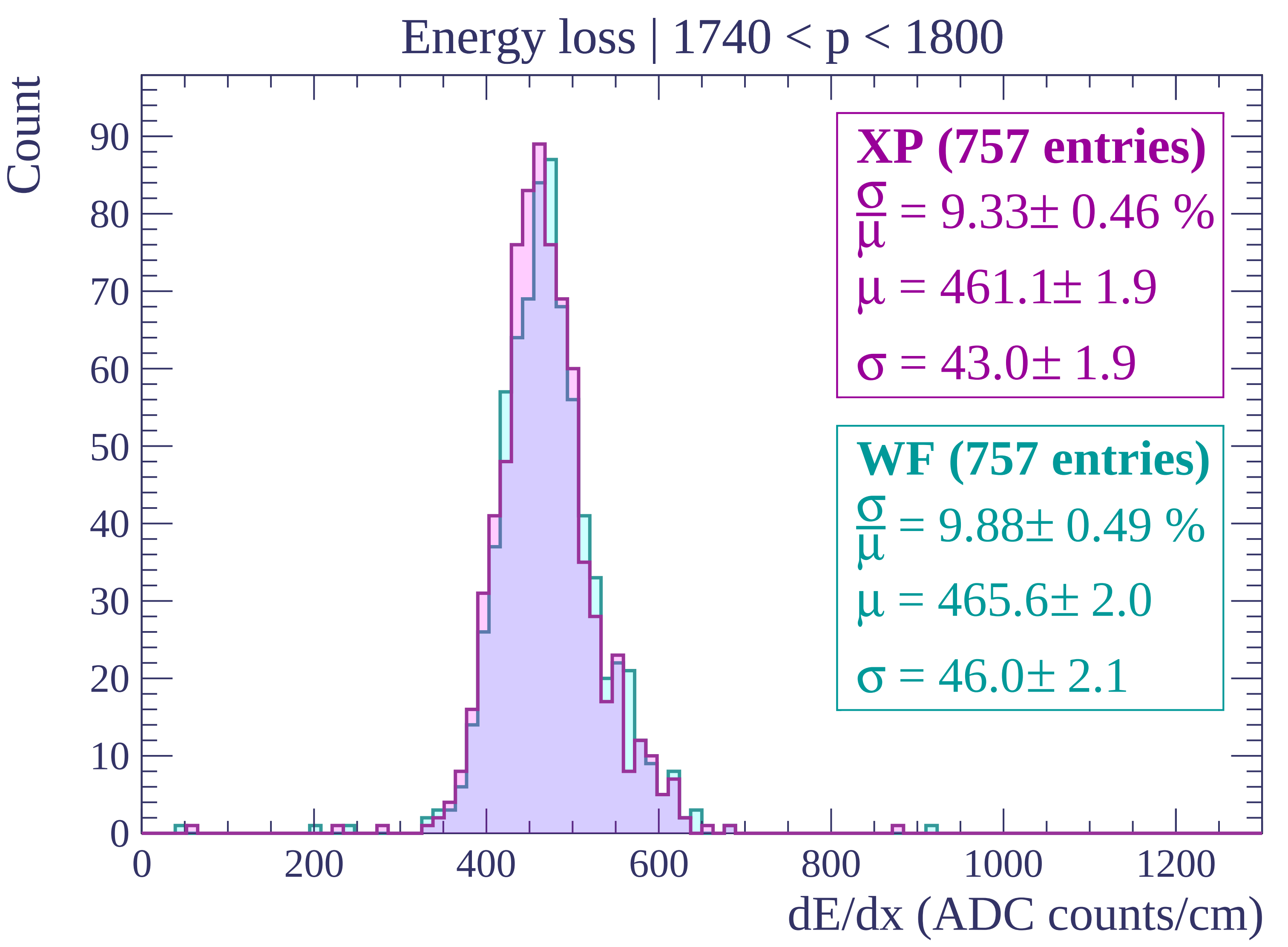

Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
XP (781 entries)
σ/μ = 9.30± 0.45 %
μ = 457.0± 1.9
σ = 42.5± 1.9
WF (781 entries)
σ/μ = 9.57± 0.50 %
μ = 461.7± 2.0
σ = 44.2± 2.1
0
20
40
60
80
100
0
200
400
600
800
1000
1200



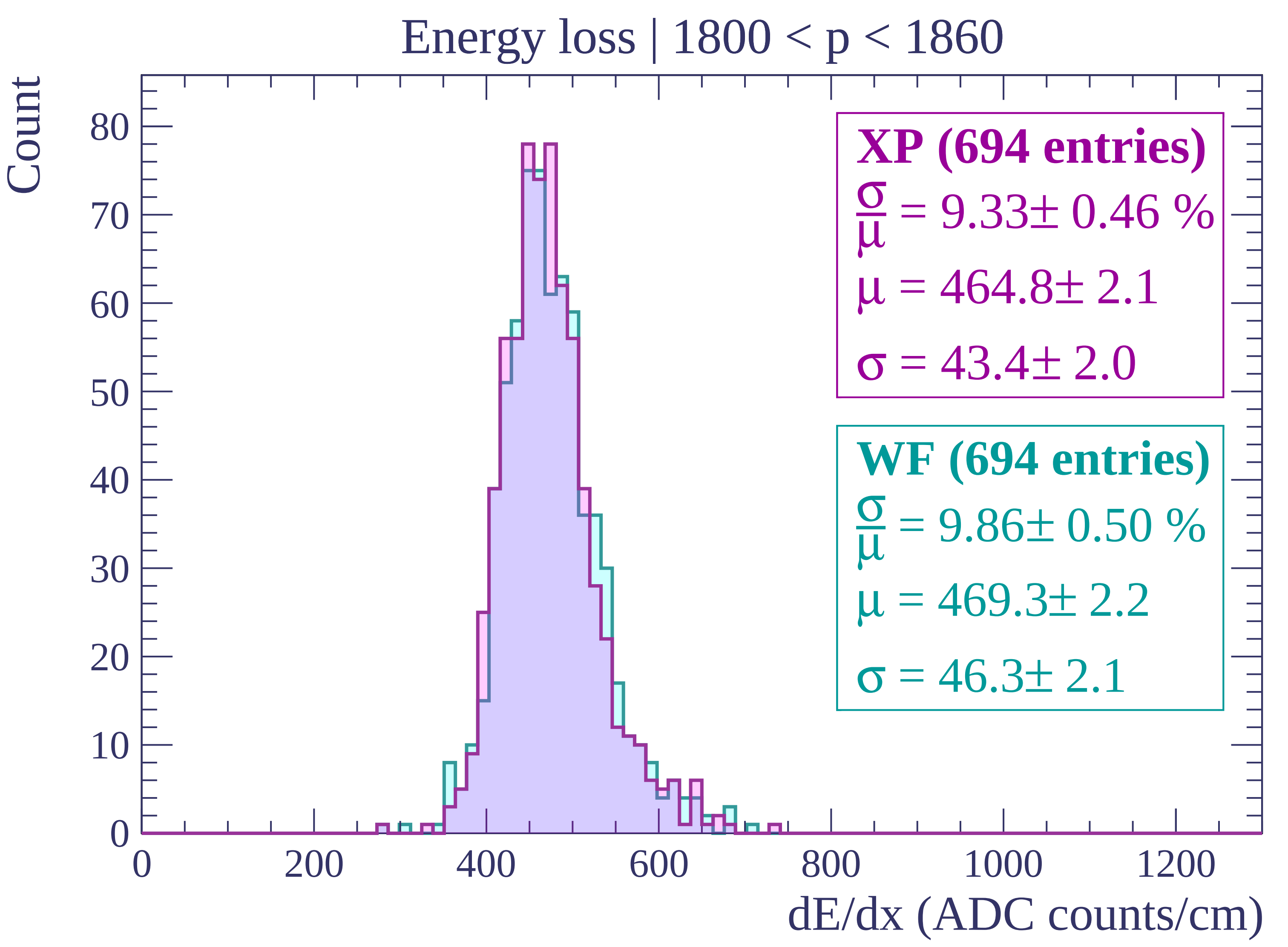
Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (848 entries)
σ/μ = 9.33± 0.42 %
μ = 459.6± 1.8
σ = 42.9± 1.7
WF (848 entries)
σ/μ = 9.82± 0.51 %
μ = 463.1± 2.0
σ = 45.5± 2.2



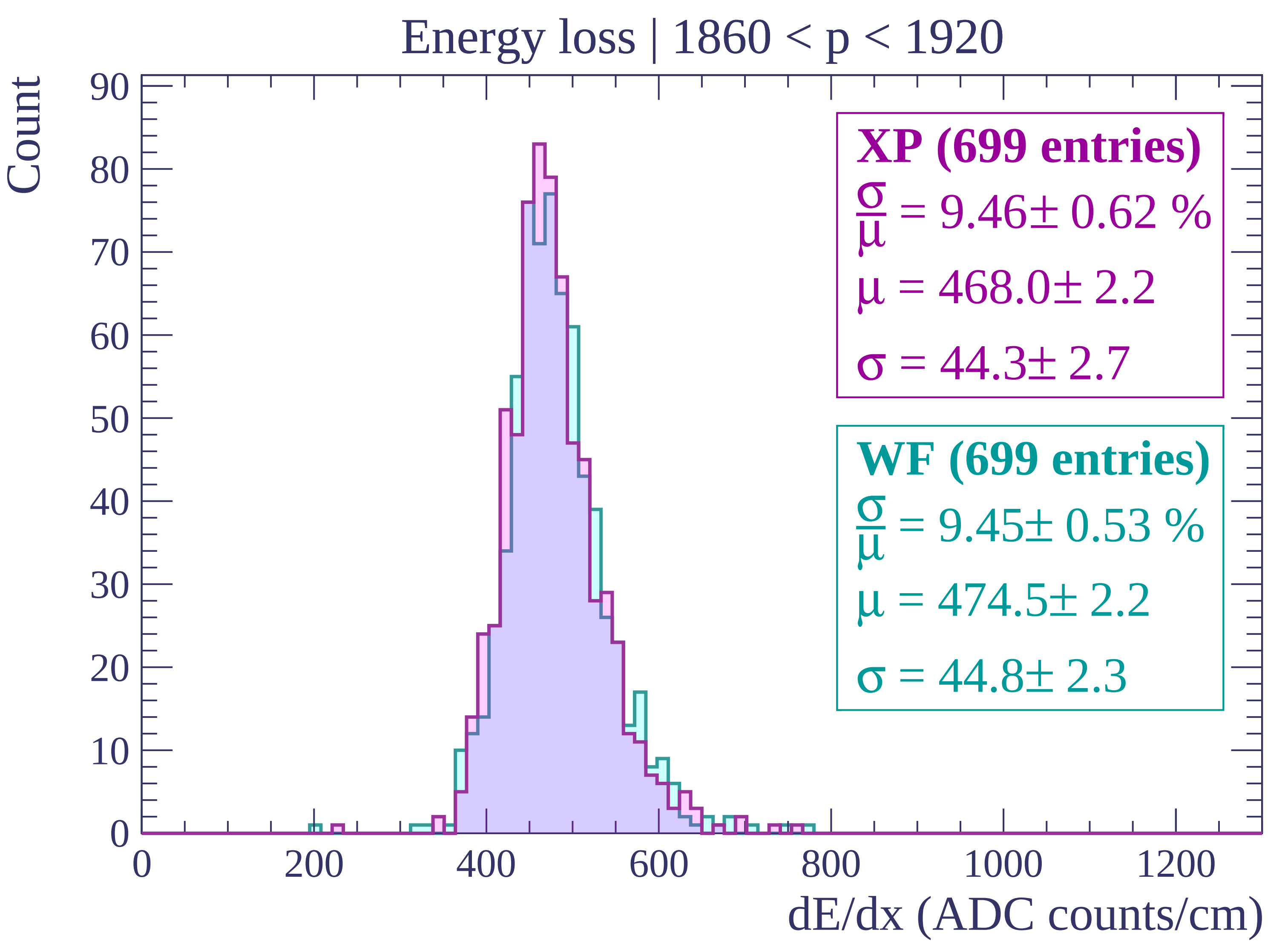
Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (751 entries)
σ/μ = 9.14± 0.40 %
μ = 463.4± 2.1
σ = 42.4± 1.7
WF (751 entries)
σ/μ = 9.67± 0.43 %
μ = 467.3± 2.1
σ = 45.2± 1.8



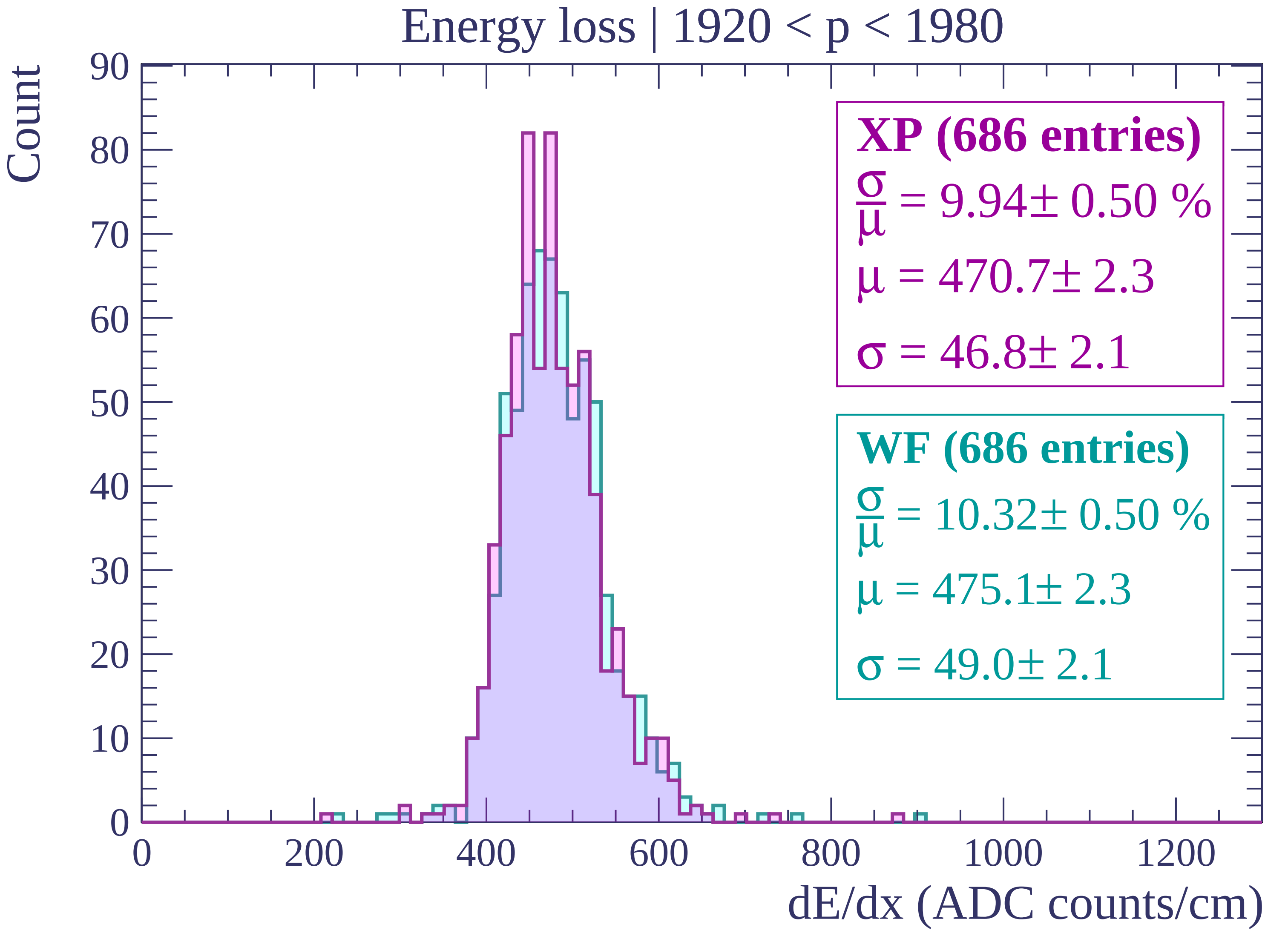
Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (757 entries)
σ/μ = 9.33± 0.46 %
μ = 461.1± 1.9
σ = 43.0± 1.9
WF (757 entries)
σ/μ = 9.88± 0.49 %
μ = 465.6± 2.0
σ = 46.0± 2.1



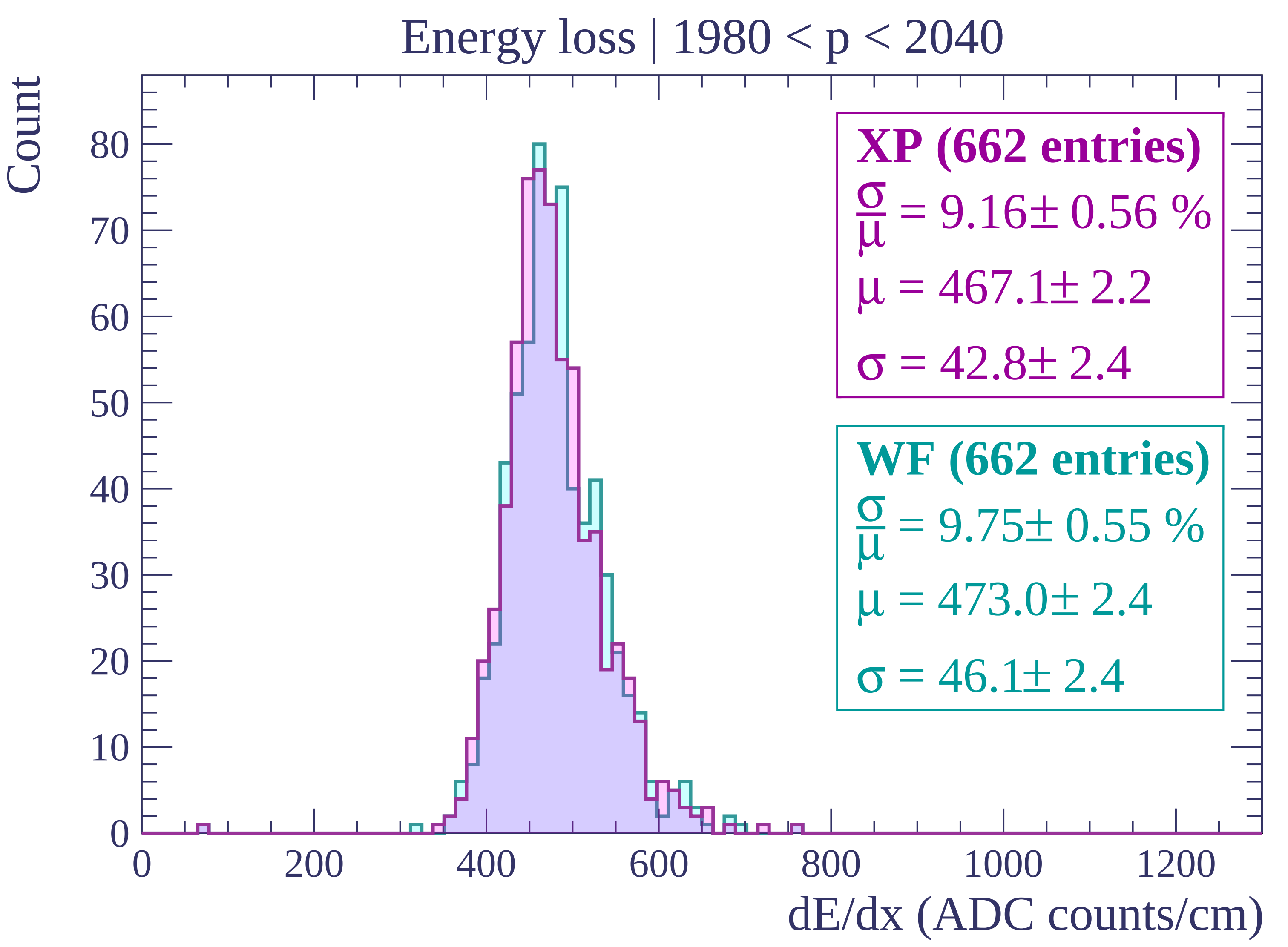
Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (694 entries)
σ/μ = 9.33± 0.46 %
μ = 464.8± 2.1
σ = 43.4± 2.0
WF (694 entries)
σ/μ = 9.86± 0.50 %
μ = 469.3± 2.2
σ = 46.3± 2.1



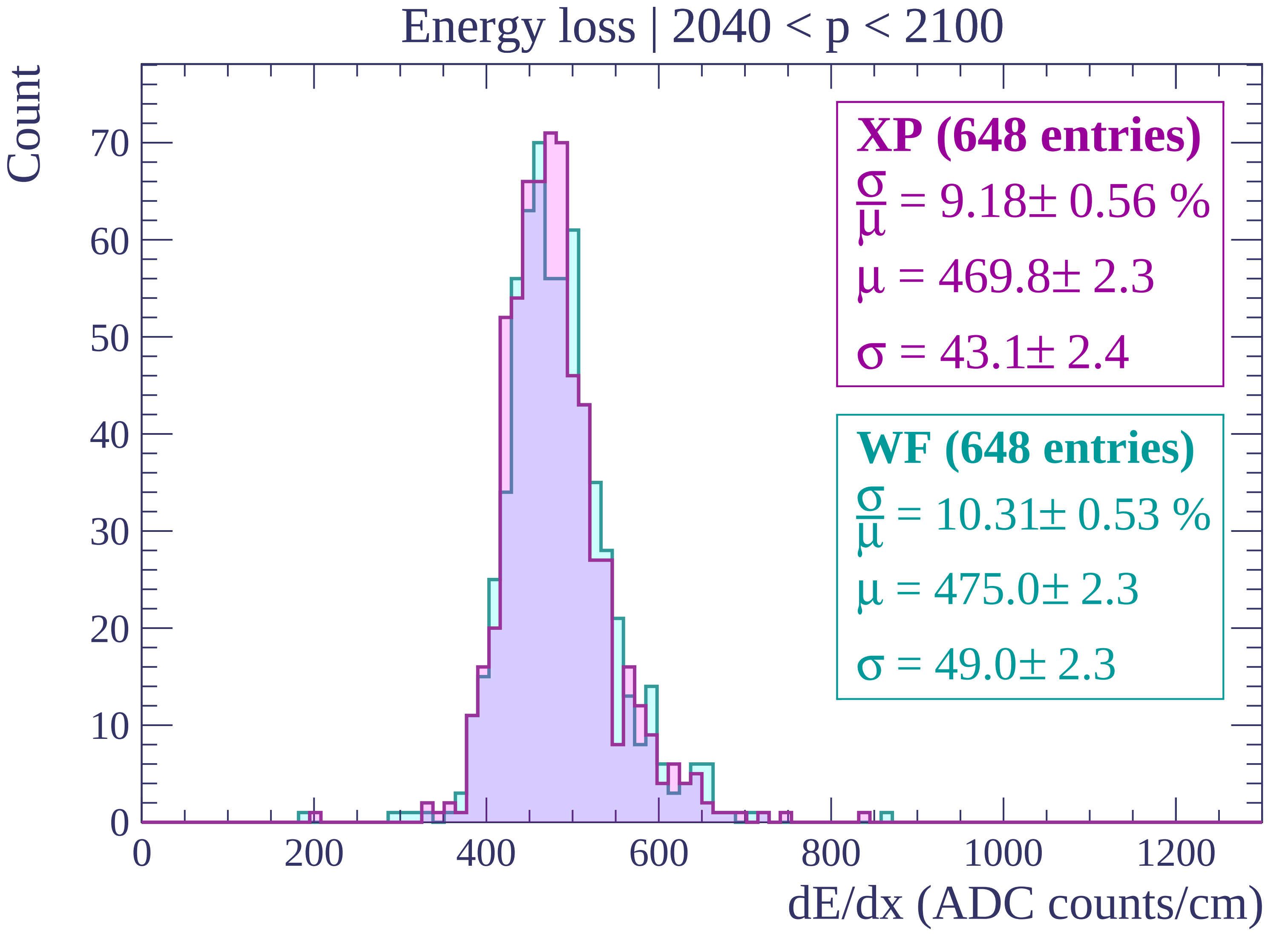
Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (699 entries)
σ/μ = 9.46± 0.62 %
μ = 468.0± 2.2
σ = 44.3± 2.7
WF (699 entries)
σ/μ = 9.45± 0.53 %
μ = 474.5± 2.2
σ = 44.8± 2.3

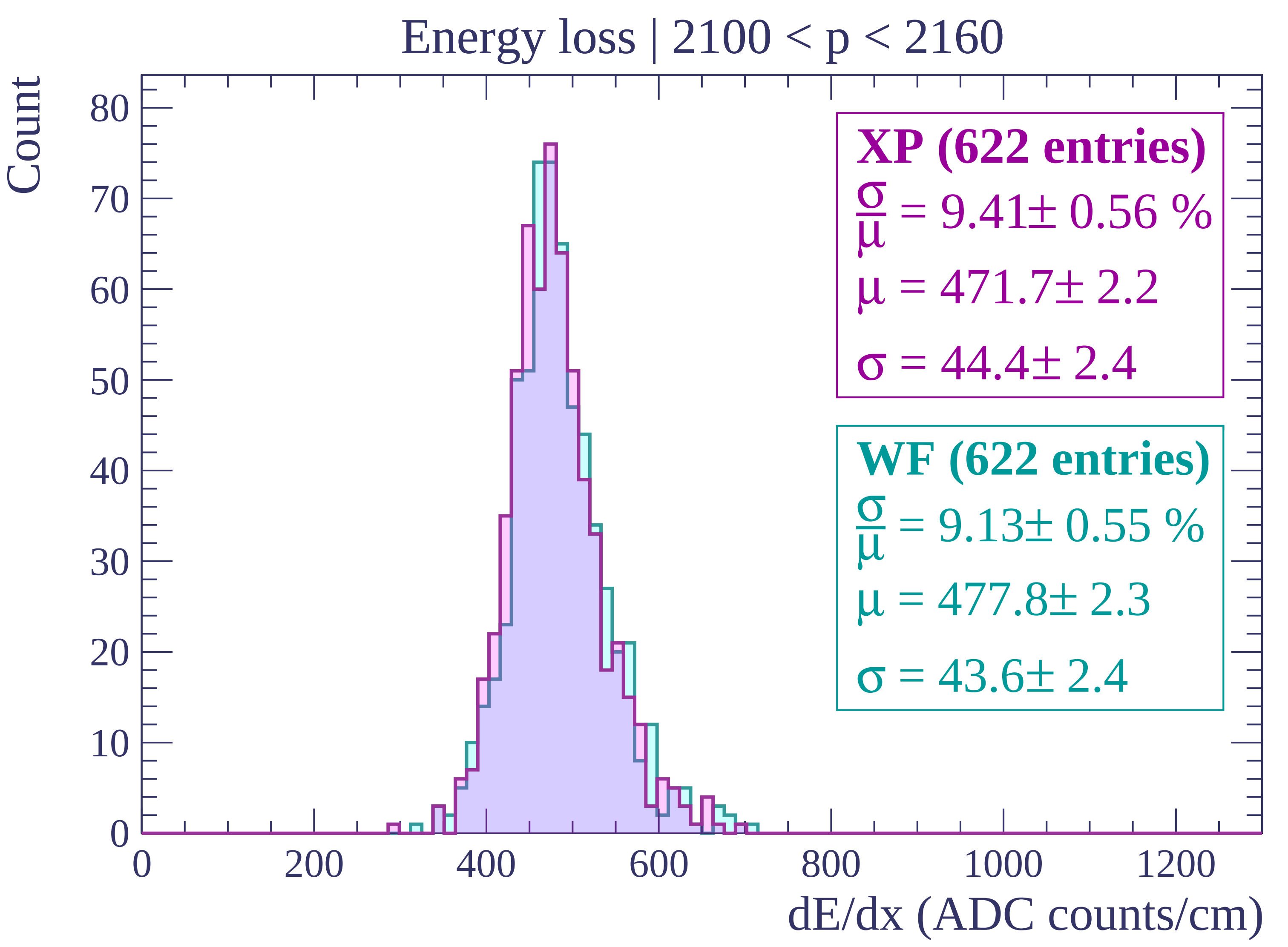


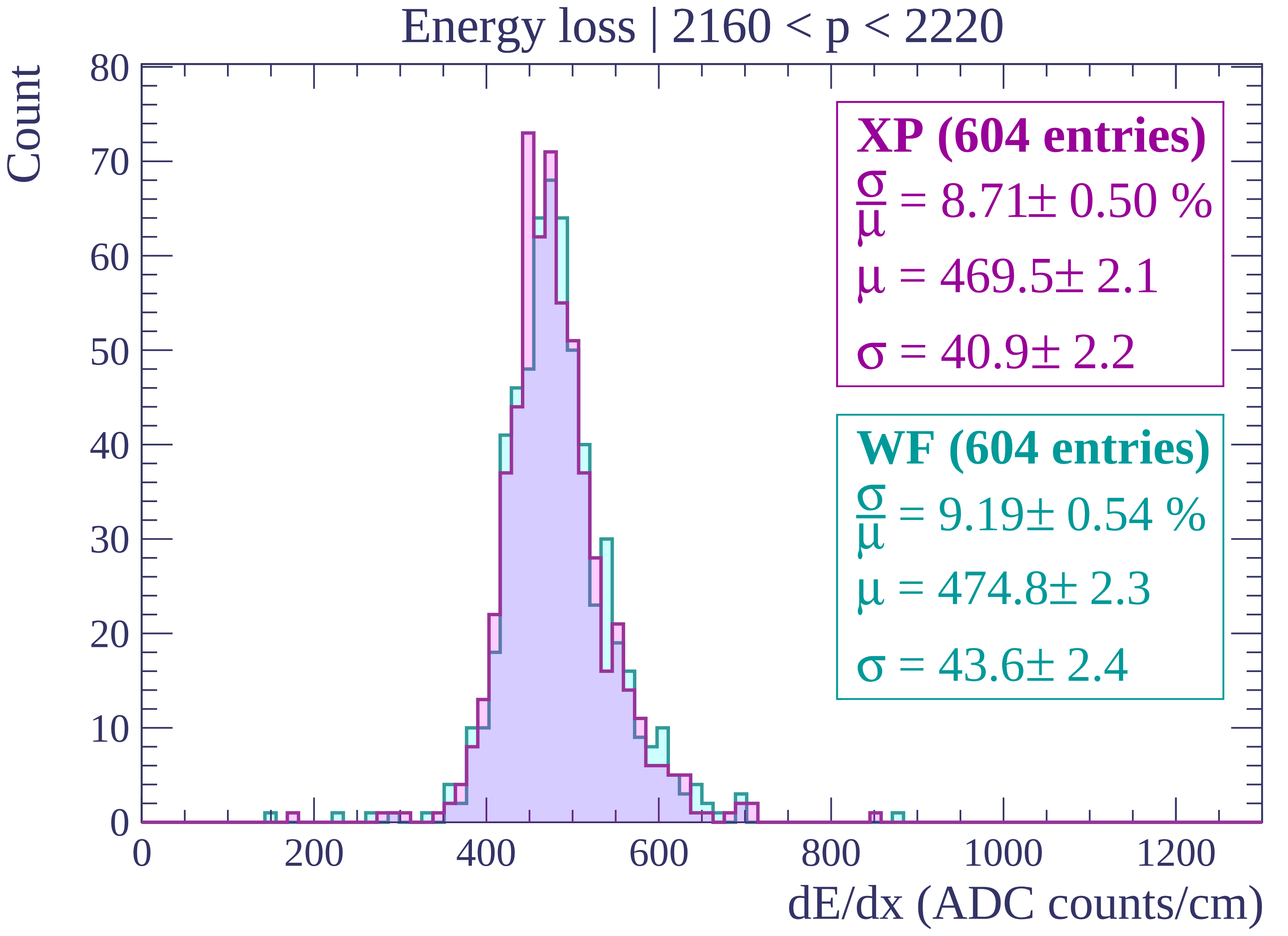
Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (686 entries)
σ/μ = 9.94± 0.50 %
μ = 470.7± 2.3
σ = 46.8± 2.1
WF (686 entries)
σ/μ = 10.32± 0.50 %
μ = 475.1± 2.3
σ = 49.0± 2.1



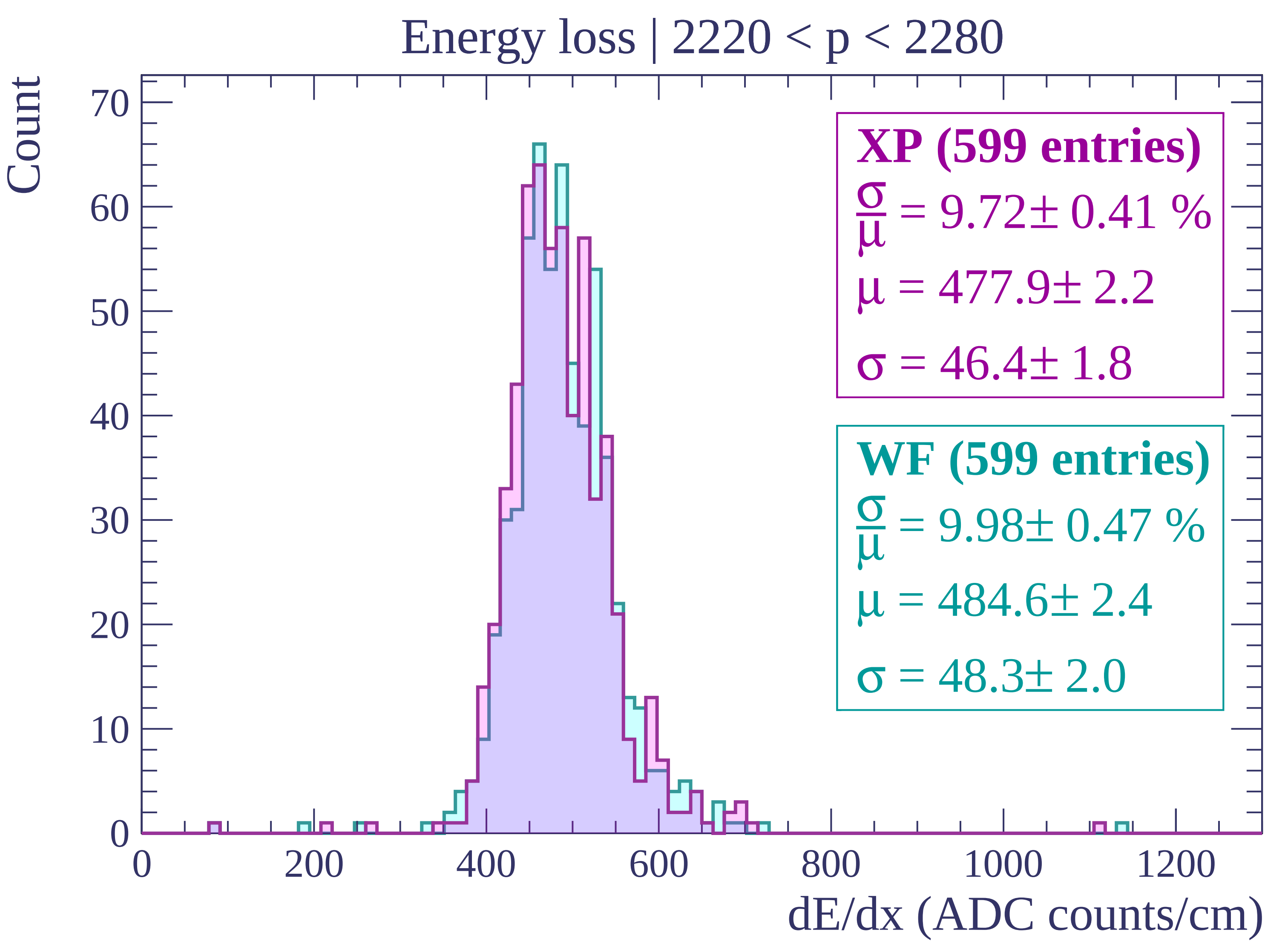
Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (662 entries)
$\frac{\sigma}{\mu} = 9.16 \pm 0.56$ %
$\mu = 467.1 \pm 2.2$
$\sigma = 42.8 \pm 2.4$
WF (662 entries)
$\frac{\sigma}{\mu} = 9.75 \pm 0.55$ %
$\mu = 473.0 \pm 2.4$
$\sigma = 46.1 \pm 2.4$


Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP (648 entries)
$\frac{\sigma}{\mu}$ = 9.18± 0.56 %
μ = 469.8± 2.3
σ = 43.1± 2.4
WF (648 entries)
$\frac{\sigma}{\mu}$ = 10.31± 0.53 %
μ = 475.0± 2.3
σ = 49.0± 2.3

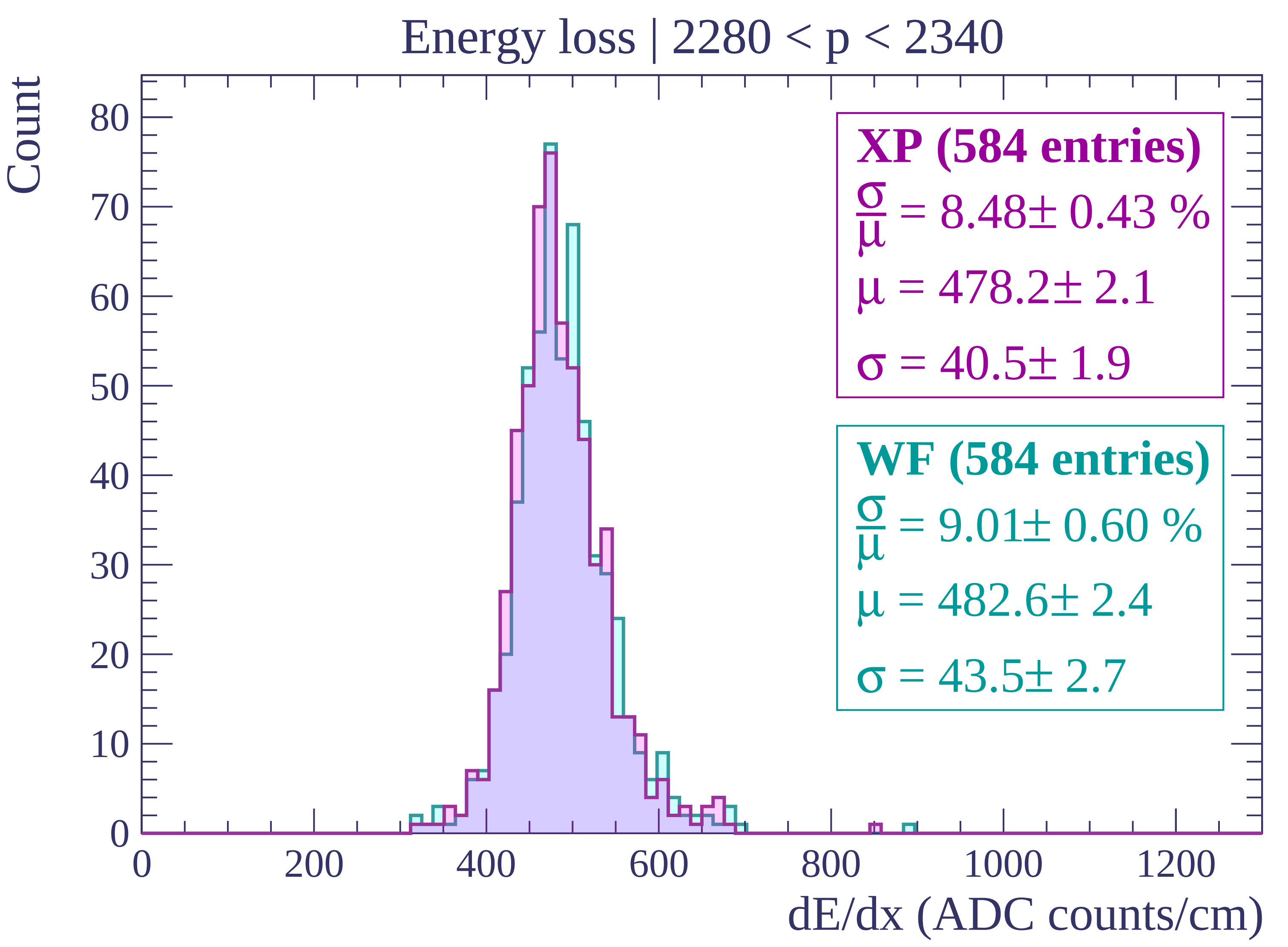

Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (622 entries)
σ/μ = 9.41± 0.56 %
μ = 471.7± 2.2
σ = 44.4± 2.4
WF (622 entries)
σ/μ = 9.13± 0.55 %
μ = 477.8± 2.3
σ = 43.6± 2.4



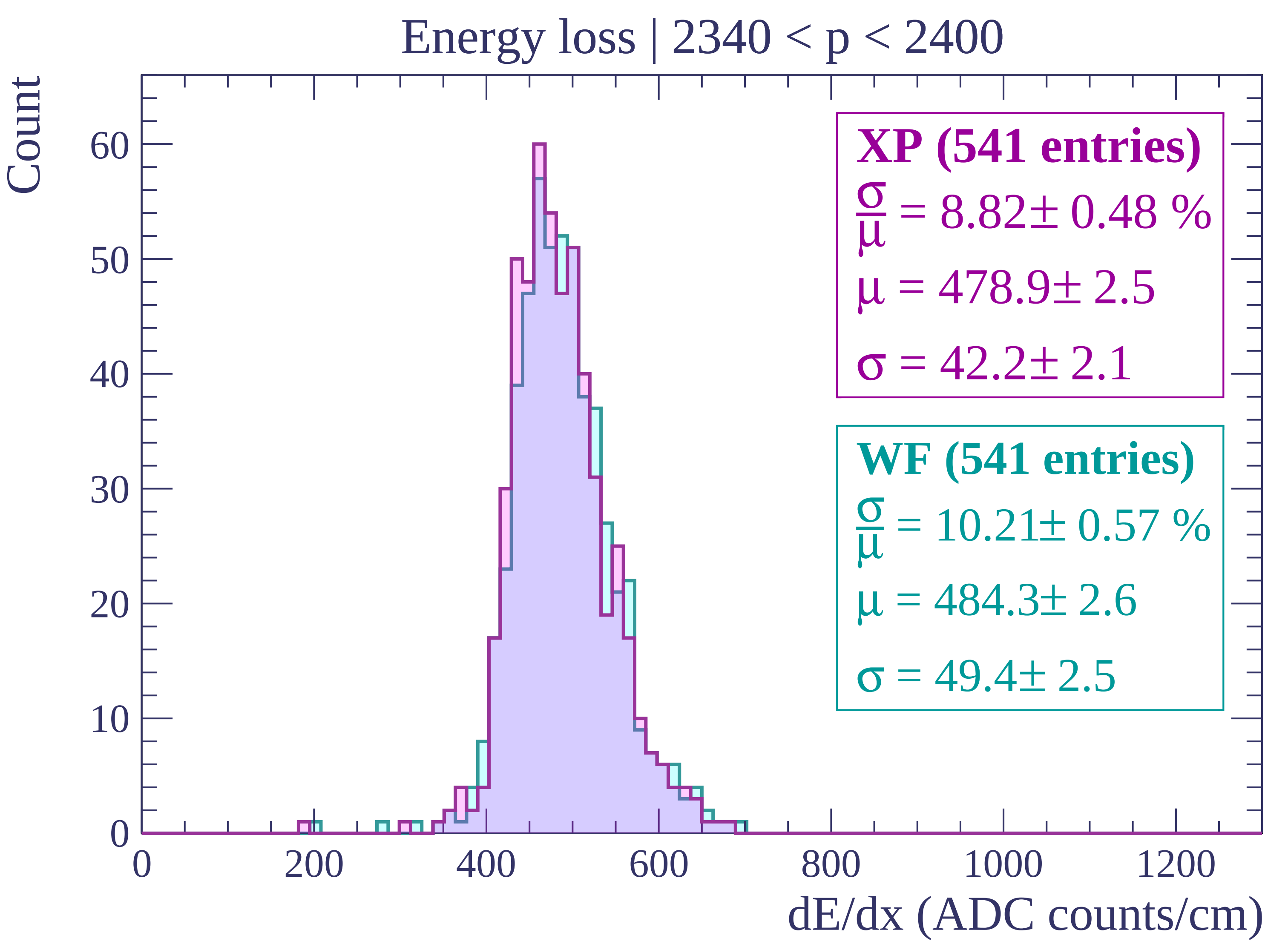
Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (604 entries)
σ/μ = 8.71± 0.50 %
μ = 469.5± 2.1
σ = 40.9± 2.2
WF (604 entries)
σ/μ = 9.19± 0.54 %
μ = 474.8± 2.3
σ = 43.6± 2.4



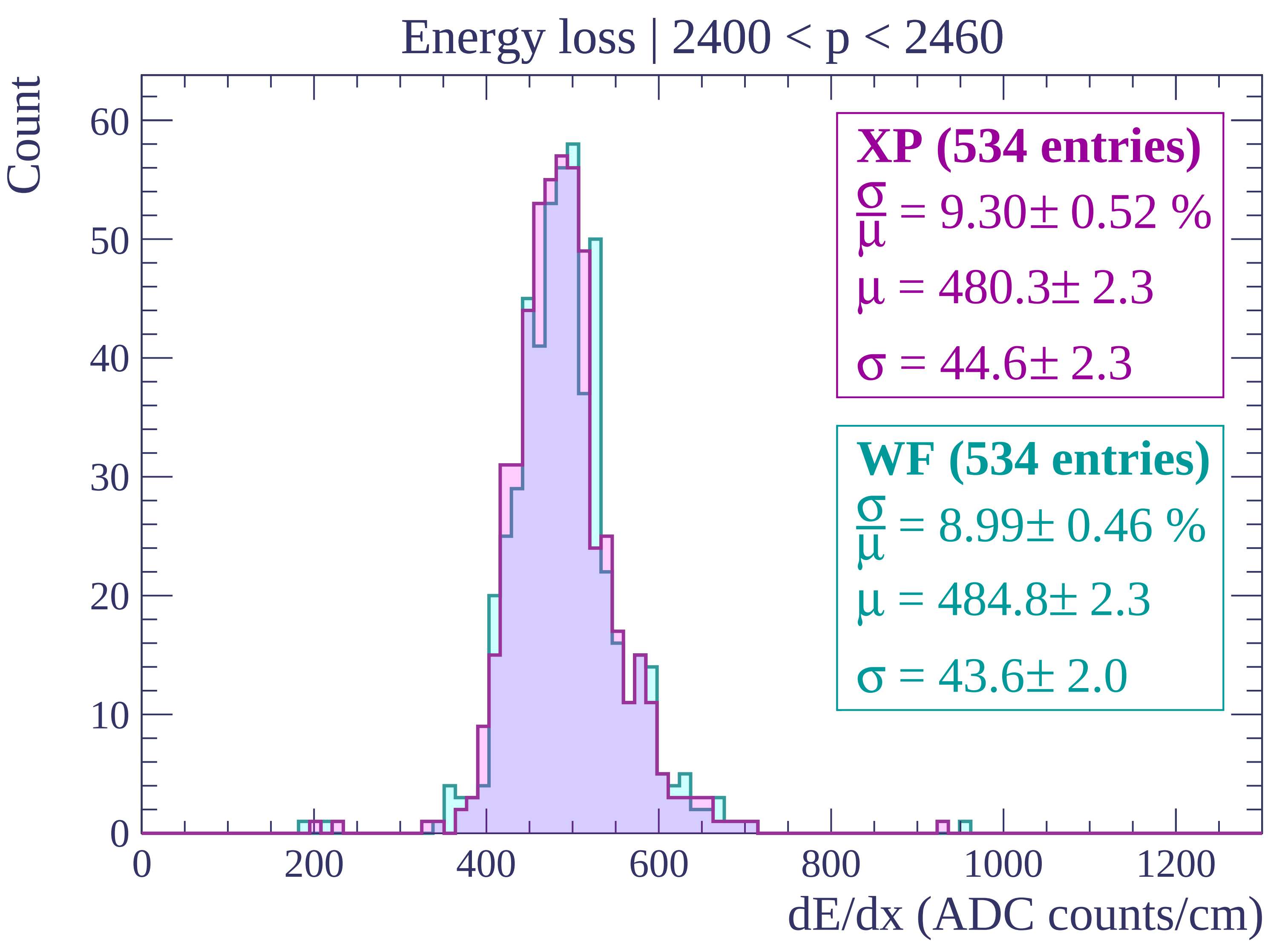
Energy loss | 2220 < p < 2280
Count
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (599 entries)
$\frac{\sigma}{\mu}$ = 9.72± 0.41 %
μ = 477.9± 2.2
σ = 46.4± 1.8
WF (599 entries)
$\frac{\sigma}{\mu}$ = 9.98± 0.47 %
μ = 484.6± 2.4
σ = 48.3± 2.0

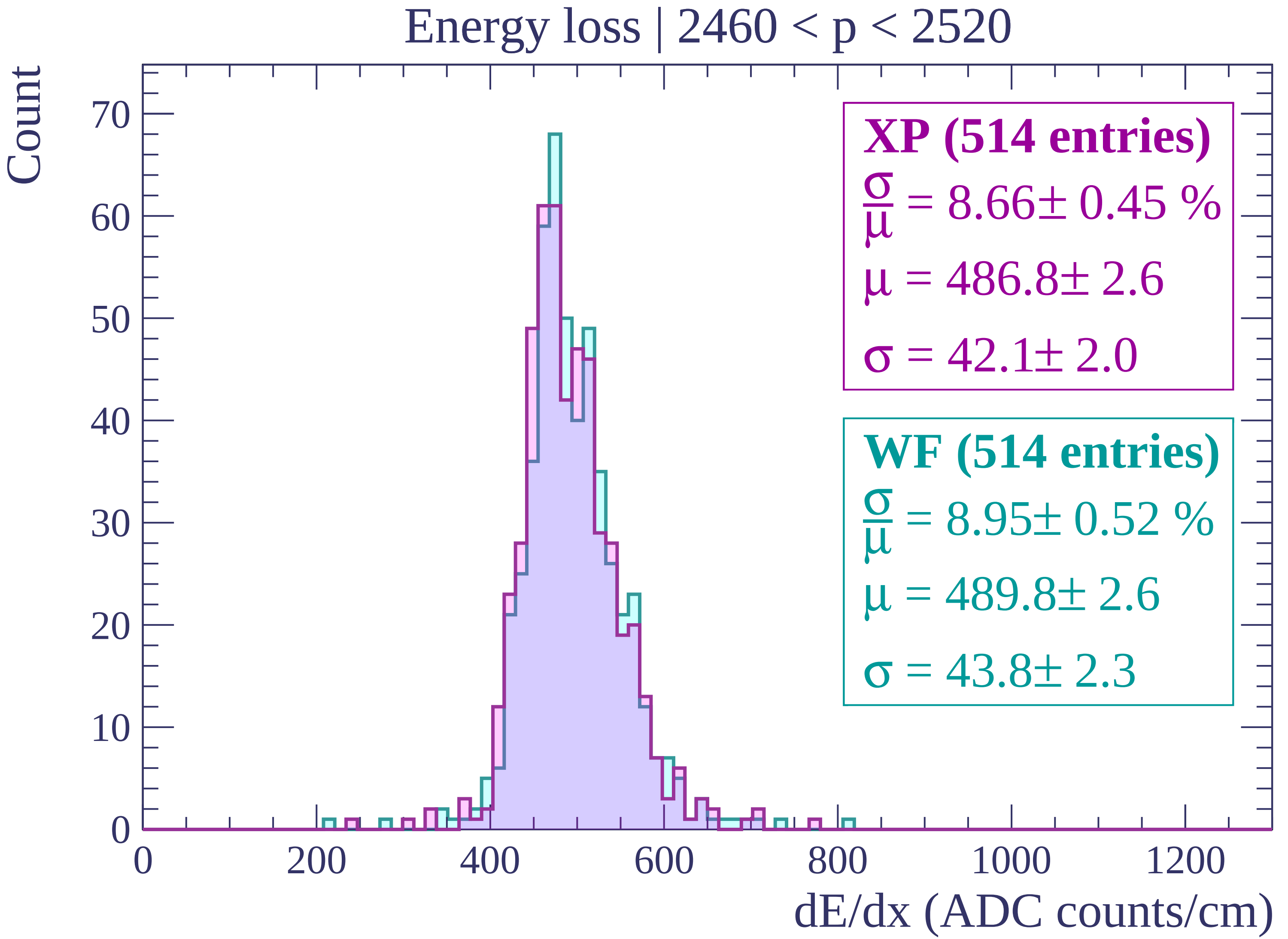


Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (584 entries)
σ/μ = 8.48± 0.43 %
μ = 478.2± 2.1
σ = 40.5± 1.9
WF (584 entries)
σ/μ = 9.01± 0.60 %
μ = 482.6± 2.4
σ = 43.5± 2.7

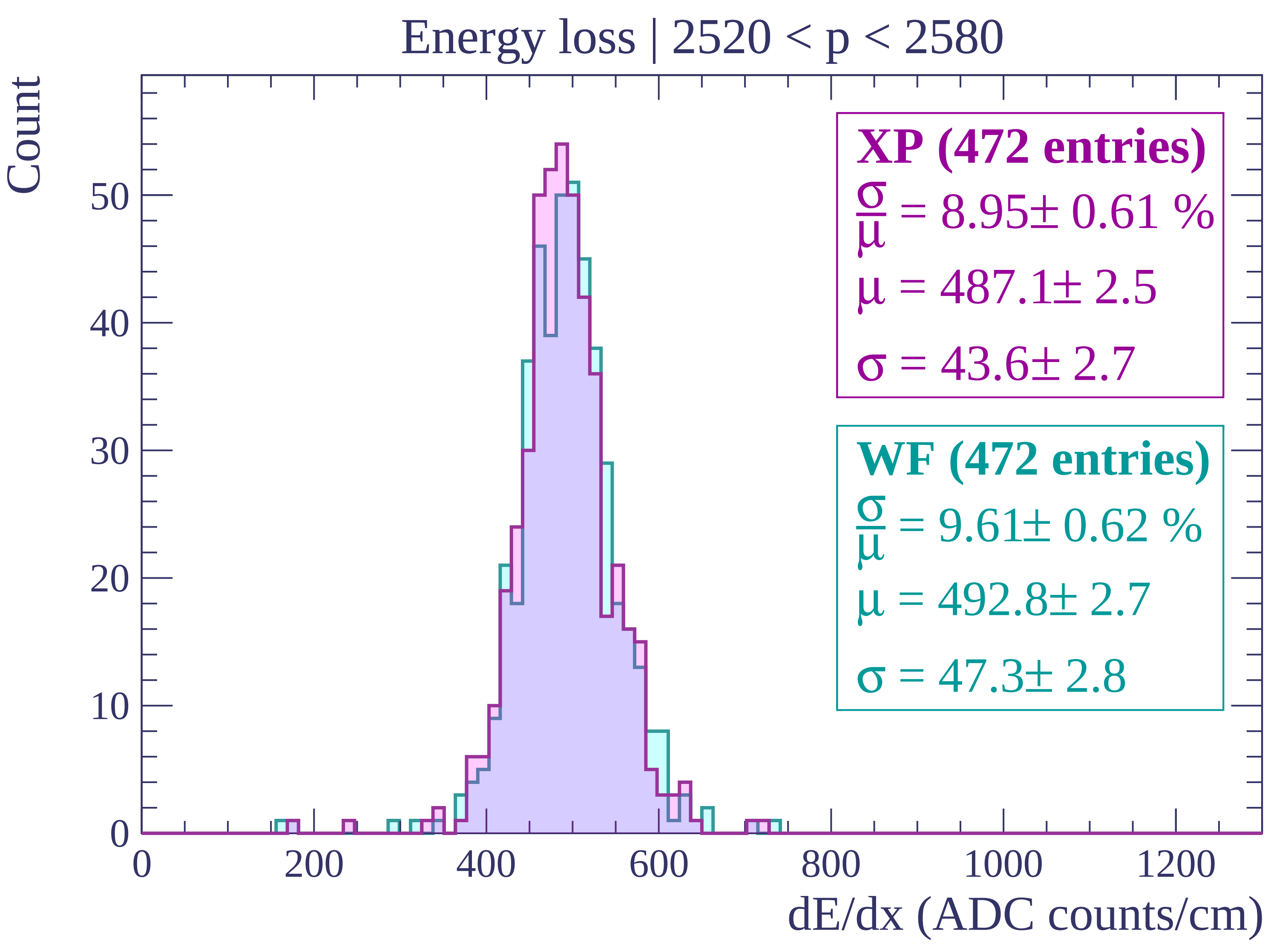


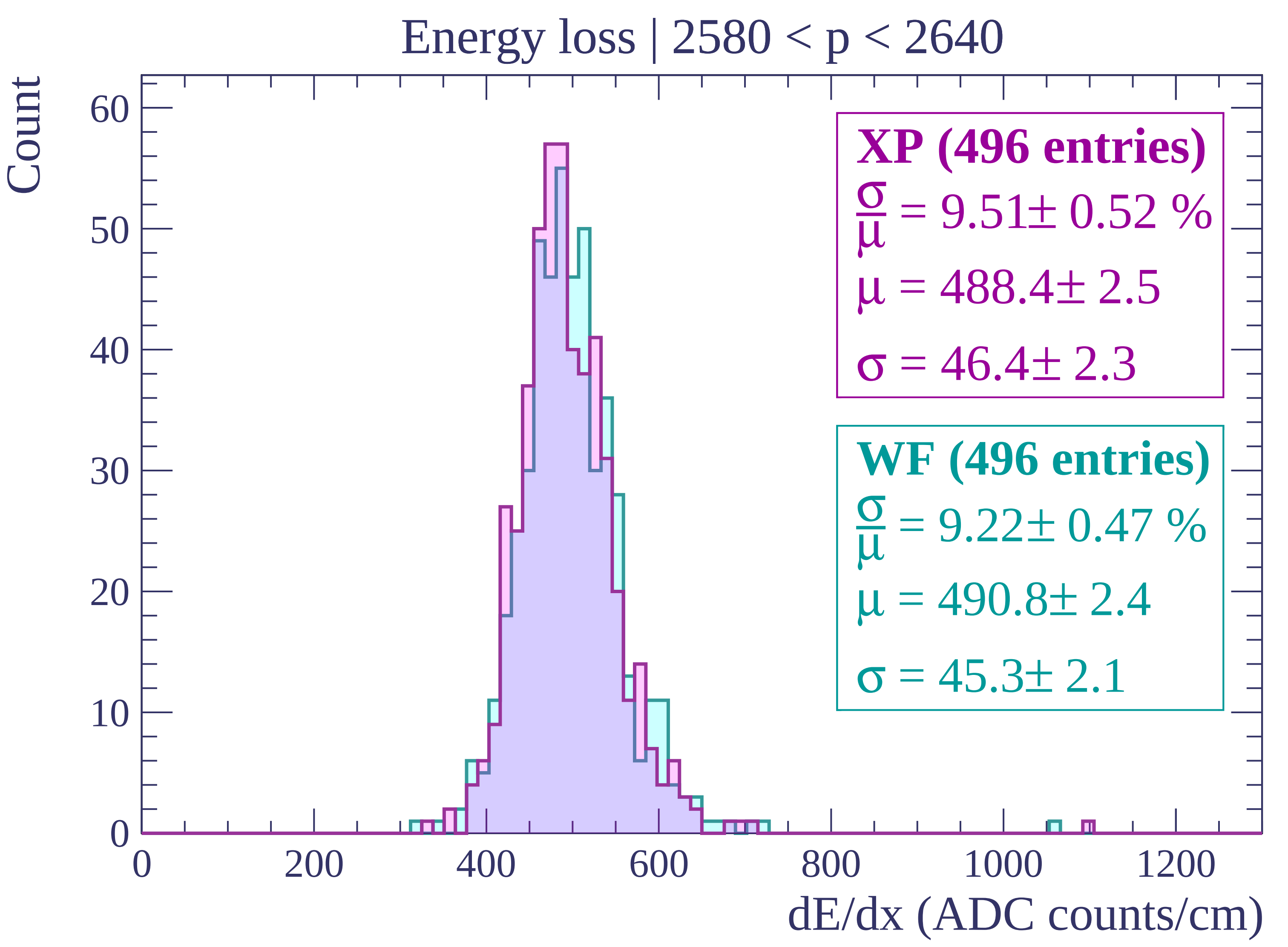
Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
XP (541 entries)
$\frac{\sigma}{\mu}$ = 8.82± 0.48 %
μ = 478.9± 2.5
σ = 42.2± 2.1
WF (541 entries)
$\frac{\sigma}{\mu}$ = 10.21± 0.57 %
μ = 484.3± 2.6
σ = 49.4± 2.5


Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
XP (534 entries)
σ/μ = 9.30± 0.52 %
μ = 480.3± 2.3
σ = 44.6± 2.3
WF (534 entries)
σ/μ = 8.99± 0.46 %
μ = 484.8± 2.3
σ = 43.6± 2.0

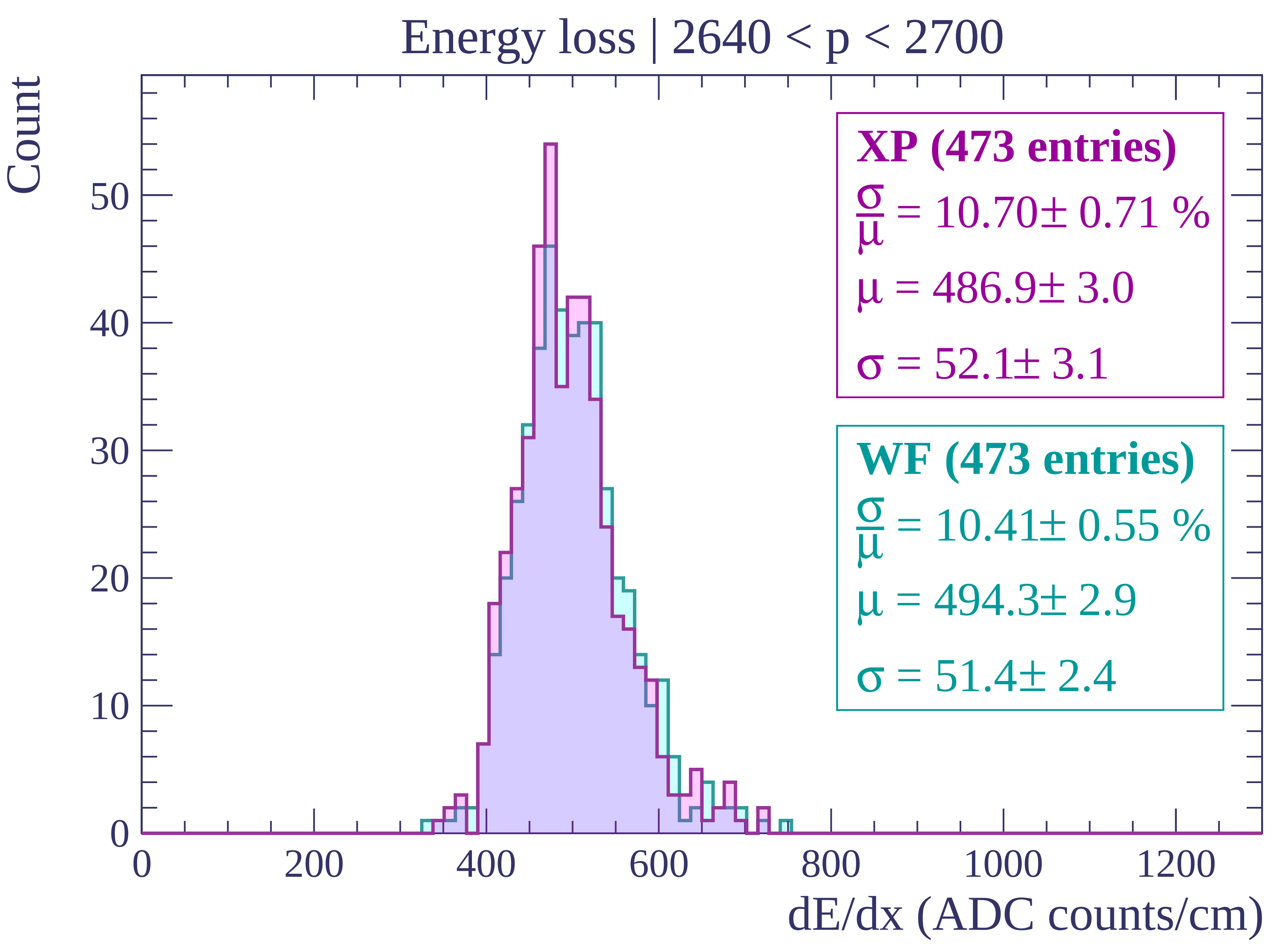


Energy loss | 2460 < p < 2520
Count
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (514 entries)
σ/μ = 8.66± 0.45 %
μ = 486.8± 2.6
σ = 42.1± 2.0
WF (514 entries)
σ/μ = 8.95± 0.52 %
μ = 489.8± 2.6
σ = 43.8± 2.3

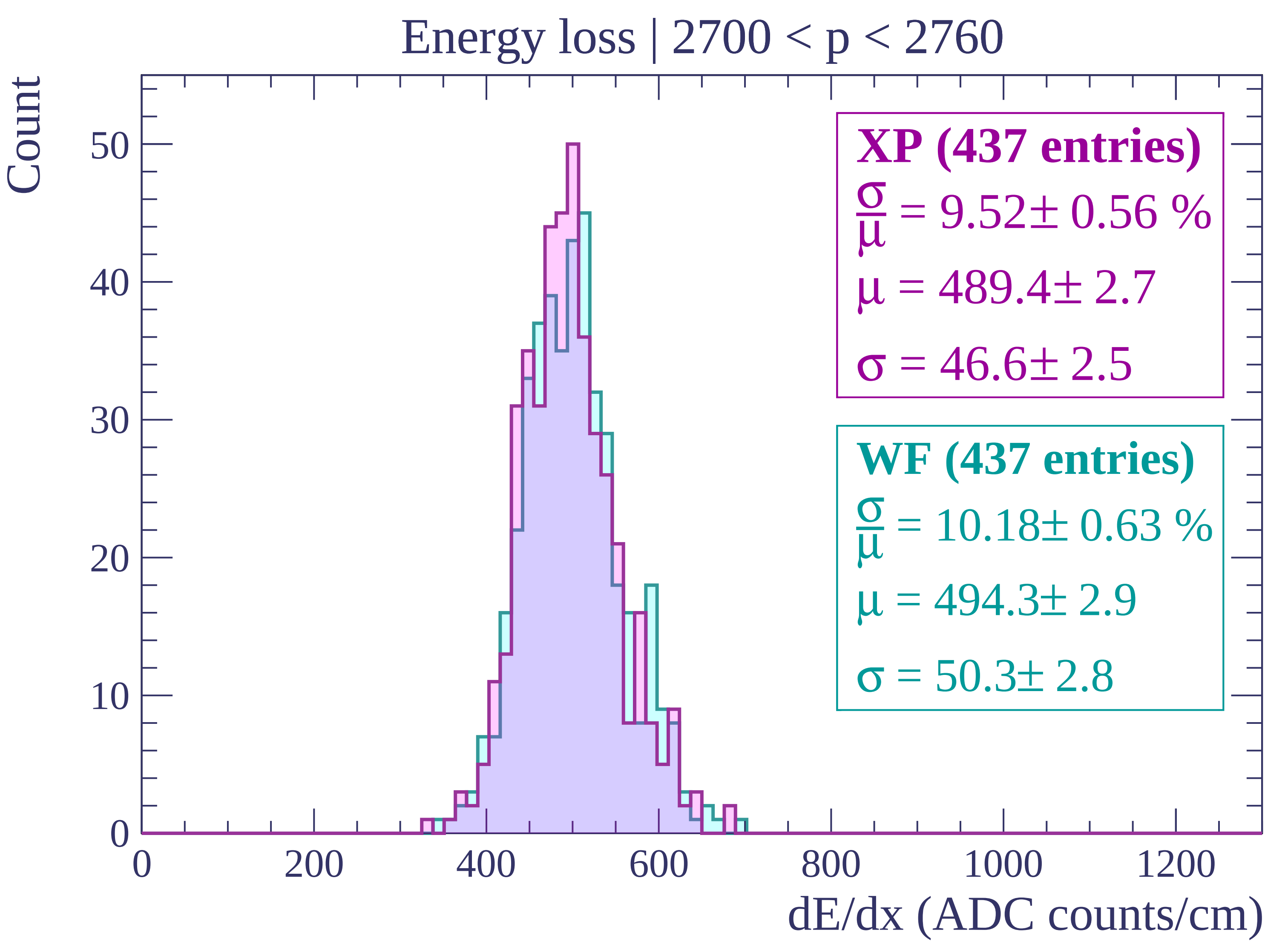

Energy loss | 2520 < p < 2580
Count
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (472 entries)
σ/μ = 8.95± 0.61 %
μ = 487.1± 2.5
σ = 43.6± 2.7
WF (472 entries)
σ/μ = 9.61± 0.62 %
μ = 492.8± 2.7
σ = 47.3± 2.8



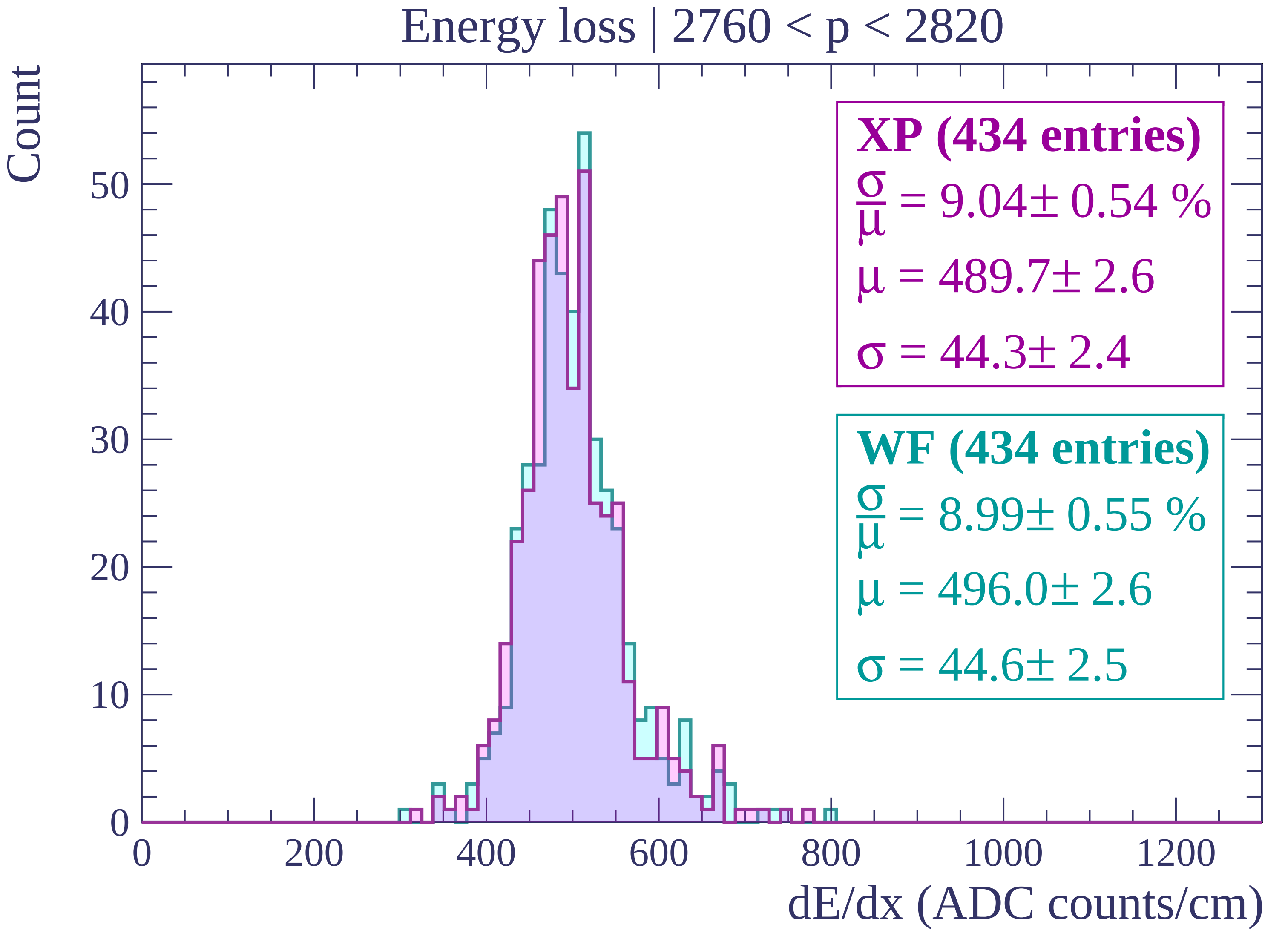
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (496 entries)
σ/μ = 9.51± 0.52 %
μ = 488.4± 2.5
σ = 46.4± 2.3
WF (496 entries)
σ/μ = 9.22± 0.47 %
μ = 490.8± 2.4
σ = 45.3± 2.1

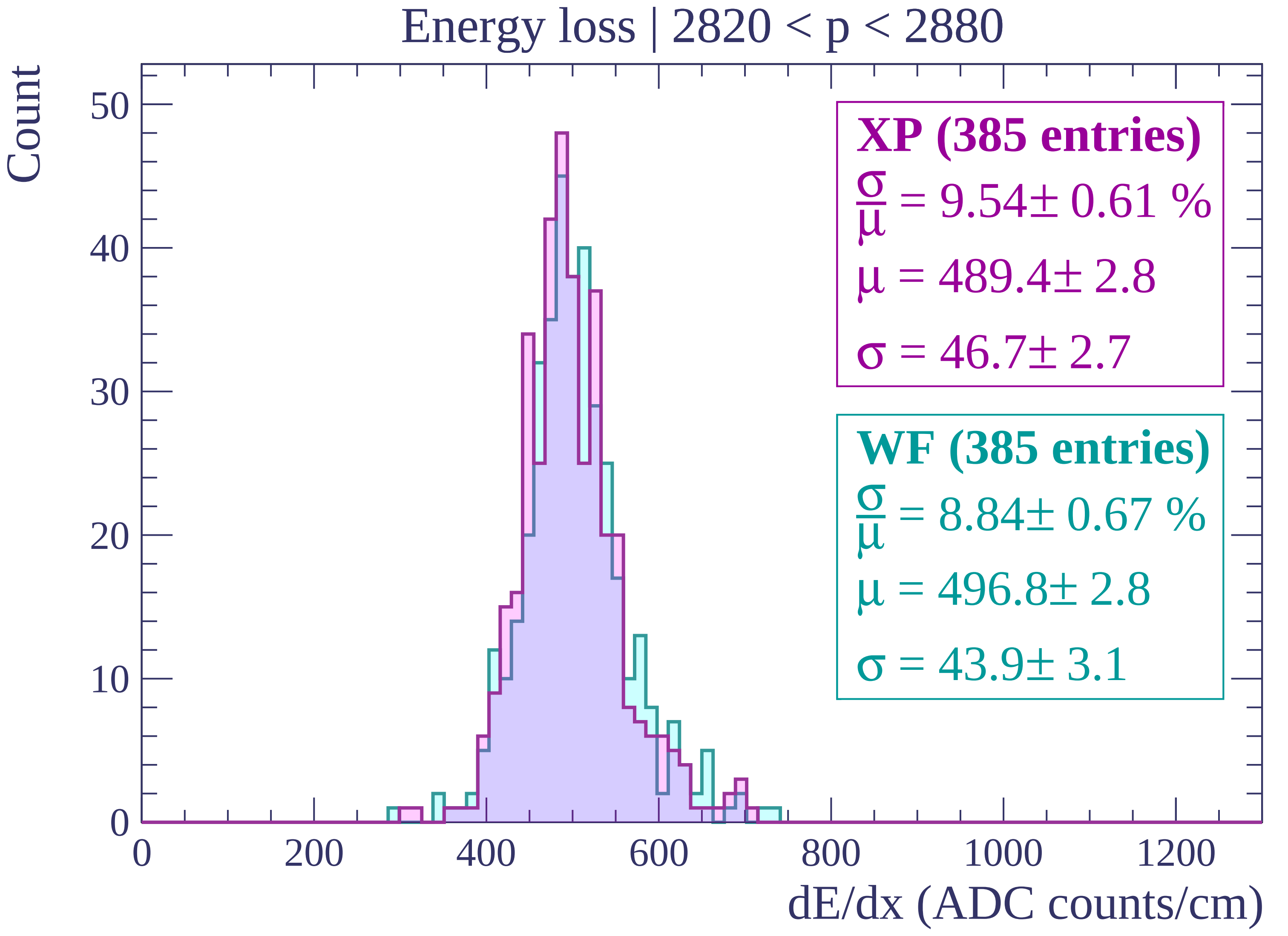


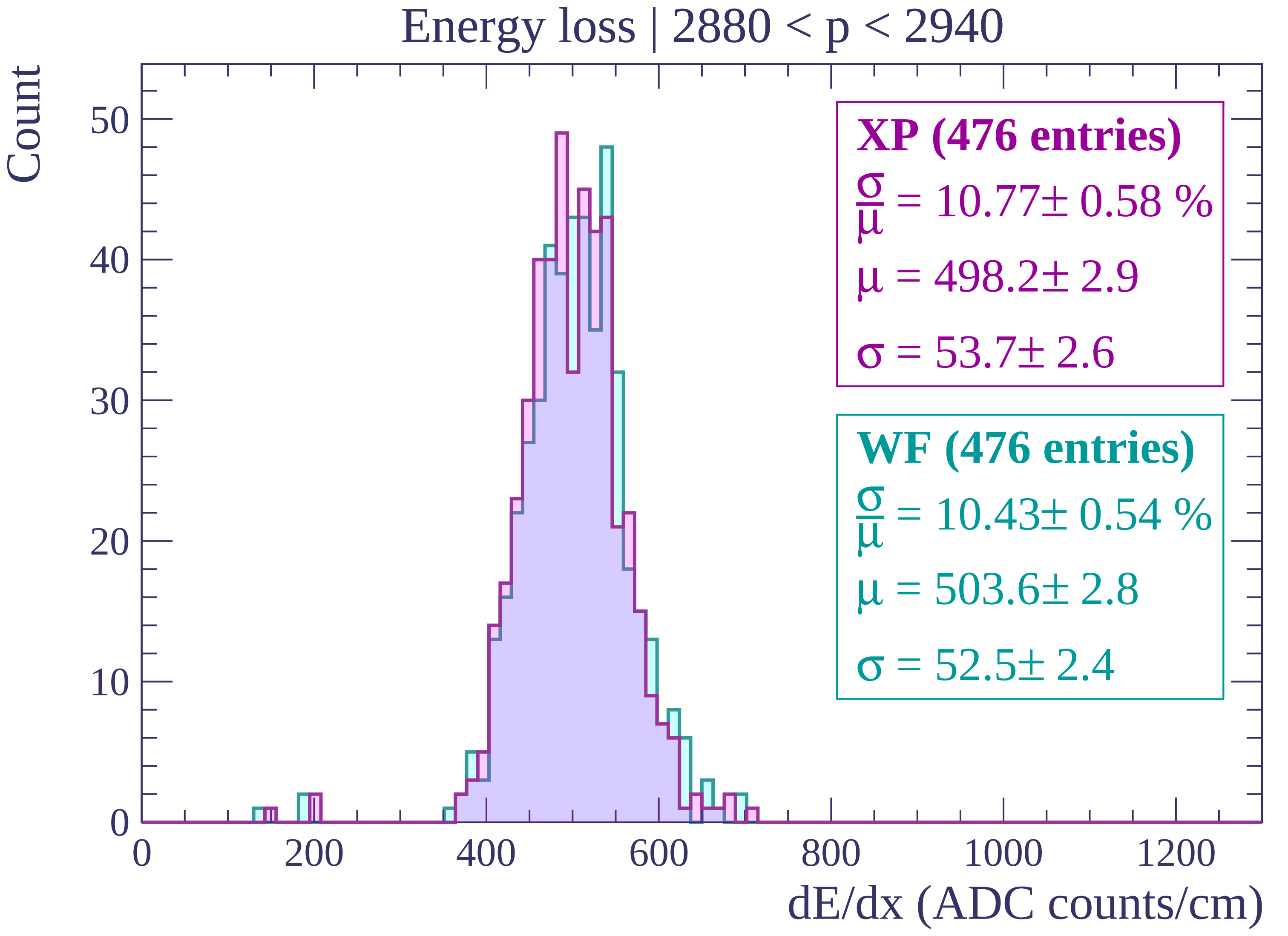
Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
XP (473 entries)
σ/μ = 10.70± 0.71 %
μ = 486.9± 3.0
σ = 52.1± 3.1
WF (473 entries)
σ/μ = 10.41± 0.55 %
μ = 494.3± 2.9
σ = 51.4± 2.4
0
10
20
30
40
50
0
200
400
600
800
1000
1200



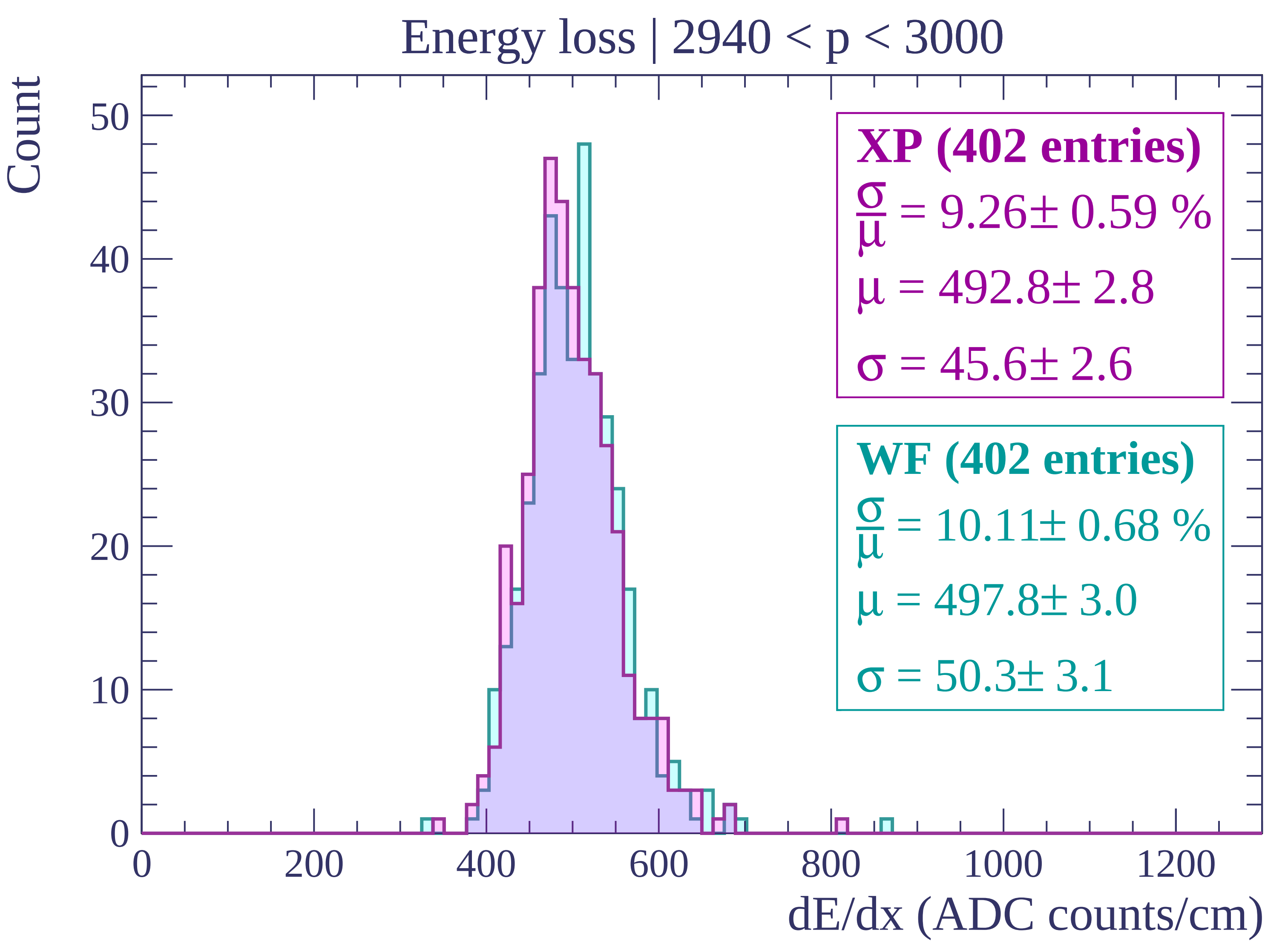
Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (437 entries)
σ/μ = 9.52± 0.56 %
μ = 489.4± 2.7
σ = 46.6± 2.5
WF (437 entries)
σ/μ = 10.18± 0.63 %
μ = 494.3± 2.9
σ = 50.3± 2.8

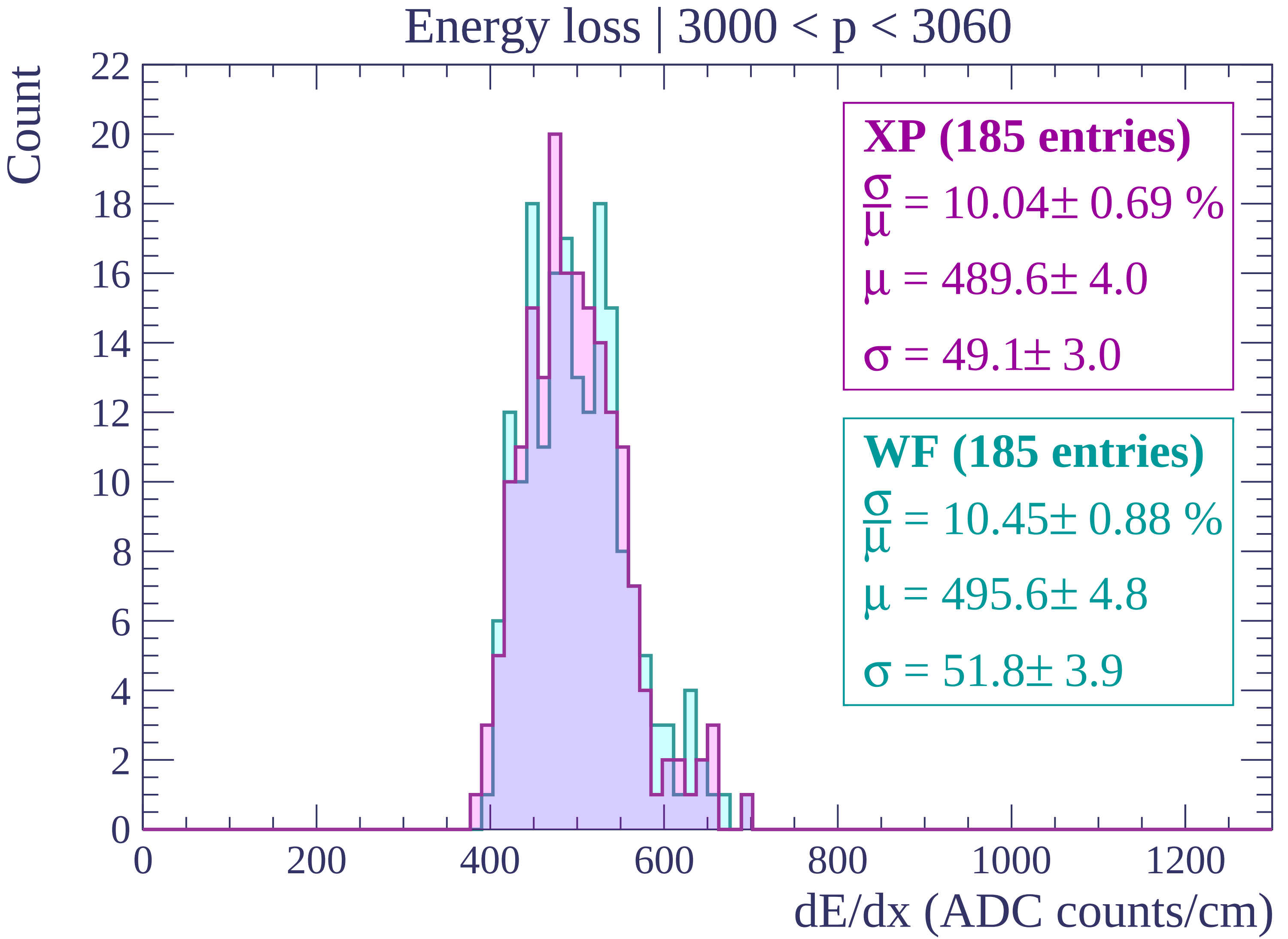

Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP (434 entries)
$\frac{\sigma}{\mu}$ = 9.04± 0.54 %
μ = 489.7± 2.6
σ = 44.3± 2.4
WF (434 entries)
$\frac{\sigma}{\mu}$ = 8.99± 0.55 %
μ = 496.0± 2.6
σ = 44.6± 2.5


Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (385 entries)
$\frac{\sigma}{\mu}$ = 9.54± 0.61 %
μ = 489.4± 2.8
σ = 46.7± 2.7
WF (385 entries)
$\frac{\sigma}{\mu}$ = 8.84± 0.67 %
μ = 496.8± 2.8
σ = 43.9± 3.1

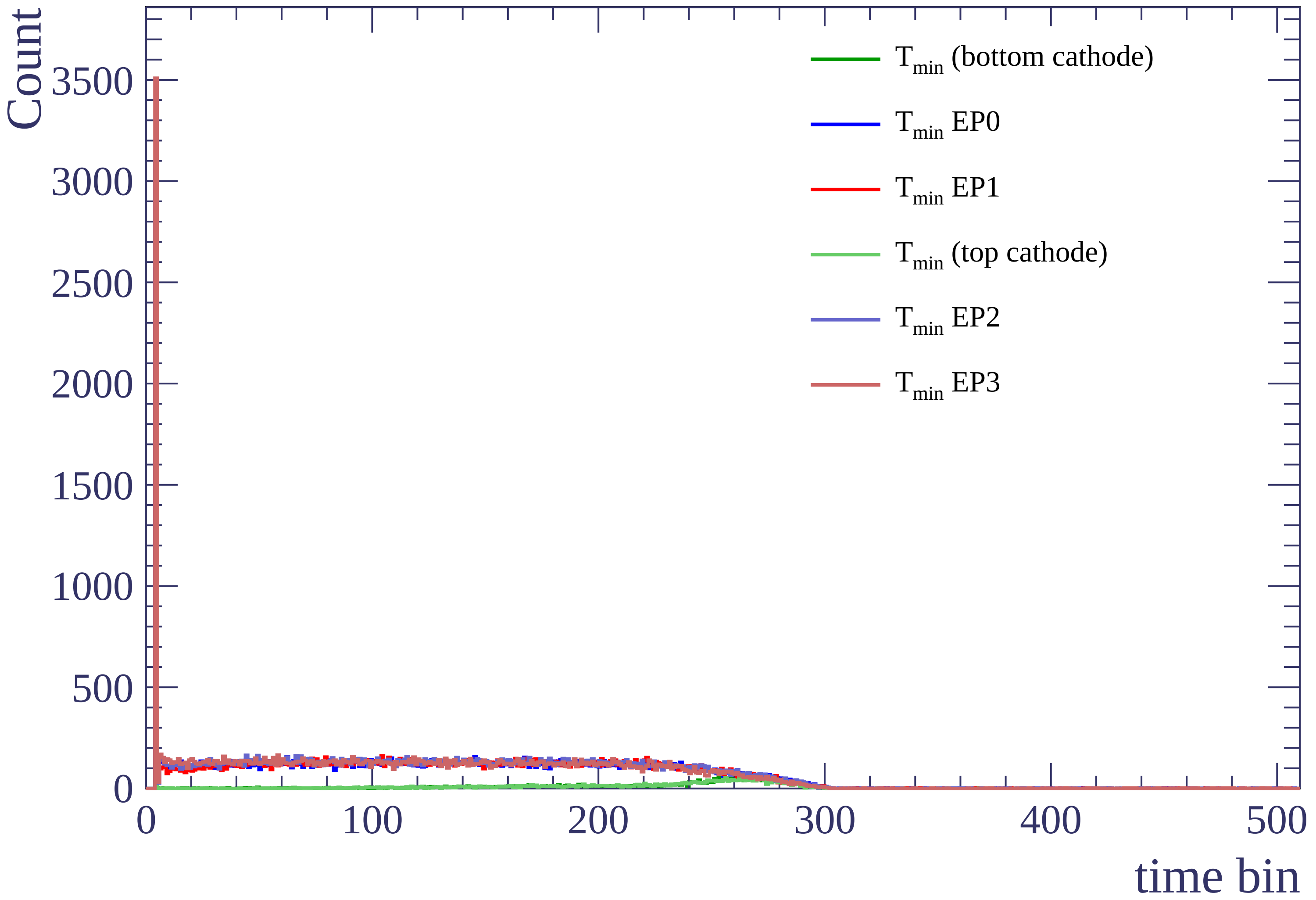


Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
XP (476 entries)
σ/μ = 10.77± 0.58 %
μ = 498.2± 2.9
σ = 53.7± 2.6
WF (476 entries)
σ/μ = 10.43± 0.54 %
μ = 503.6± 2.8
σ = 52.5± 2.4
0
10
20
30
40
50
0
200
400
600
800
1000
1200

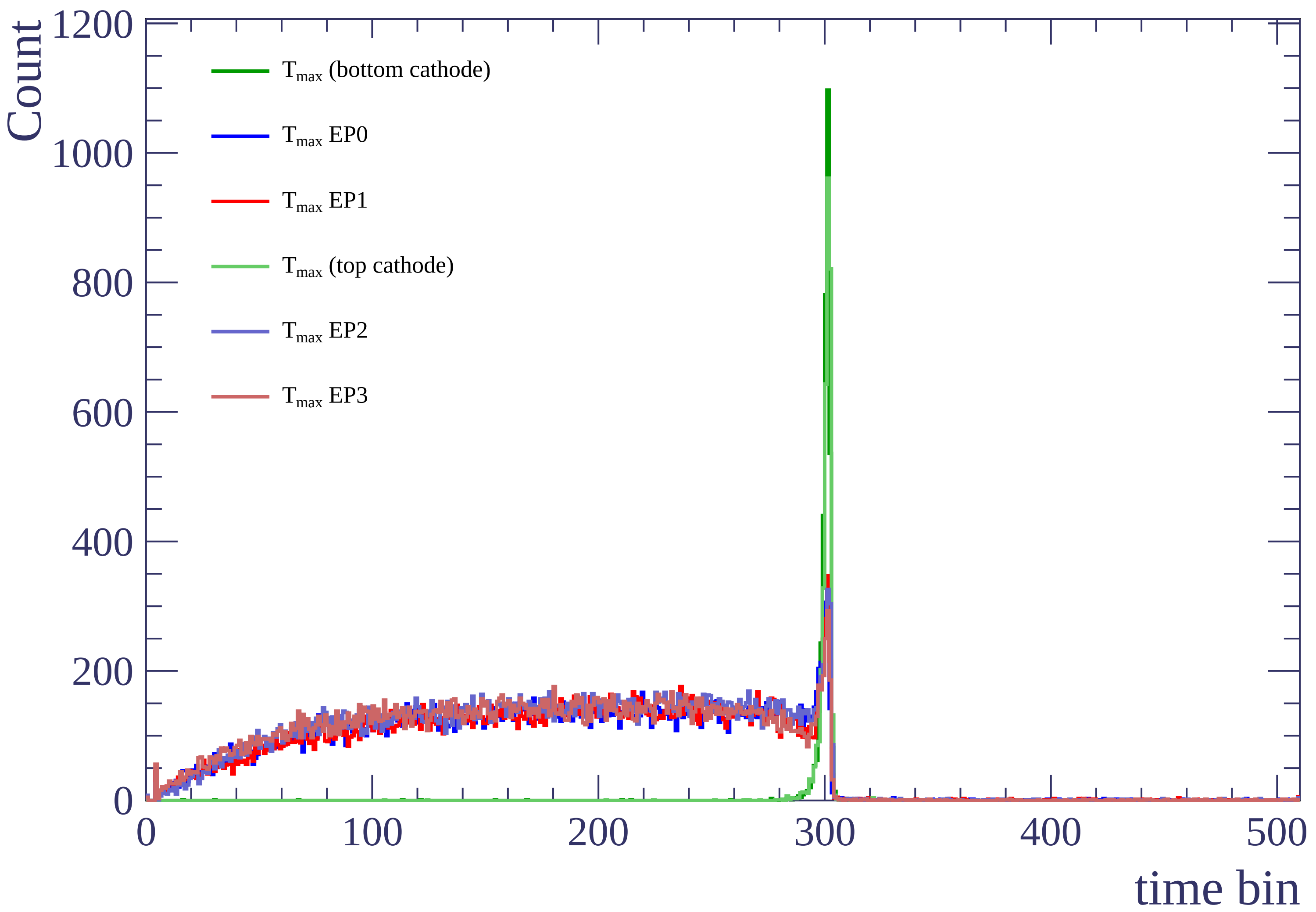

Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (402 entries)
σ/μ = 9.26± 0.59 %
μ = 492.8± 2.8
σ = 45.6± 2.6
WF (402 entries)
σ/μ = 10.11± 0.68 %
μ = 497.8± 3.0
σ = 50.3± 3.1



Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
XP (185 entries)
σ/μ = 10.04± 0.69 %
μ = 489.6± 4.0
σ = 49.1± 3.0
WF (185 entries)
σ/μ = 10.45± 0.88 %
μ = 495.6± 4.8
σ = 51.8± 3.9



Count
time bin
0
100
200
300
400
500
500
1000
1500
2000
2500
3000
3500
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



Count
1200
1000
800
600
400
200
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3